HAAGERUP–IZUMI $\mathbb{Z}_n$: TRANSPARENT F-SYMBOLS, CLASS FIELDS, AND STARK-UNIT ROUTE

Abstract. These notes summarize the current working picture for cyclic Haagerup–Izumi fusion categories in the transparent gauge, with emphasis on (1) reducing the F-symbol system to one core matrix A; (2) the exact algebraic reconstruction for the $\mathbb{Z}_{17}$ Q-system branch; (3) evidence that the generic F-symbol field is a ray class field of $K_p = \mathbb{Q}\left(\sqrt{p^2+4}\right)$; (4) the candidate relation between normalized F-symbols and Stark units; (5) reproducible PARI/GP tests and the remaining analytic construction problem; and (6) exact construction without Haagerup–Izumi seeds, uniform cocycle and sign reductions, and an infinite family of arithmetic inputs over one quadratic field. Some statements below are exact consequences of the transparent-gauge identities or of exact algebraic manipulations. Others are computational conjectures based on reconstruction from published Evans–Gannon numerical seeds and should be independently verified.

Contents

Revision notes 5
1. Fusion rules and notation 9
2. Transparent gauge: all noninvertible F-symbols from one matrix 9
3. Identification with the Izumi / Evans–Gannon A-matrix 9
4. Q-system phase parametrization 10
5. The explicit $\mathbb{Z}_{17}$ setup 10
6. Exact algebraic reconstruction for $\mathbb{Z}_{17}$ 11
7. Class-field hypothesis 11
8. Computational class-field evidence 12
8.1. $p = 7, 13, 17$ 12
9. The $p = 11$ exception 12
10. The $p = 19$ class-group phenomenon 13
11. Stark-unit candidate 14
12. PARI/GP Stark-unit test 15
13. Independent Magma replication 16
14. Higher-priority structural problem: derive all generic entries from one Stark orbit 16
15. Fourier / cocycle route 17
16. Potential Shintani–Faddeev route 17
17. Long-term existence strategy 18
17.1. Strategy A: direct ray-class / Stark formula 18
17.2. Strategy B: analytic Shintani–Faddeev formula 18
17.3. Strategy C: finite field + Hensel lifting 18
18. Concrete next-step checklist 18
18.1. First priority: verify the Stark comparison at $p = 17$ 18
18.2. Second priority: repeat at $p = 7, 13$ 19
18.3. Third priority: $p = 19$ 19
18.4. Fourth priority: $p = 11$ 19
18.5. Fifth priority: recover all $A_{g,h}$ from Stark units 19
18.6. Sixth priority: general symbolic formula 19
19. Main conjectural picture 19

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20. References to keep open 20
21. Completed exact $p = 17$ Stark comparison 20
21.1. Correct ray modulus 20
21.2. Exact HI unit polynomial 21
21.3. Comparison with PARI's Stark orbit 21
22. Completed $p = 11$ conductor analysis 22
22.1. Secondary branch: exact quadratic ray field 22
22.2. Principal branch: the missing prime over 5 23
23. Explicit Shintani–Faddeev parameters for the HI ray classes 23
24. Exact $p = 7$ comparison 24
25. Completed $p = 13$ comparison and the Frobenius-2 law 25
26. The two $p = 19$ branches and the Hilbert class field 26
26.1. The residue-(2) subgroup and the class-group quotient 27
26.2. Reconstruction of the two HI branches 27
26.3. Status of the Stark quotient at $p = 19$ 28
27. Group-ring and Shintani–Faddeev reformulation 28
27.1. Why the operator loses no field information 28
27.2. A sharper analytic candidate 29
28. A diagonal-determination theorem for the full fourth-power core 30
28.1. Correct cyclic phase normalization 30
28.2. Diagonal determination 30
28.3. Correction to the proposed higher-multiplier formula 31
29. Exact construction from Stark polynomials without HI seeds 31
29.1. Completed exact certificates 32
29.2. Removal of numerical-recognition assumptions 32
29.3. Small explicit inputs for the fixed-field family 33
29.4. The earlier class-number-two checkpoint, now completed 33
30. Why the sign problem has only one binary degree of freedom 33
31. One sequence and a Fourier formulation of the quadratic equations 34
32. An unconditional infinite family of arithmetic inputs over one field 35
32.1. Exact ray group and exact unit index 36
32.2. A fixed maximal-order RM point and explicit characteristics 36
33. What an infinitude proof now needs 37
34. One balanced finite beta identity encodes the entire system 38
34.1. Change of variables for the interior cubic equations 38
34.2. Every collision case follows from (Q) 39
34.3. The boundary cubic equations also follow from (Q) 40
34.4. Symmetry and exact verification 40
35. Exact ideal evidence and the next finite orders 40
35.1. Quadratic equations imply cubic equations at orders 3, 5, 7, 9 40
35.2. The order-19 certificate is now complete 41
35.3. The fixed-field experiment at order 199 42
36. Sign descent at composite orders and a prime-power theorem 42
37. The sharpened infinitude target 43
38. All fourth-root lifts collapse to one Kummer root 44
38.1. An explicit inverse to the doubling-root map 44
38.2. Odd-degree norm descent of the root 45
38.3. Nonunit labels need their remaining numerator ideal 45
38.4. A more economical sufficient target for prime powers 46
39. Complete exact existence at the composite order 25 46
40. Prime-power sign selection becomes rational linear algebra 47

40.1. One necessary quadratic equation is linear in new signs 47
40.2. Exact norm compatibility leaves only a global sign 48
41. A weighted-design formulation and an obstruction to Alltop deformation 48
41.1. The Weyl orbit plus one basis is a weighted 2-design 48
41.2. Explicit Alltop solutions occur at the wrong amplitude 49
42. Exact amplitude branches at orders 5 and 7 50
43. A uniform quotient by the cyclic character gauge 51
44. Diagonal quadratics and Galois covariance are not sufficient 52
45. Exact quadratic-to-cubic containment at order 11 53
46. A degree obstruction to fixed-order cyclic hypergeometric formulas 55
47. Unconditional norm elements in every prescribed odd-order ray field 55
47.1. A local–global norm theorem 56
47.2. Physical Galois-covariant phase data for every odd order 56
47.3. What remains after the norm construction 57
48. An explicit integral norm unit for every order congruent to 3 modulo 4 57
48.1. The conductor calculation 58
48.2. Primitive unit choices also exist in the full ray field 58
48.3. The explicit seed itself fails the quadratic equation uniformly 59
49. An exact half-lattice reduction and a new unit-selection experiment 59
49.1. The norm-unit coset is an exact arithmetic object 59
49.2. Exact finite Izumi certificates from short norm units 60
49.3. A precise remaining uniform target 61
50. The cubic equations are a fifth-order condition on one unitary operator 61
50.1. A transfer operator with a universal three-dimensional part 61
50.2. An exact identity for the complete cubic residual 62
50.3. An equivalent polynomial test and the remaining gap 63
51. An explicit analytic candidate at every odd order 63
51.1. The continued-fraction cycle and its phase cancellation 63
51.2. Exact inverse symmetry and the proposed signs 64
51.3. Numerical checks and a failed route to the physical embedding 65
52. Cubic reconstruction without first assuming Stark algebraicity 66
52.1. A uniform residual identity 66
52.2. Conditional categorification and automatic algebraicity 66
53. A uniform convex construction of algebraic one-column solutions 67
53.1. The domain and its unique center 67
53.2. Why the center solves a complete quadratic column 68
53.3. Computations, a counterexample, and the remaining implication 69
54. The entire one-column algebra is finite, reduced, and real 69
54.1. Positive logarithmic weights exclude every nonreal solution 69
54.2. An exact formula at every odd prime 70
54.3. Consequences for the remaining existence problem 71
55. A scalar completion condition and its prime-order field consequence 71
55.1. The condition depends on the phase lift 72
55.2. The scalar equation forces the one-root property at primes 72
55.3. The canonical trace has a conjectural period-40 formula 73
56. The two-value quadratic-character ansatz stops at orders 3 and 11 74
57. Explicit commuting matrices and scalar completion at orders 11 and 13 75
57.1. Multiplication from the hyperplane circuits 75
57.2. An integral matrix model gives a rigorous upper bound 75
57.3. Matching exact lower bounds 76
58. Positive-characteristic certificates and a Frobenius construction 76

58.1. Reconstruction and lifting from a complete finite-field solution 77
58.2. A fixed characteristic works at every odd order 78
58.3. An exact search along the infinite family $n = 3^k$ 79
59. The same hyperbolic monodromy in the arithmetic and S-fold routes 80
60. An exact fifth-order criterion in characteristic 2 81
60.1. A characteristic-free residual identity 81
60.2. The universal rank-one and nilpotent parts 82
60.3. Exact checks and a restriction on an adjoint construction 83
60.4. The consecutive-ratio bound also survives in characteristic 2 83
61. A fixed elliptic curve parametrizes the characteristic-2 phase space 83
61.1. Frobenius, inverse norms, and rational interpolation 84
61.2. A recurrence fitting three orders fails at the fourth 85
62. Every order-13 quadratic phase solution satisfies the cubics 85
63. The norm recursion permutes Frobenius branches 86
63.1. A uniform class-order and Frobenius relation 86
63.2. The saved phases obey the branch-permuting norm laws 88
64. A uniform obstruction to small-field and quotient inflation constructions 88
64.1. No Frobenius-covariant quadratic solution over the four-element field 88
64.2. No inflation constant on nonzero quotient cosets 90
64.3. Exact checks and the next-smallest field 90
65. The first phase forces the full field, and every relative norm has cubic lifts 91
65.1. A single quadratic detects a collapsed top orbit 91
65.2. Exact field degree at every power of three 91
65.3. A uniform cubic lift for every prescribed normalized norm 92
65.4. Exact verification 93
66. Two quadratics, full field degree, and the obstruction to local inflation 93
66.1. Exact completion from two quadratics at the first three orders 93
66.2. A uniform origin-patched inflation identity 94
66.3. The counts do not give a degree-six rational iteration 95
67. Patching the origin does not permit bounded rational ratios 96
68. The first quadratic is linear in all lower Frobenius seeds 97
69. Relative traces and a linear number of quadratic orbits 98
69.1. Pairing each quadratic into a relative trace 98
69.2. Exactly $(n-1)/2$ orbit representatives suffice for Q 99
69.3. An exact Boolean search, with independent acceptance 100
70. Complete characteristic-two quadratic-to-cubic ideals through 13 100
71. The cubics force a unique Frobenius gauge 101
71.1. A convolution identity from the diagonal cubics 101
71.2. Unique normalization of the phases 102
71.3. An explicit field bound and finite etaleness 102
71.4. The remaining nonemptiness problem 103
72. General doubling-orbit searches and complete finite counts through 31 103
72.1. Arbitrary odd orders: paired orbits and norm orbits 103
72.2. Completed searches and independent verification 104
72.3. Completion of the full quadratic ideal at order 17 104
72.4. The full quadratic ideal also completes at order 15 105
73. The count must include quadratic overorders 105
73.1. A positive arithmetic count, proved independently of HI 105
73.2. Exact finite comparison and the corrected prediction 106
74. The scalar action, a split quotient, and a stronger finite ray comparison 107
74.1. The quotient by scalar reindexing is split over $\mathbb{F}_4$ 107

74.2. The arithmetic action and its exact finite match 108
75. A fixed-period lens product for the uniform analytic candidate 108
75.1. The product and its origin value 109
75.2. An explicit sign and the remaining integral problem 110
76. The Shintani product is the square of one double sine 111
76.1. A product identity away from the RM specialization 111
76.2. The formula with one factor per phase 111
76.3. A meromorphic interpolation and an exact integer jump 112
77. The full lens product factors, and its contour boundary terms 112
77.1. Two factors at every continuous argument 113
77.2. Integer values and derivatives have different shift defects 113
77.3. A growth obstruction for the direct vertical integral 114
77.4. Repairing the beta balance with four explicit shifts 114
78. Rational monodromy of the quadratic and cubic kernels 116
78.1. A rational function of degree r 116
78.2. A nonsingular matrix of size two or three 116
79. Canonical balanced lifts remove all signs from the cross-free targets 117
79.1. The canonical-lift lemma 117
79.2. Exact cancellation of the interpolation signs 118
79.3. Two explicit unweighted identities suffice 119
79.4. The exact scope of a period-shift recurrence 119
79.5. Available mathematical software 120
Files from this exploration 120

Revision notes

Canonical finite-sum targets (2026-09-06). Section 79 proves that every cross-free quadratic or cubic has balanced integer lifts for which all summands agree with their canonical cyclic samples, with no origin corrections. The cubic Gaussian signs cancel exactly. The full existence problem for the analytic candidate therefore reduces to two unweighted finite identities involving the ordinary double sine, with no remaining sign choices or collision equations. Those two identities remain unproved uniformly; infinitude is not established.

Rational shift formula (2026-09-06). Section 78 expresses the n-shift of the quadratic and cubic double-sine kernels using rational functions of degree two and three, respectively. Their divided differences are controlled by nonsingular matrices of sizes two and three whenever the external residues are cross-free. These sizes are independent of n. The resulting weighted identities do not yet give the required unweighted finite sums; infinitude remains unproved.

One double sine per phase (2026-09-06). Sections 75–77 prove that the analytic candidate is exactly $W_a = -(-1)^{a(a+n)/2} d^{-1/2} \operatorname{Sin}_2(1 + da/(d+1); d, 1)$. This removes the square root of a special-function value. Its meromorphic interpolation has a single exceptional jump on integer translates. The full lens product also factors into two ordinary double sines away from the sampling line, with an explicit origin residue. Four period shifts give a correctly balanced, convergent beta integral containing the desired six sine factors, with explicit sine denominators and the same normalization as the finite cubic target. Converting that integral to the finite sum, including its origin and derivative corrections, remains unresolved. Infinitude is not proved.

Complete finite counts and overorders (2026-09-06). Sections 72–73 give exhaustive characteristic-two phase classifications at every odd order through 31, with new exact workspace certificates at 15, 21, 23, and 31. Every candidate orbit passes all Q and C independently. Magma also completes the full characteristic-two Q-to-C implication at 17. The counts agree with the sum of ray-class degrees over all quadratic orders containing $\mathbb{Z}[d]$, including the extra contributions at

11 and 29. This corrects the nonfundamental-discriminant prediction: at 3^8 the proposed count is $b_8 = 480$, not 384. The uniform count or any sufficient positive lower bound remains unproved.

Forced Frobenius gauge (2026-09-06). Section 71 proves uniformly that Q plus the diagonal cubics forces $A_{2a,2b} = A^4_{a,b}$, and that every full characteristic-two phase solution has a unique character normalization with $W^4_a = W_{2a}$. The full phase scheme is finite etale. At $n = 3^k$, every solution has matrix field exactly $\mathbb{F}_{2^{4n/3}}$; its canonical coordinate algebra is a product of b_k copies of this field. Earlier searches now give the complete phase counts $b_1 = 1, b_2 = 2, b_3 = 3$. Proving $b_k > 0$ for unbounded k, and hence infinitude by this route, remains open.

Trace reduction and characteristic-two completion (2026-09-06). Section 69 proves that at every order $n = 3^k$, only $(n-1)/2$ quadratic orbits need checking and each quadratic is a relative trace with $(n-1)/2$ summands. Exact off-solution checks reach order 81. Section 70 proves full $Q \Rightarrow C$ over characteristic two at all odd orders through 13 by ideal membership, without imposing Frobenius covariance or selecting saved solutions. The smaller SAT searches have not produced an order-81 certificate. Uniform nonemptiness and infinitude remain unproved.

Linear elimination of lower phases (2026-09-06). Section 68 proves that, at every $n = 3^k$, fixing the top Frobenius seed turns the first quadratic into an affine binary linear system with $2n/3 - 2$ unknowns and $2n/3$ target coordinates. For each of the six saved certificates, this system together with the lower inverse norms recovers exactly the saved lower phases. The first-Q count 24 at order 9 is independently reproduced by linear elimination. At order 81 the same construction has 52 unknowns; sampled top seeds yield affine solutions but no norm-compatible full candidate. This supplies a uniform elimination step, not the missing infinitude proof.

Patched rational formulas (2026-09-06). Section 67 proves that even after omitting both exceptional consecutive ratios, a characteristic-two Frobenius-covariant quadratic solution requires rational ratio degree at least $(n-3)/4$. Thus a fixed number of rational cyclic factors cannot give an infinite family by separately patching the origin. Exact interpolation gives minimum degrees 4 at order 9 and 12 at order 27. Section 66.3 also excludes every stationary cubic rational recursion on a single known branch. These are restrictions on proposed constructions; infinitude remains unproved.

Two-quadratic reduction and false inflations (2026-09-06). Section 66 proves, by exhaustive finite-field searches, that the first two phase quadratics plus the full first-phase field degree imply every cubic at orders 3, 9 and 27. At 27 the two quadratics have 72 solutions: 54 of full degree, all valid, and 18 of smaller degree. A uniform inflation identity explains 12 of the false solutions and produces false two-quadratic candidates at infinitely many larger orders. The counts $2, 12, 72$ do not arise from a single degree-six rational recursion over $\mathbb{F}_4$: all 20 possible maps are rejected exactly. Uniform completion and nonemptiness remain unproved.

Forced field degree and cubic norm lifts (2026-09-06). Section 65 proves that at every $n = 3^k$, a Frobenius-covariant quadratic solution has W_1, and already R_2, of degree exactly $2n/3$ over $\mathbb{F}_4$. One designated quadratic excludes every smaller field. A second uniform theorem gives an explicit cubic lift of any prescribed normalized relative norm and counts its irreducible trace choices. Thus the proposed coupled norm towers can always be constructed with the required field growth; making their additive equations vanish is still unresolved.

Uniform small-field obstruction (2026-09-06). Section 64 proves that no nontrivial odd abelian group has a Frobenius-covariant phase solution of all quadratics over $\mathbb{F}_4$. The proof turns a quadratic trace identity into an impossible additive closure. It also excludes, over every characteristic-two extension, constructions whose phases are constant on the nonzero cosets of a proper nontrivial subgroup. The next-smallest-field search at order 43 exhausts 125 assignments over $\mathbb{F}_{16}$ and finds none. These statements close two construction routes; they do not establish infinitude.

Twisted norm branches (2026-09-06). Section 63 proves that at $n = 3^k$ the conjugate prime above 3 has class order exactly k in the quadratic order of discriminant n^2+4, and its k-th ray-class power is Frobenius at (2). The saved order-9 and order-27 phases satisfy the corresponding norm laws across different branches, including the required Frobenius twist. This corrects the failed same-branch norm prescription. At order 81 the arithmetic gives two cycles of four branches; existence of HI solutions on those cycles is not proved.

Full quadratic completion at 13 (2026-09-06). Section 62 proves $(Q) \Rightarrow (C)$ for every order-13 phase solution, using a modular upper bound of 12 on the full row quotient and a matching exact cubic orbit. No scalar H condition is imposed. The row algebra is one degree-12 field over $\mathbb{Q}(\delta)$; the phase algebra has dimension 156. This extends the finite implication theorem, without proving it in unbounded order.

Elliptic parametrization (2026-09-06). Section 61 realizes every norm-normalized, Frobenius-covariant phase sequence at $n = 3^k$ using a rational function on the single curve $y^2 + y = x^3 + \omega$ over $\mathbb{F}_4$. The curve's Frobenius is exactly doubling. A concrete rational-function recurrence supplies complete certificates at 3, 9 and 27, but exhaustive testing of its three order-81 candidates rejects all of them at the first quadratic. The parametrization is uniform; the recurrence is not a construction of an infinite HI family.

Characteristic-2 pentagon criterion (2026-09-06). Section 60 proves that, for any full-Q phase candidate, an explicit rank-three correction $\mathcal{P} = R(I + D)$ satisfies $\mathcal{P}^5 = I$ exactly when every cubic equation holds. This uses polynomial identities, without positivity. At a full characteristic-2 solution the uncorrected transfer operator has order exactly 30, with a prescribed rank-two square-zero part in its fifth-power defect. Exact checks cover both order-9 branches, and a spectral obstruction excludes one branch from any adjoint-matrix construction. The corrected pentagon still needs a construction for unbounded orders.

Finite-field construction (2026-09-06). Section 58 proves that a complete normalized cubic certificate in characteristic $\ell \nmid n(n^2+4)$ implies complex categorification by algebraic reconstruction and categorical lifting. Characteristic 2 works at every odd order. In that characteristic, inverse norms and Frobenius covariance imply the scalar H equation uniformly. A search with no HI or Stark seeds finds complete certificates at 3, 9 and 27; all three order-27 Frobenius orbits pass every cubic. The six independent JSON certificates are saved. Nonemptiness at unbounded orders, and hence infinitude, remains unproved.

Scalar completion (2026-09-06). Section 55 introduces one scalar equation involving the phase sum and $\operatorname{Tr}\mathcal{R}$. Exact finite-algebra certificates in Sections 55 and 57 show that it completes one quadratic column to the full system at orders 3, 5, 7, 11, and 13. At every odd prime, a real phase lift satisfying the scalar equation already lies in its row field; this implication is proved uniformly. The canonical phase sum has a numerical period-40 formula, and 508 sign-chamber tests show no disagreement between scalar acceptance and full-Q acceptance. Neither scalar completion in unbounded order nor the canonical scalar identity is proved.

Uniform one-column classification (2026-09-06). Section 54 proves that the entire one-column algebra is finite and reduced at every odd order, with all its points real above the negative dimension embedding. At each odd prime p, its exact length modulo character gauge is $\binom{p-1}{(p-1)/2}$; composite counts include 58 at order 9 and 2492 at order 15. Consequently the full quadratic and cubic schemes in the phase ansatz are also finite and reduced, but their nonemptiness is still unproved uniformly. This strengthens the algebraic setting for the convex construction without asserting infinitude.

Convex construction (2026-09-06). Section 53 proves that a strictly concave logarithmic optimization problem has a unique interior maximizer for every odd n, and that this maximizer gives real algebraic phase data satisfying one complete quadratic column. The prescribed chamber recovers the signs of Section 51. This construction numerically reproduces the double-sine candidates and passes all quadratics at orders 51 and 101 as well; only sampled cubics were checked

at those two larger orders. An exact counterexample shows why one column by itself is insufficient. The remaining conjecture is that the uniquely selected maximizer satisfies the full cubic system. **Infinitude remains unproved.**

Analytic continuation (2026-09-06). Sections 51–52 give an explicit product of n double sines and a fixed quadratic sign formula, with no fitted roots or signs. Its real, negative-dimension matrices pass numerical Izumi checks through order 31. The continued-fraction formula, cancellation of its modular phases, and inverse-pair relation are proved uniformly. A separate exact residual identity shows that the cubic equations imply the quadratic and quartic reconstruction equations in the normalized $\omega = 1$ setting. Consequently a proof of the proposed analytic cubic identity would already produce nonunitary fusion categories; algebraicity would then follow from rigidity, without assuming Stark algebraicity first. **Neither this analytic identity nor infinitude is proved.** Unitary existence additionally requires the appropriate positive-embedding condition.

Further continuation (2026-09-06). Sections 41–50 give a weighted-design formulation, uniform isolation of the flat Alltop solutions, exact HI amplitude components at orders 5 and 7, and a uniform quotient of the phase equations by the cyclic character gauge. An exact physical counterexample shows that diagonal quadratics plus Galois covariance are insufficient; its arithmetic field fails both the one-root and prescribed conductor conditions. The quadratic-to-cubic ideal theorem now also holds at $n = 11$, certified by an independent homogeneous Hilbert comparison. A further uniform degree bound excludes bounded-order nonsingular cyclic hypergeometric phase formulas. New norm theorems give ray-field norm elements at every odd order and explicit integral norm units whenever $n \equiv 3 \pmod 4$, including infinitely many primes. An exact half-lattice reduction produces a second construction route, with complete finite Izumi checks at 3, 5, 7, 11, 13, 17, 29, and 199. A uniform operator theorem identifies the full cubic residual norm with a fifth-order defect of a unitary operator. Infinitude remains unproved: physical quadratic solutions and the cubic implication still need a construction or proof for unbounded orders.

Further continuation (2026-09-05). Sections 34–40 prove that the full phase-ansatz Izumi system is equivalent to one balanced finite beta identity, including its collision corrections. They extend sign descent to composite orders and show that its stabilizer obstruction always vanishes at odd prime powers under the stated root/phase hypotheses. Complete exact constructions now also pass at $n = 19, 25, 199$; the latter belongs to the fixed-$\mathbb{Q}(\sqrt{5})$ Lucas family. The order-19 polynomial recognition is discharged as well. Saved coefficient data and independent verifiers accompany all three cases. The prime-power construction now uses one Kummer root, twisted norm descent, and a rational linear system for the signs. The quadratic-to-cubic ideal containment is verified at orders 3, 5, 7, and 9. Infinitude remains unproved; Sections 37–40 specify what a uniform proof still requires.

Existence continuation (2026-09-04). Sections 28–33 go beyond field comparisons. A cocycle argument determines every generic fourth power from the labeled doubling orbit. The new `hi_stark_seedless.gp` constructs complete matrices directly from PARI Stark polynomials, with no numerical HI seeds, and verifies every Izumi equation exactly for $n = 7, 11, 13, 17, 29$. It also identifies the previously recognized $n = 13$ polynomial and all three $n = 17$ polynomials with the certified matrices. Their numerical-recognition qualifications in the earlier historical sections are therefore removed by this continuation. Infinitude is **not** proved: Section 32 supplies an unconditional infinite family of arithmetic inputs over the fixed field $\mathbb{Q}(\sqrt{5})$, and Section 33 specifies the uniform identities still needed.

Update (2026-09-04). The PARI tests have now been completed for $p = 7, 11, 13, 17, 19$. For $p = 7$, and conditional on the initial numerical recognition for $p = 13, 17$, the normalized HI square generates the same ray field as PARI's distinguished Stark unit and is the exact group-ring transform $s^{2-\mathrm{Frob}_{(2)}}$, rather than the same distinguished generator. At $p = 19$, the ray group is $C_{18} \times C_2$, and the two published HI branches are the two Hilbert-class conjugate halves of its full degree-36 orbit. See Sections 21–27 and the reproducible GP scripts listed at the end.

1. Fusion rules and notation

Let

$$G = \mathbb{Z}_N, \qquad N \text{ odd},$$

with invertible simple objects α^a, $a \in \mathbb{Z}_N$, and noninvertibles

$$\rho_a := \alpha^a \rho.$$

The fusion rules are

$$\alpha^a \alpha^b = \alpha^{a+b},$$
$$\alpha^a \rho_b = \rho_{a+b}, \qquad \rho_a \alpha^b = \rho_{a-b},$$

and

$$\rho_a \rho_b = \alpha^{a-b} + \sum_{c \in \mathbb{Z}_N} \rho_c.$$

The Frobenius–Perron dimension of ρ is

$$d = \frac{N + \sqrt{N^2+4}}{2}, \qquad d^2 = Nd + 1.$$

For the prime-order discussion below I write $N = p$.

2. Transparent gauge: all noninvertible F-symbols from one matrix

The transparent-gauge structure strongly suggests that all F-symbols are determined by:

- universal values $1, d^{-1/2}, d^{-1}$ when invertible channels occur;
- one $N \times N$ core matrix

$$A_{a,b} := \left(F^{\rho,\rho,\rho}_{\rho_a}\right)_{\rho,\rho_b}.$$

A useful reconstruction formula obtained from the transparent translation identities is

$$\boxed{\left(F^{\rho_{a_1},\rho_{a_2},\rho_{a_3}}_{\rho_{a_4}}\right)_{\rho_{a_5},\rho_{a_6}} = A_{a_4+a_2-a_1-a_3,\, a_5+a_6-a_1-a_3}}$$

with all indices modulo N.

In particular,

$$\boxed{\left(F^{\rho,\rho,\rho}_{\rho_a}\right)_{\rho_e,\rho_f} = A_{a,e+f}.}$$

Thus for fixed a, the noninvertible block of

$$F^{\rho,\rho,\rho}_{\rho_a}$$

is anti-circulant.

This is the main structural reason the enormous pentagon system should reduce to a much smaller algebraic problem.

3. Identification with the Izumi / Evans–Gannon A-matrix

For known examples, the transparent F-core appears to coincide with the Izumi/Evans–Gannon structure matrix $A_{g,h}$.

The standard symmetries include

$$A_{g,h} = A_{-h,g-h} = A_{h-g,-g}.$$

For a unitary solution,

$$A_{h,g} = \overline{A_{g,h}}.$$

The maps

$$R(g,h) = (h-g,-g), \qquad S(g,h) = (h,g)$$

generate

$$S_3 \simeq W(A_2).$$

For real solutions, A is constant on the S_3-orbits in $\mathbb{Z}_N^2$. For complex unitary solutions, reflections can act by complex conjugation.

For $3 \nmid N$, the number of S_3-orbits in $\mathbb{Z}_N^2$ is

$$\frac{(N+1)(N+2)}{6}.$$

4. Q-system phase parametrization

For the unitary Q-system branch, the generic entries can be written in terms of phases j_g. Set

$$z_g = e^{ij_g}.$$

For generic g, h,

$$A_{g,h} = \frac{\sqrt{d}}{d-1} e^{i(j_h - j_g - j_{h-g})} = \frac{\sqrt{d}}{d-1} \frac{z_h}{z_g z_{h-g}}.$$

The exceptional values are

$$A_{0,0} = 1 - \frac{1}{d-1},$$

and

$$A_{g,0} = A_{0,g} = A_{g,g} = -\frac{1}{d-1} \qquad (g \neq 0).$$

This phase form should be viewed as a discrete multiplicative second derivative of the one-dimensional sequence z_g.

A major possible route to a uniform construction is to Fourier-transform the Izumi equations in the z_g variables.

5. The explicit $\mathbb{Z}_{17}$ setup

For $p = 17$,

$$d = \delta = \frac{17 + \sqrt{293}}{2}.$$

Define

$$B_{g,h} := (\delta - 1) A_{g,h}.$$

Then

$$B_{0,0} = \delta - 2,$$

$$B_{g,0} = B_{0,g} = B_{g,g} = -1 \qquad (g \neq 0),$$

and for generic entries

$$|B_{g,h}| = \sqrt{\delta}.$$

The ratios $t = h/g \in \mathbb{F}_{17} \setminus \{0, 1\}$ split into three S_3-orbits:

$$\mathcal{O}_2 = \{2, 9, 16\},$$

$$\mathcal{O}_3 = \{3, 6, 8, 10, 12, 15\},$$

$$\mathcal{O}_4 = \{4, 5, 7, 11, 13, 14\}.$$

Therefore the 289 core positions split into only five algebraic types:

- $A_{0,0}$;
- the common axis/diagonal value;
- generic type $\mathcal{O}_2$;
- generic type $\mathcal{O}_3$;
- generic type $\mathcal{O}_4$.

6. Exact algebraic reconstruction for $\mathbb{Z}_{17}$

Starting from the published numerical Evans–Gannon phase seed and refining it to high precision, one obtains degree-16 orbit polynomials over

$$K_{17} = \mathbb{Q}(\delta) = \mathbb{Q}(\sqrt{293})$$

for the scaled generic entries B.

For the three ratio classes there are irreducible degree-16 polynomials

$$P_2, \quad P_3, \quad P_4 \in K_{17}[X].$$

The corresponding actual A-entries have minimal polynomials over K_{17}

$$Q_t(X) = \frac{P_t((\delta-1)X)}{(\delta-1)^{16}}.$$

After eliminating δ, the generic A-entries have irreducible degree-32 polynomials over $\mathbb{Q}$.

The special values satisfy:

$$A_{g,0} = A_{0,g} = A_{g,g} \quad \Longrightarrow \quad 17X^2 - 15X - 1 = 0,$$

and

$$A_{0,0} \quad \Longrightarrow \quad 17X^2 - 49X + 31 = 0.$$

The exact Sage data from the previous reconstruction is in:
`z17_haagerup_izumi_minpolys.sage`
This includes:

- the three P_t over K_{17};
- the three degree-32 rational polynomials;
- irreducibility checks;
- the special transparent values.

7. Class-field hypothesis

For prime p, define

$$K_p = \mathbb{Q}(\delta), \qquad \delta = \frac{p + \sqrt{p^2+4}}{2}.$$

Since

$$N_{K_p/\mathbb{Q}}(\delta) = -1,$$

one has

$$N(\delta - 1) = -p, \qquad N(\delta + 1) = p.$$

Define the degree-one prime

$$\mathfrak{p}_+ = (\delta - 1)$$

above p.

The natural narrow ray modulus suggested by the physical embedding is

$$\boxed{\mathfrak{m}_p = \mathfrak{p}_+ \infty_+,}$$

where ∞_+ is the real place with

$$\delta > 0.$$

If:

(1) $h(K_p) = 1$;
(2) δ is a fundamental unit;

then the ray class group should have order

$$p-1$$

and in favorable cases

$$\mathrm{Cl}_{\mathfrak{m}_p}(K_p) \simeq \mathbb{F}_p^\times \simeq C_{p-1}.$$

This matches the observed generic entry-field degrees for $p = 7, 13, 17$.

8. Computational class-field evidence

8.1. $p = 7, 13, 17$**.** For the principal branch, the reconstructed generic entry field has degree

$$p-1$$

over K_p.

A polynomial transformation implementing

$$u \mapsto gu$$

for a primitive root g mod p was reconstructed and checked algebraically to permute the roots.

This indicates

$$\mathrm{Gal}(L_p/K_p) \simeq C_{p-1} \simeq \mathbb{F}_p^\times.$$

A Frobenius / Artin comparison was then performed at many split primes, using the expected ray character arising from

$$\mathfrak{m}_p = (\delta - 1)\infty_+.$$

The observed Frobenius permutations matched the predicted ray-class action in all tested cases for $p = 7, 13, 17$.

The polynomial discriminants also fit the ray-class prediction after dividing by perfect-square index factors.

Expected relative field discriminant:

$$\boxed{\mathfrak{D}_{L_p/K_p} = \mathfrak{p}_+^{p-2}.}$$

This is consistent with a conductor-$\mathfrak{p}_+$ abelian extension.

Status. The ray groups, conductors, discriminants, and field identifications have now been reproduced with exact PARI class-field routines for $p = 7, 13, 17$. The $p = 7$ HI input is exact; the $p = 13, 17$ conclusions remain conditional only on their numerical-to-exact HI polynomial reconstructions. Independent Magma replication would still be useful.

9. The $p = 11$ exception

Here

$$K_{11} = \mathbb{Q}(\sqrt{5}).$$

Let

$$\epsilon = \frac{1+\sqrt{5}}{2}.$$

Then

$$\delta = \frac{11+5\sqrt{5}}{2} = \epsilon^5.$$

Thus δ is **not** the fundamental unit.

This changes the ray-class calculation.

The simple modulus

$$(\delta - 1)\infty_+$$

has ray class group of order 2, not 10.

Evans–Gannon have two $\mathbb{Z}_{11}$ Q-system branches.

For the second branch, the apparent degree-10 orbit collapses to five copies of one quadratic polynomial over K_{11}:

$$Q(X) = X^2 + \frac{5+3\sqrt{5}}{2}X + \frac{11+5\sqrt{5}}{2}.$$

Its discriminant is

$$\operatorname{disc} Q = -(\delta - 1).$$

This looks exactly like the quadratic ray field of conductor

$$(\delta - 1)\infty_+.$$

The principal degree-10 $p = 11$ branch is arithmetically different. The continuation in Section 22.2 identifies the conductor of its reconstructed unit field as

$$(\delta - 1)\mathfrak{q}_5^2\infty_+ = 5(\delta - 1)\infty_+,$$

where $\mathfrak{q}_5$ is the ramified prime above 5.

Important lesson. The correct global statement is probably **not** "every generic HI entry field is always the simple $(\delta - 1)\infty_+$ ray class field."

Instead, HI branches seem naturally organized by ray class arithmetic, with unit-index and class-group effects causing exceptional splitting and alternate conductors.

10. The $p = 19$ class-group phenomenon

For

$$K_{19} = \mathbb{Q}(\sqrt{365}),$$

the class group is exactly C_2, and $19 - \delta = -\delta^{-1}$ is a fundamental unit.

The simple ray modulus

$$(\delta - 1)\infty_+$$

has ray class group

$$\mathrm{Cl}_{(\delta-1)\infty_+}(K_{19}) \simeq C_{18} \times C_2,$$

of order $2(19 - 1) = 36$.

Evans–Gannon give two $\mathbb{Z}_{19}$ Q-system branches.

For either branch separately, the 18-element set

$$\{B_{u,2u} : u \in \mathbb{F}_{19}^{\times}\}$$

does not form a degree-18 polynomial over K_{19}.

However, combining the 18 values from **both branches** gives an irreducible degree-36 polynomial over K_{19}.

Numerical reconstruction followed by exact conductor and Hilbert-class calculations proves, conditional only on recognition of the phase polynomial, that

the two $p = 19$ HI branches are two halves of one ray-class Galois orbit,

with the nontrivial ideal class exchanging the branches. Section 26 gives the exact field data and isolates the remaining numerical recognition step; an independent Magma replication would still be useful.

11. Stark-unit candidate

This is the most promising new observation.

For $p = 17$, define

$$U_{g,h} = \frac{B_{g,h}}{\sqrt{\delta}}.$$

For generic entries,

$$|U_{g,h}| = 1$$

under the physical embedding.

On the $t = 2$ orbit, define

$$\boxed{\varepsilon_u = -U_{u,2u}^2 = -\frac{B_{u,2u}^2}{\delta}.}$$

Observed properties of ε_u:

(1) ε_u is an algebraic unit.
(2) It has degree 16 over K_{17}.
(3) It generates the same field as $B_{u,2u}$.
(4) Its minimal polynomial over K_{17} is reciprocal.
(5) One has
$$\varepsilon_{-u} = \varepsilon_u^{-1}.$$
(6) The Galois action is
$$\sigma_a(\varepsilon_u) = \varepsilon_{au}.$$
(7) Under the conjugate real embedding of K_{17}, all 16 Galois conjugates of ε_1 are positive real numbers.

These are exactly the qualitative features expected from a rank-one Stark / Shintani unit orbit for the ray extension

$$K_{17}^{(\delta-1)\infty_+}/K_{17}.$$

The initial working conjecture was:

$$\boxed{-\frac{B_{u,2u}^2}{\delta} \quad \text{is a Stark unit, or a fixed integral power of a canonical Stark unit.}}$$

Equivalently,

$$\boxed{U_{u,2u} = \pm i\sqrt{\varepsilon_u}.}$$

The appearance of the square root is notable because recent SIC-POVM constructions from real quadratic ray class fields also express unit-modulus overlap phases in terms of square roots / powers of Stark units.

The exact comparison in Section 21 refines this statement. The HI unit is not itself a root of PARI's Stark polynomial, nor a pure power orbit of such a root. It does, however, lie in the same ray field and satisfies, after compatible choices of roots and a cyclic Galois generator σ,

$$\boxed{\varepsilon_{\mathrm{HI}}^2 = \frac{s^2}{\sigma^2(s)},}$$

where s is a root of PARI's `bnrstarkunit` polynomial. Thus the robust conjecture is the broader one: normalized HI phases are roots of short products or quotients of Stark conjugates.

12. PARI/GP Stark-unit test

This test was completed with PARI/GP 2.17.4; full results are in Section 21.
For $p = 17$, construct

$$K = \mathbb{Q}(\delta), \qquad \delta^2 - 17\delta - 1 = 0.$$

Then construct the ray class field for

$$\mathfrak{m} = (\delta - 1)\infty_+.$$

Schematic GP:

```
bnf = bnfinit(y^2 - 17*y - 1, 1);

\\ inspect real embeddings first
polrootsreal(y^2 - 17*y - 1)

\\ prime ideal generated by delta - 1
pp = idealhnf(bnf, y - 1);

\\ PARI orders the roots increasingly, so [0,1] selects delta > 0
bnr = bnrinit(bnf, [pp, [0,1]], 1);

bnr.clgp
S = bnrstarkunit(bnr);
lift(S)
```

Important implementation details:

- `[0,1]` is the infinite-place bit for the positive embedding $\delta > 0$; `[1,0]` gives a ray group of order 8, not 16;
- construct $\mathfrak{p}_+$ from `delta-1`, rather than relying on the ordering returned by `idealprimedec`;
- inspect the ray class group;
- compare the returned Stark polynomial with the exact minimal polynomial of

$$\varepsilon_1 = -B_{1,2}^2/\delta.$$

Possible outcomes:

A. Exact polynomial match. Then the normalized HI phase is literally a square root of the canonical Stark unit, up to inversion and a root of unity.

B. Same field, polynomial related by a power map. Then

$$\varepsilon_{\mathrm{HI}} = \varepsilon_{\mathrm{Stark}}^k$$

for some integer k, which is still excellent.

C. Same field but unrelated unit-lattice class. Then class-field theory remains relevant, but the strong Stark-unit hypothesis is false.

Actual outcome: same field and a short Stark-orbit quotient. The two polynomials are unequal and there is no pure-power orbit relation, but the fields are K_{17}-isomorphic and an exact two-conjugate relation exists:

$$\varepsilon_{\mathrm{HI}}^2 = \frac{s^2}{\sigma^2(s)}.$$

This is stronger than outcome C and points toward a multiplicative-difference or cocycle formula.

13. Independent Magma replication

The analogous PARI calculations are now complete for $p = 7, 11, 13, 17, 19$. The following remains a useful independent verification plan.

For each $p = 7, 13, 17, 19$:

(1) Construct
$$K_p = \mathbb{Q}(\sqrt{p^2+4}).$$
(2) Compute:
   - ring of integers;
   - unit group;
   - class group;
   - factorization of p;
   - prime ideal $\mathfrak{p}_+ = (\delta - 1)$.
(3) Construct the ray class group for
$$\mathfrak{p}_+\infty_+.$$
(4) Construct the corresponding class field.
(5) Compare:
   - degree;
   - discriminant;
   - ramification;
   - Galois group;
   - Artin action at small primes.
(6) For $p = 17$, reproduce the exact identification of the reconstructed $B_{1,2}$ field with the class field.
(7) Reproduce the test that
$$\varepsilon = -B_{1,2}^2/\delta$$
is not a pure power of one Stark conjugate but satisfies the exact Frobenius quotient relation.

For $p = 19$, independently reproduce the $C_{18} \times C_2$ ray group, the Hilbert class field $\mathbb{Q}(\sqrt{5}, \sqrt{73})$, and the exact Hilbert-conjugate factorization of the recognized HI polynomial.

For $p = 11$, independently verify the PARI result that the principal degree-10 branch has conductor $5(\delta - 1)\infty_+$.

14. Higher-priority structural problem: derive all generic entries from one Stark orbit

The exact relation for
$$\varepsilon_u = -B_{u,2u}^2/\delta$$
as a Frobenius quotient of Stark conjugates gives only the ratio class $t = 2$.

For $p = 17$, we still have the other two generic orbit types:
$$t \in \mathcal{O}_3, \qquad t \in \mathcal{O}_4.$$

The next task is to ask whether these can be expressed using products or quotients of Stark conjugates.

Try relations of the form
$$U_{g,h}^2 \stackrel{?}{=} -\varepsilon_a^m \varepsilon_b^n \varepsilon_c^r,$$
where a, b, c are simple linear or cross-ratio functions of g, h.

More natural tests:
$$\frac{U_{1,t}^2}{U_{1,2}^2},$$

$$\frac{U_{1,t}U_{1,1-t}}{U_{1,2}},$$
$$U_{g,h}U_{h,g},$$
$$U_{g,h}U_{-h,g-h}U_{h-g,-g}.$$

Search for multiplicative relations in the unit group of the ray field using:

- Magma `UnitGroup`;
- PARI `bnfisunit`;
- integer relations among logarithmic embeddings.

If all generic $U_{g,h}$ can be expressed as products of square roots of one Stark-unit Galois orbit, then the entire transparent associator may admit a direct class-field construction.

15. Fourier / cocycle route

The phase parametrization

$$U_{g,h} = \frac{z_h}{z_g z_{h-g}}$$

suggests that U is a multiplicative second difference.

A major goal is to find z_g directly in class-field terms.

Possible ansatz:

$$z_g = \frac{Q_p(\zeta_p^g)}{Q_p(\zeta_p^{-g})},$$

or a variant built from:

- Gauss sums;
- Jacobi sums;
- Gaussian periods;
- Stark/Shintani units;
- Shintani–Faddeev cocycle values at real multiplication points.

If z_g can be produced directly, all $U_{g,h}$ follow automatically.

Then the Izumi equations should be Fourier transformed. Since they contain cyclic convolution sums, Fourier space may turn them into lower-dimensional multiplicative identities.

16. Potential Shintani–Faddeev route

The real quadratic number

$$\delta = \frac{p + \sqrt{p^2+4}}{2}$$

is the fixed point of

$$x \mapsto p + \frac{1}{x},$$

represented by

$$\begin{pmatrix} p & 1 \\ 1 & 0 \end{pmatrix}.$$

This makes δ a natural real-multiplication point.

The modulus

$$(\delta - 1)$$

has norm p, suggesting rational characteristics of denominator p.

A first plausible formula was schematically

$$U_{u,2u} = \zeta^{\,\mathbf{r}_u}[\delta],$$

where

$$\mathbf{r}_u \in \frac{1}{p}\mathbb{Z}^2/\mathbb{Z}^2$$

indexes the ray class and the right-hand side is a Shintani–Faddeev / real-multiplication cocycle value.

The exact comparisons show that a single Stark value is too restrictive. The revised target is a quotient of a value and its Frobenius-at-(2) translate, whose square matches

$$U_{u,2u}^4 = \frac{s_u^2}{s_{2u}}$$

up to the fixed signs and normalizations described in Sections 21–27.

This could eventually provide an explicit nonnumerical construction.

17. Long-term existence strategy

The numerical-to-algebraic reconstruction should be treated as a discovery tool only.

A viable infinite-family proof would ideally look like one of the following.

17.1. **Strategy A: direct ray-class / Stark formula.** Construct canonical Stark representatives s_u in

$$K_p^{(\delta-1)\infty_+}$$

and define the HI units by the Frobenius quotient

$$e_u^2 = s_u^2/s_{2u},$$

then define F-symbols algebraically from suitable roots and products of these units.

Then prove the Izumi equations by class-field or cocycle identities.

17.2. **Strategy B: analytic Shintani–Faddeev formula.** Define the F-symbols from explicit special-function values at the RM point δ.

Prove the Izumi identities analytically.

Since an isolated solution of polynomial equations over a number field has algebraic coordinates, algebraicity could then follow from rigidity rather than from direct Stark algebraicity.

17.3. **Strategy C: finite field + Hensel lifting.** Define the Izumi solution modulo suitable primes using character sums.

Prove the mod-$\mathfrak{q}$ solution is smooth.

Prove a lifting criterion for the full system over the local integer ring, then Hensel-lift. Smoothness of the special fiber alone does not exclude a vertical component: relative smoothness, or an equivalent flatness and Jacobian argument, is needed. A zero-dimensional lift has algebraic coordinates; a positive-dimensional p-adic point need not.

For unitary existence, also prove that a complex embedding realizes the formal involution as complex conjugation and satisfies positivity. A formal norm relation over a p-adic field does not supply this real condition. With these additional steps, the approach could prove existence without explicitly computing complex minimal polynomials.

18. Concrete next-step checklist

18.1. **First priority: verify the Stark comparison at $p = 17$.**

- ☒ Reconstruct / load exact $P_2(B)$.
- ☒ Compute exact minimal polynomial of
$$\varepsilon = -B^2/\delta.$$
- ☒ Run `bnrstarkunit` for
$$K_{17}, \quad (\delta - 1)\infty_+.$$
- ☒ Compare polynomials exactly: unequal, but the relative fields are exactly K_{17}-isomorphic.

☒ Test power relations

$$\varepsilon_{\mathrm{HI}} = \sigma(\varepsilon_{\mathrm{Stark}})^k.$$

No pure-power relation occurs; an exact short orbit relation does.

☒ Identify the quotient automorphism as arithmetic $\mathrm{Frob}_{(2)}$ by exact reduction modulo 2.

18.2. **Second priority: repeat at $p = 7, 13$.**

☒ Complete the $p = 7$ comparison from Evans–Gannon's exact algebraic solution.
☒ Complete the $p = 13$ comparison from its reconstructed numerical branch.
☒ Formulate the sharpened Stark-orbit quotient conjecture suggested by all three cases.
☒ Verify the Frobenius-at-(2) identification in both cases.

18.3. **Third priority: $p = 19$.**

☒ Identify the degree-36 ray class field and its group $C_{18} \times C_2$ exactly.
☒ Verify, after numerical polynomial recognition, that the combined HI orbit defines this full ray field.
☒ Identify the Hilbert class field and prove that its involution interchanges the exact branch factors.
☒ Compare with the full Stark orbit: the fields agree exactly, and a separated numerical search finds a branchwise generator-of-$\langle \mathrm{Frob}_{(2)} \rangle$ quotient law.

18.4. **Fourth priority: $p = 11$.**

☒ Confirm the second branch equals the quadratic simple ray field, including its affine normalization to PARI's Stark unit.
☒ Determine the apparent conductor of the reconstructed principal degree-10 branch.
☒ Check whether the prime above 5 enters the conductor: it enters as $\mathfrak{q}_5^2 = (5)$.
☐ Compare its modular-data distinction with conductor arithmetic.

18.5. **Fifth priority: recover all $A_{g,h}$ from Stark units.**

☒ Obtain a uniform fourth-power expression for every generic entry (Section 28), bypassing separate unit-group searches.
☒ Lift it to complete exact matrices for $n = 7, 11, 13, 17, 29$, including all three $\mathbb{Z}_{17}$ ratio types (Section 29).
☒ Determine coherent signs by linear equations over $\mathbb{F}_2$, leaving only a global generic sign (Section 30).

18.6. **Sixth priority: general symbolic formula.**

☒ Give a single phase-sequence formula including exceptional entries (Section 31).
☒ Fourier transform the entire quadratic system for arbitrary odd n (Section 31).
☐ Prove the Fourier identities and the cubic system for a uniform analytic construction.
☐ Search for a class-field / character-sum ansatz.

19. Main conjectural picture

The current working conjecture is:

Cyclic Haagerup–Izumi Q-system F-symbols are
class-field-theoretic objects over $K_p = \mathbb{Q}(\sqrt{p^2+4})$.

More specifically:

(1) the relevant algebraic fields are narrow ray class fields, or closely related sub/extensions, with modulus involving

$$(\delta - 1)\infty_+;$$

(2) the normalized unit-circle generic F-symbols are roots of short products or quotients of Stark conjugates; for the exact $p = 7$ solution and the reconstructed $p = 13, 17$ candidates, the same exact quotient form holds, and reduction at the inert prime (2) identifies the relevant field automorphism as arithmetic Frobenius at (2), up to the global Artin-reciprocity inversion convention;
(3) the relabeling symmetry
$$g \mapsto ug$$
is the Artin action of
$$\mathbb{F}_p^\times;$$
(4) class-group effects explain multiple HI branches: at $p = 19$, the two reconstructed branches are exactly the Hilbert-class conjugate factors of the full ray-field polynomial once the initial phase recognition is accepted;
(5) exceptional unit structure explains anomalies such as $p = 11$;
(6) a direct Stark/Shintani construction may provide a route to proving existence for infinitely many cyclic Haagerup–Izumi categories without numerical algebraic reconstruction.

20. References to keep open

Primary HI / Izumi references used in this exploration:

- Huang–Lin, arXiv:2007.00670 Transparent gauge and explicit F-symbol calculations.
- Evans–Gannon, arXiv:1506.03546 Non-unitary fusion categories and their doubles via endomorphisms; a later generalization of the HI/Q-system framework.
- Evans–Gannon, arXiv:1006.1326 Twisted Haagerup–Izumi modular data; Section 4.1 contains equations (4.1)–(4.3), the exact $p = 7$ construction, and the numerical phase seeds for $p = 11, 13, 17, 19$.

Stark / real quadratic references to inspect:

- PARI/GP documentation for `bnrstarkunit`.
- Kopp, *The Shintani–Faddeev modular cocycle: Stark units from q-Pochhammer ratios*, arXiv:2411.06763.
- Kopp, *SIC-POVMs and the Stark conjectures*. Its quadratic base field is different from the HI field here, so this is a structural analogy rather than a direct identification.

21. Completed exact $p = 17$ Stark comparison

All computations in this section were run with PARI/GP 2.17.4 and are reproduced by `z17_stark_test.gp`.

21.1. **Correct ray modulus.** PARI orders the roots of
$$y^2 - 17y - 1$$
as
$$\delta_- = -0.058621\ldots, \qquad \delta_+ = 17.058621\ldots.$$
Consequently the physical place $\delta > 0$ is the second real place, and the intended input is

```
pp  = idealhnf(bnf, delta - 1);
bnr = bnrinit(bnf, [pp, [0,1]], 1);
```

The relevant ray groups are

infinite part	ray group
none	C_8
∞_-	C_8, with redundant infinite part
∞_+	C_{16}
$\infty_-\infty_+$	C_{16}, with redundant ∞_-

Thus

$$\mathrm{Cl}_{(\delta-1)\infty_+}(K_{17}) \simeq C_{16}.$$

21.2. **Exact HI unit polynomial.** Write

$$P_2(z) = P_e(z^2) + zP_o(z^2)$$

and let $e = -B^2/\delta$. Its monic polynomial is obtained exactly as

$$E(X) = \delta^{-16}\,\mathrm{Res}_z(P_2(z), z^2 + \delta X) = \delta^{-16}\left(P_e(-\delta X)^2 + \delta X P_o(-\delta X)^2\right).$$

It has the reciprocal form

$$E(X) = X^{16} + \sum_{j=1}^{7} c_j(X^{16-j} + X^j) + c_8X^8 + 1,$$

with

$$\begin{aligned}
c_1 &= 5161\delta - 88037,\\
c_2 &= -1626816688\delta + 2775125324,\\
c_3 &= 22242858033398\delta - 37943249370668,\\
c_4 &= -131690666316286026\delta + 224646121653727220,\\
c_5 &= 276878876953660039250\delta - 472317193126595318329,\\
c_6 &= -178159796060899640043100\delta + 303916050690909978642141,\\
c_7 &= 391829337338055291983379990\delta - 668406831291568961410242240,\\
c_8 &= -175532701751128886825808423\delta + 299434589973784044953723115 4.
\end{aligned}$$

Exact checks give:

- E is irreducible over K_{17};
- E is reciprocal with constant term 1, so e is a relative norm-one algebraic unit;
- $K_{17}(e) = K_{17}(B)$;
- the relative field discriminant is

$$\mathfrak{D}_{K_{17}(e)/K_{17}} = (\delta - 1)^{15};$$

- the absolute discriminant is

$$293^{16}17^{15} = 844531292848242462671957546179648804922975123799686807339 3;$$

- the absolute signature is $(16, 8)$: the physical-place roots lie on the unit circle, while all sixteen roots at the conjugate real place are positive.

21.3. **Comparison with PARI's Stark orbit.** `bnrstarkunit` returns a different irreducible reciprocal polynomial S of degree 16. Nevertheless:

(1) S has the same relative discriminant $(\delta - 1)^{15}$.
(2) `rnfpolredabs` reduces E and S to the same relative field polynomial.
(3) Independently, `nfisisom` finds sixteen isomorphisms between their absolute fields, and every one fixes K_{17}.

Therefore the HI unit and PARI's Stark unit generate exactly the same relative ray field.

They are not related by a pure power. An absolute unit-group computation shows that the HI unit vector and every individual Stark-conjugate unit vector have rank 2; hence there are no nonzero integers a, b with

$$e^a = \pm\sigma(s)^b.$$

There is, however, a particularly short exact orbit relation. Compatible roots can be chosen so that, for a generator σ of $G \simeq C_{16}$,

$$\boxed{e^2 = \frac{s^2}{\sigma^2(s)}.}$$

Equivalently, if $\rho = \sigma^2$ generates the order-eight subgroup G^2, then

$$e^2 = s^2/\rho(s).$$

This automorphism has an intrinsic ray-class label. The rational prime 2 is inert in K_{17}, so

$$\mathfrak{q}_2 = (2), \qquad N\mathfrak{q}_2 = 4.$$

In the common absolute field, exact reduction gives

$$\rho(s) \equiv s^4 \pmod{\mathfrak{q}_2}.$$

Consequently $\rho = \mathrm{Frob}_{\mathfrak{q}_2}$, not merely an automorphism of the same order. Since the positive generator 2 has residue 2 mod $(\delta - 1)$, this is multiplication by 2 in the natural ray labeling, with the usual possible inversion depending on the Artin-map convention.

This exact quotient is the strongest evidence so far for a multiplicative-difference or cocycle construction of the HI phases. The algebraic comparison is exact. PARI's interpretation of S as the characteristic polynomial of an analytic Stark unit remains conditional on the Stark algebraicity conjecture, as documented for `bnrstarkunit`.

22. Completed $p = 11$ conductor analysis

Let

$$\delta^2 - 11\delta - 1 = 0, \qquad \epsilon = \frac{\delta - 3}{5} = \frac{1 + \sqrt{5}}{2}, \qquad \delta = \epsilon^5.$$

22.1. **Secondary branch: exact quadratic ray field.** For $\mathfrak{p}_+ = (\delta - 1)$, the physical modulus $\mathfrak{p}_+\infty_+$ has ray group C_2. The polynomial from Section 9 is

$$Q(t) = t^2 + \frac{3\delta - 4}{5}t + \delta, \qquad \operatorname{disc} Q = 1 - \delta.$$

PARI returns the Stark polynomial

$$S(u) = u^2 + \frac{3\delta - 24}{5}u + 1.$$

These generators are related exactly by

$$\boxed{Q(1 + \epsilon^3 u) = \epsilon^6 S(u).}$$

Moreover, Q and PARI's relative class-field polynomial define the same quadratic extension, and

$$\mathfrak{D}_{K_{11}(t)/K_{11}} = \mathfrak{p}_+, \qquad \mathfrak{f} = \mathfrak{p}_+\infty_+.$$

Thus the secondary branch is exactly the simple quadratic ray field, and its displayed HI generator is an affine transform of PARI's norm-one Stark unit.

22.2. **Principal branch: the missing prime over 5.** Refining the published Evans–Gannon phase seed and recognizing the orbit of

$$\eta = -B_{g,2g}^2/\delta$$

gives the reciprocal polynomial

$$P_\eta(X) = X^{10} + c_1X^9 + c_2X^8 + c_3X^7 + c_4X^6 + c_5X^5 + c_4X^4 + c_3X^3 + c_2X^2 + c_1X + 1,$$

where

$$\begin{aligned} c_1 &= 144\delta - 1597, \\ c_2 &= 854377 - 77039\delta, \\ c_3 &= 15771418\delta - 174907704, \\ c_4 &= 13808071707 - 1245073049\delta, \\ c_5 &= 27511525729\delta - 305107495744. \end{aligned}$$

The reproducible 100-digit refinement in `z11_stark_test.gp` gives residuals below 4.2×10^{-100} over every instance of (4.2) and all 11^4 instances of (4.3), and a maximum coefficient error below 6.4×10^{-99}. A separate 180-digit run gives corresponding bounds 3.7×10^{-181} and 3.6×10^{-170}. Polynomial recognition is the sole numerical-to-exact step; once this reconstructed polynomial is accepted as input, all following statements are exact PARI calculations:

- P_η is irreducible over K_{11};
- if $\mathfrak{q}_5$ is the ramified prime above 5, then

$$\boxed{\mathfrak{f} = \mathfrak{p}_+\mathfrak{q}_5^2\infty_+ = 5\mathfrak{p}_+\infty_+;}$$

- the full ray group for this conductor is C_{10}, and P_η defines the full ray field rather than a proper quotient;
- the relative discriminant is

$$\mathfrak{D} = \mathfrak{q}_5^{16}\mathfrak{p}_+^9;$$

- the absolute discriminant is

$$-5^{26}11^9 = -351361586302518844604492 1875.$$

This confirms the earlier guess that the ramified prime over 5 distinguishes the principal branch. The identification is still conditional on the numerical-to-exact reconstruction of P_η, so an independent reconstruction or direct verification of the Izumi equations remains desirable.

23. Explicit Shintani–Faddeev parameters for the HI ray classes

Kopp's ray-class/RM-point correspondence can be specialized explicitly to the HI fields. This is a derivation from Theorem 3.14 and Theorem 8.2 of arXiv:2411.06763, not a formula stated there specifically for Haagerup–Izumi data.

View K_p through the real embedding $\rho_1(\delta) = \delta_- < 0$, leaving the physical place as Kopp's distinguished second embedding, and put

$$\beta = -\delta = \delta_+^{-1} > 0, \qquad M = \begin{pmatrix} 1 & p \\ p & p^2+1 \end{pmatrix}, \qquad \mathbf{r}_u = -\frac{u}{p}\begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Then

$$M\beta = \beta, \qquad j_M(\beta) = \beta^{-2} > 1, \qquad M \equiv I \pmod{p}.$$

Kopp's denominator parameter becomes

$$[[\mathbf{r}_u, \beta]] = \frac{u(1-\beta)}{p} = \frac{u(1+\delta)}{p} = \frac{u\delta}{\delta - 1},$$

so its denominator ideal is exactly $\mathfrak{p}_+ = (\delta - 1)$. Taking the totally positive generator

$$\alpha = \delta(\delta - 1)$$

of $\mathfrak{p}_+$, the correspondence produces

$$\gamma_u = \alpha[[\mathbf{r}_u, \beta]] = u\delta^2 \equiv u \pmod{\mathfrak{p}_+}.$$

Thus the characteristics $\mathbf{r}_u$, for $u \in \mathbb{F}_p^\times$, carry exactly the desired residue-class labeling. For $p = 17$, they exhaust the sixteen ray classes.

For this specialization the theta and eta multiplier factors simplify to 1. Since the two-infinite-place to one-infinite-place ray-class map has trivial fiber at $p = 17$, Kopp's formula becomes

$$\exp(2Z'(0, \mathfrak{A}_u)) = \mathrm{shin}_M^{\mathbf{r}_u}(\beta)^2.$$

This also explains why the sign in

$$\varepsilon_{\mathrm{HI}} = -U^2$$

is natural: at ρ_1, $\delta < 0$, so $U = B/\sqrt{\delta}$ is imaginary and $(iU)^2 = -U^2$.

The exact algebraic comparisons at $p = 7, 13, 17$ now identify the quotient automorphism by reduction at (2): after a compatible Galois shift of the Stark root,

$$\boxed{e_u^2 = \frac{s_u^2}{\mathrm{Frob}_{(2)}(s_u)}.}$$

After a compatible identification of the HI multiplier orbit with the displayed residue-class labeling, this reads $e_u^2 = s_u^2/s_{2u}$, up to the global arithmetic-versus-geometric Artin convention. That individual-root alignment is not encoded by the present minimal-polynomial tests. The next analytic computation is therefore sharper: evaluate Kopp's Shintani–Faddeev values for the displayed $(M, \beta, \mathbf{r}_u)$, fix their root normalization and Artin convention, and determine whether this exact Frobenius quotient is a standard cocycle or distribution relation.

24. Exact $p = 7$ comparison

The $p = 7$ case can be derived without numerical reconstruction. Evans–Gannon's proof of Theorem 5(d) in arXiv:1006.1326 gives an exact construction of the entry

$$a = B_{1,2}.$$

Eliminating their auxiliary quadratic element b from the displayed cubic for a gives

$$\begin{aligned} P_a(W) = W^6 - W^5 + (\delta - 7)W^4 + (9 - 3\delta)W^3 + W^2 \\ - (7\delta + 1)W + (50\delta + 7). \end{aligned}$$

For $e = -a^2/\delta$, exact elimination yields

$$\boxed{\begin{aligned} E_7(X) = X^6 &+ (15\delta - 107)(X^5 + X) \\ &+ (-503\delta + 3591)(X^4 + X^2) \\ &+ (5153\delta - 36794)X^3 + 1. \end{aligned}}$$

For

$$K_7 = \mathbb{Q}(\delta), \qquad \delta^2 - 7\delta - 1 = 0, \qquad \mathfrak{p}_+ = (\delta - 1),$$

PARI gives

$$\mathrm{Cl}_{\mathfrak{p}_+\infty_+}(K_7) \simeq C_6.$$

Its `bnrstarkunit` polynomial is

$$\begin{aligned} S_7(X) = X^6 &+ (19\delta - 137)(X^5 + X) \\ &+ (-503\delta + 3591)(X^4 + X^2) \\ &+ (1635\delta - 11672)X^3 + 1. \end{aligned}$$

As at $p = 17$:

– $E_7 \neq S_7$, and no HI root is pure-power-related to an individual Stark conjugate;

– both polynomials are irreducible and define exactly the same relative field;
– both fields have relative discriminant $\mathfrak{p}_+^5$;
– for compatible roots e, s and a generator $\sigma \in C_6$,

$$\boxed{e^2 = \frac{s^2}{\sigma^2(s)}.}$$

Equivalently, $e^2 = s^2/\rho(s)$ for a generator ρ of the order-three subgroup G^2. Because the HI input here is exact, this proves that the quotient pattern is not an artifact of the $p = 17$ numerical reconstruction.

Here too the quotient automorphism is determined intrinsically. The prime (2) is inert with norm 4, and exact reduction in the common field gives

$$\rho(s) \equiv s^4 \pmod 2.$$

Thus

$$\rho = \mathrm{Frob}_{(2)}, \qquad \mathrm{ord}(\rho) = 3 = \mathrm{ord}_7(2).$$

All statements are reproduced by `z7_stark_test.gp`, including a certified base-field class/unit computation.

25. Completed $p = 13$ comparison and the Frobenius-2 law

The $p = 13$ input begins with the Evans–Gannon numerical seed from Section 4.1 of arXiv:1006.1326,

$$j^{(13)} = (-3.1050384,\ 0.5993399,\ -0.111708,\ -0.969766,\ 1.336848,\ 1.00483129).$$

Six independent instances of their equation (4.2) refine these phases to 180 digits. All instances of (4.2), and all 13^4 instances of (4.3), then have numerical residual below 3.7×10^{-181}. The script records 100 digits of the refined solution and recomputes the complete systems with residual below 3.1×10^{-100}; its maximum coefficient error is 1.1×10^{-96}. Thus polynomial recognition is the only numerical-to-exact step below.

For

$$K_{13} = \mathbb{Q}(\delta), \qquad \delta^2 - 13\delta - 1 = 0, \qquad e = -B_{g,2g}^2/\delta,$$

the recognized orbit polynomial is

$$E_{13}(X) = X^{12} + \sum_{j=1}^{5} c_j (X^{12-j} + X^j) + c_6 X^6 + 1,$$

where

$$\begin{aligned}
c_1 &= 467\delta - 6097,\\
c_2 &= 12814191 - 979939\delta,\\
c_3 &= 825835784\delta - 10799019382,\\
c_4 &= 3392673833885 - 259448689016\delta,\\
c_5 &= 30448109649290\delta - 398153890397194,\\
c_6 &= 11724621314790962 - 8966195332981 69\delta.
\end{aligned}$$

The following statements are then exact:

– the physical ray group is

$$\mathrm{Cl}_{(\delta-1)\infty_+}(K_{13}) \simeq C_{12};$$

– E_{13} is irreducible and reciprocal, and its field has conductor exactly $(\delta-1)\infty_+$, so it is the full ray field;
– the relative discriminant is $(\delta-1)^{11}$;
– the absolute discriminant and signature are

$$173^{12}13^{11} = 128804226203453529041353607 1230037719077, \qquad (r_1, r_2) = (12, 6);$$

– PARI's degree-twelve Stark polynomial is different from E_{13}, but the two absolute fields admit twelve K_{13}-preserving isomorphisms.

For a compatible HI root e, Stark root s, and one order-twelve generator τ, the exhaustive exact two-conjugate search finds one relation per HI conjugate, always with the same displacement. One representative is

$$e^2 = \frac{\tau^8(s)^2}{\tau(s)}.$$

Putting $s_0 = \tau^8(s)$ and $\rho = \tau^5$ gives

$$\boxed{e^2 = \frac{s_0^2}{\rho(s_0)}.}$$

Finally, 2 is inert in K_{13}, the prime $\mathfrak{q}_2 = (2)$ has norm 4, and its ray class generates C_{12}. Exact reduction gives

$$\rho(s) \equiv s^4 \pmod{\mathfrak{q}_2},$$

so

$$\rho = \mathrm{Frob}_{\mathfrak{q}_2}, \qquad \mathrm{ord}(\rho) = 12 = \mathrm{ord}_{13}(2).$$

Together, the three completed cases give

p	$\mathrm{ord}_p(2)$	$\mathrm{ord}(\mathrm{Frob}_{(2)})$
7	3	3
13	12	12
17	8	8

and, more strongly than this order coincidence, the reduction calculations identify the field automorphism itself. After choosing a compatible identification between the HI multiplier orbit and the ray-class labels, the resulting sharpened conjecture is

$$\boxed{e_u^2 = \frac{s_u^2}{s_{2u}}.}$$

The field-theoretic statement using $\mathrm{Frob}_{(2)}$ is exact. Two labeling points remain to be fixed analytically: whether a chosen reciprocity map labels arithmetic Frobenius by 2 or 2^{-1}, and which individual HI root carries the ray label $u = 1$.

All exact checks, together with the recorded 100-digit reconstruction test, are reproduced by `z13_stark_test.gp`.

26. The two $p = 19$ branches and the Hilbert class field

The class-number-two case now admits a precise explanation. Put

$$K = K_{19} = \mathbb{Q}(\delta), \qquad \delta^2 - 19\delta - 1 = 0, \qquad \mathfrak{p} = (\delta - 1).$$

PARI certifies

$$\mathrm{Cl}(K) \simeq C_2,$$

and the physical ray group is

$$\boxed{\mathrm{Cl}_{\mathfrak{p}\infty_+}(K) \simeq C_{18} \times C_2.}$$

Thus the order is 36, as predicted earlier, but the group is not cyclic.

26.1. **The residue-(2) subgroup and the class-group quotient.** The prime $\mathfrak{q}_2 = (2)$ is inert in K, with norm 4. In PARI's displayed $C_{18} \times C_2$ coordinates its ray class is $[10, 1]$, of order

$$18 = \mathrm{ord}_{19}(2).$$

Because (2) is principal, its class lies in the kernel of the natural map

$$\mathrm{Cl}_{\mathfrak{p}\infty_+}(K) \longrightarrow \mathrm{Cl}(K).$$

Both that kernel and $\langle \mathrm{Frob}_{(2)} \rangle$ have order 18, so they are equal. Moreover,

$$\mathrm{Frob}_{(2)}^9 = c,$$

where c is complex conjugation; this is the class-field version of $2^9 \equiv -1 \pmod{19}$.

The fixed field of this order-eighteen subgroup is the Hilbert class field

$$H = K\Big(\sqrt{58 - 3\delta}\Big) = \mathbb{Q}(\sqrt{5}, \sqrt{73}).$$

PARI gives the relative and absolute equations

$$X^2 + 3\delta - 58, \qquad X^4 - 59X^2 + 49,$$

and $\mathrm{disc}(H) = 365^2$, confirming that H/K is everywhere unramified.

This makes the two-branch mechanism nearly forced: the two cosets of the subgroup generated by Frobenius at 2 are exactly the two ordinary ideal classes. The nontrivial class involution exchanges the cosets, while Frobenius at 2 moves transitively inside each one.

The full ray field L/K has

$$\mathfrak{D}_{L/K} = \mathfrak{p}^{34}, \qquad |\operatorname{disc} L| = 365^{36}19^{34}, \qquad (r_1, r_2) = (36, 18).$$

PARI's full `bnrstarkunit` polynomial is irreducible, reciprocal, and of degree 36. If s is one of its roots, the trace $s + s^{-1}$ has an irreducible degree-eighteen polynomial whose field is exactly the finite-ray, maximal totally real subfield; its relative discriminant is $\mathfrak{p}^{16}$.

26.2. **Reconstruction of the two HI branches.** Both Evans–Gannon $p = 19$ phase seeds were Newton-refined to 450 digits. The regression script retains about 115 digits and checks every instance of equations (4.2) and (4.3) for both branches. From the two $t = 2$ orbits, set

$$e = -B_{u,2u}^2/\delta.$$

Their combined orbit has the reciprocal polynomial

$$E_{19}(X) = X^{36} + \sum_{j=1}^{17} c_j(X^{36-j} + X^j) + c_{18}X^{18} + 1.$$

The full coefficient list is recorded in `z19_stark_test.gp`; useful fingerprints are

$$\begin{aligned} c_1 &= -333455 + 17502\delta, \\ c_{18} &= 31345410510927098827288799476234826508033029 3 \\ &\quad - 16452136243396368439735966221635671877278307\delta. \end{aligned}$$

Polynomial recognition is the only numerical-to-exact step. Once E_{19} is accepted, exact PARI calculations show that it is irreducible over K, has conductor exactly $\mathfrak{p}\infty_+$, has trivial norm subgroup in $C_{18} \times C_2$, and has relative discriminant $\mathfrak{p}^{34}$. It therefore defines the full ray field. It differs from PARI's Stark polynomial, but both generators define the unique full ray extension attached to the same conductor and norm subgroup.

Each branch separately becomes algebraic over H, not over K. If w, w' are the first nontrivial coefficients of the two conjugate degree-eighteen branch polynomials, then

$$\begin{aligned} w + w' &= -333455 + 17502\delta, \\ ww' &= 61207710 - 3212584\delta. \end{aligned}$$

The resulting quadratic coefficient field is unramified over K and reduces to

$$X^2 + \delta - 29;$$

it is the same Hilbert class field H. All remaining branch coefficients were recognized in $K(w)$, and their exact Hilbert conjugate norm is E_{19}.

This is a direct arithmetic realization of the old branch hypothesis:

the two HI branches are the two Hilbert-class conjugate halves of one ray-class orbit.

It also explains why Evans–Gannon's second modular-data label involves the pair $\{5, 73\}$: these are precisely the two prime-discriminant factors whose genus field is $\mathbb{Q}(\sqrt{5}, \sqrt{73})$.

26.3. **Status of the Stark quotient at $p = 19$.** A high-precision all-pairs root search finds exactly 36 matches of the form

$$e_k^2 = \frac{s_i^2}{s_j}.$$

The selected edges form a permutation of the Stark roots with cycle lengths $[18, 18]$, and its ninth power sends every root to its inverse. This is exactly the cycle signature of a generator of

$$\langle \mathrm{Frob}_{(2)} \rangle = \ker\big(\mathrm{Cl}_{\mathfrak{p}\infty_+}(K) \to \mathrm{Cl}(K)\big)\,.$$

This comparison is numerical, unlike the exact reductions modulo 2 at $p = 7, 13, 17$. It does not yet distinguish $\mathrm{Frob}_{(2)}$ from another generator of the same cyclic subgroup. The branchwise conjecture is therefore

$$\boxed{e_{b,u}^2 = \frac{s_{b,u}^2}{s_{b,2u}}, \qquad b \in \mathrm{Cl}(K) \simeq C_2,}$$

subject to exact Artin/root alignment.

The exact ray arithmetic is reproduced by `z19_ray_test.gp`; the numerical reconstruction and exact downstream HI-field checks are in `z19_stark_test.gp`.

27. Group-ring and Shintani–Faddeev reformulation

The quotient law now has a useful intrinsic form. Let

$$G = \mathrm{Gal}(L/K), \qquad \phi = \mathrm{Frob}_{(2)}, \qquad H = \langle \phi \rangle, \qquad t = e^2.$$

With multiplicative group-ring notation, the exact $p = 7, 13, 17$ relation is

$$\boxed{t = s^{\Theta_2}, \qquad \Theta_2 = 2 - \phi.}$$

The opposite Artin convention replaces ϕ by ϕ^{-1}.

27.1. **Why the operator loses no field information.** If $d = \mathrm{ord}(\phi)$, define

$$\Psi_d = 2^{d-1} + 2^{d-2}\phi + \cdots + \phi^{d-1}.$$

Then

$$(2 - \phi)\Psi_d = 2^d - \phi^d = 2^d - 1,$$

and hence

$$t^{\Psi_d} = s^{2^d - 1}.$$

On each H-coset,

$$\mathrm{coker}\Big(\mathbb{Z}[H] \xrightarrow{2-\phi} \mathbb{Z}[H]\Big) \simeq \mathbb{Z}/(2^d - 1),$$

while characterwise the eigenvalues are $2 - \chi(\phi)$, none of which vanish. Because the ray fields here have a real embedding, their only roots of unity are $\{\pm 1\}$; since $2^d - 1$ is odd, the relation implies

$$K(s) = K(e^2).$$

This gives a conceptual reason, independent of polynomial reduction, that the Frobenius quotient still generates the full Stark field.

It is important not to call t uniformly a Hilbert–90 coboundary. What is always a coboundary is

$$\frac{t}{s} = \frac{s}{\phi(s)} = s^{1-\phi},$$

so $N_H(t) = N_H(s)$. At $p = 13$ and 17 this subgroup norm is 1. At $p = 7$ it is not: the exact absolute polynomial of $N_H(s)$ is

$$X^4 - 932X^3 + 70X^2 - 932X + 1.$$

These norm assertions are now included in the corresponding GP scripts.

27.2. **A sharper analytic candidate.** Kopp defines the Shintani–Faddeev modular cocycle on the upper half-plane by

$$\operatorname{shin}_A^{\mathbf{r}}(\tau) = \frac{\varpi_{\mathbf{r}}(A\tau)}{\varpi_{\mathbf{r}}(\tau)}.$$

For the odd-denominator characteristics $\mathbf{r}_u$ of Section 23,

$$\Gamma_{\mathbf{r}_u} = \Gamma_{2\mathbf{r}_u}.$$

Therefore

$$D_{2,A}^{\mathbf{r}}(\tau) := \frac{\operatorname{shin}_A^{\mathbf{r}}(\tau)^2}{\operatorname{shin}_A^{2\mathbf{r}}(\tau)}$$

is itself a modular 1-cocycle. On the upper half-plane it is the literal coboundary of

$$F_2^{\mathbf{r}}(\tau) = \frac{\varpi_{\mathbf{r}}(\tau)^2}{\varpi_{2\mathbf{r}}(\tau)}.$$

As with Kopp's original cocycle, extension to the larger domain containing an RM boundary point can leave a nontrivial cohomology class.

Under the Stark normalization in Section 23,

$$s_u = \operatorname{shin}_M^{\mathbf{r}_u}[\beta]^2$$

up to the already noted inversion and multiplier conventions. Consequently

$$\left(D_{2,M}^{\mathbf{r}_u}[\beta]\right)^2 = \frac{s_u^2}{s_{2u}}.$$

The algebraic result therefore suggests the unsquared formula

$$\boxed{e_u = \pm\frac{\operatorname{shin}_M^{\mathbf{r}_u}[\beta]^2}{\operatorname{shin}_M^{2\mathbf{r}_u}[\beta]}.}$$

This characteristic-difference cocycle is a substantially sharper analytic target than a literal Stark-unit equality.

It is not Kopp's conductor-lowering relation in disguise. Theorem 4.46 applied to doubling uses a determinant-four matrix and a product over four half-characteristics; standard arithmetic smoothing at the inert ideal (2) likewise has weight $N(2) = 4$. The observed operator $2 - \phi$ is instead a square-root, or weight-one, analogue. Establishing the algebraicity, sign, and Shimura-reciprocity law of this unsquared RM value appears to be the genuinely new theorem required.

More generally, for c coprime to p, put

$$D_{c,u} = \frac{q_u^c}{q_{cu}}, \qquad \Theta_c = c - \sigma_c.$$

Then the formal identities

$$\Theta_{ab} = b\Theta_a + \sigma_a\Theta_b, \qquad D_{ab,u} = D_{a,u}^b D_{b,au}$$

give a weighted semigroup distribution law. Whether the other HI ratio types $B_{u,cu}$ realize these D_c is now a concrete test rather than a vague search for unit relations.

The best first analytic experiment is the exact $p = 7$ case: evaluate $D^{\mathbf{r}_1}_{2,M}[\beta]$, fix its sign and Artin orientation against the known algebraic HI root, and only then attempt a proof from quantum-dilogarithm or theta identities.

28. A diagonal-determination theorem for the full fourth-power core

The short Frobenius law contains more information than Section 14 suggested. The following argument is algebraic and works for every finite abelian group of odd order. It requires a **labeled** doubling law in a common field; equality of orbit polynomials alone does not supply that hypothesis.

28.1. **Correct cyclic phase normalization.** The Evans–Gannon phases in Section 4 are initially given for integer representatives $0 < g < h < n$. Put $r = (n-1)/2$ and $C = j_r + j_{r+1}$. Their extension rule says $j_{n-g} = C - j_g$. The periodic phases to use in identities on $\mathbb{Z}_n$ are

$$z_0 = 1, \qquad z_g = \exp(i(j_g - Cg/n)), \quad 1 \le g < n.$$

Then $z_{-g} = z_g^{-1}$, and the generic formula $U_{g,h} = z_h/(z_g z_{h-g})$ holds with indices reduced modulo n. Using the raw published e^{ij_g} as a periodic sequence would miss the correction at wraparound.

Define an auxiliary normalized cocycle

$$C(a,b) = \frac{z_{a+b}}{z_a z_b}, \qquad V(a,b) = C(a,b)^4.$$

Thus $C(a,b) = U_{a,a+b}$ when $a, b, a+b \neq 0$, while $C(a,0) = C(0,a) = C(a,-a) = 1$. These auxiliary boundary values are **not** the actual exceptional scaled matrix entries.

Both cocycles are symmetric and satisfy

$$V(a,b)V(a+b,c) = V(b,c)V(a,b+c).$$

28.2. **Diagonal determination.**

Proposition 28.1. *Let V be a normalized symmetric multiplicative 2-cocycle on a finite abelian group G of odd order, taking values in a number field L with a real embedding. Suppose $s_a \in L^\times$, $s_0 = 1$, and*

$$V(a,a) = s_a^2/s_{2a} \quad (a \in G).$$

Then

$$\boxed{V(a,b) = s_a s_b/s_{a+b} \quad (a,b \in G).}$$

Proof. Divide V by the symmetric cocycle $R(a,b) = s_a s_b/s_{a+b}$, obtaining W with $W(a,a) = 1$. For any symmetric cocycle, comparing the two groupings of (a,a,b,b) gives

$$W(2a,2b) = W(a,b)^2 \frac{W(a+b,a+b)}{W(a,a)W(b,b)} = W(a,b)^2.$$

Choose d such that multiplication by 2^d is the identity on G. Iteration yields $W(a,b)^{2^d-1} = 1$. A field with a real embedding has only the roots of unity $\{\pm 1\}$, and $2^d - 1$ is odd. Hence $W(a,b) = 1$. □

In particular, assuming the common-field and labeling hypotheses, the doubling law in Section 27 implies the **entire** generic formula

$$\boxed{U^4_{g,h} = \frac{s_g s_{h-g}}{s_h}, \qquad B^4_{g,h} = \delta^2 \frac{s_g s_{h-g}}{s_h}.} \tag{28.1}$$

For $h = 2g$, this is precisely $e_g^2 = s_g^2/s_{2g}$. The formula is compatible with rotation and with reflection/conjugation when $s_{-g} = s_g^{-1}$. Section 29 verifies it with exact labels, for all entries, in the five completed cases.

This argument neither assumes a primitive root 2 nor assumes that δ is fundamental or that the base class number is one. It also explains why a field hypothesis matters: in $\mathbb{C}^\times$ alone there can be odd-order torsion ambiguities.

28.3. **Correction to the proposed higher-multiplier formula.** For an integer $2 \le c < n$ and a unit g mod n, (28.1) gives

$$U^4_{g,cg} = \frac{s_g s_{(c-1)g}}{s_{cg}}.$$

In contrast, telescoping gives

$$\boxed{\prod_{j=1}^{c-1} U^4_{g,(j+1)g} = \frac{s_g^c}{s_{cg}}.} \tag{28.2}$$

Thus the weighted operator $c - \sigma_c$ at the end of Section 27 describes a product of adjacent generic entries. Its identification with a single ratio type is appropriate only at $c = 2$, unless an additional special identity holds. Formula (28.1), with three Stark factors, is the uniform expression for individual entries.

29. Exact construction from Stark polynomials without HI seeds

The new file `hi_stark_seedless.gp` starts with

$$S = \text{bnrstarkunit}\,(K_n,\ (\delta - 1)\infty_+)$$

and constructs a full matrix over its field. There is no numerical HI seed, Newton iteration, coefficient recognition, or numerical root selection in this construction.

PARI's general analytic Stark interpretation remains conjectural. That does not make these **finite existence certificates** conditional: the returned polynomial supplies exact algebraic input, and the constructed matrix is checked against the complete defining equations. An incorrect analytic guess would simply fail those checks.

The precise construction is as follows.

(1) Certify the quadratic class/unit computation and check the returned polynomial's irreducibility, conductor, and full ray degree. Express its root s and δ in an absolute number field L.

(2) Choose a small prime ℓ inert in K_n that generates $\mathbb{F}_n^\times$. Identify arithmetic Frobenius at (ℓ) by reduction of **every integral-basis element** at a prime above ℓ. This avoids relying on a particular polynomial having index coprime to ℓ.

(3) Set $s_{\ell j} = \phi^j(s)$ and $s_0 = 1$. Repeated ray labels are permitted: at $n = 11, 29$, ϕ has smaller order than $n - 1$.

(4) For $2 \le h \le (n+1)/2$, find the roots **inside** L of

$$X^4 - \delta^2 s_1 s_{h-1}/s_h.$$

There are exactly two, differing by sign, in each completed case. Initially select either one. Set $B_{1,n+1-h} = B_{1,h}$.

(5) Define $P_1 = 1$, $P_h = \prod_{j=2}^{h} B_{1,j}$. Reconstruct

$$\boxed{B_{g,h} = P_h/(P_g P_{h-g}) \quad (0 < g < h < n).} \tag{29.1}$$

Extend by conjugation, and insert the exceptional entries from Section 4. No square root of δ is needed in (29.1).

(6) Use the sign-transport theorem in Section 30 to impose $\phi(B_{g,h}) = B_{\ell g,\ell h}$, and check the resulting linear system over $\mathbb{F}_2$. Exactly one global generic sign remains. Test its two choices against the quadratic equations.

(7) Verify rotation, conjugation, generic norms, every formula (28.1), and the full quadratic and cubic Izumi systems, with exact field arithmetic.

To justify physical unitarity rather than just a formal involution, the script proves that $s + s^{-1}$ generates a totally real subfield of index two, that $\phi^{(n-1)/2}(s) = s^{-1}$, and that every real embedding of L has $\delta < 0$. Therefore this involution is actual complex conjugation at the physical embeddings $\delta > 0$. The checked identities $B_{g,h}\overline{B_{g,h}} = \delta$ give the required positive physical norms.

29.1. **Completed exact certificates.** The full results are recorded in `hi_stark_seedless_results.txt`.

n	ray group	$[L:\mathbb{Q}]$	rank of sign system	cubic representatives checked
7	C_6	12	2	401
11	C_2	4	4	1465
13	C_{12}	24	5	2381
17	C_{16}	32	7	5221
29	C_4	8	13	25261

The cubic counts cover **every** quadruple: after verifying multiplier covariance, each nonzero (g,h,k,l) has a unique scalar multiple whose first nonzero coordinate is 1. Include the zero quadruple. The count is

$$1 + \frac{n^4 - 1}{n - 1} = n^3 + n^2 + n + 2.$$

The script uses the scaled equations

$$\sum_j B_{g+j,h}\overline{B_{j,h}} = (\delta - 1)^2 \left(\mathbf{1}_{g=0} - \frac{\mathbf{1}_{h=0}}{\delta} \right), \tag{29.2}$$

$$\begin{aligned} \sum_j B_{j,g+h}\overline{B_{j+k,g}B_{j+l,h}} &= (\delta - 1)B_{g+l,k}B_{h+k,l} \\ &\quad - \frac{(\delta - 1)^3}{\delta}\mathbf{1}_{g=0}\mathbf{1}_{h=0}. \end{aligned} \tag{29.3}$$

Together with the checked symmetries, these are exactly the Evans–Gannon/Izumi conditions. Their sufficiency for the corresponding subfactor and HI fusion rules is the result used from Evans–Gannon, Section 4.1 and Theorem 6.

These are certificates of the construction in this repository, not a claim that existence at these five orders is new to the literature.

29.2. **Removal of numerical-recognition assumptions.** As **downstream** checks, the script proves:

- at $n = 11$, its $B_{1,2}$ satisfies $X^2 + (3\delta - 4)X/5 + \delta$, identifying the simple-ray secondary branch of Section 22.1;
- at $n = 13$, the exact product $\prod_{g \neq 0}(X + B_{g,2g}^2/\delta)$ is exactly E_{13} of Section 25;
- at $n = 17$, all three exact products $\prod_{g \neq 0}(X - B_{g,tg})$, $t = 2, 3, 4$, agree with `z17_haagerup_izumi_minpolys.sage`.

Thus the old recognized polynomials really occur in complete exact HI solutions. Their downstream field identifications no longer require an assumption that the numerical phases were correctly recognized. The principal $n = 11$ branch retains its previously stated qualification. The later exact order-19 certificate and polynomial assertions in Section 35.2 discharge the order-19 recognition issue.

29.3. **Small explicit inputs for the fixed-field family.** The two certificates over $K = \mathbb{Q}(\sqrt{5})$ start from particularly small polynomials. With $\epsilon = (1+\sqrt{5})/2$, PARI returns

$$S_{11}(X) = X^2 + (3\epsilon - 3)X + 1,$$

$$S_{29}(X) = X^4 + (14\epsilon - 21)(X^3 + X) + (45 - 27\epsilon)X^2 + 1.$$

Here $\delta_{11} = \epsilon^5$ and $\delta_{29} = \epsilon^7$. Applying the construction above yields complete exact matrices over fields of absolute degrees 4 and 8. These two cases show concretely that a nonfundamental δ and the resulting collapse of residue labels are compatible with the full fourth-power formula and exact sum identities.

29.4. **The earlier class-number-two checkpoint, now completed.** A bounded $n = 19$ run completed the exact field, Frobenius-label, and physical-conjugation checks, and found both roots inside the degree-72 absolute ray field for each of the nine independent row quartics, $2 \le h \le 10$. The transcript is `hi_stark19_partial_results.txt`. The run reached its ten-minute limit before completing matrix verification. That bounded run alone was not a full certificate. Section 35.2 records its completed continuation, including every cubic equation and the exact identification of the earlier branch polynomial.

30. Why the sign problem has only one binary degree of freedom

The observed ranks in Section 29 have a structural explanation.

Proposition 30.1. *Let p be odd prime. Suppose two normalized phase cocycles C, C' on $\mathbb{Z}_p$ have $C(a,-a) = C'(a,-a) = 1$. Suppose their corresponding generic B-entries lie in a common number field with a real embedding, have the same fourth powers, and are equivariant for one automorphism implementing a primitive scalar $\ell \bmod p$. Then their generic entries agree everywhere, or all differ by the same sign.*

Proof. Their ratio $W = C'/C$ is a normalized symmetric cocycle with values in $\{\pm 1\}$. Write $D(a) = W(a,a)$. The duplication identity from Section 28 gives

$$W(2a, 2b) = D(a+b)D(a)D(b).$$

Since doubling is invertible, the function $f(a) = D(a/2)$ satisfies

$$W(a,b) = f(a)f(b)/f(a+b).$$

Equivariance fixes the signs, so $W(\ell a, \ell b) = W(a,b)$. Consequently $f(\ell a)/f(a)$ is a homomorphism $\mathbb{Z}_p \to \{\pm 1\}$, which is trivial. The scalar ℓ is transitive on nonzero indices; hence f is constant there. Its value is either 1 or -1. The associated coboundary equals that value on every generic pair. □

This first proposition is an **at-most-two** statement. Under the quartic-lift hypotheses of the construction, consistency also has a uniform proof.

Proposition 30.2 (Sign-transport)**.** *Suppose the arbitrary initial row roots in Section 29 exist in L, the labeled s_a have physical modulus one, L has a real embedding with $\delta < 0$, and $\phi^{(p-1)/2}$ is physical complex conjugation. Then signs making the reconstructed matrix scalar-equivariant exist. They are determined up to the global generic sign above.*

Proof. The row roots have physical modulus $\sqrt{\delta}$. Their reflection symmetry $B_{1,h} = B_{1,p+1-h}$ permits a unit phase sequence with $z_{-a} = z_a^{-1}$: apply the prefix recursion and choose z_1 to close the cyclic sequence. Thus the initial matrix defines a normalized phase cocycle C.

Extend ϕ to fix $\sqrt{\delta}$. This is legitimate because $\sqrt{\delta} \notin L$: at a real embedding of L the element δ is negative. In particular the extension still satisfies $\phi^{(p-1)/2} =$ physical conjugation. Put

$$E(a,b) = \frac{\phi(C(a,b))}{C(\ell a, \ell b)}.$$

The fourth-power formula shows $E(a,b)^4 = 1$; the square roots of δ cancel in this quotient, so $E(a,b) \in L$ and hence $E(a,b) \in \{\pm 1\}$. It is a symmetric normalized cocycle. As above,

$$E(a,b) = f(a)f(b)/f(a+b), \qquad f(a) = E(a/2, a/2).$$

Write $r = (p-1)/2$. Iterating the error equation r times gives

$$\prod_{j=0}^{r-1} E(\ell^j a, \ell^j b) = \frac{\overline{C(a,b)}}{C(-a,-b)} = 1.$$

Therefore $F(a) = \prod_{j=0}^{r-1} f(\ell^j a)$ is a homomorphism $\mathbb{Z}_p \to \{\pm 1\}$, so $F(a) = 1$. Define $t_0 = 1$, choose $t_1 = 1$, and transport signs by

$$\boxed{t_{\ell a} = f(a)t_a.}$$

The half-orbit product just proved ensures $t_{-a} = t_a$ and closure of the full orbit. Replacing $C(a,b)$ by $C(a,b)t_a t_b/t_{a+b}$ kills its covariance error and preserves the normalization and conjugation conditions. □

Thus root-sign compatibility is not a further conjecture in prime order once these field and fourth-root hypotheses are available. Neither proposition asserts that the quadratic or cubic sum identities hold; those remain separate requirements.

The row algorithm also makes the constraints explicit. There are $(p-1)/2$ initial row signs. Formula (29.1) expresses the sign of every entry as a linear expression in their bits. Comparing scalar translates gives a linear system over $\mathbb{F}_2$, of rank $(p-3)/2$ in every completed case. The script constructs its solution by the transport formula, checks every linear equation, and verifies the rank. This replaces a search over $2^{(p-1)/2}$ phase choices with sign transport and two final tests.

For composite n, multiplication by units is not transitive on all nonzero residues. The prime-order conclusion must not be used unchanged: additional sign freedoms can occur on the different gcd strata.

31. One sequence and a Fourier formulation of the quadratic equations

There is a useful uniform formula that includes the exceptional entries. For any odd n, choose b_g with

$$b_0 = -\delta^{-1/2}, \qquad |b_g| = 1 \ (g \neq 0), \qquad b_{-g} = \overline{b_g}.$$

For a phase solution, take $b_g = z_g^{-1}$ for $g \neq 0$. Put

$$\lambda = \frac{\sqrt{\delta}}{\delta - 1}, \qquad \mu = 1 - \delta^{-1}.$$

Then the whole matrix is

$$\boxed{A_{g,h} = \lambda b_g b_{h-g} b_{-h} + \mu\, \mathbf{1}_{g=0}\mathbf{1}_{h=0}.} \tag{31.1}$$

Exactly one zero index gives $-1/(\delta-1)$. At $(0,0)$, the right-hand side is $-1/(\delta(\delta-1))+1-1/\delta = (\delta-2)/(\delta-1)$. Rotation and Hermitian symmetry follow directly from the three factors. Also

$$\sum_g |b_g|^2 = n - 1 + 1/\delta = \delta - 1.$$

For $h \neq 0$, let

$$f_h(m) = b_m b_{h-m} = b_m \overline{b_{m-h}}, \qquad \widehat{f_h}(k) = \sum_{m=0}^{n-1} f_h(m) e^{-2\pi i k m/n}.$$

The quadratic Izumi equation becomes

$$\lambda^2 \sum_m f_h(m+g)\overline{f_h(m)} = \mathbf{1}_{g=0}.$$

Taking the discrete Fourier transform in g proves the equivalence

$$\boxed{|\widehat{f_h}(k)|^2 = \frac{(\delta-1)^2}{\delta} \quad (h \neq 0,\ k \in \mathbb{Z}_n).} \tag{Q}$$

The entire $h = 0$ quadratic system holds automatically for the exceptional column, using $\delta^2 = n\delta + 1$.

Equivalently, the unit vector $v = b/\sqrt{\delta - 1}$ has a prescribed discrete ambiguity function:

$$\left|\sum_m v_m \overline{v_{m-h}} e^{-2\pi i k m/n}\right|^2 = \begin{cases} 1, & (h,k) = (0,0), \\ \delta^{-2}, & h = 0,\ k \neq 0, \\ \delta^{-1}, & h \neq 0. \end{cases} \tag{31.2}$$

The $h = 0, k \neq 0$ overlap itself is $-1/\delta$. This formulation provides a concrete target for a character-sum or special-function identity. The two nontrivial magnitudes differ, so this is not the usual equal-overlap SIC condition.

Under the Stark construction, the natural phase target is

$$b_g^4 = s_g \quad (g \neq 0),$$

with coherent root choices and the prescribed b_0. Formula (31.1) then gives (28.1) immediately. If a phase reconstruction initially gives $b_g^4 = s_g \chi(g)$ for an additive character χ, its unique fourth-root character removes χ without changing the generic entries, since n is odd.

Remaining equation. Let $\mathcal{A}_{g,h}(b)$ denote the explicit right-hand side of (31.1). Besides (Q), one must prove

$$\sum_m \mathcal{A}_{m,g+h}(b)\overline{\mathcal{A}_{m+k,g}(b)\mathcal{A}_{m+l,h}(b)} = \mathcal{A}_{g+l,k}(b)\mathcal{A}_{h+k,l}(b) - \delta^{-1}\mathbf{1}_{g=0}\mathbf{1}_{h=0} \tag{C}$$

for all indices. Away from the explicit exceptional terms this is a six-factor finite-sum identity in b. No implication from (Q) to (C) is proved here. They are verified independently in Section 29. Neither the multiplicative cocycle law nor Stark algebraicity by itself proves these additive identities.

`hi_phase_reductions.py` checks the rational cocycle identities and binary sign ranks exactly, checks the Fourier equivalence on arbitrary phases as well as a solution, and verifies (Q) for the recorded $n = 13$ phases at 80-digit precision. Its numerical checks supplement the derivation; the independent GP certificates supply the finite existence proofs.

32. An unconditional infinite family of arithmetic inputs over one field

Restricting an infinitude strategy to primes p with $h(\mathbb{Q}(\sqrt{p^2+4})) = 1$ would introduce a separate unproved infinitude requirement. There is an explicit alternative if the target is infinitely many **odd orders**, as in the present question.

Let

$$K = \mathbb{Q}(\sqrt{5}), \qquad \epsilon = (1+\sqrt{5})/2, \qquad m > 1, \quad m \equiv 1, 5 \pmod 6,$$

and set

$$\delta_m = \epsilon^m, \qquad n_m = L_m = \epsilon^m - \epsilon^{-m}.$$

Here L_m is the usual Lucas number for odd m, and F_m below is the Fibonacci number. Then

$$\delta_m^2 - n_m\delta_m - 1 = 0, \qquad n_m^2 + 4 = 5F_m^2.$$

These give infinitely many distinct odd n_m, all with $K_{n_m} = K$. No assertion that infinitely many of them are prime is needed.

32.1. **Exact ray group and exact unit index.**

Proposition 32.1. *Put* $\mathfrak{a}_m = (\epsilon^m - 1)$. *Then*

$$\mathcal{O}_K/\mathfrak{a}_m \simeq \mathbb{Z}/n_m\mathbb{Z},$$

and the residue of ϵ *has exact order* m. *Consequently*

$$\boxed{\mathrm{Cl}_{\mathfrak{a}_m\infty_+}(K) \simeq (\mathbb{Z}/n_m\mathbb{Z})^\times/\langle\bar{\epsilon}\rangle, \qquad [K^{\mathfrak{a}_m\infty_+} : K] = \frac{\varphi(n_m)}{m}.} \tag{32.1}$$

Proof. The expansion $\epsilon^m = F_m\epsilon + F_{m-1}$ and the identity $n_m^2 - 5F_m^2 = -4$ show $\gcd(F_m, n_m) = 1$, because n_m is odd. Hence in the quotient

$$\bar{\epsilon} = (1 - F_{m-1})F_m^{-1} \pmod{n_m}.$$

The quotient has size $|N(\epsilon^m - 1)| = n_m$, proving the ring isomorphism. Its unit order divides m. If $0 < r < m$ and $\epsilon^r \equiv 1 \pmod{\mathfrak{a}_m}$, then n_m divides $N(\epsilon^r - 1)$. For odd r this norm is $-L_r$; for even r it is $2 - L_r$. Its absolute value is nonzero and strictly less than n_m, a contradiction.

The field K has class number one: its Minkowski bound is $\sqrt{5}/2 < 2$. Its units are generated by -1 and ϵ. In the one-infinite-place ray exact sequence, -1 removes the sign factor and the remaining positive-unit image is generated by $\bar{\epsilon}$. This proves (32.1). □

The examples are reproduced by `hi_pell_ray.gp`, with results in `hi_pell_ray_results.txt`.

m	n_m	ray group	ray degree
5	11	C_2	2
7	29	C_4	4
11	199	C_{18}	18
13	521	C_{40}	40
17	3571	C_{210}	210
19	9349	C_{492}	492
23	$64079 = 139 \cdot 461$	$C_{1380} \times C_2$	2760
25	$167761 = 11 \cdot 101 \cdot 151$	$C_{300} \times C_{10} \times C_2$	6000

Only $n = 11, 29$ in this table have the new HI certificates of Section 29. The later rows are arithmetic computations, not claimed categorifications.

32.2. **A fixed maximal-order RM point and explicit characteristics.** There is a necessary correction when specializing the analytic route to this family:

$$[\mathcal{O}_K : \mathbb{Z}[\delta_m]] = F_m.$$

Thus the lattice $\mathbb{Z} + \mathbb{Z}(-\delta_m)$ used naively in Section 23 is a nonmaximal order. It cannot silently be identified with the maximal-order ray problem in (32.1).

Instead, use the fixed element $\beta = -\epsilon$, viewed under the nonphysical real embedding, so numerically $\beta = (\sqrt{5} - 1)/2 > 0$. Its lattice $\mathbb{Z} + \mathbb{Z}\beta$ is $\mathcal{O}_K$. Put

$$M_0 = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}, \qquad M_m = M_0^m, \qquad \mathbf{r}_{m,u} = -\frac{u}{n_m}\begin{pmatrix} F_{m-1} + 1 \\ F_m \end{pmatrix}. \tag{32.2}$$

Then, exactly,

$$M_m\beta = \beta, \qquad j_{M_m}(\beta) = \delta_m^{-2} > 1 \quad \text{at the chosen embedding.}$$

For odd m, the Fibonacci identities give

$$M_m - I = n_m \begin{pmatrix} F_{m-1} & F_m \\ F_m & F_{m+1} \end{pmatrix},$$

so $M_m \equiv I \pmod{n_m}$ and it stabilizes every displayed characteristic. In the convention $[[\mathbf{r}, \beta]] = r_2\beta - r_1$,

$$[[\mathbf{r}_{m,u}, \beta]] = \frac{u(1+\delta_m)}{n_m} = \frac{u\delta_m}{\delta_m - 1}.$$

For u coprime to n_m, its denominator ideal is precisely $\mathfrak{a}_m$. With the totally positive generator $\alpha = \delta_m(\delta_m - 1)$, the numerator becomes

$$\alpha[[\mathbf{r}_{m,u}, \beta]] = u\delta_m^2 \equiv u \pmod{\mathfrak{a}_m}.$$

This realizes the desired residue labels at **one fixed RM point**, with stabilizer matrices that are powers of a single matrix. It uses the RM/ray correspondence of Kopp, Sections 3 and 8, but the displayed specialization is derived here. This section does not assert that the multiplier normalization from the $p = 17$ case carries over unchanged.

When n_m is composite, the full HI matrix also needs nonunit residue labels. These require the ray class monoid and values for smaller conductors, as in Kopp's Proposition 6.11; the invertible ray orbit alone supplies too few labels. The prime-only sign algorithm also needs an extension to the gcd strata.

The subsequence $m = 5^k$, $k \geq 1$, even gives nested moduli, since $\epsilon^a - 1$ divides $\epsilon^{ab} - 1$. It is therefore a possible setting for a distribution argument. A distribution law for units would still need a proof that it preserves (Q) and (C) as the additive group changes size.

33. What an infinitude proof now needs

This records the target after the first continuation. Sections 34–37 sharpen it to one balanced identity and extend the sign theorem.

The work above proves finite existence and removes two general obstacles: recovering higher ratio types and managing the numerous root signs in prime order. It does not prove existence for infinitely many n.

A precise sufficient target is now available. For infinitely many $m \equiv 1, 5 \pmod 6$, construct physical phases b_g, $g \in \mathbb{Z}_{L_m}$, with

$$b_0 = -\delta_m^{-1/2}, \qquad b_{-g} = \overline{b_g}, \qquad |b_g| = 1 \ (g \neq 0),$$

and prove **(Q) and (C)** of Section 31. Formula (31.1) and the Izumi construction then give the desired infinite family.

The candidate is to take coherent fourth roots of ray/monoid Stark values at the explicit parameters (32.2). A proof through this candidate must establish the following substantive points:

(1) Define and normalize values for all required residues, including nonunits at composite L_m, and produce the physical phases. If using the algebraic route, prove the necessary fourth-root lifts and sign compatibility; algebraicity of the Stark units alone is weaker than these requirements.
(2) Prove the finite Fourier identity (Q) for those phases, uniformly in the chosen exponents m.
(3) Prove the cubic finite-sum identity (C), or prove a theorem that makes it follow from the particular construction and (Q).

These requirements are not supplied by the multiplicative $2 - \mathrm{Frob}_{(2)}$ law, the usual Stark conjecture, or class-field existence. They are the remaining analytic/algebraic existence problem. In an analytic construction that directly produces the physical phases and proves the two identities, prior Stark algebraicity would not be necessary for existence.

The suggested next experiment, $n = 199$, is now complete in Section 35.3. It stays over $\mathbb{Q}(\sqrt{5})$ with ray degree

(1) The beta reduction and exact integer batches made its full cubic verification feasible. This tests the fixed-field formula at a larger order; an infinitude theorem still requires a uniform construction.

34. One balanced finite beta identity encodes the entire system

There is a uniform reduction beyond Section 31: **within its phase ansatz, (Q) and (C) together are equivalent to a single balanced finite beta identity**. This is an algebraic change of formulation, proved below; it does not assert that arbitrary Stark values satisfy the identity. It applies to every odd order, including composite orders.

Write $d = \delta$, $d^2 = nd + 1$, and retain

$$b_0 = -d^{-1/2}, \quad b_{-u} = \overline{b_u}, \quad |b_u| = 1 \quad (u \neq 0), \qquad \lambda = \frac{\sqrt{d}}{d-1}, \quad \mu = \frac{d-1}{d}.$$

All indices below lie in $\mathbb{Z}_n$. For two triples $\mathbf{a} = (a_1, a_2, a_3)$, $\mathbf{c} = (c_1, c_2, c_3)$ satisfying $\sum_i a_i + \sum_j c_j = 0$, define

$$S(\mathbf{a}, \mathbf{c}) = \sum_m \prod_{i=1}^{3} b_{m+a_i} \prod_{j=1}^{3} b_{-m+c_j}, \qquad P(\mathbf{a}, \mathbf{c}) = \prod_{i,j=1}^{3} b_{a_i+c_j}.$$

Put $q = d^{-1}$, $[r]_q! = \prod_{s=1}^{r}(1 + q + \cdots + q^{s-1})$, and let m_x be the multiplicity of x in $\mathbf{a}$. The identity is

$$\boxed{S(\mathbf{a}, \mathbf{c}) = \lambda^{-1} P(\mathbf{a}, \mathbf{c}) + \frac{(d-1)^3}{d^2} \mathbf{1}_{\{c_1,c_2,c_3\}=\{-a_1,-a_2,-a_3\}} \prod_x [m_x]_{1/d}!.} \tag{34.1}$$

The braces in the indicator denote **multisets**. Repeated entries matter: the three possible correction factors are

$$1, \qquad 1 + d^{-1}, \qquad (1 + d^{-1})(1 + d^{-1} + d^{-2}),$$

for multiplicities $1 + 1 + 1$, $2 + 1$, and 3, respectively.

A matrix in the exact certificates really has the required physical phase form. To see this without numerical phase recognition, let $r_h = B_{1,h}/\sqrt{d}$, $2 \leq h < n$, put $z_0 = z_1 = 1$, and set $z_h = \prod_{t=2}^{h} r_t$. The exact row norms give $|z_h| = 1$, and the mirror row identity gives $z_g z_{n-g} = z_{n-1}$ for $g \neq 0$. Choose any unit-modulus u with $u^n = z_{n-1}$, and put $b_g = u^g / z_g$ for $g \neq 0$, with the prescribed exceptional b_0. Then $b_{-g} = \overline{b_g}$. The checked prefix-product formula for the upper triangle gives $B_{g,h} = \sqrt{d}\, b_g b_{h-g} b_{-h}$; Hermitian symmetry gives the lower triangle. The exceptional entries supply (31.1) everywhere. The choice of u changes the sequence only by a cyclic character.

34.1. Change of variables for the interior cubic equations.

For (g, h, k, l), set

$$\mathbf{a} = (0, k - g, l - h), \qquad \mathbf{c} = (g + h, -k, -l). \tag{34.2}$$

Their cross-sum matrix is

$$(a_i + c_j) = \begin{pmatrix} g+h & -k & -l \\ h+k & -g & k-g-l \\ g+l & l-h-k & -h \end{pmatrix}. \tag{34.3}$$

Conversely, shift $a_i \mapsto a_i - a_1$, $c_j \mapsto c_j + a_1$, then set $k = -c_2$, $l = -c_3$, $g = -c_2 - a_2$, $h = -c_3 - a_3$. Thus (34.2) parametrizes every balanced pair, up to a harmless reindexing of the sum.

If $g, h, g + h \neq 0$, substitute $A_{u,v} = \lambda b_u b_{v-u} b_{-v} + \mu \mathbf{1}_{u=v=0}$ into (C). The three factors on its left have no origin correction. After dividing its residual by $\lambda^3 b_{-(g+h)} b_g b_h$, the result is precisely the residual of (34.1).

Indeed, the two generic factors on its right give $\lambda^{-1}P$. An origin correction on the right occurs only at $(k,l)=(0,-g)$ or $(-h,0)$. Each contributes $\mu/\lambda^2=(d-1)^3/d^2$ after division. These are exactly the possible multiset matches when $g,h,g+h\neq 0$; then all three entries of $\mathbf{a}$ are distinct. To see the last assertion directly, a matching permutation must avoid the three nonzero cross entries corresponding to $g,h,g+h$. Its only possibilities are the two derangements. A repeated entry would give a forbidden matching permutation as well.

In particular, when **all nine cross sums are nonzero**, the remaining cubic equation has the simple form

$$\sum_m \prod_i b_{m+a_i} \prod_j b_{-m+c_j} = \frac{d-1}{\sqrt{d}} \prod_{i,j} b_{a_i+c_j}. \tag{34.4}$$

34.2. **Every collision case follows from (Q).** For a full multiset match, the three sums and product terms in (34.1) are determined by norms alone. They are as follows.

Multiplicity in $\mathbf{a}$	S	$\lambda^{-1}P$
$1+1+1$	$n-3+3/d$	$-(d-1)/d^2$
$2+1$	$n-2+d^{-1}+d^{-2}$	$-(d-1)/d^3$
3	$n-1+d^{-3}$	$-(d-1)/d^5$

Subtracting the last column and using $n=d-d^{-1}$ gives exactly the three correction coefficients in (34.1).

For a collision without a full multiset match, permute the triples so that $a_1+c_1=0$. The corresponding pair of factors in the summand is

$$b_{m+a_1}b_{-m-a_1} = 1+(d^{-1}-1)\mathbf{1}_{m=-a_1}.$$

The unrestricted sum of the other four factors is a quadratic correlation. In fact, set $H=a_2+c_2$, $V=a_3+c_2$. The balancing condition identifies that sum with

$$\sum_t b_t b_{H-t} b_{-t-V} b_{t+V-H}. \tag{34.5}$$

This is exactly a correlation from (Q). Both relevant shifts are nonzero: if either H or V vanishes, the remaining two pairs match, contrary to the assumption. Hence (34.5) vanishes by (Q).

For clarity, one can instead normalize $a_2=0$ and write the remaining two triples as $(0,V-H)$ and $(H,-V)$. Their four-factor sum is $\sum_t b_t b_{t+V-H} b_{-t+H} b_{-t-V}$, which is exactly (34.5). The surviving term of S is therefore

$$(d^{-1}-1)\prod_{i=2}^{3} b_{a_i-a_1} \prod_{j=2}^{3} b_{c_j+a_1}.$$

The product P is b_0 times these four boundary factors, since its other four factors pair to $|b_H|^2|b_V|^2=1$. Finally $\lambda^{-1}b_0=d^{-1}-1$, proving (34.1) in this case.

Conversely, (34.1) implies every nontrivial quadratic correlation. Given nonzero g,h, take

$$\mathbf{a}=(t,0,g-h), \qquad \mathbf{c}=(-t,h,-g).$$

There is a cross-zero pair but no full multiset match. The calculation just made subtracts the exceptional term from (34.1) and leaves $\sum_m b_m b_{h-m} b_{-m-g} b_{m+g-h}=0$. The zero-shift correlations follow from the norms and $d^2=nd+1$, and the $h=0$ column of (Q) is automatic. Thus the collision part of (34.1) recovers all of (Q).

34.3. **The boundary cubic equations also follow from (Q).** Here are explicit reductions, including the exceptional origin. Write $c_0 = (d-1)^{-1}$, so $A_{u,0} = \mathbf{1}_{u=0} - c_0$. For $g = 0$, $h \neq 0$, the left side of (C), using (Q), becomes

$$A_{-k,h}\overline{A_{l-k,h}} - c_0\mathbf{1}_{l=0}.$$

For $l \neq 0$ this equals $A_{l,k}A_{h+k,l}$ directly from the phase formula and the unit norms at nonzero indices. For $l = 0$, use $|A_{u,h}|^2 = \lambda^2$ when $u \neq 0, h$, and $|A_{u,h}|^2 = c_0^2$ otherwise. The case $h = 0, g \neq 0$ follows by exchanging (g,k) with (h,l).

For $h = -g \neq 0$, the left side similarly reduces to

$$\overline{A_{k,g}}A_{-l,g} - c_0\mathbf{1}_{k=g+l}.$$

If $k \neq g + l$, the phase products equal the right side of (C). If $k = g + l$, the assertion reduces to the same two norm values; use $\lambda^2 - c_0 = c_0^2$.

For $g = h = 0$, the left side is

$$\mathbf{1}_{k=l=0} - c_0(\mathbf{1}_{k=0} + \mathbf{1}_{l=0} + \mathbf{1}_{k=l}) + 3c_0^2 - nc_0^3.$$

The right side is $|A_{l,k}|^2 - d^{-1}$. The cases $k = l = 0$, exactly one of $k = 0, l = 0, k = l$, and none of them agree by $d^2 = nd + 1$.

Combining these observations proves the equivalences

$(Q) + (C) \iff (34.1) \iff (Q) + (34.4)$ for all cross-zero-free balanced pairs.

No implication from (Q) alone to (C) in arbitrary order is proved here.

34.4. **Symmetry and exact verification.** The two sides of (34.1) are invariant under independent permutations of the triples, interchange of the triples, and opposite translation of the triples. The first two operations form $(S_3 \times S_3) \rtimes C_2$, of order 72. For prime order, checked multiplier covariance supplies the scalar action as well. It is enough to check (C) at one representative of each resulting orbit of cross-zero-free quadruples; all other equations follow by the preceding proof. Scalar covariance also reduces (Q) to the columns $h = 0, 1$.

Prime	Earlier cubic scalar representatives	New beta representatives
7	401	6
11	1,465	17
13	2,381	30
17	5,221	63
19	7,241	91
29	25,261	320
199	7,920,401	108,406

At $n = 199$, this changes the cubic summand count from 1,576,159,799 to 21,572,794. These are exact enumeration counts, not estimates of runtime. It remains a substantial computation.

`hi_beta_reductions.py` independently checks the orbit coverage, the rational collision coefficients, the change of variables on arbitrary nonsolutions at orders 3, 5, 7, 9, and every balanced identity for the recorded order-13 solution. `hi_stark_seedless.gp` uses the same proven reduction for exact number-field certificates. The optional `HI_STARK_FULL_CUBIC=1` runs the older exhaustive scalar check and the full quadratic check too; both methods agree exactly at orders 7, 11, 13, 17, 29.

35. Exact ideal evidence and the next finite orders

35.1. **Quadratic equations imply cubic equations at orders 3, 5, 7, 9.** There is now an exact algebraic result supporting the possible redundancy of (C), beyond the earlier numerical

experiments. Put $c_g = \sqrt{d}\, b_g$, so that $c_0 = -1$ and $c_g c_{-g} = d$ for $g \neq 0$. Over $K = \mathbb{Q}(d)$, $d^2 = nd + 1$, consider the ideal

$$I_n = \left(c_g c_{-g} - d, \quad \sum_m c_m c_{h-m} c_{-m-g} c_{m+g-h} \;\middle|\; g, h \neq 0 \right) \subset K[c_1, \ldots, c_{n-1}]. \tag{35.1}$$

The second set of generators is precisely the nontrivial part of (Q) in these coordinates. For each balanced cross-zero-free pair, set

$$F_{\mathbf{a},\mathbf{c}} = d^2 \sum_m \prod_i c_{m+a_i} \prod_j c_{-m+c_j} - (d-1) \prod_{i,j} c_{a_i+c_j}. \tag{35.2}$$

Here the bold $\mathbf{c}$ denotes the second triple of indices, whereas c_g denotes a variable. The different degrees, six and nine, are intentional: clearing the phase normalization multiplies (34.4) by d^5.

`hi_quadratic_ideal.py` computes a reduced Gröbner basis in Singular and verifies the actual ideal memberships

$$\boxed{F_{\mathbf{a},\mathbf{c}} \in I_n \quad \text{for all such pairs when } n = 3, 5, 7, 9.}$$

The ideal is invariant under scalar permutations of its variables. Together with the exact beta symmetries, this makes the finite representative checks exhaustive even without assuming Galois covariance of an individual solution. Section 34 then proves $(Q) \Rightarrow (C)$ for every phase solution at these four orders. This is ideal membership, stronger than numerical vanishing or membership merely in the radical.

n	Reduced Gröbner basis size	$\dim_K K[c_1, \ldots, c_{n-1}]/I_n$	Beta representatives tested
3	3	6	1
5	14	20	2
7	41	42	6
9	132	108	14

The dimensions in this table count scheme length; reducedness or a classification of physical solutions is not inferred from these numbers. The same direct Gröbner computations at orders 11 and 13 reached a 300-second limit without completing. The later row-coordinate certificate in Section 45 resolves order 11. The earlier timed-out runs establish no implication at those orders. A **uniform proof of (35.2) lying in (35.1)** would remove (C) as a separate requirement in the phase ansatz. Existence of physical solutions of (Q) would still need proof.

35.2. **The order-19 certificate is now complete.** The earlier partial checkpoint in Section 29.4 has been completed. The construction from the full ray Stark polynomial gives exact fourth-root lifts, sign transport, every instance of (Q), all norm, rotation and multiplier identities, and all of (C), the last by 91 balanced-beta representatives. The field has absolute degree 72 and ray group $C_{18} \times C_2$.

Two further exact assertions identify the formerly recognized data. If E_{19} is the degree-36 polynomial of Section 26, the matrix satisfies

$$E_{19}(-B_{1,2}^2/d) = 0.$$

Its principal-orbit coefficient

$$\tau = \sum_{g=1}^{18} B_{g,2g}^2/d$$

satisfies

$$\tau^2 - (-333455 + 17502d)\tau + 61207710 - 3212584d = 0.$$

These checks occur **after** the construction and use no numerical HI seeds. The earlier exact irreducibility and Hilbert-class-field computations therefore identify the full relative minimal polynomial and its two branch factors without a phase-recognition assumption. The other

Hilbert-class conjugate of the matrix also satisfies every equation: the ray Galois group is abelian, so its action commutes with the verified physical conjugation and fixes d.

The logs `hi_stark19_results.txt` record the exact initial field and root stages, the completed matrix verification, and these downstream polynomial assertions. The old partial log is retained as history.

35.3. **The fixed-field experiment at order 199.** For $n = L_{11} = 199$, the base field is $\mathbb{Q}(\sqrt{5})$, the ray group is C_{18}, and the absolute field degree is 36. The exact construction has passed all field and physical-involution checks, every fourth-root lift, sign transport of rank 98, every quadratic equation, rotation, scalar covariance, and all generic norms and fourth-power Stark identities. In particular, the fixed-field candidate satisfies the complete Fourier identity (Q) at this much larger order.

All 108,406 cubic representatives have now passed exactly, in 16 disjoint batches. Together with the preceding checks, this is a complete existence certificate at order 199. The construction and coverage logs are `hi_stark199_construction_results.txt` and `hi_stark199_cubic_results.txt`.

The saved `hi_stark199_certificate.gp` contains 458 distinct entries and their index matrix, plus the defining field and involutions. `hi_stark19_certificate.gp` has the analogous data with 62 distinct entries. These are exact rational expressions, with no phase seeds. `hi_stark_verify_certificate.py` verifies the fields, physical involutions, coherent phase form, symmetries and quadratic equations directly from the saved data, then runs the cubic batches. For example:

```
python hi_stark_verify_certificate.py hi_stark199_certificate.gp --jobs 16
```

The `-structure-only` option stops before the separately required cubic stage. It does not report a complete existence certificate.

The separate final-stage verifier `hi_stark_cubic_batches.py` clears a common denominator D of B and d. With $X = DB$ and $Y = D(d-1)$, it checks the integer-polynomial congruences

$$\sum_j X_{j,g+h} X_{g,j+k} X_{h,j+l} = Y X_{g+l,k} X_{h+k,l} \pmod{P}, \tag{35.3}$$

where P is the monic absolute defining polynomial. Hermitian symmetry converts conjugation to transposition. The representatives are cross-zero-free, so the exceptional $g = h = 0$ term is absent. All operations take place in $\mathbb{Z}[y]/(P)$; there is no floating-point tolerance or probabilistic modular test. Disjoint batches together cover every representative, and the log records the SHA-256 of the matrix checkpoint to identify their common input. The previously completed field, phase-form, covariance and quadratic checks remain essential parts of this certificate.

36. Sign descent at composite orders and a prime-power theorem

The prime-order sign argument extends to a useful general criterion. Let G be a finite abelian group of odd order and let a finite group H act on G and on the coefficient field. Write σ_h for the field action. Assume a normalized symmetric phase cocycle C has been constructed and that its covariance errors

$$E_h(a,b) = \frac{\sigma_h(C(a,b))}{C(ha,hb)}$$

lie in $\{\pm 1\}$. The fourth-power identities and the real embedding argument of Section 30 give this hypothesis in the prime construction; it must be checked for a proposed composite construction. Also assume $-1 \in H$ acts as physical conjugation, so $E_{-1} = 1$.

Define

$$f_h(a) = E_h(a/2, a/2).$$

The normalized sign-valued cocycle calculation gives, uniquely,

$$E_h(a,b) = \frac{f_h(a)f_h(b)}{f_h(a+b)}, \qquad f_{hk}(a) = f_h(ka)f_k(a). \tag{36.1}$$

Uniqueness uses the absence of nontrivial homomorphisms $G \to \{\pm 1\}$. For $a \in G$, let $H_a = \{h : ha = a\}$. Then

$$\chi_a : H_a \longrightarrow \{\pm 1\}, \qquad \chi_a(h) = f_h(a),$$

is a character.

Proposition 36.1 (Sign descent criterion)**.** *A correction $C'(a,b) = C(a,b)t_at_b/t_{a+b}$, $t_0 = 1$, $t_a \in \{\pm 1\}$, with exact H-covariance exists if and only if every χ_a is trivial. When it exists, the remaining freedom is one sign per nonzero H-orbit in G. The correction respects $t_{-a} = t_a$.*

To prove the criterion, exact covariance is equivalent, by uniqueness in (36.1), to

$$t_{ha} = f_h(a)t_a. \tag{36.2}$$

For $h \in H_a$ this requires $\chi_a(h) = 1$. Conversely, choose one sign at a representative of each orbit and extend by (36.2). The cocycle identity makes this well-defined precisely when the stabilizer characters vanish. Finally $f_{-1} = 1$ gives the negation condition. Two choices differ by an orbit-constant sign function. Distinct such functions give distinct corrected cocycles, because their quotient cannot be a nontrivial sign character of the odd group G.

Corollary 36.2 (Prime-power)**.** *For $G = \mathbb{Z}_{p^k}$, p odd, and $H = (\mathbb{Z}/p^k)^\times$, all stabilizers of nonzero elements have odd order. An element of additive order p^j has stabilizer of order p^{k-j}, the kernel of reduction to $(\mathbb{Z}/p^j)^\times$. Thus every stabilizer character is trivial, and the sign correction always exists under the stated phase and covariance-error hypotheses. There are exactly k remaining orbit signs, one for each additive order $p, p^2, \ldots, p^k$, giving 2^k choices before imposing (Q). This removes a sign obstruction uniformly on an infinite family of candidate orders. It does not supply their fourth-root lifts or (Q).*

For general odd cyclic n, the number of nonzero scalar orbits is $\tau(n) - 1$, but their stabilizers can have even order. Their characters cannot simply be ignored. The script `hi_composite_signs.py` checks the prime-power orbit counts and constructs an abstract example at $n = 15$ with $f_{-1} = 1$, yet $\chi_3(11) = -1$. This is an example of sign-error data, not a claimed HI or Stark counterexample.

There is a concrete class-field sufficient condition for general n. If a has additive order q, require its diagonal fourth-root lift $B_{a/2,a}$ to lie in the field fixed by σ_h for every $h \equiv 1 \pmod q$. Then

$$f_h(a) = \sigma_h(B_{a/2,a})/B_{a/2,a} = 1 \quad (h \in H_a),$$

and the obstruction vanishes. In a ray-monoid construction this is the expected lower-conductor field condition on the **root lift**; having only its fourth power in that field is insufficient. Once these lower-conductor lifts and the global phase cocycle are constructed, there is no further composite sign compatibility problem beyond the orbit signs.

37. The sharpened infinitude target

The strongest uniform target from this continuation is now explicit: construct, for infinitely many odd orders, the physical sequence of Section 34 and prove **the single balanced identity (34.1)**. That identity includes unitarity through its collision cases and gives the remaining Izumi equations through its cross-zero-free cases. This would be an existence proof without needing prior Stark algebraicity.

There are two concrete arithmetic settings. The fixed-field Lucas family of Section 32 supplies explicit RM characteristics and ray moduli; its composite labels require lower-conductor root control as in Sections 36 and 38. Alternatively, the orders p^k are already an unconditional

infinite sequence, and Section 36 makes sign descent automatic once the phase/root hypotheses hold. For that route the quadratic fields vary with k, and the phase construction and finite identities remain unproved. Neither setting requires an unproved infinitude of special primes.

The beta reformulation suggests a more specific analytic comparison than the general appeal to quantum dilogarithms. The integral identity (1.11) in Bazhanov–Sergeev, *A distant descendant of the six-vertex model* has three numerator and three denominator factors, a balancing condition, and nine pairwise factors on the other side. The authors relate it to the constant beta pentagon identity of Garoufalidis and Kashaev. Our finite identity has a comparable pattern after using $b_{-x} = b_x^{-1}$ at nonzero indices. This comparison is a research direction, not a specialization theorem.

A successful specialization would have to establish the finite $\mathbb{Z}_n$ sum, the normalization $(d-1)/\sqrt{d}$, and all three collision coefficients in (34.1). The continuous integral in that paper does not by itself provide them. The explicit fixed-RM characteristics from Section 32 are candidates for such a descent; no descent or limiting argument accomplishing it has been proved here.

A second precise route is to prove the uniform ideal containment (35.2) in (35.1), then construct physical solutions of (Q) uniformly. The small-order containments and the larger exact Stark-derived quadratic solutions support this approach, but they do not replace those two required theorems. Existence for infinitely many n remains open in these notes.

38. All fourth-root lifts collapse to one Kummer root

There is a stronger simplification of Sections 28, 30 and 36. The many row-root computations can be replaced by **one fourth-root condition**, followed by Galois transport and, at prime powers, odd-degree norms. This does not assert the remaining sum identities.

38.1. **An explicit inverse to the doubling-root map.** Let L be the ray field, let $\alpha = \sigma_2$ be its residue-2 automorphism, and suppose $\alpha(d) = d$. Then the two root problems

$$D^4 = d^2 \frac{s^2}{\alpha(s)} \qquad \text{and} \qquad W^4 = d^2 s \tag{38.1}$$

are equivalent **inside the same field** L. The mutually inverse maps are

$$\boxed{D = \frac{W^2}{\alpha(W)}, \qquad W = \frac{d^2 \alpha^{-2}(s)}{\alpha^{-2}(D)^2 \alpha^{-1}(D)}.} \tag{38.2}$$

For example, raising the second expression to the fourth power and using the first equation in (38.1) gives $d^2 s$; substituting it into $W^2/\alpha(W)$ returns D exactly. No extraction of additional roots occurs in this inverse formula.

Thus even the single known doubling root $B_{1,2}$ already gives a fourth root of $d^2 s$. The apparently weaker doubling quotient condition is not weaker at the level of fourth-root existence. This is an algebraic equivalence, without any Stark conjecture.

For prime order, define $W_a = \sigma_a(W)$ for $a \neq 0$. At a physical embedding, $\overline{s} = s^{-1}$ gives $(W\overline{W})^4 = d^4$. Positivity of $W\overline{W}$ therefore forces $W\overline{W} = d$. Consequently

$$b_0 = -d^{-1/2}, \qquad b_a = W_a/\sqrt{d} \quad (a \neq 0)$$

has the required physical norms and negation symmetry, and

$$\boxed{B_{g,h} = \frac{W_g W_{h-g}}{W_h} \quad (g, h, h-g \neq 0).} \tag{38.3}$$

The exceptional entries are prescribed as before. Every generic fourth-power formula, rotation symmetry, Hermitian symmetry and scalar covariance now follows directly. Since L has a real embedding, a solvable fourth-root problem has exactly two roots in L; these give the two global generic sign choices. No separate row-sign linear system is necessary for this construction.

The formula also recovers an already certified matrix, not just some matrix with the same fourth powers. Write $D_a = B_{a,2a}$. Then

$$W_a = \frac{d^2 s_{a/4}}{D_{a/4}^2 D_{a/2}}, \qquad W_a^4 = d^2 s_a, \qquad W_a^2 / W_{2a} = D_a.$$

The quotient of the two normalized phase cocycles has values in $\{\pm 1\}$ and trivial diagonal, hence is identically one by the odd-order sign-cocycle lemma. Direct exact computation confirms (38.3) for every generic entry of both saved matrices at orders 19 and 199. In this gauge the phase values themselves satisfy $b_a^4 = s_a$; the cyclic-character ambiguity of the prefix-product construction has been removed.

38.2. **Odd-degree norm descent of the root.** Let L/M have odd degree r, with $d \in M$, and suppose $N_{L/M}(s_L) = s_M$. If $W_L^4 = d^2 s_L$, set

$$\boxed{W_M = d^{(1-r)/2} N_{L/M}(W_L).} \tag{38.4}$$

Then

$$W_M^4 = d^{2(1-r)} d^{2r} s_M = d^2 s_M.$$

If physical conjugation commutes with the norm and $W_L \overline{W}_L = d$, it also gives $W_M \overline{W}_M = d$. The exponent in (38.4) is an integer; this is why odd degree removes the apparent extra root extraction.

For $n = p^k$, p odd, write $(d-1) = \mathfrak{p}^k$ in the maximal quadratic order. The prime p splits since $n^2 + 4 \equiv 4 \pmod{p}$. The ray extensions from conductor $\mathfrak{p}^k$ to $\mathfrak{p}^j$, with the same infinite place, have degree a power of p. Thus (38.4) applies whenever the distinguished units have the required norm compatibility. One top-level Kummer root then supplies roots in every smaller nontrivial conductor. There are initially up to 2^k stratum signs in the phase ansatz. Section 40 makes one necessary quadratic equation linear in transformed signs; imposing the exact norm law leaves only a global sign.

38.3. **Nonunit labels need their remaining numerator ideal.** The lower-conductor values must be assigned to the correct ray-monoid class. They cannot simply be copied from the identity class at the smaller conductor. For $g = p^j u$, $p \nmid u$, cancelling $\mathfrak{p}^j$ from the numerator ideal gives

$$(g)\mathfrak{p}^{-j} = (u)\overline{\mathfrak{p}}^{\,j}.$$

Therefore the label at additive order p^{k-j} includes the Artin action of $\overline{\mathfrak{p}}^{\,j}$. In compatible normalizations, the intended formula is

$$s_{p^j u} = \sigma_u \sigma_{\overline{\mathfrak{p}}}^j(s_{k-j}), \qquad W_{p^j u} = \sigma_u \sigma_{\overline{\mathfrak{p}}}^j(W_{k-j}). \tag{38.5}$$

Here s_{k-j} and W_{k-j} are the lower-conductor unit and root obtained before applying that class label. The ideal calculation is the specialization of the conductor-reduction formula in Kopp, Proposition 6.11, equation (6.35). Its norm factor contributes no derivative correction because the differenced zeta function vanishes at zero. The assertion identifying canonical analytic values with algebraic Stark units remains subject to the hypotheses stated in that proposition.

The need for this numerator ideal is visible at $n = 25$. Over $\mathbb{Q}(\sqrt{629})$, the ray groups at $\mathfrak{p}^2$ and $\mathfrak{p}$ are C_{40} and C_8. The exact degree-five norm of the full Stark generator is a root of the lower-conductor Stark polynomial. Its untwisted cochain gives all quartic lifts and physical norms, but **none of its four stratum-sign choices satisfies (Q)**. This rules out that untwisted assignment; it does not rule out order-25 existence.

For the corrected assignment, the conjugate-prime Artin map can be identified exactly without reducing a poorly conditioned power basis at the ramified rational prime. An auxiliary prime above 23 has base root $d \equiv 6 \pmod{23}$, and its ray coordinate is 29 in C_{40}, while that of $\overline{\mathfrak{p}}$ is 33. The auxiliary Frobenius is identified modulo a squarefree factor of P mod 23, and

its 33/29 mod 40 power gives the desired map. Its square is exactly the rational-2 Frobenius, as also follows from the positive generator $(d+1)/d$ of $\overline{\mathfrak{p}}^2$, which is congruent to 2 modulo $\mathfrak{p}^2$.

38.4. **A more economical sufficient target for prime powers.** For a fixed odd prime p, an infinitude proof along p^k can now be organized around the following substantive assertions:

(1) Construct compatible ray units with the physical involution and the same-support norm law across the nontrivial conductors.
(2) Prove just **one** Kummer condition $d_k^2 s_k \in L_k^{\times 4}$ at the top conductor. Equations (38.4)–(38.5) then produce all nonzero residue phases with their correct class labels.
(3) Prove the balanced identity (34.1) for some choice of stratum signs, uniformly for infinitely many k.

Section 40 replaces the sign search in the third item by a rational linear system. If the roots are required to obey (38.4), only one global sign remains, and one necessary quadratic equation uniquely selects it whenever it can be selected at all.

The quadratic base field varies with k. Class-field existence, Stark algebraicity, and the norm formulas do not establish the last two assertions. Nevertheless, this removes the separate collection of mixed-index fourth-root problems and turns lower-conductor sign compatibility into an explicit construction.

39. Complete exact existence at the composite order 25

The corrected construction of Section 38 passes **every** Izumi equation at $n = 25$. It uses no numerical HI seed. The arithmetic data are

$$K = \mathbb{Q}(\sqrt{629}), \quad h_K = 2, \quad \mathrm{Gal}(L/K) = C_{40}, \quad [L : \mathbb{Q}] = 80,$$

with lower ray group C_8 and relative conductor-descent degree five. One exact root $W^4 = d^2 s$, its normalized norm $d^{-2}N(W)$, and the conjugate-prime Artin twist produce the cochain on both nonzero residue strata. The untwisted assignment fails (Q), while the corrected one passes it.

The independently verified coefficient artifact is `hi_stark25_certificate.gp`, with SHA-256

```
e19474b7d332a306ab2d13009d526a7bd484011bdbeeb7fa77eef7d10d73477b
```

It contains exact rational coefficients in a degree-80 number field. The verifier checks the field and physical involution, every matrix norm, the full coherent phase form, Hermitian and rotation symmetries, scalar covariance, and all quadratic equations. For the last check, the columns $h = 0, 1, 5$ cover all scalar strata.

The cubic enumeration for composite orders uses actual unit-group orbits; it never normalizes an index by division by a nonunit. There are 267,720 cross-zero-free quadruples and 246 representatives under the balanced-beta symmetries. All 246 exact cubic checks pass in eight disjoint batches. Section 34 covers every remaining cubic equation. The complete log is `hi_stark25_verification_results.txt`.

To verify the saved artifact without rerunning Stark computations:

```
python hi_stark_verify_certificate.py hi_stark25_certificate.gp --jobs 8
```

To reproduce the construction in a fresh temporary directory:

```
hi_work=$(mktemp -d /tmp/hi25_reproduce.XXXXXX)
export HI_STARK_LIBRARY=1
export HI_STARK_CHECKPOINT="$hi_work/rows.bin"
export HI_SINGLE_ROW_CHECKPOINT="$HI_STARK_CHECKPOINT"
export HI_SINGLE_CONJUGATE_FROBENIUS="$hi_work/conjugate_prime.bin"
export HI_SINGLE_OUTPUT="$hi_work/certificate.gp"
gp -q -f hi_stark_seedless.gp hi_stark_prime_power.gp
gp -q -f hi_stark_seedless.gp hi_stark25_artin.gp
```

```
gp -q -f hi_stark_seedless.gp hi_stark_single_root.gp
python hi_stark_verify_certificate.py "$HI_SINGLE_OUTPUT" --jobs 8
```

The first three stages are construction stages; only the last command certifies all equations. `hi_stark25_artin.gp` deliberately implements the exact cyclic ray-group calculation for this order, without claiming to handle every prime power. The single-root and sign formulas are uniform. This is a finite existence theorem, not an induction step from order 5 to order 25: the dimension and quadratic base field differ.

40. Prime-power sign selection becomes rational linear algebra

There is a further uniform reduction of the apparent 2^k sign search. Let $n = p^k$, with p odd, and suppose a physical cochain and its phase matrix B have already been constructed. No quadratic or cubic equation is assumed. Changing the sign of W_a on the valuation stratum $v_p(a) = j$ multiplies it by $\epsilon_j \in \{\pm 1\}$, for $0 \le j < k$.

40.1. **One necessary quadratic equation is linear in new signs.** Set

$$D = B_{1,2}, \qquad T = D + \overline{D}, \qquad R_m = B_{m+1,1}B_{1,m} \quad (2 \le m \le n-2).$$

Partition the generic terms by $j = v_p(m^2-1)$, and define

$$C_0 = 1 + \sum_{\substack{2\le m\le n-2\\ p\mid m^2-1}} R_m, \qquad C_j = \sum_{\substack{2\le m\le n-2\\ v_p(m^2-1)=j}} R_m \quad (1 \le j < k).$$

The valuations in these sums are less than k: the only square roots of 1 modulo an odd prime power are ± 1, and those indices have been excluded. For the sign-changed matrix, the single quadratic equation with $g = h = 1$ is exactly

$$\boxed{C_0 - \epsilon_0 T + \sum_{j=1}^{k-1} \epsilon_0\epsilon_j C_j = 0.} \tag{40.1}$$

Here is the sign calculation. The $m = 0$ term is 1. The two terms at $m = 1, -1$ sum to $-\epsilon_0 T$. In every other term, the cochain signs occur as $\epsilon^2_{v_p(m)}\epsilon_{v_p(m-1)}\epsilon_{v_p(m+1)}$. If $p \mid m$, both neighbors are units and this product is 1. If $m, m-1, m+1$ are all units it is again 1. Otherwise exactly one neighbor has positive valuation j, giving $\epsilon_0\epsilon_j$. Oddness of p is essential here.

The substitution

$$y_0 = \epsilon_0, \qquad y_j = \epsilon_0\epsilon_j \ (j > 0)$$

is a bijection of the sign cube, with inverse $\epsilon_0 = y_0$, $\epsilon_j = y_0 y_j$. Expanding (40.1) in any rational basis of the coefficient field gives the exact linear system

$$\boxed{[-T\ C_1\ \cdots\ C_{k-1}]\, y = -C_0, \qquad y_j \in \{\pm 1\}.} \tag{40.2}$$

If the k coefficient columns are rationally independent, there is at most one sign choice even for this one necessary equation. Solve the rational system and check that all coordinates are signs. An inconsistent system, or a unique solution with a coordinate other than ± 1, rejects every stratum-sign choice for the proposed roots. In deficient rank, intersect the rational affine solution space with the sign cube before testing the other quadratic equations.

For the saved order-25 matrix, the rational rank is exactly two and the unique solution is $y = (1, 1)$. Directly testing the other three stratum-sign choices independently confirms that **each fails already at** $g = h = 1$. The routine `hi_prime_power_sign_candidates` implements (40.2); the one-root constructor now uses it before the full quadratic check. Full rank for all prime powers has not been proved, and this one equation does not replace the remaining quadratic or cubic equations.

40.2. **Exact norm compatibility leaves only a global sign.** Under the norm hypotheses of Section 38, requiring the roots themselves to satisfy (38.4) is stronger than independently choosing a root on each stratum. If two compatible choices differ by signs η_r at conductor level r, an odd-degree norm gives $\eta_{r-1} = \eta_r^{[L_r:L_{r-1}]} = \eta_r$. The Artin twists fix signs. Thus **all norm-compatible root choices differ by one global sign**. This uses that the fields have a real embedding, so their fourth roots of unity are only ± 1. It does not assert that every solution of (Q) must obey these norm laws.

Put $U = C_0 + \sum_{j>0} C_j$. For a global change $\epsilon_j = \eta$, equation (40.1) becomes

$$U = \eta T. \tag{40.3}$$

For the Stark construction, $T \neq 0$ whenever $s \neq 1$. Indeed, $T = 0$ and $D\overline{D} = d$ would imply $D^2 = -d$, hence $D^4 = d^2$. The doubling formula then gives $\alpha(s) = s^2$, where $\alpha = \sigma_2$. If α has order r, this forces $s^{2^r-1} = 1$. A field with a real embedding has only the roots of unity ± 1, so the odd exponent forces $s = 1$, a contradiction.

Consequently the necessary and sufficient condition that one of the two global choices satisfy this single quadratic equation is

$$U^2 = T^2,$$

and its sign is then determined algebraically by $\eta = U/T$. The selected cochain $(U/T)W$ is unchanged if the initially extracted fourth root is replaced by its negative. There is no remaining global choice after this equation is imposed.

The saved matrices at orders 19, 199 and 25 pass exact reconstruction from the single root and the test $U = T \neq 0$. At order 25 the reconstructed nonunit root is exactly $d^{-2}\sigma_{\overline{\mathfrak{p}}}N(W)$, with no additional relative sign. `hi_single_root_checks.gp` reproduces these assertions; `hi_prime_power_signs.py` separately checks (40.1) on arbitrary phases and all sign choices at several prime powers.

The prime-power route to infinitude now has an explicit sign procedure. What remains is a uniform source of the top Kummer root and a proof of the balanced identity for the selected phases, or a construction of physical quadratic solutions together with a uniform proof that (Q) implies (C). Neither finite rank tests nor norm compatibility establish those missing statements.

41. A weighted-design formulation and an obstruction to Alltop deformation

The quadratic system has a uniform formulation in finite-dimensional frame theory. This gives another possible route to constructing its physical solutions, independent of Stark algebraicity. It also permits a rigorous test of a tempting deformation argument.

41.1. **The Weyl orbit plus one basis is a weighted 2-design.** First allow a variable exceptional amplitude:

$$b_0 = -t, \qquad |b_j| = 1 \ (j \neq 0), \qquad b_{-j} = \overline{b_j}, \qquad t > 0.$$

Set $u = t^2$, $s = n - 1 + u$, and $v = b/\sqrt{s}$. Consider the zero-correlation equations

$$\sum_m b_m b_{h-m} b_{-m-g} b_{m+g-h} = 0 \quad (g, h \neq 0). \tag{41.1}$$

At $t = d^{-1/2}$, these are exactly the nontrivial equations (Q). Define

$$a = \frac{(u-1)^2}{s^2}, \qquad \beta = \frac{n-2+2u}{s^2}, \qquad w = n(\beta - a).$$

The identity $a + n\beta = 1$ holds for every u.

Let X, Z be the cyclic shift and modulation on $\mathbb{C}^n$, with $Ze_j = e^{2\pi ij/n}e_j$, and put

$$P_{h,k} = |X^h Z^k v\rangle\langle X^h Z^k v|, \qquad E_j = |e_j\rangle\langle e_j|.$$

Then (41.1) is equivalent to the tensor identity

$$\boxed{\sum_{h,k} P_{h,k} \otimes P_{h,k} + w \sum_j E_j \otimes E_j = n\beta\,(I + \mathsf{Swap}).} \tag{41.2}$$

When $w > 0$, this says that the Weyl orbit, with weight 1 per vector, together with the coordinate basis, with weight w per vector, forms a weighted complex projective 2-design. The weights in (41.2) are not normalized to sum to 1. Weighted 2-designs and their relation to mutually unbiased bases are discussed in Roy–Scott, *Weighted complex projective 2-designs from bases*.

Proof. The norm pattern gives squared Weyl overlaps 1 at the identity and a at every nonidentity diagonal Weyl operator. Fourier transforming (41.1) gives squared overlap β for each Weyl operator with nonzero shift, and conversely. The frame operator

$$\mathcal{F}(Y) = \sum_{h,k} \operatorname{tr}(P_{h,k} Y) P_{h,k}$$

is diagonal on the orthogonal Weyl basis of matrices, with eigenvalues n, na, and $n\beta$ on the identity, traceless diagonal matrices, and off-diagonal Weyl operators, respectively. The coordinate-basis frame operator is diagonal projection. Adding w times this projection gives

$$\mathcal{F}(Y) + w\,\operatorname{diag}(Y) = n\beta(Y + \operatorname{tr}(Y) I),$$

since $n + w = n(n+1)\beta$. Writing this equality in matrix entries gives (41.2). Each step is reversible. □

At the HI amplitude, $s = d - 1$, $a = d^{-2}$, $\beta = d^{-1}$, and

$$\boxed{w = \frac{n(d-1)}{d^2}, \qquad \sum_{h,k} P_{h,k}^{\otimes 2} + w \sum_j E_j^{\otimes 2} = \frac{n}{d}(I + \mathsf{Swap}).} \tag{41.3}$$

Thus constructing these particular weighted designs would solve (Q). A general weighted-design existence theorem is insufficient: the single cyclic Weyl orbit and the exceptional coordinate amplitudes are essential restrictions. Equation (41.3) does not establish (C).

41.2. **Explicit Alltop solutions occur at the wrong amplitude.** For a prime $p \geq 5$, write $\zeta = e^{2\pi i/p}$ and choose $c \in \mathbb{F}_p^\times$. The sequence

$$b_m = -\zeta^{cm^3}$$

satisfies (41.1) at $t = 1$, since its summand is

$$\zeta^{-3cgh(g-h+2m)}$$

and its nonconstant character sums to zero. Here $w = 1$, and the design is the familiar Alltop construction from mutually unbiased bases; see Blanchfield, *Orbits of mutually unbiased bases*. The required HI amplitude is $t = d^{-1/2} < 1$.

Theorem 41.1 (Rigidity)**.** *For every prime $p \geq 5$, each of these Alltop solutions is an isolated solution of (41.1) in the enlarged real phase space in which t may vary and $b_{-m} = \overline{b_m}$ is retained. In particular, a continuous deformation within this space cannot start at an Alltop solution and reach the HI amplitude.*

Proof. Write a first-order variation as

$$\dot{b}_m = i\theta_m b_m \quad (m \neq 0), \qquad \theta_{-m} = -\theta_m, \quad \theta_0 = 0, \qquad \dot{b}_0 = -\dot{t}.$$

Let $F_\ell = \sum_m \theta_m \zeta^{\ell m}$. For a nonzero pair g, h, put

$$\ell = -6cgh, \quad A = \zeta^{-3cgh(g-h)}, \quad R = \zeta^{\ell h}, \quad S = \zeta^{-\ell g}.$$

Differentiating the corresponding equation (41.1) gives exactly

$$\dot{Q}_{g,h} = A\left(i(1-R)(1-S)F_\ell + (1+R)(1+S)\dot{t}\right). \tag{41.4}$$

The four origin contributions are counted with multiplicity when two origins coincide. All factors $1-R$ and $1-S$ are nonzero.

Suppose the full differential vanishes and $\dot{t} \neq 0$. Fix $gh = 1$, hence $\ell = -6c$. Equation (41.4) would require

$$\frac{1+\zeta^{\ell h}}{1-\zeta^{\ell h}}\frac{1+\zeta^{-\ell g}}{1-\zeta^{-\ell g}}$$

to be independent of the choice of (g,h). This is an equality in $\mathbb{Q}(\zeta)$. Applying the automorphism $\zeta \mapsto \zeta^{\ell^{-1}}$, and comparing $(g,h) = (1,1)$ with $(2,2^{-1})$, would give

$$\cot^2(\pi/p) = \cot(2\pi/p)\cot(\pi\,[2^{-1}]_p/p).$$

This is impossible in absolute value: for residues other than ± 1, the absolute cotangent is strictly smaller than $\cot(\pi/p)$, and both 2 and 2^{-1} are such residues when $p \geq 5$. Hence $\dot{t} = 0$.

Every nonzero ℓ occurs as $-6cgh$, so (41.4) now forces $F_\ell = 0$ for every $\ell \neq 0$. Oddness gives $F_0 = 0$, and Fourier inversion yields $\theta_m = 0$. The real Jacobian therefore has full column rank. Selecting an invertible square minor and applying the inverse function theorem proves local isolation, not merely absence of one chosen tangent. □

`hi_design_alltop.py` checks the weight identities symbolically, the cotangent obstruction by exact reduction modulo cyclotomic polynomials, and the differential formula on arbitrary variations. It also checks (41.2) directly as a tensor equation for Alltop examples and for the recorded HI solution at order 13. These computations support the displayed proofs; they do not assert infinitude.

The design formulation remains available as a construction target. The rigidity theorem rules out one specific way of reaching it: locally varying the exceptional amplitude of a cubic Alltop sequence while preserving all quadratic correlations.

42. Exact amplitude branches at orders 5 and 7

The varying-amplitude system in Section 41 also permits an exact algebraic comparison of three different design problems. This comparison does not assume the HI value of the amplitude at the outset.

Work in the polynomial ring

$$R_n = \mathbb{Q}[x_1, \ldots, x_{n-1}, v, t], \qquad b_0 = -t, \quad b_j = x_j \ (j \neq 0),$$

and impose

$$I_n = \left(x_j x_{-j} - 1, \quad vt - 1, \quad \sum_m b_m b_{h-m} b_{-m-g} b_{m+g-h} \ (g,h \neq 0)\right). \tag{42.1}$$

The variable v excludes zero amplitude. This is a complex algebraic scheme: the equations $x_j x_{-j} = 1$ do not by themselves require $x_{-j} = \overline{x_j}$. Physical points are the points with that additional conjugation condition and $t > 0$.

Exact elimination certificates. Singular's rational-arithmetic `std` calculation, followed by elimination, gives

$$\boxed{I_5 \cap \mathbb{Q}[t] = ((t^2-1)(t^4+5t^2-1)),} \tag{42.2}$$

$$\boxed{I_7 \cap \mathbb{Q}[t] = ((t^2-1)(t^4+7t^2-1)(t^4-4t^2-4)).} \tag{42.3}$$

Both quotient rings are zero-dimensional. The complete scheme lengths and the lengths after adjoining each indicated factor are:

Order	Total	t^2-1	t^4+nt^2-1	$t^4-4t^2-(n-3)$
5	120	40	80	0
7	308	84	168	56

These are scheme lengths over $\mathbb{Q}$, including every algebraic amplitude conjugate. They are not counts of physical solutions. The coprime factors give disjoint algebraic branches, and their lengths sum to the full length. In particular, the HI factor represents an actual nonempty component in both calculations, rather than merely an annihilating polynomial which could have extraneous factors.

To interpret the three factors, set $u = t^2$:

- $u = 1$ is the flat Alltop amplitude.
- $u^2 + nu - 1 = 0$ has the unique positive root $u = 1/d$, the HI amplitude.
- $u^2 - 4u - (n-3) = 0$ is exactly the condition $w = 0$ in (41.2). At a physical point, the Weyl orbit alone is then a SIC: indeed $a = \beta = 1/(n+1)$. Its positive amplitude is $u = 2 + \sqrt{n+1}$.

The third interpretation follows directly from the formulas of Section 41 and does not use a conjecture about SIC existence. The order-7 algebraic branch has positive length; the table alone does not certify which of its points satisfy physical conjugation.

`hi_amplitude_ideal.py` generates the equations, computes the exact Gröbner basis, checks both inclusions of the displayed elimination ideal, and computes all branch lengths. Its saved output is `hi_amplitude_ideal_results.txt`.

This gives a concrete algebraic question for a uniform construction: prove that the fiber

$$\operatorname{Spec}(R_n/(I_n + (t^4 + nt^2 - 1))) \tag{42.4}$$

has a physical point for unbounded odd n. The two small elimination certificates do not prove a general amplitude-factorization formula, nor does their nonemptiness imply nonemptiness at larger orders. Even a uniform construction of physical points in (42.4) would still require the cubic identity, or a uniform quadratic-to-cubic theorem. The isolation result in Section 41 explains why simply deforming the flat component is not enough to establish (42.4).

43. A uniform quotient by the cyclic character gauge

The prefix formula (29.1) can be strengthened to an equivalence of algebraic schemes, without any Stark-unit hypotheses. This gives a smaller coordinate ring for the remaining quadratic-to-cubic problem.

Let $n = 2m+1$, $K = \mathbb{Q}(d)$, and

$$\mathcal{C}_n = K[c_1, \dots, c_{n-1}]/(c_j c_{n-j} - d).$$

All the c_j are units. For $2 \le h \le n-1$, put

$$r_h = \frac{c_1 c_{h-1}}{c_h} = B_{1,h}.$$

Then $r_h = r_{n+1-h}$, so there are just m independent row variables $r_2, \dots, r_{m+1}$. Extend any choice of these nonzero variables by that reflection and set

$$P_1 = 1, \qquad P_h = \prod_{j=2}^{h} r_j, \qquad \mathcal{R}_n = K[r_2^{\pm 1}, \dots, r_{m+1}^{\pm 1}].$$

Proposition 43.1. *There is an explicit isomorphism*

$$\boxed{\mathcal{C}_n \cong \mathcal{R}_n[a]/(a^n - dP_{n-1}), \qquad c_h = \frac{a^h}{P_h}.} \tag{43.1}$$

In particular, $\mathcal{C}_n$ is a finite free, faithfully flat, étale algebra of rank n over $\mathcal{R}_n$.

Proof. In the forward direction, the recurrence for r_h gives $c_h = c_1^h/P_h$. At $h = n-1$, the norm relation $c_1 c_{n-1} = d$ gives $c_1^n = dP_{n-1}$. Conversely, reflection of the row variables gives $P_h P_{n-h} = P_{n-1}$. Thus the displayed formulas yield $c_h c_{n-h} = a^n/P_{n-1} = d$, and the two constructions are inverse. The monic relation gives the free basis $1, a, \dots, a^{n-1}$. Both a and n are invertible, so its derivative na^{n-1} is a unit, proving étaleness. □

Over an algebraic closure, the n points over one row are exactly $c_h \mapsto \zeta^{jh} c_h$, $0 \le j < n$, where $\zeta^n = 1$. This is the cyclic character gauge. The invariant ring is $\mathcal{R}_n$, and the entire matrix is recovered by

$$B_{g,h} = \frac{P_h}{P_g P_{h-g}} \quad (0 < g < h < n), \qquad B_{h,g} = \frac{d}{B_{g,h}}, \tag{43.2}$$

together with the fixed axes, diagonal, and origin.

For the physical embedding $d > 0$, the physical row conditions are exactly $|r_h| = \sqrt{d}$. They imply $|P_h| = d^{(h-1)/2}$ and $|a| = \sqrt{d}$, hence $|c_h| = \sqrt{d}$ and $c_{n-h} = \overline{c_h}$. Conversely, physical phase data give those row norms. Thus (43.1) also describes the physical phase space as a finite covering, with no omitted sign or continuous parameter.

Let $J_n \subset \mathcal{R}_n$ be the ideal of the scaled quadratic equations after (43.2). For $g, h \neq 0$, direct substitution gives

$$\sum_j B_{g+j,h} B_{h,j} = \frac{1}{d} \sum_m c_m c_{h-m} c_{-m+g} c_{m-g-h}.$$

Thus the phase quadratic ideal of Section 35, after the already imposed norm relations, is exactly the extension of J_n to $\mathcal{C}_n$. Faithful flatness therefore gives

$$f \in J_n \quad \Longleftrightarrow \quad f \in J_n \mathcal{C}_n \qquad (f \in \mathcal{R}_n). \tag{43.3}$$

In particular, the cubic ideal-membership problem can be carried out in the smaller row ring without weakening it. Whenever these quotients are finite-dimensional,

$$\dim_K(\mathcal{C}_n/J_n\mathcal{C}_n) = n \, \dim_K(\mathcal{R}_n/J_n). \tag{43.4}$$

`hi_row_ideal.py` implements this reduction. It clears Laurent monomials and adds one equation in an auxiliary variable to invert the product of the row variables. Its exact computations at orders 3, 5, 7, and 9 give row scheme lengths 2, 4, 6, and 12, respectively, with all cubic remainders zero. Multiplication by n reproduces the phase scheme lengths 6, 20, 42, and 108 from Section 35.

The proposition applies to every odd order. It reduces the size of the uniform existence problem but does not prove that its physical quadratic locus is nonempty or that its cubic ideal is contained in its quadratic ideal for unbounded orders.

44. Diagonal quadratics and Galois covariance are not sufficient

There is an exact physical counterexample to a further tempting reduction. At $n = 5$, even the phase cocycle, every generic norm, full multiplier Galois covariance, and all the diagonal equations $Q_{g,g} = 0$ together do **not** imply the remaining quadratics. This does not contradict Section 40: that section uses one quadratic to select signs of an already specified arithmetic candidate, and then still checks all of (Q).

Put $d = (5 + \sqrt{29})/2$ and let r be a root of

$$f(X) = X^4 + (d-2)X^3 + dX^2 + (3d+1)X + d^2. \tag{44.1}$$

This polynomial is irreducible over $K = \mathbb{Q}(d)$. Define

$$R(X) = \frac{(d-6)X^3 + (6-d)X^2 + (13-3d)X + 3 - 3d}{5}, \qquad r_2 = R(r).$$

Exact reduction modulo f proves

$$f(R(r)) = 0, \qquad R(R(r)) = d/r.$$

Thus $L = K(r)$ is cyclic quartic over K, with generator $\phi(r) = R(r)$, and $\phi^2(r) = d/r$.

Take the reflected row $(B_{1,2}, B_{1,3}, B_{1,4}) = (r, r_2, r)$ and reconstruct the matrix by (43.2). Exact reduction gives

$$\phi(B_{g,h}) = B_{2g,2h}, \qquad \phi^2(B_{g,h}) = B_{h,g}.$$

These are genuine physical conjugation identities. Indeed, dividing (44.1) by r^2 shows that the two possible traces $v = r + d/r$ satisfy

$$v^2 + (d-2)v - d = 0.$$

Both roots lie strictly between $-2\sqrt{d}$ and $2\sqrt{d}$, so every physical root of f has modulus $\sqrt{d}$. The real-coefficient formula for R and the formal norm identity give $|r_2| = \sqrt{d}$ as well. Section 43 then supplies physical phases.

Nevertheless, writing the scaled quadratic residual as

$$\mathcal{Q}_{g,h} = \sum_{j \in \mathbb{Z}_5} B_{g+j,h} B_{h,j}, \qquad g, h \neq 0,$$

we get exactly

$$\boxed{\mathcal{Q}_{g,h} = \begin{cases} 0, & h = \pm g, \\ 2(d-1), & h \neq \pm g. \end{cases}} \tag{44.2}$$

Thus there are eight nonzero residuals despite every diagonal one vanishing and full Galois covariance holding. All these identities, irreducibility, and the exact inequalities on the traces are checked by `hi_diagonal_counterexample.py`.

The one-root condition does distinguish this counterexample. By (43.1), a root $W = c_1$ inside its row field would have to satisfy

$$W^5 = d\, r^2 R(r).$$

There is **no** such root in L. The exact `nfroots` calculation in `hi_diagonal_kummer.gp` proves this using the absolute polynomial

$$X^8 + X^7 - 2X^6 + 5X^5 - 9X^4 - 60X^3 + 2X^2 + 12X + 1$$

and the embedding

$$d = -\frac{r^4 - 2r^3 + r + 1}{r^3 + r^2 + 3r + 5}.$$

The absolute field has degree 8, signature $(4, 2)$, and discriminant $5^5\, 29^4$. It also fails the prescribed ray conductor condition: the prime with $d \equiv 1 \pmod 5$ has ramification index 4, while the conjugate prime with $d \equiv -1 \pmod 5$ has ramification index 2. Both primes ramify, whereas the intended modulus contains only the first. These local assertions are checked in the same GP script. Consequently the arithmetic requirement that the phase root already lie in the prescribed abelian field is substantive. This example suggests testing that condition together with the specified conductor as a way to exclude spurious diagonal components. It does not prove that the condition excludes every such component in arbitrary order, nor replace the missing uniform quadratic identity.

45. Exact quadratic-to-cubic containment at order 11

The row reduction in Section 43 now extends the ideal theorem to $n = 11$. This certifies a statement about **every** phase solution of the quadratic equations at that order, not just the previously constructed secondary Stark branch.

Theorem 45.1 (exact computation)**.** *At $n = 11$, every cross-zero-free cubic polynomial lies in the quadratic phase ideal (35.1). Hence $(Q) \Rightarrow (C)$ for every phase solution. The row quotient has K-dimension 12, and the phase quotient has dimension 132.*

The proof uses a homogeneous certificate to avoid an issue with unverified modular reconstruction. A candidate Gröbner basis of a nonhomogeneous ideal can pass finite-field comparisons and exact forward inclusion without certifying the reverse inclusion. Zero cubic remainders against that candidate alone would not prove the theorem. The following exact argument supplies the missing step.

Lemma 45.2 (Hilbert comparison)**.** *Let I be a homogeneous ideal over a number field K, specified by homogeneous generators integral at a prime $\mathfrak{q}$, and let H be another homogeneous ideal over K. Suppose that*

(1) $I \subseteq H$ is verified exactly;
(2) the Hilbert series of $K[\mathbf{x}]/H$, computed from a certified Gröbner basis, equals that of $k_{\mathfrak{q}}[\mathbf{x}]/\overline{I}$.

Then $I = H$.

Proof. In each degree, form the matrix whose rows are the appropriate monomial multiples of the specified generators of I. Its rank can only decrease on reduction modulo $\mathfrak{q}$. Therefore

$$\dim_K(K[\mathbf{x}]/H)_a \le \dim_K(K[\mathbf{x}]/I)_a \le \dim_{k_{\mathfrak{q}}}(k_{\mathfrak{q}}[\mathbf{x}]/\overline{I})_a.$$

Equality of the outer Hilbert functions forces equality throughout in every degree. The inclusion $I \subseteq H$ is therefore an equality. □

For the order-11 calculation, homogenize the 15 distinct nonzero row quadratic generators and the product-inversion equation with a new variable z. A saved homogeneous candidate has 816 generators. Its standard basis is certified by an **exact rational-arithmetic** `std` computation, treating d as a final block variable and adjoining $d^2 - 11d - 1$. Passing back to the coefficient field $K = \mathbb{Q}(d) = \mathbb{Q}(\sqrt{5})$ preserves the standard-basis property for the ordinary variables in this block order.

The independent reduction uses the prime 32003. The polynomial $d^2 - 11d - 1$ is irreducible modulo this prime, so the residue field has 32003^2 elements. Exact `std` of the reduced **original** homogeneous generators gives the same Hilbert-series numerator:

$$\begin{aligned} N(T) = 1 &- 6T^6 - 4T^7 - 3T^8 - 2T^9 - T^{10} + 93T^{11} + 391T^{12} \\ &- 1728T^{13} + 2404T^{14} - 1587T^{15} + 500T^{16} - 58T^{17}. \end{aligned}$$

Both Hilbert series are $N(T)/(1-T)^7$. The exact input inclusion and the lemma prove equality of the homogeneous ideals.

After $z = 1$, exact inclusions identify the affine ideal with the saved 21-element row basis. Its scheme length is 12. All 17 beta representatives have zero exact remainder. The beta orbit symmetries and invariance of the full quadratic ideal under index multipliers cover the remaining cubic equations, as in Section 35. Faithful flatness from (43.3) transfers membership back to the phase ideal, and (43.4) gives its length $11 \cdot 12 = 132$.

The independent verifier is `hi_row11_verify.py`. It reads `hi_row11_homogeneous_basis.txt.gz` and `hi_row11_basis.txt`, regenerates the original equations, and performs all the checks above. Its saved output is `hi_row11_verification_results.txt`. The compressed large basis is an input to the verifier; its origin in modular reconstruction is not relied on for ideal equality.

The certified orders for the quadratic-to-cubic implication are now

$$\boxed{n = 3, 5, 7, 9, 11.}$$

There is still no proof of this implication for unbounded orders, and proving it uniformly would still leave physical existence of quadratic solutions to be established.

46. A degree obstruction to fixed-order cyclic hypergeometric formulas

A uniform analytic construction cannot use a bounded-degree nonsingular rational formula for consecutive phase ratios. This follows already from the exceptional amplitude, independently of (Q) or (C).

Proposition 46.1. *Let $b_j \neq 0$, $j \in \mathbb{Z}_n$, satisfy*

$$|b_0| = t \neq 1, \qquad |b_j| = 1 \quad (j \neq 0).$$

Let ζ be a primitive n-th root of unity. Suppose

$$\frac{b_j}{b_{j-1}} = \frac{P(\zeta^j)}{Q(\zeta^j)} \quad \text{for every } j \in \mathbb{Z}_n, \qquad Q(\zeta^j) \neq 0,$$

where $P, Q \in \mathbb{C}[z]$ have degrees at most r. Then

$$\boxed{2r \geq n - 2.}$$

In particular, for odd n, $r \geq (n-1)/2$.

Proof. Define the reflected polynomials

$$P^*(z) = z^r \overline{P(1/\overline{z})}, \qquad Q^*(z) = z^r \overline{Q(1/\overline{z})}.$$

The polynomial $F = PP^* - QQ^*$ has degree at most $2r$. For $j = 2, \dots, n-1$, the consecutive ratio has modulus 1, so $F(\zeta^j) = 0$. These are $n-2$ distinct roots. But

$$F(1) = |P(1)|^2 - |Q(1)|^2 = (t^2 - 1)|Q(1)|^2 \neq 0,$$

so F is not the zero polynomial. The degree bound follows. □

For comparison, a standard nonsingular cyclic hypergeometric weight

$$W(j) = \gamma^j \frac{(\alpha;\zeta)_j}{(\beta;\zeta)_j}$$

has a consecutive ratio of degree at most 1. Products of k such weights have ratio degree at most k. The standard weight, its periodicity condition, and the distinction between cyclic sums and unrestricted root-of-unity limits are given in Au-Yang–Perk, *About 30 Years of Integrable Chiral Potts Model*, Section 4.2. Thus a product with a fixed number of nonsingular factors cannot produce HI phases for unbounded orders. A formula with a growing number of factors, a different special function, or a limit requiring separate regularization at the exceptional indices is outside the proposition. Such a limit would still need its own proof of the finite beta identity and the collision corrections.

47. Unconditional norm elements in every prescribed odd-order ray field

The multiplicative and physical part of the arithmetic ansatz can be made nonempty uniformly, without assuming Stark algebraicity. This is an existence theorem for **norm elements**, not for solutions of the additive Izumi equations or for roots of a prescribed Stark unit.

Fix any odd $n \geq 3$, put

$$d = (n + \sqrt{n^2+4})/2, \quad K = \mathbb{Q}(d), \quad \mathfrak{f} = (d-1),$$

and let v_+ be the embedding with $d > 0$. For any nontrivial ideal divisor $\mathfrak{a} \mid \mathfrak{f}$, let $L_\mathfrak{a}$ be the full ray class field of modulus $\mathfrak{a}v_+$.

47.1. **A local–global norm theorem.**

Theorem 47.1. *Physical complex conjugation defines a nontrivial involution $c \in \mathrm{Gal}(L_{\mathfrak{a}}/K)$. Its fixed field $M_{\mathfrak{a}} = L_{\mathfrak{a}}^{c}$ is totally real, and*

$$\boxed{\exists W \in L_{\mathfrak{a}}^{\times} \quad Wc(W) = d.} \tag{47.1}$$

In particular, every embedding above v_+ has $|W| = \sqrt{d}$. The element W need not be an algebraic unit.

Proof. The unit d has norm -1. Choose the fundamental unit $\epsilon > 1$ at v_+; it has norm -1, and $d = \epsilon^m$ for an odd integer m. Since $d \equiv 1$ modulo $\mathfrak{a}$, the residue of ϵ has odd order. No unit negative at v_+ can be congruent to 1 modulo $\mathfrak{a}$: such a unit is $-\epsilon^j$, which would require an odd-order element to have a power equal to -1. Here $-1 \neq 1$ because $N\mathfrak{a} > 1$ is odd.

Consequently the sign part of the ray-class exact sequence survives, and v_+ ramifies in the full ray field. The other real place, which is omitted from the modulus, splits into real places. Abelianity over K makes conjugation at every embedding above v_+ the same nontrivial involution c. Its fixed field is totally real.

Consider the quadratic extension $L_{\mathfrak{a}}/M_{\mathfrak{a}}$. The element d is a norm at every completion:

- Above v_+, the extension is $\mathbb{C}/\mathbb{R}$, and $d > 0$.
- Above the other real place, the local algebra is $\mathbb{R} \times \mathbb{R}$, whose norm takes every nonzero real value.
- Away from $\mathfrak{a}$, every finite local extension is unramified or split. The element d is a unit, and units are norms in an unramified quadratic local extension.
- At a finite place above $\mathfrak{a}$, the residue characteristic is odd and $d \equiv 1$. Hensel's lemma makes d a square in the base completion. A base-field square is a norm from any quadratic extension.

The cyclic Hasse norm theorem now gives (47.1); see Milne, *Class Field Theory*, the Hasse norm theorem in VIII.3. At the physical embeddings, c is actual complex conjugation, so the displayed norm equation proves the asserted modulus. □

The crucial distinction is that (47.1) does not prescribe W^4/d^2. It therefore does not prove that the distinguished Stark unit satisfies the Kummer-root condition in Section 38.

47.2. **Physical Galois-covariant phase data for every odd order.** The theorem supplies an unconditional input space for all the multiplicative matrix constraints, including at composite orders. First observe that $\mathfrak{f}\bar{\mathfrak{f}} = (n)$ and that these two ideals are coprime: a prime dividing both would force $d \equiv \bar{d} \equiv 1$ and hence $1 \equiv -1$, so would lie above 2. Thus every prime dividing n splits in K, and $\mathcal{O}_K/\mathfrak{f} \cong \mathbb{Z}/n\mathbb{Z}$. For each divisor $q > 1$ of n, there is a unique ideal $\mathfrak{f}_q \mid \mathfrak{f}$ of norm q. Write $L_q = L_{\mathfrak{f}_q} \subseteq L_n$.

Choose $w_q \in L_q^{\times}$ with $w_q c(w_q) = d$ by (47.1). For $a \neq 0$ modulo n, write

$$q = n/\gcd(a, n), \qquad a = (n/q)u, \qquad (u, q) = 1,$$

and define

$$W_a = \sigma_u(w_q), \qquad b_a = W_a/\sqrt{d}, \qquad b_0 = -1/\sqrt{d}. \tag{47.2}$$

Here σ_u is the ray automorphism labeled by a positive integer lift of $u \pmod q$. This convention includes $\sigma_{-1} = c$; it does not mean the Artin symbol of the unit ideal (-1). The definition is independent of the lift.

For every unit k modulo n, restriction of ray automorphisms and (47.1) give

$$\sigma_k(W_a) = W_{ka}, \qquad W_{-a} = c(W_a) = d/W_a, \qquad |b_a| = 1.$$

Therefore the matrix

$$B_{g,h} = W_g W_{h-g}/W_h \quad (g, h, h-g \neq 0),$$

with the usual exceptional entries, has the physical norms, Hermitian and rotation symmetries, phase cocycle, and every multiplier covariance identity. All its entries belong to the prescribed full ray field L_n. If $s_a = W_a^4/d^2$, then it also satisfies

$$B^4_{g,h} = d^2 s_g s_{h-g}/s_h.$$

These s_a are norm-one elements; integrality and their identification with analytic Stark units have not been asserted. This constructs the multiplicative data at **every odd order**. It makes no assertion that (Q) or (C) vanishes for the chosen data.

For prime powers, one can additionally impose compatible norms throughout the conductor tower. If $n = p^k$, let $L_j = L_{\mathfrak{p}^j}$ and choose W_k by (47.1). The degrees $r_j = [L_k : L_j]$ are odd powers of p. Set

$$W_j = d^{(1-r_j)/2} N_{L_k/L_j}(W_k).$$

Then $W_j c(W_j) = d$, the normalized norm descents are transitive, and the corresponding $s_j = W_j^4/d^2$ satisfy $N_{L_j/L_i}(s_j) = s_i$. Applying the conjugate-prime twists from Section 39 gives the monoid labeling there. This is an unconditional multiplicative norm system, still with no specified Stark units and no additive identity.

47.3. **What remains after the norm construction.** Given one W_0 satisfying (47.1), Hilbert's theorem 90 gives all such elements as

$$W = W_0 \frac{U}{c(U)}, \qquad U \in L_{\mathfrak{a}}^{\times}. \tag{47.3}$$

Equivalently, choose a nonzero $v \in L_{\mathfrak{a}}$ with $c(v) = -v$. Apart from one point, the norm conic has the rational parameterization

$$W = W_0 \frac{1 + zv}{1 - zv}, \qquad z \in M_{\mathfrak{a}}.$$

This follows directly by solving for z, or from Milne, *Fields and Galois Theory*, Hilbert's theorem 90.

The permitted algebraic norm elements are dense in their possible physical phase tuples. Indeed, the Minkowski embedding identifies the rational vector space $L_{\mathfrak{a}}$ with a dense rational subspace of its real archimedean vector space. Approximating the chosen complex values of U in (47.3) approximates any prescribed phases, with opposite embeddings paired by conjugation. This density does **not** imply an exact zero of the quadratic or cubic equations. It emphasizes that those additive identities remain substantive.

For example, at the limiting phase tuple $b_a = 1$ for $a \neq 0$, with $t = d^{-1/2}$, the diagonal correlation is

$$\sum_m b_m^2 b_{1-m} b_{-1-m} = n - 3 + t^2 - 2t \neq 0.$$

It is negative when $n = 3$ and positive when $n \geq 5$. Hence nearby admissible algebraic norm elements also fail (Q). The norm theorem cannot be promoted to HI existence by omitting this step.

`hi_ray_norm.gp` constructs the ray fields using `bnrclassfield`, finds the totally real fixed subfield, solves its relative quadratic norm equation, and checks $Wc(W) = d$ exactly. It uses neither Stark polynomials nor HI seeds. The completed checks at orders 3, 5, 7, 9, and 11 are recorded in `hi_ray_norm_results.txt`; they illustrate the uniform theorem rather than replace its local–global proof.

48. An explicit integral norm unit for every order congruent to 3 modulo 4

The preceding norm theorem can be strengthened to **unit** existence on an unconditional infinite family. This supplies more of the arithmetic input, while leaving the additive Izumi identities to be proved.

48.1. **The conductor calculation.**

Theorem 48.1. *Let $n \geq 3$ and $n \equiv 3 \pmod 4$, and retain $K = \mathbb{Q}(d)$, $\mathfrak{f} = (d-1)$, and the full ray field $L = L_{\mathfrak{f}v_+}$. Then*

$$L_0 = K(\sqrt{1-d}) \subseteq L, \qquad W_0 = 1 + \sqrt{1-d} \in \mathcal{O}_L^\times, \qquad W_0 c(W_0) = d. \tag{48.1}$$

In particular, integral norm elements exist at every prime $p \equiv 3 \pmod 4$, an infinite set of orders.

Proof. The element

$$\theta = \frac{d + \sqrt{1-d}}{2}$$

satisfies the monic integral polynomial

$$X^2 - dX + \frac{n+1}{4}d.$$

Its discriminant is $d^2 - (n+1)d = 1 - d$. Thus the relative field discriminant of L_0/K divides $\mathfrak{f}$; in particular there is no extra ramification at 2. The extension is quadratic, since $1 - d < 0$ at v_+, and has precisely the permitted archimedean ramification: at the other real place, $1 - \overline{d} = 1 + 1/d > 0$.

For a quadratic extension the finite conductor is its relative discriminant. Its full conductor therefore divides $\mathfrak{f}v_+$, proving the inclusion by class field theory. See Milne, *Class Field Theory*, V.3.27 and Example 3.29. Finally W_0 is integral and

$$N_{L_0/K}(W_0) = (1 + \sqrt{1-d})(1 - \sqrt{1-d}) = d.$$

Since d is a unit, W_0 is a unit. Physical conjugation restricts to the nontrivial automorphism of L_0/K, giving (48.1). □

This proof works for composite $n \equiv 3 \pmod 4$ as well. At a composite order it supplies a unit in the top conductor field; it does not assert unit-valued descent at every proper divisor stratum. For prime orders there is only one nonzero stratum.

The absolute polynomial of this particular unit is

$$X^4 - 4X^3 + (n+4)X^2 - 2nX - 1.$$

It follows by substituting $d = 2W_0 - W_0^2$ in $d^2 - nd - 1 = 0$. Its constant term also confirms the absolute unit norm -1.

48.2. **Primitive unit choices also exist in the full ray field.** The low degree of the explicit seed is not a restriction on the available norm units. Write $[L : K] = 2h$, and put

$$E^- = \{u \in \mathcal{O}_L^\times : uc(u) = 1\}.$$

The signature of L is $(2h, h)$, and L^c is totally real of degree $2h$. Dirichlet's unit theorem therefore gives $\operatorname{rank} E^- = h$. More concretely, its logarithms at the $2h$ real embeddings span the subspace on which c acts by -1; its logarithms at the complex embeddings vanish. The unit-lattice statement used here is the usual logarithmic form of Dirichlet's theorem, Milne, *Algebraic Number Theory*, Section 5.

The real embeddings over the other embedding of K form a regular $\mathrm{Gal}(L/K)$-set. Its anti-invariant subspace is faithful: the vector supported with values $1, -1$ at the positions $1, c$ is fixed by no nonidentity translation. Consequently every nonidentity σ acts nontrivially on $E^- \otimes \mathbb{R}$.

For any fixed $\sigma \neq 1$, each condition

$$\sigma(W_0 u) = W_0 u \quad \text{or} \quad \sigma(W_0 u) = -W_0 u$$

restricts $u \in E^-$ to an empty set or a coset of a subgroup of rank less than h. A finite union of these cannot cover the rank-h lattice. Choosing u outside it gives

$$K(W_0 u) = L, \qquad K\big((W_0 u)^4/d^2\big) = L. \tag{48.2}$$

For the second assertion, an automorphism fixing the fourth power would have $\sigma(W)/W \in \mu_4(L) = \{1, -1\}$, since L has real embeddings. These possibilities were both excluded.

Thus, at infinitely many prime orders, one can unconditionally choose **primitive integral units** satisfying all the norm and fourth-root multiplicative relations in the prescribed full ray field. The corresponding fourth powers have not been identified with distinguished analytic Stark units, and additive identities are not part of this conclusion.

48.3. **The explicit seed itself fails the quadratic equation uniformly.** It is possible to locate the missing step exactly even for the simple seed (48.1). Suppose $p \equiv 3 \pmod 4$ is prime and let χ be the quadratic character modulo p. The quadratic ray character restricts to χ on the integer residue classes, so its labeled phases in the scaled variables of Section 34 are

$$c_0 = -1, \qquad c_a = 1 + \chi(a) r \ (a \neq 0), \qquad r^2 = 1 - d.$$

Let $e = \chi(2)$. Then the necessary diagonal quadratic residual is

$$\boxed{\sum_{m \bmod p} c_m^2 c_{1-m} c_{-1-m} = \begin{cases} 2p + 2 - (p+6)d, & e = 1, \\ 2p - 14 + (10-p)d, & e = -1. \end{cases}} \tag{48.3}$$

Both expressions are nonzero. In the first case the coefficient of d is nonzero; in the second case $10 - p \neq 0$. Irrationality of d then excludes vanishing.

For completeness, pair m with $-m$ to cancel all odd powers of r. On the complement of $0, 1, -1$, the two character sums needed for the even part are

$$\sum \chi((m-1)(m+1)) = 0, \qquad \sum \chi(m)\big(\chi(m-1) + \chi(m+1)\big) = -2 - 2e.$$

They follow from the complete quadratic-character sums -1 by removing those three indices. The remaining sum is $(p-3)(2-d) + 4(1+e)(1-d)$. The three excluded terms sum to $-d$ when $e = 1$, and to $-8 + 7d$ when $e = -1$, giving (48.3).

Changing the overall sign of the nonzero phases changes only the terms with exactly one zero index. The residual for $-W_0$ is

$$2p + 2 - (p+2)d \neq 0,$$

independently of $\chi(2)$. Thus neither sign of the explicit seed solves even this one quadratic equation.

`hi_explicit_ray_unit.gp` independently verifies the integral-order discriminant, the unit norm, and (48.3) exactly at every prime $p \equiv 3 \pmod 4$ below 100. Its saved results accompany the uniform proof. This calculation shows why the explicit unit theorem alone is not an existence proof for HI categories.

49. An exact half-lattice reduction and a new unit-selection experiment

49.1. **The norm-unit coset is an exact arithmetic object.** Let $L, M = L^c$ be as above, and suppose a unit W_0 with $W_0 c(W_0) = d$ is available. Section 48 proves this hypothesis for every $n \equiv 3 \pmod 4$. All unit solutions are

$$\{W \in \mathcal{O}_L^\times : W c(W) = d\} = W_0 E^-. \tag{49.1}$$

There is also an exact test for the hypothesis at other orders. Choose fundamental units $u_1, \dots, u_r$ of L, let $C \in \mathrm{GL}_r(\mathbb{Z})$ describe c on their free exponent lattice, and let $\delta \in \mathbb{Z}^r$ be the free exponent vector of d. Then

$$d \in N_{L/M}(\mathcal{O}_L^\times) \quad \Longleftrightarrow \quad (I + C)x = \delta \text{ has } x \in \mathbb{Z}^r. \tag{49.2}$$

Indeed a solution makes the relative norm equal to d up to a root of unity. The only roots of unity are ± 1, and positivity of both the norm and d at a physical embedding forces the plus sign. The same argument shows that a basis of $\ker_{\mathbb{Z}}(I + C)$ gives norm-one units, with no missed negative-norm correction.

Choose a basis $v_1, \ldots, v_h$ of $E^-/\{1, -1\}$. Let $\tau_1, \ldots, \tau_{2h}$ be all the real embeddings of L, and write

$$\mathcal{L}_{ij} = \log|\tau_i(v_j)|, \qquad a_i(W) = \log|\tau_i(W)| + \tfrac{1}{2}\log d_+,$$

where $d_+ > 0$ denotes the physical value of d. Since $W_0^2/d \in E^-$, there is an **integer** vector κ with

$$W_0^2/d = \pm \prod_j v_j^{\kappa_j}.$$

At a real embedding $\tau_i(d) = -1/d_+$, and therefore

$$2a(W_0) = \mathcal{L}\kappa, \qquad \boxed{\{a(W) : Wc(W) = d,\ W \text{ a unit}\} = \mathcal{L}(\mathbb{Z}^h + \tfrac{1}{2}\kappa).} \tag{49.3}$$

The parity class κ mod 2 is independent of the initial norm unit. Both signs of W give the same logarithms. Thus minimizing the centered squared logarithmic size is exactly the shortest-vector problem

$$\min_{z \in \mathbb{Z}^h,\, z \equiv \kappa \ (2)} z^{\mathsf{T}}(\mathcal{L}^{\mathsf{T}}\mathcal{L})z. \tag{49.4}$$

The reported score in the computation below is this doubled-log score, four times $\|a(W)\|^2$.

49.2. **Exact finite Izumi certificates from short norm units.** `hi_ray_unit_search.gp` constructs the prescribed ray field, solves its norm equation, computes the action of physical conjugation on units, and searches (49.4). It uses no HI seeds and calls no Stark unit constructor. Logarithms order the candidates; acceptance requires **exact** full quadratic identities, every remaining balanced-beta cubic representative, and multiplier covariance. Candidates in the same exactly checked Galois orbit share that certificate.

The completed results are:

Order	Relative unit rank	Accepted vectors modulo conjugation in the observed lowest shell	Exact Q and C
3	1	1	pass
5	2	2	pass
7	3	3	pass
11	1	1	pass
13	6	6	pass
17	8	8	pass
29	2	2	pass
199	9	9	exact transport of the saved complete certificate

At each order the accepted vectors form one Galois orbit, after choosing their signs. The initially returned norm element failed both sign choices at 11, 13, and 17; the unit-lattice search found complete solutions at all three. This demonstrates a selection mechanism different from specifying a distinguished Stark unit.

At order 199, the new construction yields nine vectors in its observed lowest shell. Each one, with the appropriate sign, exactly matches the previous complete certificate through a field isomorphism that preserves d and the Artin generator. The saved certificate is used only after the unit-lattice construction, to identify and verify its output. This avoids repeating the 108,406 cubic representatives. The absolute unit group completeness and global logarithmic minimality are not certified at this order either; the transported finite Izumi certificate is exact. The result is saved in `hi_ray_unit199_results.txt`.

The optional explicit-unit mode now starts directly from (48.1), without a general norm solver. Its completed exact Izumi checks at 3, 7, and 11 are in `hi_ray_unit_explicit_results.txt`.

There are two separate limits on the shortest-shell claim. The logarithmic enumeration uses floating arithmetic and is not an interval certificate of global minimality. Also, completeness of the absolute fundamental-unit group was certified at 3, 5, 7, 11, and 29, while the expensive completeness certificates at 13 and 17 were not completed. At those two orders all units, conjugation relations, norm identities, and accepted Izumi identities were still verified exactly. Consequently the **finite existence certificates are unconditional**, while global shortest-shell classification is not claimed by these logs.

At rank one the geometric minimization itself is elementary once the full unit group and the parity class are certified: an odd coset has just two shortest integer vectors, 1 and -1. At order 11 this explains the correction of the initial candidate. For the unit basis used in the calculation, its exact twice-offset is 5, so the two nearest exponents relative to it are -2 and -3. These are precisely the two translates passing (Q), with opposite overall signs.

49.3. **A precise remaining uniform target.** For every prime $p \equiv 3 \pmod 4$, Sections 48–49 now provide an unconditional nonempty norm-unit coset and hence a nonempty finite shortest shell. A sufficient next theorem would be:

> For every such prime, some shortest norm unit, with one of its two signs, gives a matrix satisfying (Q) and (C) under the Artin labeling of Section 47.

This statement is **unproved**. Even a proof for an unbounded subset of these primes would establish the requested infinitude. The arithmetic construction, integrality, nonempty unit coset, and availability of primitive generators no longer need Stark conjectures on this prime family. What is still missing is a theorem connecting unit selection to the additive identities. The finite checks above do not establish that connection for unbounded orders.

50. The cubic equations are a fifth-order condition on one unitary operator

There is a uniform spectral formulation of the remaining cubic problem. It applies at every odd order and does not assume that quadratic equations imply cubic equations.

50.1. **A transfer operator with a universal three-dimensional part.** Let A be a physical matrix in the present phase ansatz satisfying all of (Q), and put $t = 1/d$. On $\mathcal{H} = \mathbb{C}^{\mathbb{Z}_n \times \mathbb{Z}_n}$, define

$$(Rf)(a,b) = \sum_{c \in \mathbb{Z}_n} A_{b,a+c} f(b,c). \tag{50.1}$$

Let u, v be unit vectors with

$$u(a,b) = n^{-1/2}\mathbf{1}_{b=0}, \qquad v(a,b) = n^{-1/2}\mathbf{1}_{a=0}, \qquad w = Ru.$$

The column correlations (Q), together with Hermitian symmetry, give

$$RR^* = I - (1-t^2)uu^*, \qquad R^*R = I - (1-t^2)vv^*. \tag{50.2}$$

Here $nt = 1 - t^2$. The fixed exceptional row also gives $\sum_a A_{0,a} = -t$, and direct multiplication yields

$$Rv = -tu, \qquad Rw = v - tu. \tag{50.3}$$

For the second identity, one uses

$$(R^2u)(a,b) = n^{-1/2} \sum_c A_{b,a+c} A_{c,b} = n^{-1/2}(\delta_{a,0} - t\delta_{b,0}).$$

The space $\mathcal{K} = \operatorname{span}(u, v, w)$ has dimension three. Its Gram matrix is

$$\begin{pmatrix} 1 & 1/n & -t/n \\ 1/n & 1 & -t/n \\ -t/n & -t/n & 1 - t/n \end{pmatrix},$$

whose determinant is

$$\frac{(2t+1)(t^2+t-1)^2}{(1-t^2)^3} > 0 \qquad (n \geq 3).$$

In this basis the restriction of R to $\mathcal{K}$ is

$$R_{\mathcal{K}} = \begin{pmatrix} 0 & -t & -t \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \qquad \det(XI - R_{\mathcal{K}}) = X^3 + tX + t. \tag{50.4}$$

This restriction is invertible. Equations (50.2) and invariance of $\mathcal{K}$ then show that $R^*\mathcal{K} \subseteq \mathcal{K}$: write $y = Rx$ with $x \in \mathcal{K}$, and use $R^*Rx = x - (1-t^2)v\langle v, x\rangle$. Thus $\mathcal{K}$ reduces R, and

$$U = R|_{\mathcal{K}^\perp} \quad \text{is unitary of dimension } n^2 - 3. \tag{50.5}$$

All of these statements use (Q) and the fixed matrix symmetries, with no use of (C).

50.2. **An exact identity for the complete cubic residual.** Write the residual of (C) in the A normalization as

$$\Phi_{g,h,k,l} = \sum_m A_{m,g+h}\overline{A_{m+k,g}A_{m+l,h}} - A_{g+l,k}A_{h+k,l} + t\delta_{g,0}\delta_{h,0}.$$

Then

$$\boxed{\sum_{g,h,k,l} |\Phi_{g,h,k,l}|^2 = 2\left(d^2 - 5 + 6d^{-2} - \operatorname{Re}\operatorname{Tr} R^5\right) = \|U^5 - I\|_{\mathrm{HS}}^2.} \tag{50.6}$$

In particular, under (Q), the following are equivalent:

$$\boxed{\text{all of (C)} \iff U^5 = I \iff \operatorname{Re}\operatorname{Tr} R^5 = d^2 - 5 + 6d^{-2}.} \tag{50.7}$$

Thus all the cubic equations can be replaced by one scalar fifth-trace identity, or by a fixed fifth-order condition on the unitary part.

Proof of the residual identity. Write $\Phi = X - Y + Z$, with X the summed term, Y the product of two entries, and $Z = t\delta_{g,0}\delta_{h,0}$. Column orthogonality and the exceptional column give

$$\begin{aligned} \|X\|^2 &= n^2 - 3nt + 2t^2 + t^4, \\ \|Y\|^2 &= (n-t)^2, \\ \|Z\|^2 &= n^2t^2, \\ \sum X\overline{Z} &= -t^4, \\ \sum Y\overline{Z} &= t(n-t). \end{aligned}$$

For example, when summing $|X|^2$ over k, l, (Q) gives $(\delta_{m,m'} - t\delta_{g,0})(\delta_{m,m'} - t\delta_{h,0})$. Summing the remaining indices produces the first displayed expression.

The remaining cross term is a cyclic contraction of five entries:

$$\sum X\overline{Y} = \sum_{x_1,\ldots,x_5} \prod_{i=1}^{5} A_{x_i, x_{i-1}+x_{i+1}} = \operatorname{Tr} R^5, \tag{50.8}$$

with indices i taken modulo 5. The equality follows by Hermitian symmetry and the cyclic ordering (m, g, k, l, h). Combining the six terms and substituting $n = d - d^{-1}$ gives the first equality in (50.6).

The cubic in (50.4) gives $\operatorname{Tr} R_{\mathcal{K}}^5 = 5t^2$, either by Newton's identities or direct multiplication. Therefore unitarity of U gives

$$\|U^5 - I\|_{\mathrm{HS}}^2 = 2(n^2 - 3 - \operatorname{Re}(\operatorname{Tr} R^5 - 5t^2)),$$

which is the same expression. Positivity of the two sums of squares proves (50.7). □

For the phase ansatz, $\overline{A_{g,h}} = A_{-g,-h}$. Conjugating every index by negation gives $\overline{R} = JRJ^{-1}$, so $\operatorname{Tr} R^5$ is real. Hence the scalar equality itself, without a real-part operation, is an exact number-field certificate in this setting.

50.3. **An equivalent polynomial test and the remaining gap.** The cubic $f(X) = X^3 + tX + t$ has no root on the unit circle: $|tX + t| \le 2t < 1 = |X^3|$. Hence $f(U)$ is invertible. Together with (50.4), this proves one more equivalent condition:

$$\boxed{(R^5 - I)(R^3 + d^{-1}R + d^{-1}I) = 0.} \tag{50.9}$$

This is a degree-eight matrix polynomial, with a degree independent of n. It eliminates the need to construct the orthogonal complement explicitly when checking an exact candidate.

Alternatively, the Hermitian operator $H = U + U^*$ has spectrum in $[-2, 2]$, and (C) is equivalent to

$$(H - 2I)(H^2 + H - I) = 0.$$

Thus its permitted eigenvalues are exactly $2, (\sqrt{5} - 1)/2, -(\sqrt{5} + 1)/2$. The same defect has the nonnegative spectral expression

$$\sum |\Phi|^2 = \operatorname{Tr}((2I - H)(H^2 + H - I)^2).$$

These formulas follow from $U^* = U^{-1}$ and the fifth Chebyshev identity. They provide an equivalent formulation using a Hermitian operator and three allowed eigenvalues.

The operator theorem does **not** assert that (50.7) follows from (Q) alone. It isolates the missing implication as a finite-order spectral constraint. A uniform proof that the arithmetic candidates satisfy both (Q) and this fifth-order condition would establish HI existence on the infinite prime family of Section 48.

`hi_pentagon_operator.gp` verifies (50.2), (50.3), the exact scalar trace certificate, and the polynomial test at orders 3, 5, and 7. It uses the unit construction and does not evaluate the original cubic loops. The companion Python file checks the indexing of the complete residual-norm calculation on a random phase input and the known order-13 solution; these latter checks are explicitly numerical. The proof above supplies the uniform statement.

51. An explicit analytic candidate at every odd order

The boundary-value experiment proposed in Sections 23 and 27 can now be replaced by a finite, directly evaluable double-sine formula. Throughout this section,

$$d = \frac{n + \sqrt{n^2 + 4}}{2} > 0, \qquad \delta = -d^{-1} < 0.$$

The resulting real matrices use δ, not the physical dimension d. They will be called **ghost candidates**. Numerical agreement with their equations is not an existence certificate.

51.1. **The continued-fraction cycle and its phase cancellation.**

Proposition 51.1. *For every odd $n \ge 3$, the reduced quadratic irrational $\alpha = d + 1$ has Hirzebruch–Jung cycle*

$$[\overline{n+2, \underbrace{2, \dots, 2}_{n-1}}], \qquad \beta_0 = d + 1, \qquad \beta_j = \frac{d - j + 1}{d - j} \quad (1 \le j \le n). \tag{51.1}$$

Here $\beta_n = \beta_0$, and $\prod_{j=1}^{n} \beta_j = d^2$. For the residue label $1 \le a < n$, set

$$v_j = \left\{ \frac{(j-1)a}{n} \right\}, \qquad r_{j,1} = v_{j-1} - 1, \quad r_{j,2} = v_j, \quad z_j = 1 - v_{j-1} + \beta_j v_j. \tag{51.2}$$

Indices on v are periodic modulo n. These are exactly the normalized characteristics in $[-1, 0) \times [0, 1)$.

Proof. The recurrence $\beta_{j+1} = 1/(b_j - \beta_j)$, with $b_0 = n + 2$ and $b_j = 2$ otherwise, gives (51.1) directly using $d - n = d^{-1}$. If $T = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ and $S = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, its word is

$$P = (T^{n+2}S)(T^2S)^{n-1} = \begin{pmatrix} n^2 + n + 1 & -n^2 \\ n & 1 - n \end{pmatrix} = T^{n+1} \begin{pmatrix} 1 & n \\ n & n^2 + 1 \end{pmatrix} T^{-(n+1)}.$$

The characteristic of Section 23 becomes $(-2a/n, -a/n)$ after this conjugation. Each inverse step acts by $(r_1, r_2) \mapsto (r_2, -r_1 + b_j r_2)$. Since $b_j \equiv 2 \pmod{n}$, the unnormalized characteristic at step j is $((j-2)a/n, (j-1)a/n)$ modulo integers, proving (51.2). □

Write $\mathrm{Sin}_2(z; \beta, 1) = \Gamma_2(\beta + 1 - z; \beta, 1)/\Gamma_2(z; \beta, 1)$. The explicit positive candidate is

$$\boxed{q_a = \prod_{j=0}^{n-1} \mathrm{Sin}_2(z_j; \beta_j, 1)^{-1}.} \tag{51.3}$$

All factors lie in their positive-period strip $0 < z_j < \beta_j + 1$. Kopp's double-sine cocycle formula identifies this product with the normalized RM value; the construction also applies to the order $\mathbb{Z}[d]$, which need not be maximal. Kopp, Sections 7.2–7.3.

In this specialization even the phase normalization is elementary. The two sums occurring in the double-sine transformation formula vanish:

$$\sum_j (\beta_j - 3 + \beta_j^{-1}) = 0, \qquad \sum_j \frac{(\beta_j - z_j)(1 - z_j)}{\beta_j} = 0. \tag{51.4}$$

For the first use $\sum_j b_j = 3n$. For the second, substitute (51.2) and pair β_j with β_{j+1}^{-1}. The sum becomes

$$\sum_j (v_j - v_{j-1})^2 - n v_0 (1 - v_0).$$

The differences are a/n on $n - a$ steps and $a/n - 1$ on a steps, including when a is not a unit modulo n. Their squared sum is $a(n-a)/n = n v_0 (1 - v_0)$. Thus the cocycle value in the convention of Section 23 is the positive number q_a itself, with no unresolved sign in this evaluation.

The order qualification matters. At $n = 11$, for example, $\mathbb{Z}[d]$ has conductor 5 in $\mathbb{Q}(\sqrt{5})$. Formula (51.3) must not silently be identified with the smaller maximal-order ray field used for the secondary branch. The formula defines its analytic numbers without requiring any such identification.

51.2. **Exact inverse symmetry and the proposed signs.**

Proposition 51.2. *The numbers in (51.3) satisfy*

$$q_a > 0, \qquad q_a q_{n-a} = 1. \tag{51.5}$$

Proof. For a step with $v_{j-1}, v_j \neq 0$, negating a replaces z_j by $\beta_j + 1 - z_j$, so reflection cancels the two double sines. A boundary step $v_j = 0$ is followed by the first-coordinate boundary $r_{j+1,1} = -1$. Put $x = 1 - v_{j-1} \in (0, 1)$. At the first step the product of the two double sines is $2\sin(\pi x)$; at the next it is $(2\sin(\pi x))^{-1}$. These follow from reflection and the two shifts $\mathrm{Sin}_2(z + \beta; \beta, 1) = \mathrm{Sin}_2(z; \beta, 1)/(2 \sin \pi z)$ and $\mathrm{Sin}_2(z + 1; \beta, 1) = \mathrm{Sin}_2(z; \beta, 1)/(2\sin(\pi z/\beta))$. The boundary steps pair disjointly because two successive v_j cannot both vanish for $0 < a < n$. This proves (51.5). □

The sign search at small orders suggests the fixed, elementary rule

$$\varepsilon_a = -(-1)^{a(a+n)/2} = -i^{a(a+n)}, \qquad W_0 = -1, \qquad W_a = \varepsilon_a\sqrt{q_a/d} \quad (1 \le a < n). \tag{51.6}$$

There is no fitted parameter in this proposal. The signs satisfy $\varepsilon_{n-a} = -\varepsilon_a$, hence $W_a W_{n-a} = \delta$ exactly. Define

$$B_{0,0} = \delta - 2, \qquad B_{g,0} = B_{0,g} = B_{g,g} = -1 \quad (g \neq 0), \qquad B_{g,h} = \frac{W_g W_{h-g}}{W_h}$$

at the remaining positions, and $A = B/(\delta-1)$. This establishes the multiplicative normalization and order-three symmetry $A_{g,h} = A_{-h,g-h}$ uniformly. The claim that (51.6) gives the correct signs for the **additive** equations remains conjectural.

Conjecture 51.3 (Explicit analytic)**.** *For every odd $n \ge 3$, the matrix defined by (51.1)–(51.6) satisfies*

$$\sum_m A_{m,g+h}A_{g,m+k}A_{h,m+l} = A_{g+l,k}A_{h+k,l} - \delta^{-1}\delta_{g,0}\delta_{h,0}. \tag{51.7}$$

Every quantity here is given by a convergent real integral and a finite product. No Stark algebraicity assumption is part of the statement. Section 52 explains its precise categorical consequence.

51.3. **Numerical checks and a failed route to the physical embedding.** The implementation `hi_shintani_ghost.py` uses

$$\log \mathrm{Sin}_2(z;\beta,1) = \int_0^\infty \left[\frac{\beta+1-2z}{\beta t} - \frac{\sinh((\beta+1-2z)t/2)}{2\sinh(\beta t/2)\sinh(t/2)}\right]\frac{dt}{t}. \tag{51.8}$$

The subtraction at zero is evaluated by a convergent series. The program independently checks reflection, both shifts, and period scaling. It evaluates both halves of (51.5), rather than imposing that equality numerically.

Initial screening passed the quadratic equations and numerical cubic checks for every odd order from 3 through 31. The saved higher-precision run checks all quadratics and all cross-free cubic orbits at $3, 7, 11, 13, 19, 25, 31$, using 60 requested digits and 15 guard digits. The orbit calculation uses only the proved balanced-beta symmetries; it makes **no scalar/Galois quotient**, since reciprocity of the analytic numbers is not assumed. Collision equations follow algebraically from the quadratics as in Section 34. Independent full cubic loops are available with `-full-cubic`.

For example, direct analytic evaluation gives

$$q_1\big|_{n=7} = 10.307869678080459005009439702 4\ldots,$$
$$q_2\big|_{n=7} = 5.448461720572675697251802026 8\ldots.$$

These agree numerically with the positive square roots of the earlier real Stark roots. They were computed without reading those roots. At order 31 the saved quadratic and representative cubic errors are 1.10×10^{-68} and 1.39×10^{-64}, respectively. An independent full cubic loop at order 7, rerun with 80 requested digits and 15 guard digits, has maximum error below 8.3×10^{-95}. The residuals and the precision checks remain numerical, not rigorous error bounds for exact zeros.

Replacing d by $-1/d$ inside the positive-period double-sine product does **not** implement the desired algebraic conjugation to a physical matrix. The corresponding normalized product is 1: the unit $\delta \equiv 1 \pmod{\delta-1}$ has the sign that kills the relevant second-place class. This is the fiber-one case of Kopp's Theorem 7.15. Kopp, proof of Theorem 7.15. Direct evaluation checks this behavior too. An analytic substitution at the conjugate quadratic point therefore supplies no unitarity proof.

52. Cubic reconstruction without first assuming Stark algebraicity

The negative-dimension calculation need not wait for an algebraicity theorem before it can have a categorical consequence. The following reduction is entirely polynomial and does not use positivity.

52.1. **A uniform residual identity.** Let G be a finite abelian group, $t = \delta^{-1}$, and assume only the order-three symmetry $A_{g,h} = A_{-h,g-h}$. Define

$$Q(g,h) = \sum_m A_{m+g,h}A_{h,m} - \delta_{g,0} + t\delta_{h,0},$$

$$C(g,h,k,l) = \sum_m A_{m,g+h}A_{g,m+k}A_{h,m+l} - A_{g+l,k}A_{h+k,l} + t\delta_{g,0}\delta_{h,0},$$

and the quartic residual

$$\begin{aligned} K(g,h,i,k) = &\sum_{l,m} A_{l,m}A_{l+g,h}A_{h+m,l+i}A_{i,k+m} \\ &- A_{h-g,i-g}\delta_{k,g} + t\delta_{h,0}A_{i,k} + tA_{g,h}\delta_{i,0}. \end{aligned}$$

Then, identically even away from solutions,

$$\boxed{K(g,h,i,k) = \sum_m A_{i,k+m}C(-h,h+m,g-h,i) + A_{i-h,g-h}Q(g-k,i).} \tag{52.1}$$

Proof. In the inner sum over l, replace $A_{l+g,h}$ by $A_{-h,l+g-h}$. The definition of C makes that inner sum

$$C(-h,h+m,g-h,i) + A_{i-h,g-h}A_{m+g,i} - t\delta_{h,0}\delta_{m,0}.$$

Multiply by $A_{i,k+m}$ and sum over m. The remaining quadratic sum is $Q(g-k,i) + \delta_{g,k} - t\delta_{i,0}$. Finally use $A_{i-h,g-h} = A_{h-g,i-g}$, and, when $i = 0$, $A_{-h,g-h} = A_{g,h}$. This gives (52.1). □

With the exceptional row and column $A_{g,0} = A_{0,g} = a + \delta_{g,0}$, where $a = -1/(\delta-1)$, there is also the identity

$$\boxed{C(g,0,k,0) = a\,Q(-k,g).} \tag{52.2}$$

To see it, split $A_{0,m} = a + \delta_{m,0}$ in the cubic sum; the constants cancel using $a(1-t) + t = 0$. Thus **all cubics imply all quadratics and quartics** in this normalization, over either quadratic embedding. This is distinct from the still unresolved converse $Q \Rightarrow C$.

`hi_cubic_quartic.py` checks (52.1) exactly on rational order-three symmetric matrices that deliberately fail Q and C, and checks (52.2) after imposing the exceptional row and column. All positions at orders 3, 5, and 7 pass. The proof above, not the finite tests, supplies the uniform ideal containment.

52.2. **Conditional categorification and automatic algebraicity.**

Theorem 52.1. *Fix an odd $n \geq 3$. If the explicit matrix of Section 51 satisfies (51.7), it gives a spherical fusion category with the cyclic HI fusion rules and categorical dimension $\delta < 0$. All of its entries, and all the numbers q_a, are algebraic. No Stark algebraicity hypothesis is required for these conclusions.*

Proof. The order-three symmetry and exceptional values are already proved. Their column sum is $-\delta^{-1}$, since $\delta^2 = n\delta + 1$. Equations (52.1)–(52.2) supply the quadratic and quartic identities. These are precisely the $\omega = 1$ hypotheses of Evans–Gannon's algebraic reconstruction, which permits either sign of δ. Evans–Gannon, Theorem 2(a).

For fixed $n, \delta, \omega = 1$, their Theorem 2(b) identifies equivalent matrices exactly by the finite group $\mathrm{Aut}(G)$. Ocneanu rigidity gives finitely many equivalence classes for fixed fusion rules. Etingof–Nikshych–Ostrik, Theorem 2.28. It follows that the polynomial reconstruction system has finitely many complex points. Since it is defined over $\mathbb{Q}(\delta)$, every coordinate of each point

is algebraic. This also establishes algebraicity in the particular displayed gauge, not just after an unspecified change of basis.

Finally let $R_h = B_{1,h}$ for $2 \le h < n$; these entries are nonzero. The multiplicative formula gives

$$W_h = \frac{W_1^h}{\prod_{j=2}^{h} R_j}, \qquad W_1^n = \delta \prod_{j=2}^{n-1} R_j.$$

Consequently every W_h, and then $q_h = dW_h^2$, is algebraic. □

After algebraicity, a field embedding taking δ to $d > 0$ also yields a spherical categorification with positive categorical dimensions, hence a pseudo-unitary one. This does not establish that the conjugated matrix is Hermitian. Evans–Gannon's Theorem 2(c) requires that additional condition for unitarity.

Therefore (51.7) at infinitely many odd orders would establish infinitely many categorifications, and would simultaneously prove algebraicity for this family of RM values. The stronger unitary existence goal still needs either a Hermitian positive-embedding argument or an independent physical construction. At present the analytic identity itself is supported by the recorded computations, not proved. Integrality, the precise class field, and reciprocity do not follow merely from the algebraicity argument.

53. A uniform convex construction of algebraic one-column solutions

The signs in Section 51 also define a bounded convex domain. Its logarithmic center gives an unconditional algebraic candidate at every odd order, without evaluating a Stark invariant. The theorem below proves one complete quadratic column, not the full HI system.

53.1. **The domain and its unique center.** Fix $n = 2m + 1 \ge 3$, let $h = (n + 1)/2 \in \mathbb{Z}_n$, and retain $d > 0$, $\delta = -1/d$. Set

$$L = (\delta - 1)^2, \qquad \gamma = \frac{d-1}{d+1} = -\frac{n\delta}{L} \in (0, 1). \tag{53.1}$$

Consider real sequences u with

$$u_0 = u_1 = -1, \qquad u_a = u_{1-a}, \qquad (-1)^{a+1} u_a > 0 \quad (2 \le a < n). \tag{53.2}$$

There are m free coordinates, represented by $u_2, \dots, u_h$. Center the reflection by putting $x_j = u_{h+j}$; then $x_j = x_{-j}$. The real eigenvalues of its symmetric circulant matrix T are

$$\lambda_k = \sum_{j \in \mathbb{Z}_n} x_j e^{2\pi i j k/n}, \qquad \lambda_{n-k} = \lambda_k.$$

Impose the additional linear inequalities

$$-(-1)^k \lambda_k > 0 \qquad (0 \le k \le m). \tag{53.3}$$

Denote the resulting open polytope by $\mathcal{P}_n$.

Theorem 53.1. *The polytope $\mathcal{P}_n$ is nonempty and bounded. The function*

$$\boxed{\mathcal{F}(u) = \log|\det T| + \gamma \sum_{a=2}^{n-1} \log|u_a|} \tag{53.4}$$

has a unique critical point u^* in $\mathcal{P}_n$, its global maximum. This point is nondegenerate and all its coordinates are algebraic over $\mathbb{Q}(\delta)$.

Proof. Start at $u_0 = u_1 = -1$, with all free coordinates zero. Its Fourier eigenvalues are

$$\lambda_k^{(0)} = -2\cos(2\pi k h/n) = -2(-1)^k \cos(\pi k/n).$$

They are nonzero and have the signs in (53.3). Small free coordinates with the prescribed signs therefore give points in $\mathcal{P}_n$.

Fourier inversion at $x_h = u_1 = -1$ gives the positive linear relation

$$|\lambda_0| + 2\sum_{k=1}^{m} \cos(\pi k/n)|\lambda_k| = n. \tag{53.5}$$

All coefficients are positive. Thus all eigenvalues, and by Fourier inversion all coordinates, are bounded on the domain.

On this domain, (53.4) is a sum of logarithms of positive affine functions, with positive weights. The coordinate-log terms alone have negative-definite Hessian on the m free variables. Consequently $\mathcal{F}$ is strictly concave. At every boundary point at least one coordinate or eigenvalue tends to zero, while all others remain bounded; hence $\mathcal{F} \to -\infty$. Existence of an interior maximum and its uniqueness follow.

Finally, the critical equations are rational equations over $\mathbb{Q}(\delta)$: use the derivative of $\log\det T$, where T has entries linear in u. Their Jacobian is the invertible Hessian just described. Clearing the nonzero denominators gives an isolated simple complex algebraic point, so its coordinates are algebraic. □

The Fourier signs select a chamber; no trigonometric values enter the coefficients of the critical equations. The same existence argument works for each of the 2^m choices of signs for the free coordinate orbits, while retaining (53.3). Thus there are at least 2^m distinct real algebraic solutions of the one-column problem below. Selecting the particular chamber (53.2) is substantive.

53.2. **Why the center solves a complete quadratic column.** Write T^{-1} as a symmetric circulant with coefficients c_j. Differentiating (53.4) in a free coordinate orbit gives

$$nc_{a-h} + \frac{\gamma}{u_a} = 0 \qquad (2 \le a < n). \tag{53.6}$$

The orbit multiplicity, either 1 or 2, cancels. Define $v_a = Lc_{a-h}$. Equations (53.1) and (53.6) yield

$$v_a = \frac{\delta}{u_a} \quad (2 \le a < n).$$

The two exceptional values coincide by symmetry. The coefficient of zero in $TT^{-1} = I$ gives

$$-2c_h - \frac{(n-2)\gamma}{n} = 1.$$

Using $L = (n-2)\delta + 2$, it follows that

$$v_0 = v_1 = -1.$$

Consequently inversion of the circulant is exactly the required weighted reciprocal relation:

$$\boxed{\sum_a u_{a+g} v_a = L\delta_{g,0}, \qquad v_0 = v_1 = -1, \quad v_a = \delta/u_a \ (a \ne 0, 1).} \tag{53.7}$$

This reconstructs the phase variables explicitly. Put

$$\begin{gathered} R_a = \delta/u_a \quad (2 \le a < n), \qquad W_1 = \sqrt[n]{\delta \prod_{a=2}^{n-1} R_a} \text{ (the real root)}, \\ W_a = \frac{W_1^a}{\prod_{j=2}^{a} R_j}, \quad W_0 = -1. \end{gathered} \tag{53.8}$$

Reflection of the R_a implies $W_a W_{n-a} = \delta$. Construct B and A as in Section 51. Then $B_{a,1} = u_a$, $B_{1,a} = v_a$, and (53.7) proves

$$\sum_a A_{a+g,1} A_{1,a} = \delta_{g,0} \quad \text{for every } g. \tag{53.9}$$

The exceptional zero column and all multiplicative symmetries also hold. Every W_a is real and algebraic.

Moreover $\operatorname{sgn} R_a = (-1)^a$. Taking signs in (53.8) gives exactly

$$\operatorname{sgn} W_a = -(-1)^{a(a+n)/2}. \tag{53.10}$$

Thus the signs proposed in Section 51 arise from this proved construction. This does not by itself identify its magnitudes with the double-sine product.

53.3. **Computations, a counterexample, and the remaining implication.** `hi_column_barrier.py` computes the unique center by Newton's method with a line search inside the polytope. It then reconstructs the entire matrix by (53.8). At orders $3, 7, 11, 19, 25, 31$, all quadratics and all cross-free cubic orbits pass numerically; comparison with the independently evaluated double sines agrees to their saved 60-digit precision. No arithmetic or analytic candidate is supplied to the optimization.

Larger runs at $n = 51, 101$ check **every quadratic**, but only eight sampled cubics. At order 101 the maximum all-column quadratic residual is below 2.0×10^{-126}, and the sampled cubic residuals are below 1.8×10^{-119}, at 150-digit working precision. These larger runs do not constitute complete cubic checks or exact existence certificates.

One column alone does not force the other equations, even with the formal norms and order-three symmetry. There is a small exact counterexample at $n = 5$. In the row coordinates of Section 45 take

$$R_2 = x, \qquad R_3 = \frac{(2-\delta)x - 1}{2}, \qquad R_4 = R_2, \qquad x^2 + 2\delta x + \delta = 0.$$

Its complete quadratic column $h = 1$ vanishes, whereas the four scaled residuals with $g, h \in \{2, 3\}$ equal $-31\delta - 4 \neq 0$. `hi_column_counterexample.py` verifies these identities exactly over the quadratic extension. At $\delta < 0$ both roots give real candidates, but neither has the coordinate signs in (53.2). This is an algebraic ghost counterexample; physical unit moduli are not asserted.

The resulting construction conjecture is now entirely algebraic and variational:

> For every odd n, the matrix reconstructed from the unique maximizer of (53.4) in $\mathcal{P}_n$ satisfies all of (C).

This conjecture would imply categorification for all odd orders by Section 52. Unlike the analytic approach, algebraicity and existence of the candidate itself are already proved. The missing assertion is that the chamber-selected center satisfies the additional equations. Identifying it uniformly with the double-sine product would connect the two routes, but is not a prerequisite for proving the variational conjecture. A Hermitian positive-embedding argument would still be needed for the stronger unitary conclusion.

54. The entire one-column algebra is finite, reduced, and real

Section 53 extends from the selected chamber to a classification of **all complex one-column solutions** in the phase ansatz. In particular, the isolatedness observed by elimination in small orders now has a uniform proof. This does not prove that the full cubic locus is nonempty.

54.1. **Positive logarithmic weights exclude every nonreal solution.** Fix odd $n \geq 3$, keep the negative embedding $\delta = -1/d$, and set $m = (n-1)/2$. Allow the m free coordinates of Section 53 to be complex, subject only to nonvanishing of those coordinates and of $\det T$. Factoring the circulant determinant gives the master function

$$\mathcal{F} = \sum_{k=0}^{m} \mu_k \log \lambda_k + \gamma \sum_{a=2}^{h} \nu_a \log u_a, \quad \mu_0 = 1, \quad \mu_k = 2 \ (k > 0), \quad \nu_h = 1, \quad \nu_a = 2 \ (a < h). \tag{54.1}$$

Only its logarithmic derivative is used over $\mathbb{C}$; branches of the logarithms do not affect it. All weights are strictly positive real numbers, and all linear forms have real coefficients.

The following elementary argument is the positive-weight master-function principle; compare Varchenko, *Critical points of the product of powers of linear functions*, Theorem 1.2.1. For real affine forms $f_j(z) = b_j + a_j \cdot z$ with positive weights w_j, a complex critical point $z = x + iy$ satisfies

$$0 = y \cdot \operatorname{Im} \sum_j \frac{w_j a_j}{f_j(z)} = -\sum_j \frac{w_j (a_j \cdot y)^2}{|f_j(z)|^2}. \tag{54.2}$$

Thus $a_j \cdot y = 0$ for every j. The coordinate forms alone span the dual space, so $y = 0$. Every critical point is real. At a real point the Hessian is

$$-\sum_j \frac{w_j a_j a_j^{\mathsf{T}}}{f_j(x)^2}, \tag{54.3}$$

which is negative definite. Hence every critical point is simple.

Each bounded chamber of the real hyperplane complement has exactly one critical point, by the same strict-concavity and boundary argument as in Section 53. An unbounded chamber has none: a nonzero recession direction v makes all ratios $(a_j \cdot v)/f_j(x)$ nonnegative, and at least one is positive. The directional derivative of $\mathcal{F}$ is therefore strictly positive throughout that chamber.

There are finitely many chambers. Consequently the critical scheme is finite and reduced, and its points are in bijection with the bounded chambers. The argument applies at **every odd order**, including composites.

This is exactly the one-column scheme, not an unrelated critical subsystem. The full column equations give $TV = LI$, where V is the centered circulant of the reciprocal row. Thus T is invertible and its inverse yields (53.6). Conversely, (53.6) gives $TV = LI$, including both exceptional entries, by the trace computation in Section 53. These implications use identities in the localized coordinate ring, so they also identify the schemes and exclude hidden multiplicities.

54.2. **An exact formula at every odd prime.** Let b_n denote the number of one-column solutions modulo cyclic character gauge. The preceding theorem identifies it with the number of bounded chambers of $2m + 1$ affine hyperplanes in $\mathbb{R}^m$. In particular,

$$2^m \le b_n \le \binom{2m}{m}. \tag{54.4}$$

The lower bound comes from the 2^m chambers constructed in Section 53. The upper bound is the general-position bound for bounded regions of an affine arrangement.

Theorem 54.1. *For every odd prime p,*

$$\boxed{b_p = \binom{p-1}{(p-1)/2}.} \tag{54.5}$$

To prove this, use centered homogeneous coordinates $x_0, \dots, x_m$. The arrangement before setting $x_m = -1$ consists of the $m+1$ coordinate hyperplanes and the $m+1$ Fourier hyperplanes. Up to nonzero column scalings, the latter have coefficient matrix

$$C_{k,j} = \cos(2\pi kj/p), \qquad 0 \le k, j \le m.$$

Every square minor of this matrix is nonzero by the discrete-cosine version of Chebotarev's theorem: Garcia–Karaali–Katz, *An improved uncertainty principle for functions with symmetry*, Theorem 1.2. Thus every $(m+1)$-column minor of $[I \mid C]$ is nonzero. The central arrangement

is in general position, and its decone has $2m+1$ affine hyperplanes in general position. Its characteristic polynomial is

$$\chi_n(t) = \sum_{j=0}^{m} (-1)^j \binom{2m+1}{j} t^{m-j}.$$

The bounded-region formula $b_n = (-1)^m \chi_n(1)$ and the elementary alternating-binomial identity give (54.5). The region-count formula is Zaslavsky's; see Stanley's lectures on hyperplane arrangements.

For composite n, the cosine matrix has vanishing minors, and the prime formula need not hold. Exact ranks over $\mathbb{Q}(\zeta_n)$ give the following counts:

n	3	5	7	9	11	13	15
b_n	2	6	20	58	252	924	2492

`hi_column_arrangement.gp` computes the central characteristic polynomial directly by its subset-rank formula, then divides by $t-1$. For example,

$$\chi_9(t) = t^4 - 9t^3 + 36t^2 - 83t + 113, \qquad b_9 = 58,$$

so the general-position upper bound 70 is strict. The counts and all rank decisions in this calculation are exact.

54.3. **Consequences for the remaining existence problem.** Let $K = \mathbb{Q}(\delta)$, and let $\mathcal{A}_n$ be the localized coordinate algebra of one full quadratic column in the row or column coordinates. Then

$$\boxed{\mathcal{A}_n \text{ is a finite product of number fields,} \qquad \dim_K \mathcal{A}_n = b_n.} \tag{54.6}$$

Each factor has only real embeddings above the negative embedding of K. Its embeddings above the positive place may be complex. The algebra and its reducedness are defined over K, so the same scheme length and simplicity hold after changing the dimension embedding.

All remaining quadratic and cubic residuals are elements of $\mathcal{A}_n$: the character gauge cancels from B, and the row formulas express its entries rationally in these coordinates. Adjoining more equations therefore retains some field factors and discards others. In particular, within the phase ansatz, both the full quadratic scheme and the full cubic scheme are finite and reduced for **every odd** n, even though they may a priori be empty. For the cubic scheme use (52.2), which puts the one-column equations in the full cubic ideal.

Before quotienting by the character gauge, the one-column phase scheme has length nb_n. Indeed, adjoining W_1 by (53.8) is a degree-n finite étale extension: its right side is a unit and $n \neq 0$. Over the negative embedding each character orbit has exactly one representative with all W_a real, because n is odd.

This supplies a finite, nondegenerate algebraic setting for the variational conjecture at every order. The canonical chamber selects one of its real points. Infinitude would follow if its field factor survives all cubic equations for infinitely many orders. Neither the chamber count nor reducedness proves that survival: at order 5 only four of the six one-column points survive the full quadratic system, as Sections 43 and 53 demonstrate.

55. A scalar completion condition and its prime-order field consequence

A single scalar condition now separates the tested genuine candidates from the additional one-column solutions. Its sufficiency is proved exactly at orders 3, 5, and 7, and it has a uniform arithmetic consequence at every odd prime. Its sufficiency in unbounded order and its validity at the canonical center remain conjectural.

55.1. **The condition depends on the phase lift.** For reconstructed phases W, define

$$S(W) = \frac{\sum_{a\in\mathbb{Z}_n} W_a}{\delta - 1}, \qquad T(A) = \sum_{a\in\mathbb{Z}_n} A_{a,2a} = \operatorname{Tr}\mathcal{R}.$$

The last equality follows directly from the definition of $\mathcal{R}$ in Section 50; it does not require the quadratic equations. Consider

$$\boxed{S(W)^2 + T(A) = 2.} \tag{55.1}$$

After clearing its nonzero constant denominator, this is

$$H(W) = \left(\sum_a W_a\right)^2 + (\delta - 1)\sum_a B_{a,2a} - 2(\delta - 1)^2 = 0. \tag{55.2}$$

This involves $\operatorname{Tr}\mathcal{R}$, not the fifth-power trace in Section 50. No implication from that section to (55.1) is being asserted.

Although $T(A)$ is invariant under cyclic character gauge, $S(W)$ generally is not. Thus (55.1) is imposed on the phase cover of the one-column algebra. At the negative dimension embedding, requiring every W_a to be real selects exactly one lift of each row point, as proved in Section 54.

Theorem 55.1 (exact finite computation)**.** *At $n = 3, 5, 7$, the phase norms, reflection and reconstruction identities, one complete quadratic column, and (55.1) imply every quadratic and cubic equation. The scalar-constrained phase schemes have respective lengths $2, 4, 6$ over $K = \mathbb{Q}(\delta)$.*

The proof uses a finite-algebra dimension comparison, avoiding a large direct elimination. Let E be the one-column phase algebra and $M_H : E \to E$ multiplication by (55.2), after multiplication by a Laurent monomial unit to obtain a polynomial. Then

$$\dim_K E/(H) = \dim_K E - \operatorname{rank}_K M_H. \tag{55.3}$$

At a prime where every entry of this exact matrix is integral, its rank can only decrease after reduction. This gives an upper bound on $\dim_K E/(H)$. An independent exact Gröbner computation adjoins all of Q as well and gives a lower bound, because that quotient is a quotient of $E/(H)$.

n	$\dim_K E$	Exact modular rank of M_H	Exact length after adjoining all Q
3	6	4	2
5	30	26	4
7	140	134	6

In each case the upper and lower dimension bounds coincide. Therefore adjoining the remaining quadratics changes no ideal. Exact cubic reductions in that same quotient vanish. The checks use all cross-free beta orbits without scalar quotienting, and the collision identities of Section 34 cover the other cubics.

`hi_column_trace_exact.py` regenerates the equations and the exact multiplication matrix, checks integrality of every reduced coefficient, and runs both sides of this comparison. The recorded prime is 32003, with the respective images of δ equal to 20720, 19299, and

(1) No finite-field dimension is silently treated as a characteristic-zero equality: (55.3) and the matching exact lower bound supply that equality. These certificates use no HI seeds or numerical root recognition.

55.2. **The scalar equation forces the one-root property at primes.**

Proposition 55.2. *Let $n = p$ be an odd prime. Take a real phase lift at the negative embedding, reconstructed as in (53.8), and let*

$$E = K(R_2, \ldots, R_{p-1}) \subset \mathbb{R}$$

be its row field. If (55.1) holds, then

$$\boxed{W_1, \dots, W_{p-1} \in E.} \tag{55.4}$$

The proposition does not need Q or C beyond any assumptions used to construct the row; the reconstruction and scalar equation suffice.

Proof. Put $x = W_1$ and $P_a = \prod_{j=2}^{a} R_j$, with $P_1 = 1$. Then

$$x^p = \delta P_{p-1} \in E^\times, \qquad W_a = x^a/P_a \quad (1 \le a < p).$$

For a real x with $x^p \in E$, the degree $[E(x) : E]$ is 1 or p. Indeed, if its minimal polynomial has degree $1 \le k < p$, its constant coefficient is $(-1)^k x^k \zeta_p^j \in E \subset \mathbb{R}$. Since x is real and p odd, $\zeta_p^j = 1$; hence $x^k \in E$. Bezout's identity for k, p then gives $x \in E$.

Suppose instead that $[E(x) : E] = p$. All entries of A, and therefore $T(A)$, lie in E. Equation (55.1) implies that $s = \sum_a W_a$ has $s^2 \in E$. The degree $[E(s) : E]$ divides both 2 and p, so $s \in E$. But

$$s = -1 + \sum_{a=1}^{p-1} x^a/P_a$$

is a nonconstant linear combination of the independent basis $1, x, \dots, x^{p-1}$, with all displayed coefficients nonzero. This is a contradiction. Thus $x \in E$, and the reconstruction gives all the other phases in E. □

This links the scalar condition to the arithmetic one-root requirement in Sections 38 and 44. It does not establish that the row field is abelian or has the desired ray conductor, nor does the one-root conclusion alone supply the remaining Izumi equations.

55.3. **The canonical trace has a conjectural period-40 formula.** For the unique canonical center of Section 53, the numerical data suggest the particularly small trace

$$\boxed{S(W) = \eta_n := \sqrt{2} \sin \frac{\pi n}{4} \frac{\sin(\pi n/5)}{\sin(2\pi/5)}, \qquad T(A) = 2 - \eta_n^2.} \tag{55.5}$$

The first expression is periodic modulo 40. For odd n, its values belong to

$$\{0, \pm 1, \pm(\sqrt{5} - 1)/2\}.$$

Consequently the proposed operator trace is 2 when $5 \mid n$, 1 when $n \equiv \pm 2 \pmod 5$, and $(1 + \sqrt{5})/2$ when $n \equiv \pm 1 \pmod 5$.

`hi_column_trace.py` computes the canonical center without imposing either trace identity. Its saved tests include orders 51 and 79 as well as the smaller orders. At order 79, with 120-digit working precision, the scalar defect is below 9×10^{-111}, and the first equality of (55.5) agrees within 3×10^{-115}. These are numerical observations.

The script also scans all coordinate-sign choices inside the fixed Fourier-sign chamber (53.3) for $n = 5, 7, 9, 11, 13, 15, 17$: 508 centers in total. Exactly 24 pass the full quadratic tests numerically, and these are exactly the 24 passing (55.1); no false positives or false negatives appear in this scan. It is not an enumeration of every Fourier chamber. Equation (55.5) is specific to the canonical center: other full-Q candidates can have different trace values, including $S(W) = 2$ in the secondary order-11 branch.

The proposed route to infinitude now has two explicit unresolved assertions: prove scalar completion from one column in unbounded orders, and prove (55.1) at the canonical center in those orders. The stronger trace formula (55.5) would give the latter. Only the finite scalar-completion theorem and the uniform prime-order field consequence above are proved here. Infinitude remains unproved.

56. The two-value quadratic-character ansatz stops at orders 3 and 11

The secondary order-11 branch in Section 55 has only two phase values. That feature suggests a possible uniform construction, but the quadratic equations themselves give an exact obstruction to extending it.

Theorem 56.1. *Let $p \equiv 3 \pmod 4$ be prime, let $\delta^2 = p\delta + 1$, and write χ for the quadratic character of $\mathbb{F}_p$, with $\chi(0) = 0$. Suppose*

$$W_0 = -1, \qquad W_j = a + b\chi(j) \quad (j \neq 0), \qquad a^2 - b^2 = \delta. \tag{56.1}$$

If every quadratic equation holds, then $p = 3$ or $p = 11$. At the negative embedding, the real solutions in this ansatz are precisely

p	a	b^2
3	$\delta/2$	$(1-\delta)/4$
11	$(2\delta-1)/10$	$(1-12\delta)/20$

(56.2)

with either sign of b. They satisfy all cubics as well.

The obstruction uses only two necessary consequences of Q; the scalar completion conjecture is not assumed. Put $e = \chi(2)$ and

$$Q(g,h) = \sum_j W_j W_{h-j} W_{-j-g} W_{j+g-h}.$$

The diagonal equation $Q(1,1) = 0$, after substituting $b^2 = a^2 - \delta$, is

$$D_e = (4e + 2p - 2)a^4 + (4e - 4)a^3 - \delta(4e + p + 1)a^2 + \delta(2 - 4e)a + \delta = 0. \tag{56.3}$$

This follows by expanding $W_j = a + b\chi(j) - (1 + a)\mathbf{1}_{j=0}$, treating the repeated root at zero and the simple roots at ± 1 separately. The terms odd in b cancel; the usual two-character sum gives the displayed coefficients.

A second necessary equation comes from summing Q over its first index. For $s \neq 0$, direct convolution gives

$$(W * W)(s) = (p-1)a^2 - 2a - \delta - 2b(1 + a)\chi(s).$$

Consequently

$$\sum_g Q(g,1) = (W * W)(1)(W * W)(-1), \qquad Q(0,1) = 2\delta + (p-2)\delta^2.$$

All nonzero-index quadratics therefore force

$$E = \big((p-1)a^2 - 2a - \delta\big)^2 - 4(a^2 - \delta)(a+1)^2 - 2\delta - (p-2)\delta^2 = 0. \tag{56.4}$$

Eliminating a and then δ gives the polynomial identities

$$\begin{aligned} &\operatorname{Res}_\delta\Big(\operatorname{Res}_a(E, D_{-1}), \delta^2 - p\delta - 1\Big) = p^6(p-11)^2(p-3)^{14}, \\ &\operatorname{Res}_\delta\Big(\operatorname{Res}_a(E, D_{+1}), \delta^2 - p\delta - 1\Big) = p^6(p+1)^{12}(p^4 - 76p^3 + 406p^2 + 532p - 4047). \end{aligned} \tag{56.5}$$

The first expression permits only 3 and 11. A prime root of the quartic in the second expression must divide $4047 = 3 \cdot 19 \cdot 71$. None of these three primes is a root, so that case is impossible.

At $p = 3$, the common equation factors as $(a - \delta/2)(a^2 + \delta a/2 + 1/4) = 0$; the quadratic factor has no real root when $\delta < 0$. At $p = 11$, the monic gcd of (56.3) and (56.4) over $\mathbb{Q}(\delta)$ is $a - (2\delta - 1)/10$. The norm condition then gives (56.2). For $p \equiv 1 \pmod 4$, a real ansatz (56.1) at $\delta < 0$ is already impossible, since $\chi(-1) = 1$ would require a real phase square to equal δ.

`hi_paley_obstruction.py` verifies the resultants as identities with symbolic p, checks the character expansions off solution, and checks every Q and C position in the two remaining quartic number fields. Thus the finite cases have exact certificates. The theorem excludes only the two-value quadratic-character construction; it places no upper bound on orders admitting general HI solutions.

57. Explicit commuting matrices and scalar completion at orders 11 and 13

The one-column algebra now has explicit multiplication matrices at every odd prime, obtained directly from the hyperplane arrangement. This avoids eliminating the one-column equations. Applying these matrices extends the exact scalar-completion theorem of Section 55 to orders 11 and 13.

Theorem 57.1. *At each $n \in \{3,5,7,11,13\}$, the phase norms, reflection and reconstruction identities, one complete quadratic column, and $H = 0$ imply every quadratic and cubic equation. The respective scalar-constrained phase lengths over $\mathbb{Q}(\delta)$ are $2,4,6,12,12$.*

The matrix construction below is uniform in the prime. Nonvanishing of the scalar kernel in unbounded orders, and scalar completion in those orders, remain unproved.

57.1. **Multiplication from the hyperplane circuits.** Let $p = 2m+1$ be prime. Use the centered coordinates of Section 54, with $x_m = -1$, and put $\mu_0 = 1$, $\mu_j = 2$ for $j > 0$. The p affine forms and their weights are

$$\begin{array}{l|l} f & \alpha \\ \hline x_j \quad (0 \le j < m) & \gamma\mu_j \\ \lambda_k = \displaystyle\sum_{j=0}^{m-1} \mu_j \cos(2\pi kj/p)x_j - 2\cos(2\pi km/p) \quad (0 \le k \le m) & \mu_k \end{array} \tag{57.1}$$

Index these forms by $1,\dots,p$, and write $f_i = z_i + \beta_i \cdot x$. Work over $F = \mathbb{Q}(\delta, \zeta_p + \zeta_p^{-1})$. The minor theorem used in Section 54 ensures general position of both the normals and the affine arrangement.

Here is the circuit presentation specialized to this arrangement. Take oriented symbols e_I for m-element subsets, with relations $\sum_j e_{j,J} = 0$ for every $(m-1)$-element subset J. The symbols with $I \subset \{1,\dots,p-1\}$ form a basis of size $b_p = \binom{p-1}{m}$. For each $(m+1)$-subset $C = (c_0,\dots,c_m)$, choose nonzero cofactors $\ell^C_{c_r}$ so that $\sum_{i\in C} \ell^C_i \beta_i = 0$, and set $f_C = \sum_{i\in C} \ell^C_i z_i \neq 0$. Define

$$L_C e_{C\setminus c_r} = (-1)^r \sum_{s=0}^{m} (-1)^s \alpha_{c_s} e_{C\setminus c_s}, \qquad L_C e_I = 0 \quad (I \not\subset C),$$

and

$$\boxed{K_i = \sum_{C\ni i} \frac{\ell^C_i}{f_C} L_C.} \tag{57.2}$$

These are multiplication by α_i/f_i in the critical algebra. The identification follows from Varchenko, *Critical set of the master function and characteristic variety of the associated Gauss–Manin differential equations*, Corollaries 2.17 and 3.8. The marked element corresponding to e_I is $\det(\beta_i)_{i\in I} \prod_{i\in I} \alpha_i/f_i$. The positive weights satisfy the required unbalanced-weight condition.

For a coordinate form $f_i = x_j$, this immediately gives

$$M_{x_j} = \alpha_i K_i^{-1}, \qquad M_{\delta/x_j} = \frac{\delta}{\alpha_i} K_i. \tag{57.3}$$

Reflection converts these into every row coordinate R_a. Thus reconstruction and the scalar polynomial H can be evaluated in explicit commuting matrices, without any critical-point roots or HI seeds.

57.2. **An integral matrix model gives a rigorous upper bound.** Choose a rational prime $q \equiv 1 \pmod p$ at which p^2+4 is a nonzero square. Choose images of ζ_p and δ in $\mathbb{F}_q$. Check that all normal minors, circuit constants, and weights appearing in (57.2) stay nonzero. The resulting matrices are reductions of the characteristic-zero matrices in a free local lattice. The coordinate multiplication matrices and the Laurent inverses needed for H are also checked to be integral by verifying invertibility of their reductions.

Adjoin the reconstruction root as in Section 55:

$$\mathcal{E} = E[X]/\left(X^p - \delta \prod_{a=2}^{p-1} R_a\right), \qquad \dim_F \mathcal{E} = p\binom{p-1}{m}, \tag{57.4}$$

where E is the row critical algebra. Multiplication by H on this regular module has an integral matrix. Rank specialization therefore gives

$$\dim_F \mathcal{E}/(H) \;\le\; \dim_{\mathbb{F}_q} \overline{\mathcal{E}}/(\overline{H}). \tag{57.5}$$

Extending the characteristic-zero coefficient field from $\mathbb{Q}(\delta)$ to F does not change this dimension.

The program computes the right side efficiently. A linear combination of the row matrices has a cyclic vector; its exact Krylov basis gives $\overline{E} = \mathbb{F}_q[t]/(f(t))$. The program checks that f is squarefree and factors it exactly. In each resulting finite field, the degree of

$$\gcd\left(X^p - \delta \prod R_a,\, \overline{H}(X)\right) \tag{57.6}$$

counts the surviving phase roots, with the degree of that finite field as multiplicity. This computes a quotient dimension, not a count of only the $\mathbb{F}_q$-rational points.

p	Row dimension	Phase dimension	q	Image of δ	Modular H-quotient dimension
3	2	6	32029	16171	2
5	6	30	32051	648	4
7	20	140	32089	16285	6
11	252	2772	32099	4506	12
13	924	12012	32059	1153	12

At 11 the selected row-field factor degrees are 2 and 10; at 13 there is one factor of degree 12. Each has one surviving phase root over its row field. These finite-field statements alone would not prove the characteristic-zero theorem; matching lower bounds are essential.

57.3. **Matching exact lower bounds.** At 11, the previously certified full-Q algebra of Section 45 has dimension 12 and satisfies every C. Exact FGLM converts its saved basis to a primitive-coordinate presentation. Over $\mathbb{Q}(\delta)$, the primitive polynomial factors into degrees 2 and 10. In each field, the characteristic-zero version of (57.6) has degree one. Thus the full-Q/H phase quotient has dimension 12, matching the upper bound.

At 13, the construction of Section 28 produces a full Q/C certificate in a field of absolute degree 24. Exact reconstruction finds a unique phase root in that field and verifies $H = 0$. A row coordinate has absolute degree 24, so its 12 conjugates over $\mathbb{Q}(\delta)$ are distinct. They supply 12 distinct points of the full-C/H locus, again matching the upper bound. The rank comparison therefore proves scalar completion here without computing the full order-13 Q ideal.

`hi_column_circuits.py` regenerates the circuit matrices, verifies their critical relations, commutativity and complete quadratic column, computes the modular bounds, and supplies the exact lower bounds. At 11 it uses the already verified `hi_row11_basis.txt`; at 13 it reruns the full exact construction and cubic certificate. The resulting theorems establish finite scalar completion, not the canonical trace identity or infinitude.

58. Positive-characteristic certificates and a Frobenius construction

There is a second use for finite-field calculations, distinct from the specialization upper bounds in Section 57. A **complete cubic solution** in a suitable positive characteristic already implies existence over $\mathbb{C}$, by algebraic reconstruction followed by categorical lifting. No characteristic-zero lower bound is needed for this existence statement. This section proves that implication, specializes it to characteristic 2 at every odd order, and gives new complete certificates at order 27. It does **not** establish nonemptiness for unbounded orders.

Prior work. The finite-field/lifting strategy is also explicitly reported in joint work of T. Gannon, A. Schopieray and H. Yadav, *Haagerup–Izumi categories using the lifting technique*, listed as in progress in Yadav's research description. That account reports models over a positive-density set of prime fields for odd orders through 19; it does not report unbounded-order existence. Yadav, "Exotic fusion categories in characteristic p". The reconstruction argument and certificate calculations below are self-contained here; no unpublished result from that project is used.

58.1. **Reconstruction and lifting from a complete finite-field solution.**

Theorem 58.1. *Let $n \geq 3$ be odd and let k be an algebraically closed field of characteristic $\ell > 0$, with $\ell \nmid n(n^2+4)$. Suppose $\delta^2 = n\delta + 1$ and an $n \times n$ matrix A over k has the exceptional row, column and order-three symmetry of Section 52, and satisfies every cubic equation $C(g,h,k,l) = 0$. Then there is a fusion category over $\mathbb{C}$ with the cyclic HI fusion rules at order n.*

Proof. First, $\delta \neq 0, 1$: the first assertion follows from its quadratic equation, and $\delta = 1$ would force $n = 0$ in k. The polynomial residual identities (52.1)–(52.2) therefore supply all quadratic and quartic reconstruction equations. The exceptional column also has sum $-\delta^{-1}$.

Here is the characteristic check for the reconstruction step. Choose $b \in k^\times$ with $b^2 = \delta^{-1}$. In the Leavitt algebra over k, use generators s, s', t_g, t'_g and relations

$$s's = 1, \quad t'_g t_h = \delta_{g,h}, \quad s't_g = t'_g s = 0, \qquad ss' + \sum_g t_g t'_g = 1.$$

Define

$$\begin{aligned} \rho(s) &= \delta^{-1}s + b\sum_g t_g t_g, & \rho(s') &= \delta^{-1}s' + b\sum_g t'_g t'_g, \\ \rho(t_g) &= bst'_{-g} + t_{-g}ss' + \sum_{h,j} A_{h+g,j+g}\, t_h t_{g+h+j} t'_j, \\ \rho(t'_g) &= bt_{-g}s' + ss't'_{-g} + \sum_{h,j} A_{j+g,h+g}\, t_j t'_{g+h+j} t'_h. \end{aligned} \tag{58.1}$$

Let α_g fix s, s' and translate each t_h, t'_h by $h \mapsto h + 2g$. These are the $\omega = 1$ reconstruction formulas. Their verification is by polynomial coefficient comparison: they preserve the Leavitt relations, satisfy $\alpha_g \rho = \rho \alpha_{-g}$, and give

$$\rho^2(x) = sxs' + \sum_g t_g \alpha_g \rho(x) t'_g. \tag{58.2}$$

In particular the occurrence of the integer 2 in the label translation uses that the **group order is odd**; it does not require division by 2 in the coefficient field.

For completeness, the simplicity argument is also valid in this characteristic. Reduced-word leading terms show that the simultaneous centralizer of $\rho(s), \rho(s')$ consists of scalars. Comparing $\rho(t_{-2g})$ distinguishes the $\alpha_g \rho$. The analogous centralizer of s, s' distinguishes the α_g. A nonzero intertwiner between the two types would have an uncancelled leading coefficient proportional to $1 - \delta^{-1}$, which is nonzero. Thus the simple objects have exactly the required Hom spaces over k. Iterating (58.2) gives matrix-unit bases for all Hom spaces; closing under finite sums and idempotents gives a semisimple category with exactly the $2n$ indicated simples. Evaluation and coevaluation for ρ are s'/b and s/b. The duality calculations on these generators give the identity double dual, and dimensions 1 and δ. They use linear maps and coefficient identities, with no complex conjugation or positivity assumption. This verifies the extension to k of the proof of Evans–Gannon, Theorem 2(a) and Proposition 1; the statement in that paper itself is over $\mathbb{C}$.

The resulting spherical category has global dimension

$$D = n(1 + \delta^2), \qquad (1 + \delta^2)^2 = (n^2 + 4)\delta^2. \tag{58.3}$$

Our characteristic assumption makes $D \neq 0$. The nondegenerate category therefore lifts to the Witt ring $W(k)$, preserving its fusion rules, by Etingof–Nikshych–Ostrik, Theorem 9.3. Passing to the generic fiber gives the required fusion rules over a characteristic-zero field. The finite associator, unit, rigidity and invertibility equations, in fixed bases, consequently have a solution in characteristic zero. The weak Nullstellensatz supplies a solution over $\overline{\mathbb{Q}}$, hence over $\mathbb{C}$. □

This conclusion is categorification, without a unitarity assertion. It also does not assert that the prescribed finite-field phase gauge lifts entry by entry. For existence, lifting the fusion category is sufficient. A nonzero modular H-quotient **without** a complete cubic certificate still supplies only the upper bound of Section 57.

The standalone `hi_finite_field_certificate.py -verify` accepts a small JSON file specifying a finite field, δ, and a reflected row R. It checks irreducibility of the field polynomial, the characteristic and dimension hypotheses, the row normalization, and then reconstructs B by prefix products as in Section 43. It checks the symmetry, linear sum, **every** Q and **every** C independently. Exporting the scalar-selected row factors of Section 57 gives complete finite-field certificates at 3, 5, 7, 11 and 13. The program does not read characteristic-zero HI seeds for that construction.

58.2. **A fixed characteristic works at every odd order.** Take $\ell = 2$, put $\mathbb{F}_4 = \mathbb{F}_2[\omega]/(\omega^2 + \omega + 1)$, and set $\delta = \omega$. For every odd n,

$$\delta^2 = n\delta + 1, \qquad n(1+\delta^2) = \omega \neq 0. \tag{58.4}$$

Thus characteristic 2 is admissible at **all** the target orders. Given phases

$$W_0 = 1, \qquad W_a W_{-a} = \omega \quad (a \neq 0), \tag{58.5}$$

the normalized matrix becomes

$$A_{0,0} = \omega^2, \qquad A_{g,0} = A_{0,g} = A_{g,g} = \omega \quad (g \neq 0), \qquad A_{g,h} = \omega \frac{W_g W_{h-g}}{W_h} \quad (gh(g-h) \neq 0). \tag{58.6}$$

Its full cubic target is simply

$$\boxed{\sum_m A_{m,g+h} A_{g,m+k} A_{h,m+l} = A_{g+l,k} A_{h+k,l} + \omega^2 \delta_{g,0} \delta_{h,0}.} \tag{58.7}$$

An infinite set of odd orders with solutions to this finite-field identity would prove the requested complex infinitude.

One constraint is automatic in a natural Frobenius ansatz.

Lemma 58.2. *In characteristic 2, (58.5) together with*

$$W_{2a} = W_a^4 \tag{58.8}$$

implies the scalar equation H of Section 55, without assuming any Q.

Proof. As $\delta - 1 = \omega^2$, the scalar is

$$H = \left(\sum_a W_a \right)^2 + \omega^2 \sum_a B_{a,2a}.$$

The origin contributes $1 + \omega^2 B_{0,0} = 1 + \omega^3 = 0$. For $a \neq 0$, (58.8) gives

$$W_a^2 + \omega^2 B_{a,2a} = W_a^2 + \frac{\omega^2}{W_a^2} = W_a^2 + W_{-a}^2.$$

Summing over the inverse pairs proves $H = 0$. □

This does not prove the quadratic or cubic equations. For example, the constant choice $W_a = \omega^2$ for all $a \neq 0$ satisfies (58.5) and (58.8), and hence H, but fails the first diagonal phase Q. Its residual is ω at every odd n: the terms at $m = 0, 1, -1$ are $\omega, 1, 1$, and the remaining $n - 3$ terms, all ω^2, cancel in characteristic 2.

58.3. **An exact search along the infinite family $n = 3^k$.** The integer 2 generates $(\mathbb{Z}/3^j\mathbb{Z})^\times$ for every $j \geq 1$. Consequently the nonzero labels at $n = 3^k$ split into k doubling orbits, represented by 3^s, $0 \leq s < k$. Their lengths are

$$r_s = 2 \cdot 3^{k-s-1}.$$

Put $N_s = 4^{r_s/2} + 1$, and choose a primitive N_0-th root ξ in $\mathbb{F}_{4^{r_0}}$. Since $N_s \mid N_0$, every solution of (58.5)–(58.8) is specified uniquely by

$$\begin{aligned} 0 \leq i_s < N_s, \qquad W_{3^s} &= \omega^2 \xi^{(N_0/N_s) i_s}, \\ W_{2^j 3^s} &= W_{3^s}^{4^j} \quad (0 \leq j < r_s). \end{aligned} \tag{58.9}$$

Indeed closure gives $W_{3^s} \in \mathbb{F}_{4^{r_s}}$, while $2^{r_s/2} \equiv -1 \pmod{3^{k-s}}$ turns inverse symmetry into $W_{3^s}^{1+4^{r_s/2}} = \omega$. Its solutions are exactly the coset $\omega^2 \mu_{N_s}$ in (58.9). There is no unknown field or square-root selection in this parametrization.

Frobenius covariance reduces all the nonexceptional quadratic columns to $h = 3^s$. We therefore filter (58.9) by

$$\sum_{m \in \mathbb{Z}_n} W_m W_{h-m} W_{-m-g} W_{m+g-h} = 0, \qquad 0 < g < n, \quad h \in \{1, 3, \ldots, 3^{k-1}\}. \tag{58.10}$$

Every term is $\omega^e \xi^{L(i_0, \ldots, i_{k-1})}$, with an explicit linear exponent. Binary-field addition is XOR, so tables of $\omega^e \xi^a$ make exhaustive search inexpensive at the first three orders. This is an exact finite algorithm, not an asymptotic nonemptiness argument.

n	Ambient field	Number of phase candidates	First diagonal Q survivors	Full-Q survivors	Frobenius orbit lengths
3	$\mathbb{F}_{2^4}$	5	2	2	2
9	$\mathbb{F}_{2^{12}}$	325	24	12	6, 6
27	$\mathbb{F}_{2^{36}}$	85,197,125	1,350	54	18, 18, 18

Every orbit in this table passes every cubic equation. The search program `hi_char2_frobenius.py` exports one row certificate per orbit, then invokes the independent verifier. In particular, each of the three order-27 files passes all 729 Q and all 531,441 C. The theorem in Section 58.1 therefore establishes cyclic HI categorification at order 27 without numerical HI seeds, Stark polynomials, or recognition of characteristic-zero algebraic numbers.

For a compact explicit order-27 specification, take

$$\mathbb{F}_{2^{36}} = \mathbb{F}_2[z]/(1 + z + \cdots + z^{36}).$$

The modulus is irreducible, since 2 has order 36 modulo 37. Encode $\sum_j c_j z^j$ by $\sum_j c_j 2^j$. In this encoding use

$$\omega = 41370444133, \qquad \xi = 25668351056,$$

where ξ has order $262145 = 5 \cdot 13 \cdot 37 \cdot 109$. The three representatives (i_0, i_1, i_2) in (58.9) are

$$(1509, 12, 3), \qquad (3577, 14, 2), \qquad (6763, 2, 4). \tag{58.11}$$

The JSON certificates retain the complete field and row data, so checking them does not depend on reproducing the search or the particular primitive-element choices made by PARI.

There is still no established induction from 3^k to 3^{k+1}. The independent lower-stratum phases cannot simply be omitted. For example, at order 9 the tempting normalized norm

prescription $W_3 = W_1^{13}$ excludes all 12 full-Q solutions in (58.9): with compatible generators it would require $i_1 \equiv i_0 \pmod 5$, which none of the successful pairs satisfy. Their pairs are

$$(1,0),(4,0),(5,1),(15,1),(16,0),(20,4),(45,1),(49,0),(50,4),(60,4),(61,0),(64,0).$$

This obstructs that particular finite-field recursion, without asserting a failure of the differently twisted ray-field norm formulas in Sections 38–40. Section 63 gives a correction at 9 and 27: the norm must use a different Frobenius branch, with a twist when cycling back to the first branch.

The remaining uniform target is now entirely algebraic over $\overline{\mathbb{F}}_2$: prove that (58.9) has a solution to (58.7) for infinitely many k, or construct another unbounded family of complete characteristic-2 solutions. Frobenius covariance settles H uniformly, and categorical lifting settles the passage to complex existence; neither supplies the missing nonemptiness theorem.

59. The same hyperbolic monodromy in the arithmetic and S-fold routes

A literature check identifies a precise common lattice transformation behind the arithmetic candidate and a physical construction of HI modular data. This comparison does not prove categorification, but it makes the possible connection more specific than a shared occurrence of the quantum dilogarithm.

Let $d^2 = nd + 1$, with n odd, and use the basis $(d, 1)$ of $\mathbb{Z}[d]$. Multiplication by d and d^2 has matrices

$$M_n = \begin{pmatrix} n & 1 \\ 1 & 0 \end{pmatrix}, \qquad P_n = M_n^2 = \begin{pmatrix} n^2+1 & n \\ n & 1 \end{pmatrix}. \tag{59.1}$$

The first reverses the standard alternating form, and the second preserves it. Moreover,

$$\operatorname{Tr} P_n = n^2 + 2, \qquad (\operatorname{Tr} P_n)^2 - 4 = n^2(n^2+4), \qquad P_n - I = nM_n. \tag{59.2}$$

Thus its eigenvalues are d^2, d^{-2}, with the real quadratic field of the HI dimension. Smith normal form gives

$$\operatorname{coker}(M_n - I) \cong \mathbb{Z}/n\mathbb{Z}, \qquad \operatorname{coker}(P_n - I) \cong (\mathbb{Z}/n\mathbb{Z})^2. \tag{59.3}$$

Indeed the first matrix has an entry 1 and determinant $-n$, whereas the second is n times a unimodular matrix. Modulo n, M_n exchanges the two coordinates. Its two eigensubgroups are $\{(a,a)\}$ and $\{(a,-a)\}$; as n is odd, these are complementary cyclic Lagrangian subgroups of the finite symplectic module in (59.3). These are proved lattice statements; they do not identify a category or a matrix of associators.

There is also an exact comparison with S-fold monodromies. Put

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad R = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

Then

$$P_n = T^n R^n, \qquad ST^{-n}ST^n = -R^n T^n = -R^n P_n R^{-n}. \tag{59.4}$$

In particular the two give conjugate transformations in $\mathrm{PSL}_2(\mathbb{Z})$. The central sign in $\mathrm{SL}_2(\mathbb{Z})$ should be retained when discussing physical or quantum lifts.

Gang–Kim's S-fold analysis uses levels $n^2 + 2$ to obtain nonunitary HI modular data after removing a semion factor. Gang–Kim, *Generalized non-unitary Haagerup-Izumi modular data from 3D S-fold SCFTs*. The more recent two-segment construction includes the monodromies $ST^{-n}ST^n$, and compares their Galois-transformed modular matrices with the untwisted HI data for every odd n. Jeong–Sohn, Section 4.2 and Table 1. Equation (59.4) is the direct arithmetic comparison with that family.

For the analytic side, Yalkinoglu announces approximations of Shintani invariants by cyclic quantum dilogarithms along rational approximants to a modular geodesic. The stated setting assumes a positive fundamental unit with a one-term minus continued fraction; the paper defers full proofs. Its cyclic dilogarithm orders are the growing denominators of those approximants.

Yalkinoglu, Theorems 3.1–3.2. This is not an exact finite-order HI pentagon identity. Applying such an approximation to Section 51 would require handling its actual continued-fraction cycle and proving that a suitable identity survives in the limit with exactly the HI index sum and collision corrections.

The missing categorical step on the physical side is also explicit: one needs a rigorous semisimple ribbon category realizing the proposed modular data, together with a suitable Lagrangian algebra and an identification of its module fusion rules. Equality of S and T matrices alone supplies neither these algebra structure maps nor the HI associators. Alternatively, a direct identification of the finite operator of Section 50 with a quantum-dilogarithm construction would need to prove the fifth-order relation on the complementary subspace and account for the exceptional three-dimensional block. The lattice equalities (59.1)–(59.4) do not establish either step. **Infinitude remains unresolved.**

60. An exact fifth-order criterion in characteristic 2

The positivity argument in Section 50 cannot simply be reduced modulo

(1) Its three-dimensional Gram matrix becomes singular. Nevertheless, the entire cubic system has an exact operator reformulation in this characteristic. A universal rank-three correction turns the transfer operator into an operator whose fifth power must be the identity. This gives a precise target for a finite quantum-dilogarithm approach.

60.1. **A characteristic-free residual identity.** Work over any field where $\delta(\delta-1)\neq 0$, with $\delta^2 = n\delta + 1$, $t = \delta^{-1}$, and the normalized exceptional row and column. Define R by (50.1), and use **unnormalized** vectors in the space indexed by (a, b):

$$u(a,b) = \delta_{b,0}, \quad v(a,b) = \delta_{a,0}, \quad w(a,b) = A_{b,a}, \quad z(a,b) = A_{a,b}.$$

All transposes in this section are ordinary linear transposes. Define

$$J_{(a,b),(c,e)} = \delta_{a,e}A_{b+c,a} - \delta_{a,0}\delta_{e,0}. \tag{60.1}$$

For the quadratic residual Q of Section 52, direct multiplication gives

$$\begin{aligned}(RJ - I)_{(a,b),(c,e)} &= \delta_{b,e}Q(c-a,b),\\ (JR - I)_{(a,b),(c,e)} &= \delta_{a,c}Q(b-e,a).\end{aligned} \tag{60.2}$$

The exceptional column sum $-t$ cancels the rank-one corrections. Consequently all Q imply $J = R^{-1}$, in any characteristic.

Reshape the cubic residual as $F_{(a,b),(c,e)} = C(b, c, a, e)$. There is the **off-solution** identity

$$\boxed{F = R^3 - J^2 - wu^{\mathsf T} - vz^{\mathsf T} + vu^{\mathsf T} + tuv^{\mathsf T}.} \tag{60.3}$$

Indeed the $((a, b), (c, e))$ entry of R^3 is the cubic sum in $C(b, c, a, e)$, while

$$(J^2)_{(a,b),(c,e)} = A_{b+e,a}A_{a+c,e} - \delta_{e,0}A_{b,a} - \delta_{a,0}A_{c,e} + \delta_{a,0}\delta_{e,0}.$$

Substitution proves (60.3), without assuming Q or C.

Assume Q from now on. Its direct vector consequences are

$$\begin{aligned} &Ru = w, && Rv = -tu, && Rw = v - tu,\\ &u^{\mathsf T}R = -tv^{\mathsf T}, && v^{\mathsf T}R = z^{\mathsf T}, && z^{\mathsf T}R = u^{\mathsf T} - tv^{\mathsf T}.\end{aligned} \tag{60.4}$$

Multiplying (60.3) on the right by R^2, and using these relations, proves

$$\boxed{\text{all C} \iff R^5 - I = D, \qquad D = -t(wz^{\mathsf T} + vv^{\mathsf T} + uu^{\mathsf T} - tuv^{\mathsf T}).} \tag{60.5}$$

Both directions follow since R is invertible. This is an equality of specified matrices; a rank bound on $R^5 - I$ alone is insufficient. The matrix D is defined from any full-Q candidate and commutes with R, by (60.4), before C is imposed.

In the phase ansatz, the columns u, v, w and the rows $u^{\mathsf{T}}, v^{\mathsf{T}}, z^{\mathsf{T}}$ are independent: u, v have different axis supports, and w, z are nonzero off those axes. The coefficient matrix in (60.5) is invertible, so D has rank exactly 3.

60.2. **The universal rank-one and nilpotent parts.** Now take characteristic 2, n odd and $\delta = \omega$ as in Section 58. Thus $t = \omega^2$, $t^2 + t + 1 = 0$ and $t^3 = 1$. Put

$$K = [u\ v\ w], \qquad L = [u\ v\ z]^{\mathsf{T}}.$$

The Q identities give

$$LK = \begin{pmatrix} 1 & 1 & t \\ 1 & 1 & t \\ t & t & 1+t \end{pmatrix} = rr^{\mathsf{T}}, \qquad r = (1, 1, t)^{\mathsf{T}}. \tag{60.6}$$

This matrix has rank 1, rather than rank 3. In particular the orthogonal-complement decomposition of Section 50 is unavailable.

There is instead a completely explicit algebraic decomposition. Set

$$e = u + tv + t^2 w, \qquad f^{\mathsf{T}} = tu^{\mathsf{T}} + v^{\mathsf{T}} + t^2 z^{\mathsf{T}}.$$

Using (60.5)–(60.6), without assuming C, gives

$$D^2 = ef^{\mathsf{T}}, \qquad f^{\mathsf{T}}e = t^2, \qquad D^3 = tD^2, \qquad Re = te, \qquad f^{\mathsf{T}}R = tf^{\mathsf{T}}. \tag{60.7}$$

Consequently

$$E = tD^2, \qquad N = D - tE \tag{60.8}$$

satisfy

$$\boxed{E^2 = E, \quad \operatorname{rank} E = 1, \quad RE = ER = tE, \quad N^2 = 0, \quad EN = NE = 0, \quad \operatorname{rank} N = 2.} \tag{60.9}$$

To see the last rank assertion, use $D = tE + N$, the direct sum $\operatorname{im} E \oplus \ker E$, and $\operatorname{rank} D = 3$. All of (60.7)–(60.9) hold for a full-Q phase candidate, regardless of C.

Theorem 60.1. *In characteristic 2, for any normalized full-Q phase candidate, define*

$$\boxed{\mathcal{P} = R(I + D).} \tag{60.10}$$

Then

$$\boxed{\text{all } C \iff \mathcal{P}^5 = I.} \tag{60.11}$$

The correction $\mathcal{P} - R$ has rank 3.

Proof. Write $V = I + D$. On $\operatorname{im} E$, V is multiplication by $1 + t = t^2$, of order 3. On $\ker E$, it is $I + N$, of order 2. Hence $V^6 = I$ and $V^5 = V^{-1}$. As R commutes with V,

$$\mathcal{P}^5 = R^5 V^{-1}.$$

This is the identity exactly when $R^5 = V = I + D$, which is (60.5). The rank statement follows from invertibility of R. □

At a full solution $\mathcal{P} = R^6$. The original transfer operator satisfies

$$R^{10} = I + t^2 E, \qquad R^{30} = I. \tag{60.12}$$

In fact its order is **exactly 30**. The eigenvalue t forces a factor 3 in its order, and the nonzero square-zero N forces a factor 2. On $\operatorname{span}(u, v, w)$, the characteristic polynomial is

$$X^3 + tX + t = (X + t)(X^2 + tX + 1).$$

The two roots of the quadratic are primitive fifth roots of unity, forcing a factor 5. This also shows why R itself cannot be identified with a diagonalizable fifth-order operator. The corrected operator $\mathcal{P}$, however, is diagonalizable at every full solution, since $X^5 - 1$ is separable in characteristic 2.

60.3. **Exact checks and a restriction on an adjoint construction.** `hi_finite_field_transfer.py` verifies (60.2)–(60.3) on order-5 phase matrices deliberately failing Q and C, both over $\mathbb{F}_{16}$ and over $\mathbb{Q}(\sqrt{29})$. It then checks the complete matrix identities (60.5)–(60.12) on the saved order-3 and both order-9 characteristic-2 certificates. All checks pass, including the stated ranks and the exact order 30.

One tempting construction would identify $\mathcal{P}$, up to a linear change of basis, with conjugation by an $n \times n$ cyclic quantum-dilogarithm matrix. This has a testable restriction. If $\mathcal{P} \sim \mathrm{Ad}(B)$ and $\mathcal{P}^5 = I$, then B^5 is scalar. Over an algebraically closed field we may normalize $B^5 = I$. If the multiplicities of its fifth-root eigenvalues are $m_0, \dots, m_4$, then those of $\mathrm{Ad}(B)$ are

$$M_j = \sum_{i \in \mathbb{Z}_5} m_i m_{i+j}, \qquad \sum_i m_i = n. \tag{60.13}$$

These are integer dimensions, even though the matrices have characteristic 2.

At the order-9 representative $(i_0, i_1) = (1, 0)$, the computed multiplicities of $\mathcal{P}$ are

$$(M_0, M_1, M_2, M_3, M_4) = (17, 17, 15, 15, 17)$$

for one choice of primitive fifth root. Equation (60.13) is impossible: $\sum m_i = 9$ and $\sum m_i^2 = 17$ force the multiset $\{2, 2, 2, 2, 1\}$, whose four nontrivial autocorrelations are all

(1) Thus this complete HI solution cannot arise from that adjoint construction in any linear basis.

The other order-9 representative, $(5, 1)$, has multiplicities $(17, 16, 16, 16, 16)$, so it passes this necessary test. This does not construct its phase matrix, much less an infinite family. It shows that the adjoint approach, if successful, must select suitable branches. The exact criterion (60.11) is the remaining operator target; one still needs full-Q phase candidates for unbounded orders whose corrected operators satisfy it.

60.4. **The consecutive-ratio bound also survives in characteristic 2.** The obstruction of Section 46 has an algebraic version requiring no absolute values. Let k be any field containing a primitive n-th root ζ, let $W_0^2 = 1$, and suppose $W_a W_{-a} = \delta \neq 1$ for $a \neq 0$. Assume all phases are nonzero. If

$$\frac{W_a}{W_{a-1}} = \frac{P(\zeta^a)}{Q(\zeta^a)}, \qquad \deg P, \deg Q \leq r, \qquad Q(\zeta^a) \neq 0$$

for every a, then $2r \geq n - 2$.

Indeed the inverse norms make the consecutive ratios at a and $1 - a$ equal for $a \notin \{0, 1\}$. Therefore

$$z^r \big(P(z)Q(\zeta/z) - Q(z)P(\zeta/z)\big)$$

is a polynomial of degree at most $2r$ vanishing at the $n - 2$ distinct points ζ^a, $2 \leq a < n$. Its value at 1 is nonzero: the ratios at 0 and 1 differ by the nontrivial factor δ^{-1}. This proves the bound in every characteristic.

Thus characteristic 2 does not permit a fixed number of nonsingular cyclic-hypergeometric factors to provide all the phases in unbounded order. A quantum-dilogarithm approach to (60.11) would need a different extraction of phase data, or a separately proved treatment of the exceptional indices. Neither that extraction nor infinitude is established here.

61. A fixed elliptic curve parametrizes the characteristic-2 phase space

The multiplicative parameters in (58.9) have a geometric realization on one curve, independent of k. This identifies a possible setting for addition identities, but does not make the quadratic or cubic equations automatic.

61.1. **Frobenius, inverse norms, and rational interpolation.** Work over $\mathbb{F}_4$, with $\omega^2 + \omega + 1 = 0$, and put

$$E: \quad y^2 + y = x^3 + \omega, \qquad c = \omega^2. \tag{61.1}$$

Negation sends (x, y) to $(x, y + 1)$. The tangent formula gives

$$[2](x, y) = (x^4, x^6 + x^3 + y + 1) = (x^4, y^4), \tag{61.2}$$

where the second equality uses (61.1). Thus the fourth-power Frobenius is exactly [2], as an endomorphism of the curve.

For any odd-order point P and any rational function $r \in \mathbb{F}_4(x)$, define, wherever the values are nonzero and finite,

$$W_0 = 1, \qquad W_a = c\,\frac{y(aP) + r(x(aP))}{y(aP) + r(x(aP)) + 1} \quad (a \neq 0). \tag{61.3}$$

At a pole of r, the displayed quotient is interpreted as 1, so $W_a = c$. More concretely, for $r = A/B$ use

$$W_a = c\,\frac{y(aP)B(x(aP)) + A(x(aP))}{(y(aP) + 1)B(x(aP)) + A(x(aP))}. \tag{61.4}$$

Negation and (61.2) immediately give

$$W_a W_{-a} = \omega, \qquad W_{2a} = W_a^4. \tag{61.5}$$

The prescribed value at zero is essential: the underlying rational function in (61.3) has value c, rather than 1, at the identity of the curve. Indeed y has pole order 3 there, whereas a rational function of x has even pole order if it has a pole at all. Their leading poles cannot cancel.

Proposition 61.1. *Fix $n = 3^k$ and a point P of order n. Every phase sequence satisfying (58.5) and (58.8) has a representation (61.3) for some $r \in \mathbb{F}_4(x)$. Conversely, every such rational function, written without common numerator and denominator zeros at the relevant points, gives a valid sequence satisfying those two conditions.*

Proof. Suppose aP has order 3^j, and put $s = 3^{j-1}$. Since $2^s \equiv -1 \pmod{3^j}$, (61.2) shows that

$$x(aP)^{4^s} = x(aP), \qquad y(aP)^{4^s} = y(aP) + 1. \tag{61.6}$$

The degree of $x(aP)$ over $\mathbb{F}_4$ is exactly s: an earlier Frobenius return would give $2^b \equiv \pm 1 \pmod{3^j}$ for $0 < b < s$. In particular $y(aP) \notin \mathbb{F}_{4^s}$.

For $W_a \neq c$, prescribe

$$r(x(aP)) = \frac{W_a}{W_a + c} + y(aP); \tag{61.7}$$

for $W_a = c$, prescribe the value ∞. Inverse norms make these prescriptions identical at aP and $-aP$; Frobenius covariance makes them compatible under fourth powers. By (61.6), each finite value belongs to $\mathbb{F}_{4^s}$.

The polynomial

$$\Psi_P(X) = \prod_{a=1}^{(n-1)/2} (X - x(aP)) \tag{61.8}$$

is squarefree over $\mathbb{F}_4$. It has one irreducible factor of each degree $1, 3, \ldots, 3^{k-1}$. Chinese remaindering constructs polynomials A, B over $\mathbb{F}_4$ with the prescribed projective values: on each finite-value factor set $B = 1$ and prescribe A; on each infinite-value factor set $B = 0, A = 1$. Both polynomials may be chosen with degree less than $(n - 1)/2$. They have no common zero on this finite set, and their quotient gives (61.3).

Conversely, a finite value of $r(x(aP))$ lies in $\mathbb{F}_{4^s}$, while $y(aP)$ does not. Thus neither numerator nor denominator in (61.3) can vanish. A pole gives the nonzero value c. Equations (61.5) finish the proof. □

In particular the phase space in (58.9) is, after choosing P,

$$\prod_{j=1}^{k} \mathbb{P}^1(\mathbb{F}_{4^{3^{j-1}}}). \tag{61.9}$$

Its factor sizes are exactly $4^{3^{j-1}}+1$, agreeing with the norm-torus counts. The proposition concerns the norm and Frobenius constraints only. Its interpolation degree grows with n, and it provides no nonemptiness theorem for the full HI equations.

61.2. **A recurrence fitting three orders fails at the fourth.** Rational interpolation of the known phase data suggested the following particularly short recurrence. For each $\eta \in \mathbb{F}_4^\times$, put

$$\begin{aligned} r_1^{(\eta)}(x) &= \eta^2, \\ s_k(x) &= r_k^{(\eta)}(\eta(x+c)), \\ r_{k+1}^{(\eta)}(x) &= c^2 s_k(x)^3 + \eta^2 s_k(x)^2 + \frac{c}{x} + c. \end{aligned} \tag{61.10}$$

The next two values are independent of η:

$$r_2(x) = \frac{c}{x}, \qquad r_3(x) = \frac{cx^4 + x^3 + x^2 + x + c}{x^4 + cx^3 + c^2x^2 + x}. \tag{61.11}$$

Both identities follow by direct substitution, and the program checks them symbolically. At orders 3, 9 and 27, evaluating these functions on suitable torsion points gives full HI solutions. At order 81, however, none of the three functions $r_4^{(\eta)}$ works.

`hi_char2_elliptic.py` verifies this without reading saved HI data. It constructs a basis of $E[n]$, certifies its independence with the Weil pairing, and enumerates every primitive point modulo doubling. There are $4n/3$ such orbits: the number of primitive points is $8n^2/9$, and each orbit has length $2n/3$. Since the functions are defined over $\mathbb{F}_4$, vanishing of all Q is unchanged under this orbit operation. The exact search gives:

Order	Distinct functions	Function/point-orbit evaluations	First diagonal Q survivors	Full-Q survivors
3	3	12	7	7
9	1	12	3	1
27	1	36	1	1
81	3	324	0	0

Every full-Q survivor in this table is exported and checked against **every cubic** by `hi_finite_field_certificate.py`. The order-27 candidate passes all 729 Q and all 531,441 C. At order 81, the nonzero first diagonal residual already rules out each candidate, including all 5,832 primitive torsion points for each function.

Thus (61.10) is an exact fit to three successive orders whose proposed inductive continuation is false. The geometric parametrization (61.3) remains available, but a successful rational-function construction needs additional structure beyond this cubic recurrence. Neither an unbounded family of complete certificates nor infinitude is established.

62. Every order-13 quadratic phase solution satisfies the cubics

The circuit model of Section 57 also permits imposing all Q directly, without first selecting phases by H.

Theorem 62.1 (exact computation)**.** *At $n = 13$, every complex phase solution of (Q), with the prescribed inverse norms and normalization, satisfies every (C). Its row algebra over $K = \mathbb{Q}(\delta)$ is a field of degree 12, and its phase algebra has dimension 156 over K.*

Proof. Let $\mathcal{A}_{13}$ be the full one-column row algebra of Section 54, of dimension 924. Let I_Q be the ideal generated in it by the remaining quadratic residuals. Section 57's integral circuit matrices give a specialization at

$$q = 32059, \qquad \delta \mapsto 1153.$$

All the hypotheses for the integral model are checked, including the units required in the row reconstruction. Rank specialization for the map with images generating I_Q gives

$$\dim_K \mathcal{A}_{13}/I_Q \le \dim_{\mathbb{F}_q} \overline{\mathcal{A}}_{13}/\overline{I}_Q. \tag{62.1}$$

The exact Krylov presentation of the specialized algebra is $\mathbb{F}_q[t]/(F(t))$, with F squarefree of degree 924. In every irreducible factor, reconstruct the entire matrix and test all nonexceptional Q. Exactly one factor, of degree 12, survives. This counts the full quotient over the algebraic closure, including points not rational over $\mathbb{F}_q$; the right side of (62.1) is 12.

Independently, `hi_stark_run(13)` supplies an exact full Q/C solution in a field of absolute degree 24. A row coordinate has minimal polynomial of degree 24 and the dimension generates a quadratic subfield. Thus this row point has residue field of degree 12 over K. Evaluation gives a surjection

$$\mathcal{A}_{13}/I_Q \longrightarrow L, \qquad [L : K] = 12,$$

where every cubic residual vanishes. The upper bound forces this surjection to be an isomorphism. Hence all cubics vanish identically in the full Q quotient, proving the implication for all its complex points, at either dimension embedding.

Finally the character-gauge extension $X^{13} = \delta \prod_{a=2}^{12} R_a$ is finite étale of degree 13. The phase algebra therefore has dimension $13 \cdot 12 = 156$. □

`hi_column_fullq.py` regenerates the critical matrices and the modular quotient, then reruns the independent exact cubic construction and degree checks. No H equation is imposed. The verified orders for the full phase implication are now

$$n = 3, 5, 7, 9, 11, 13.$$

The matrix construction is uniform, but neither the required quotient dimension nor a cubic lower bound is proved for unbounded orders.

63. The norm recursion permutes Frobenius branches

The failed norm prescription in Section 58 used a single branch at a time. The ray-class structure instead predicts a permutation of branches. This permutation has a uniform description, and the saved finite certificates satisfy its norm laws exactly.

63.1. **A uniform class-order and Frobenius relation.** Put

$$n = 3^k, \qquad D = n^2 + 4, \qquad d = \frac{n + \sqrt{D}}{2}, \qquad \mathcal{O} = \mathbb{Z}[d].$$

Here $\mathcal{O}$ means the order of discriminant D, even when it is not maximal. Use the ray modulus $\mathfrak{m} = (d - 1)$ together with the real place at which $d > 0$. Let G be this ray class group of $\mathcal{O}$, and write $h(D) = |\operatorname{Pic}(\mathcal{O})|$.

Theorem 63.1. *Let $\overline{\mathfrak{p}} = (3, d + 1)$, and let $\beta \in G$ be its ray class. Then*

$$\begin{gathered} 1 \longrightarrow (\mathbb{Z}/n\mathbb{Z})^\times \longrightarrow G \longrightarrow \operatorname{Pic}(\mathcal{O}) \longrightarrow 1, \\ \operatorname{ord}_{\operatorname{Pic}(\mathcal{O})}[\overline{\mathfrak{p}}] = k, \qquad \beta^k = \operatorname{Frob}_{(2)}, \qquad \operatorname{ord}(\beta) = k\varphi(n). \end{gathered} \tag{63.1}$$

In particular $k \mid h(D)$. On the $h(D)$ cosets of $\langle \operatorname{Frob}_{(2)} \rangle$, multiplication by β consists of $h(D)/k$ cycles, each of length k.

Proof. First d is a fundamental unit of $\mathcal{O}$. If a unit u satisfied $1 < u < d$, write $u - u' = b\sqrt{D}$, where b is a positive integer and $u' = \pm u^{-1}$. Since $d + d^{-1} = \sqrt{D}$,

$$0 < b \leq \frac{u + u^{-1}}{\sqrt{D}} < 1,$$

a contradiction. Thus $\mathcal{O}^\times = \{\pm d^j : j \in \mathbb{Z}\}$.

The quotient $\mathcal{O}/(d-1)$ is $\mathbb{Z}/n\mathbb{Z}$. In the ray-class exact sequence, d has image $(1, +1)$ and -1 has image $(-1, -1)$. Dividing $(\mathbb{Z}/n\mathbb{Z})^\times \times \{\pm 1\}$ by this latter diagonal subgroup gives the first line of (63.1). This uses the exact sequence for orders in Kopp–Lagarias, Corollary 6.7, so no maximal-order or squarefreeness hypothesis is being inserted.

The two primes above 3 are coprime to the conductor of $\mathcal{O}$, and

$$(d-1) = \mathfrak{p}^k, \qquad (d+1) = \overline{\mathfrak{p}}^k. \tag{63.2}$$

To prove that the class order is exactly k, suppose $\overline{\mathfrak{p}}^e = (\alpha)$ for $0 < e < k$. Then $|N\alpha| = N = 3^e \leq n/3$. Multiplying by a power of d and by a sign, arrange

$$\sqrt{N/d} \leq \alpha < \sqrt{Nd}.$$

Writing $\alpha - \alpha' = b\sqrt{D}$, now with $b \in \mathbb{Z}$, gives

$$|b|\sqrt{D} \leq \alpha + N/\alpha \leq \sqrt{N}(\sqrt{d} + 1/\sqrt{d}) < \sqrt{D}.$$

For the last inequality use $N(\sqrt{D}+2) \leq (n/3)(\sqrt{D}+2) < D$. Hence $b = 0$, so α is an integer. Its principal ideal has equal valuations at the two primes above 3, contradicting $(\alpha) = \overline{\mathfrak{p}}^e$.

In the ray group, (63.2) gives $\beta^k = [(d+1)] = [(2)]$, because $d + 1 \equiv 2 \pmod{\mathfrak{m}}$ and both generators are positive at the selected real place. The image of $[(2)]$ in the local kernel is 2, which has order $\varphi(3^k)$. Since the image of β in the ordinary class group has order k, its ray order is $k\varphi(n)$. The coset-cycle statement follows. □

The prime 2 is inert in the quadratic field and is coprime to the order conductor and the ray modulus. Thus, for the ray class field L of this order, its residue algebra at 2 has the form

$$\mathcal{O}_L/2\mathcal{O}_L \cong \prod_{h(D)} \mathbb{F}_{4^{\varphi(n)}}. \tag{63.3}$$

This is a theorem about the arithmetic field. Identifying this algebra with an algebra of full HI solutions would require an additional proof.

`hi_char2_ray_indices.gp` certifies the class groups and units with `bnfcertify`, then checks the ray groups, the prime classes, and $\beta^k = \mathrm{Frob}_{(2)}$:

n	$h(n^2+4)$	Ray group invariant factors	Frobenius order	Cycles of β on Frobenius cosets
3	1	2	2	one cycle of length 1
9	2	12	6	one cycle of length 2
27	3	54	18	one cycle of length 3
81	8	216, 2	54	two cycles of length 4
243	25	4050	162	five cycles of length 5
729	36	2916, 6	486	six cycles of length 6
2187	84	30618, 2, 2	1458	twelve cycles of length 7

The first three arithmetic Frobenius indices agree with the complete phase-orbit counts of Section 58. The later entries are arithmetic counts, not established HI orbit counts.

The nonmaximal-order distinction already matters at $n = 3^8 = 6561$: the discriminant is $43046725 = 5^2 \cdot 1721869$. The same script certifies class number 384 for $\mathbb{Z}[d]$, compared with 96 for the maximal order, and verifies that the conjugate prime has order 8 in the former class group. It obtains the order class group as the maximal-order ray group modulo the conductor,

divided by the image of the rational residue units. Thus the uniform theorem does not rely on the first seven displayed orders happening to be maximal.

63.2. **The saved phases obey the branch-permuting norm laws.** At orders 9 and 27, number the saved Frobenius representatives in their existing JSON order, $j = 0, \dots, k-1$. On tuples of field elements define

$$(Tx)_0 = x_{k-1}^4, \qquad (Tx)_j = x_{j-1} \ (j > 0). \tag{63.4}$$

Then T^k is componentwise fourth-power Frobenius, exactly the relation in (63.1). For $3^s \| a$ with $s < k-1$, set

$$r_s = 2 \cdot 3^{k-s-1}, \qquad \nu_s(x) = \omega^2 N_{\mathbb{F}_{4^{r_s}}/\mathbb{F}_{4^{r_s/3}}}(x). \tag{63.5}$$

The scalar factor is essential: an ordinary degree-three field norm sends the norm coset to the norm-one group.

Exact finite result. The complete saved phase families satisfy

$$\boxed{(W_{3a}^{(j)})_j = T\big(\nu_s(W_a^{(j)})_j\big)} \qquad (n = 9, 27; \quad 3a \neq 0). \tag{63.6}$$

In the exponent coordinates of (58.9), the order-9 top exponents $(1, 5)$ give lower exponents $(4 \cdot 5, 1) \bmod 5 = (0, 1)$. At order 27,

$$\begin{aligned} i_0 &= (1509, 3577, 6763), \\ i_1 &= (4 \cdot 6763, 1509, 3577) \bmod 65 = (12, 14, 2), \\ i_2 &= (4 \cdot 2, 12, 14) \bmod 5 = (3, 2, 4). \end{aligned} \tag{63.7}$$

These are precisely the independent stratum values found by the search.

`hi_char2_twisted_norm.py` checks (63.6) directly from the saved row certificates, without using their search exponents. Frobenius covariance gives $R_2 = W_1^{-2}$, so the unique characteristic-two square root recovers W_1; prefix reconstruction then recovers every phase. All 12 order-9 and all 72 order-27 norm comparisons pass. The same-branch prescription fails in every comparison. At order 27, omitting the fourth-power twist at the return to branch zero causes 24 failures.

This is a norm relation among branches and conductor strata at a **fixed** order. It is not an induction from order 3^k to order 3^{k+1}. At order 81, (63.1) supplies two arithmetic cycles of four branches; a construction based on it should allow that permutation and its Frobenius twist. The present data neither produce those HI branches nor prove the required cubic identities in unbounded order. Infinitude remains unresolved.

64. A uniform obstruction to small-field and quotient inflation constructions

The smallest characteristic-two field and a natural quotient induction can now be excluded uniformly. These are obstructions to particular constructions, not obstructions to HI existence at the corresponding orders. They explain why extending the finite certificates requires more than repeating their values on larger groups.

64.1. **No Frobenius-covariant quadratic solution over the four-element field.** Let G be any nontrivial finite abelian group of odd order. Work in characteristic two, put $\omega^2 + \omega + 1 = 0$ and $c = \omega^2$, and write the phase quadratic residual as

$$F(g, h) = \sum_{m \in G} W_m W_{h-m} W_{-m-g} W_{m+g-h} \qquad (g, h \neq 0). \tag{64.1}$$

The norm and Frobenius conditions are

$$W_0 = 1, \qquad W_a W_{-a} = \omega \ (a \neq 0), \qquad W_{2a} = W_a^4. \tag{64.2}$$

The nonexceptional equations (Q) are exactly $F(g, h) = 0$.

Theorem 64.1. *There is no solution of (64.1)–(64.2) with every $W_a \in \mathbb{F}_4$.*

Proof, first step: eliminate the third nonzero value. In this field, Frobenius covariance means $W_{2a} = W_a$. Put

$$T_g = \sum_m W_m W_{m-g} W_{m+g}, \qquad S_g = W_g + W_{-g}.$$

The identity $W_{-a} = \omega W_a^2 + c\delta_{a,0}$ gives, without assuming Q,

$$F(g,g) = cT_g^2 + S_g. \tag{64.3}$$

Indeed, expand the two reversed factors in (64.1); the two boundary terms are $W_g^2 W_{2g}^2 = W_g$ and W_{-g}. Pairing the remaining terms of T_g under $m \mapsto -m$ gives

$$T_g = \omega + S_g^2 + b_g, \qquad b_g \in \mathbb{F}_2. \tag{64.4}$$

Away from $m = 0, \pm g$, the paired products are inverses in $\mathbb{F}_4^\times$, so their sum lies in $\mathbb{F}_2$. The three exceptional positions contribute the displayed terms.

If $W_g = c$, its norm partner is also c, so $S_g = 0$. Then (64.4) makes T_g either ω or c, and (64.3) cannot vanish. Thus Q forces

$$W_a \in \{1, \omega\} \quad (a \neq 0). \tag{64.5}$$

Second step: an additive closure contradiction. Write $W_a = 1 + cf(a)$, where $f : G \to \mathbb{F}_2$, including $f(0) = 0$. The norm and Frobenius laws become

$$f(-a) = 1 + f(a) \quad (a \neq 0), \qquad f(2a) = f(a). \tag{64.6}$$

Let $\mu = (|G| - 1)/2 \pmod 2$. In the group algebra $\mathbb{F}_2[G]$, write f^- for index reversal and J for the sum of all group elements. Then

$$f^2 = f, \qquad f^- = J + 1 + f, \qquad ff^- = \mu J.$$

Consequently every two-point correlation is μ, including the zero shift.

For $g, h, g - h, g + h \neq 0$, the following identity holds before imposing Q:

$$\boxed{\operatorname{Tr}_{\mathbb{F}_4/\mathbb{F}_2}(\omega F(g,h)) = y + (x+y)z + (x+y+1)v,} \qquad \begin{cases} x = f(g), & y = f(h), \\ z = f(h-g), & v = f(h+g). \end{cases} \tag{64.7}$$

Here is an expansion check of this identity. Define $t(a,b) = \sum_m f(m)f(m+a)f(m+b)$. On substituting $W = 1 + cf$ and the reversed-factor formula into (64.1), the constant, linear, quadratic and quartic contributions to the trace vanish. For the quadratic terms use the constant two-point correlation. The two boundary terms also have zero trace after multiplication by ω: their sum is $u + \omega u^{-1} \in c\mathbb{F}_2$, where $u = W_g W_h W_{g+h}^2 \in \mathbb{F}_4^\times$. The remaining trace is

$$t(g,h) + t(-g,-h) + t(g,-h) + t(-g,h).$$

For nonzero distinct a and b, expanding (64.6) at the three exceptional positions gives

$$t(a,b) + t(-a,-b) = f(a)f(b) + f(b) + (f(a) + f(b))f(b-a).$$

Apply this twice to obtain (64.7).

Now let $S = \{a : f(a) = 1\}$. It contains exactly one element from each pair $\{a, -a\}$, so is nonempty and excludes zero. If distinct $g, h \in S$, they cannot be negatives of each other. Putting $x = y = 1$ in (64.7), Q forces $f(g + h) = 1$. For $g = h$, the same conclusion follows from $f(2g) = f(g)$. Thus S is closed under addition. A nonempty additive subsemigroup of a finite group contains zero, a contradiction. □

This proof applies to every odd abelian group, although the companion computations use cyclic groups. In the specified Frobenius gauge, the smallest possible finite coefficient field therefore has 16 elements: ω already requires an even extension degree over $\mathbb{F}_2$, degree two is excluded, and the saved order-3 certificate attains degree four. The theorem makes no claim about arbitrary gauges or about all solutions over a fixed larger field.

64.2. **No inflation constant on nonzero quotient cosets.**

Theorem 64.2. *Let K be a proper nontrivial subgroup of an odd finite abelian group G. Over any field of characteristic two containing ω, a solution of Q and (64.2) cannot be constant on each coset $a + K$ with $a \notin K$.*

Proof. Suppose it were, and put $V = W|_K$. For $g, h \in K \setminus \{0\}$, all summands outside K in (64.1) are ω^2, by the norm law and coset constancy. There are an even number of them. Therefore

$$F_G(g, h) = F_K(g, h),$$

so V satisfies every quadratic on K.

Next take $0 \neq g \in K$ and $h \notin K$. Splitting the sum into the cosets K, $h + K$, and their complement gives

$$\begin{aligned} F_G(g, h) &= \omega \sum_{a \in K} (V_a V_{-g-a} + V_a V_{g-a}) + \omega^2 \\ &= \omega(V_{g/2}^2 + V_{-g/2}^2) + \omega^2. \end{aligned} \tag{64.8}$$

The second equality uses cancellation of the nonfixed pairs in a characteristic-two convolution on an odd group. The complement of the two cosets has odd cardinality, giving the final ω^2.

Vanishing of (64.8), for every g, forces $V_a + V_{-a} = c$. Together with $V_a V_{-a} = \omega$, this puts every nonzero-index value among the roots of

$$X^2 + cX + \omega = (X + 1)(X + \omega).$$

Hence V takes values in $\mathbb{F}_4$, inherits (64.2), and satisfies every quadratic on K. Section 64.1 excludes this. □

Thus a successful induction through cyclic quotients must change the values within some nonzero cosets. This obstruction does not exclude the branch-permuting norm relation of Section 63.

64.3. **Exact checks and the next-smallest field.** `hi_char2_small_field_obstruction.py` checks (64.3)'s diagonal exclusion and the off-solution trace formula on norm/Frobenius inputs at twelve odd orders through 127. The larger binary assignment spaces are sampled with a fixed seed; these checks are diagnostics for the uniform proof, not exhaustive existence searches. At every tested binary assignment it also exhibits a failed additive closure and independently evaluates the corresponding nonzero quadratic. Independent off-solution checks of both identities in Section 64.2 pass at group/kernel orders $(9, 3), (15, 5), (21, 7), (27, 9), (35, 7), (63, 9)$, using general $\mathbb{F}_{16}$ norm inputs without assuming Q or Frobenius.

The script separately exhausts the norm/Frobenius ansatz over $\mathbb{F}_{16}$ at orders 3, 11 and 43. Each doubling orbit at these orders has length twice an odd number, with inversion halfway around the orbit. Thus its initial phase satisfies $W^5 = \omega$ and has precisely five choices. There is one orbit at 3 and 11 and three at 43. The exact full-Q counts are

n	assignments	full-Q assignments
3	5	2
11	5	2
43	125	0.

These counts exclude only the stated field and Frobenius gauge. They do not exclude HI categories at order 43, nor an infinite family over larger finite fields. A construction satisfying the additive equations in unbounded order is still missing; infinitude is unproved.

65. The first phase forces the full field, and every relative norm has cubic lifts

The characteristic-two tower has a forced field size, rather than merely an ambient field convenient for search. Its relative norm steps also admit an explicit cubic construction. Both results hold uniformly; neither asserts nonemptiness of the full additive system.

65.1. **A single quadratic detects a collapsed top orbit.** Let $n = 3^k$, $k \geq 2$, and put $g = n/3$. Assume the norm conditions in (64.2), and suppose only that the unit-index phases are periodic under translation by g:

$$W_{a+g} = W_a \qquad (3 \nmid a). \tag{65.1}$$

No periodicity is imposed on the nonunit indices. For every unit h, the phase residual (64.1) then satisfies

$$\boxed{F(g,h) = \omega(W_g + W_{-g})^2 + \omega^2.} \tag{65.2}$$

Proof. Split the sum into the three cosets of $H = 3\mathbb{Z}_n$. At $m \in H$, the unit factors $W_{h-m}W_{m+g-h}$ multiply to ω, using (65.1). At $m \in h + H$, the other two unit factors multiply to ω. In the third coset all four indices are units and their product is ω^2; this coset has odd size. Thus

$$F(g,h) = \omega \sum_{a \in H} (W_a W_{-g-a} + W_a W_{g-a}) + \omega^2.$$

Each convolution is the square at its unique fixed index, because all other terms pair and cancel in characteristic two. Since g has order three, those two fixed indices are g and $-g$. This is (65.2). □

If the bottom phases additionally satisfy Frobenius covariance, then $W_{-g} = W_{2g} = W_g^4$ and $W_g^5 = \omega$. The right side of (65.2) is **always nonzero**. Indeed, its vanishing would give $W_g + W_{-g} = \omega^2$. The norm would then put both phases among the roots of

$$X^2 + \omega^2 X + \omega = (X+1)(X+\omega).$$

Both lie in $\mathbb{F}_4$, where fourth powering fixes them, contradicting their sum $\omega^2 \neq 0$.

In fact the five possible bottom phases lie in $\mathbb{F}_{16}$, and (65.2) takes the three values $1, \omega, \omega^2$, with multiplicities 2, 2 and 1 respectively. Thus this obstruction needs no estimate, field search, or unknown lower-stratum data.

65.2. **Exact field degree at every power of three.**

Theorem 65.1. *At $n = 3^k$, suppose the norm and Frobenius laws (64.2) hold. For $k \geq 2$, assume the single quadratic $F(n/3, 1) = 0$; for $k = 1$, assume $F(1,1) = 0$. Then*

$$\boxed{[\mathbb{F}_4(W_1) : \mathbb{F}_4] = 2 \cdot 3^{k-1} = 2n/3.} \tag{65.3}$$

Consequently this holds for every full-Q solution in this gauge. Every phase and row coordinate belongs to this field, and already $R_2 = B_{1,2}$ generates it. The exact field of all coefficients, including ω, is therefore $\mathbb{F}_{2^{4n/3}}$.

Proof. Put $r = 2 \cdot 3^{k-1}$. Doubling generates the units modulo n. Its orbit closure and halfway inversion give

$$W_1 \in \mathbb{F}_{4^r}, \qquad W_1^{1+4^{r/2}} = \omega.$$

Let $e = [\mathbb{F}_4(W_1) : \mathbb{F}_4]$, so $e \mid r$. If e is odd, it divides $r/2$, and the norm equation becomes $W_1^2 = \omega$. Hence $W_1 = \omega^2$ and $e = 1$. Otherwise e is twice a power of three.

For $k \geq 2$, every proper possibility for e therefore divides $r/3$. The element

$$u = 2^{r/3} \pmod{n}$$

has order three and equals $1 + t(n/3)$, with $t \not\equiv 0 \pmod 3$. Because $W_1^{4^{r/3}} = W_1$, Frobenius covariance gives $W_{ua} = W_a$ for every unit a. The three labels in this u-orbit are $a, a+n/3, a-n/3$. This is exactly (65.1), which contradicts the assumed quadratic by Section 65.1.

For $k = 1$, the only proper degree is one; then both nonzero phases equal ω^2, and direct substitution gives $F(1,1) = \omega \neq 0$. This proves (65.3).

The orbit closure at every other conductor stratum puts all phases in $\mathbb{F}_{4^r}$, as in Section 58. Finally,

$$R_2 = \frac{W_1^2}{W_2} = W_1^{-2}.$$

Squaring is an automorphism of a finite field, so $\mathbb{F}_4(R_2) = \mathbb{F}_4(W_1)$. The other row entries are rational expressions in the phases and ω, proving the remaining claims. □

In particular, all full-Q Frobenius orbits have length exactly $2n/3$. Their scalar stabilizer is trivial, since every scalar unit is a power of 2. This recovers the residue-field degree predicted by Section 63 directly from the phase equations, without identifying their solution algebra with a ray class field or assuming their number is a class number. At order 81, any such solution necessarily has first phase of degree 54 over $\mathbb{F}_4$, and coefficient field $\mathbb{F}_{2^{108}}$.

65.3. **A uniform cubic lift for every prescribed normalized norm.** Let $q = 4^s$, $s \geq 1$, and fix $Y \in \mathbb{F}_{q^2}$ with $Y^{q+1} = \omega$. For $t \in \mathbb{F}_{q^2}$, define

$$\boxed{P_{Y,t}(X) = X^3 + tX^2 + Yt^qX + \omega Y.} \tag{65.4}$$

Theorem 65.2. *Precisely $q(q-1)/3$ of the q^2 choices of t make (65.4) irreducible over $\mathbb{F}_{q^2}$. Every root x of every such irreducible cubic satisfies*

$$x \in \mathbb{F}_{q^6} \setminus \mathbb{F}_{q^2}, \qquad x^{q^3+1} = \omega, \qquad \omega^2 N_{\mathbb{F}_{q^6}/\mathbb{F}_{q^2}}(x) = Y. \tag{65.5}$$

Conversely every x satisfying (65.5) is obtained in this way, with $t = \mathrm{Tr}_{\mathbb{F}_{q^6}/\mathbb{F}_{q^2}}(x)$.

Proof. Write $N = \omega Y$; then $N^{q+1} = 1$ and $Y = \omega^2 N$. The polynomial is $X^3 + tX^2 + \omega^2 N t^q X + N$. Its roots are preserved by

$$\rho(x) = \omega / x^q,$$

as is checked by the identity

$$x^{3q} P_{Y,t}(\omega/x^q) = N P_{Y,t}(x)^q.$$

For an irreducible cubic, $\rho^2(x) = x^{q^2}$ acts on its three roots as a 3-cycle. Therefore ρ itself is a 3-cycle, and $\rho^3(x) = \omega/x^{q^3} = x$. This proves the first norm in (65.5). The constant coefficient is N, proving the second.

Conversely take $x^{q^3+1} = \omega$, outside the base field, with ordinary relative norm N. Put $t = x + x^{q^2} + x^{q^4}$. The second elementary symmetric coefficient is

$$N(x^{-1} + x^{-q^2} + x^{-q^4}) = \omega^2 N t^q = Y t^q.$$

Thus its minimal polynomial is (65.4).

For the count, the top norm coset is $\omega^2 \mu_{q^3+1}$, and the ordinary relative norm maps it onto μ_{q+1}. Each fiber has size

$$(q^3+1)/(q+1) = q^2 - q + 1.$$

Exactly one element of each fiber lies in the base field. To see this, write a base-field phase as $\omega^2\eta$, with $\eta \in \mu_{q+1}$. Its ordinary relative norm is η^3, and cubing is bijective on μ_{q+1}, since $q+1 \equiv 2 \pmod 3$. The other $q^2 - q$ elements form disjoint triples of conjugate roots of distinct irreducible cubics. Their number is $q(q-1)/3$. □

There is also an explicit normalization that separates the prescribed norm from the trace choice. Put

$$H = q^2 - q + 1, \qquad x_0 = \omega^2(\omega Y)^{(q+2)/3}.$$

This is the unique base-field point of the norm fiber. Every cubic lift is

$$x = x_0\zeta, \qquad \zeta \in \mu_H \setminus \{1\}, \qquad t = x_0(\zeta + \zeta^{q^2} + \zeta^{q^4}). \tag{65.6}$$

Indeed $\gcd(H, q^2 - 1) = 1$, so only $\zeta = 1$ can leave x in the base field. On setting $X = x_0 Z$ and $u = t/x_0$, the cubic becomes

$$x_0^{-3} P_{Y,t}(x_0 Z) = Z^3 + uZ^2 + u^q Z + 1. \tag{65.7}$$

Thus the permitted normalized traces are one fixed set of cubic Gaussian periods, independent of Y. Formula (65.6), or enumeration of the irreducible cubics, constructs the lifts without an HI seed.

For the step from $\mathbb{F}_{4^{2\cdot 3^{j-1}}}$ to $\mathbb{F}_{4^{2\cdot 3^j}}$, use $q = 4^{3^{j-1}}$. These are exactly the degree-three steps between the conductor strata of Section 58. All non-base-field roots produced by (65.4) have the full degree over $\mathbb{F}_4$ at this step: the proper-divisor argument in Section 65.2 applies to their norm coset.

If the branch-permuting prescription of Section 63 is imposed on candidate tuples, take the target normalized norm to be the corresponding component of $T^{-1}W_{3a}$. Equation (65.4) then provides a nonempty, explicit choice of cubic lifts at every step, including the required Frobenius twist. Iterating this procedure constructs candidate norm towers of arbitrary height. The theorem does **not** say that those towers satisfy Q or C, or that the norm prescription has been proved necessary for every HI solution. The remaining problem is to select the trace parameters t so that the additive equations hold in unbounded order.

65.4. **Exact verification.** `hi_char2_field_tower.py` checks the off-solution identity (65.2) at orders 9, 27, 81 and 243, with general nonunit norm data and all five possible bottom phases. The unit h is exhaustive through 81 and sampled at 243. Every resulting residual is nonzero. It also checks directly that R_2 in every saved order-3, order-9 and order-27 Frobenius representative has degree 2, 6 and 18 over $\mathbb{F}_4$, respectively.

For the cubic theorem, the script exhausts all 16 traces for each of the five normalized norms at $q = 4$, finding four irreducible cubics per norm. At $q = 64$, it exhausts all 4096 traces for three specified normalized norms, finding 1344 in each case. It constructs all 20 cubic extensions from the first test and one from each of the latter three tests, and independently checks both norms in (65.5), the norm-fiber parameter, and the cubic-period trace in the resulting extension fields. All exact checks pass. Infinitude of HI categories remains unproved.

66. Two quadratics, full field degree, and the obstruction to local inflation

The searches along powers of three admit a substantially smaller finite acceptance test. Its field-degree condition is essential: omitting it permits explicitly false candidates in unbounded order.

66.1. **Exact completion from two quadratics at the first three orders.** Retain the phase residual F of (64.1), the norm and Frobenius laws (64.2), and $n = 3^k$. Consider just

$$F(1,1) = F(2,1) = 0. \tag{66.1}$$

Theorem 66.1 (Finite)**.** *For $n = 3, 9, 27$, conditions (64.2), (66.1), and*

$$[\mathbb{F}_4(W_1) : \mathbb{F}_4] = 2n/3 \tag{66.2}$$

imply every quadratic and cubic. Conversely, every full cubic phase solution in this Frobenius gauge satisfies these conditions.

Proof by exhaustive finite algebra. The parametrization (58.9) exhausts the norm/Frobenius solutions over the algebraic closure. The modified `hi_char2_frobenius.py` records the survivors of (66.1) and the first-phase degrees, before applying the remaining Q filters:

n	Solutions of (66.1)	First-phase degrees over $\mathbb{F}_4$	Full Q and C
3	2	2 points of degree 2	2
9	12	12 points of degree 6	12
27	72	18 points of degree 6; 54 of degree 18	54

At each order, the full-Q subset has exactly the cardinality of the subset satisfying (66.2); the program also checks that each full-Q point belongs to that subset. Hence the subsets coincide. All Frobenius representatives are passed afresh through the independent complete Q/C verifier. The converse follows from C implying Q and the uniform field-degree theorem of Section 65. □

At order 27, the finite étale algebra cut out by (66.1) inside the norm/Frobenius scheme is consequently

$$(\mathbb{F}_{4^{18}})^3 \times (\mathbb{F}_{4^6})^3;$$

the full-Q/C quotient retains precisely the first three factors. For a degree-six first phase, all lower-stratum phases already lie in $\mathbb{F}_{4^6}$, so the degree counts also give the degrees of the entire tuples. The norm/Frobenius algebra is a product of separable finite fields; taking equation quotients here introduces no hidden nilpotents.

By Section 65, the three equations

$$F(1,1) = F(2,1) = F(n/3,1) = 0 \tag{66.3}$$

therefore suffice at these three orders as well. This presents a concrete smaller system to investigate in unbounded order. Neither the implication from (66.1)–(66.2) to C for arbitrary k, nor the nonemptiness of that system for unbounded k, has been proved.

66.2. **A uniform origin-patched inflation identity.** Let M be odd, let $N = sM$ with odd $s > 1$, and suppose V is phase data on $\mathbb{Z}_M$ satisfying the norm law. Set $c = \omega^2$, and define

$$W_0 = 1, \qquad W_a = \begin{cases} V_{a \bmod M}, & a \not\equiv 0 \pmod{M}, \\ c, & a \equiv 0 \pmod{M},\ a \neq 0. \end{cases} \tag{66.4}$$

These phases satisfy the norm law on $\mathbb{Z}_N$, and inherit Frobenius covariance whenever V has it. Denote their residuals by F_N, and the seed residuals by F_M.

Theorem 66.2. *If reduction modulo M is injective on the distinct elements of*

$$E(g,h) = \{0, h, -g, h-g\} \subset \mathbb{Z}_N,$$

then

$$F_N(g,h) = F_M(g \bmod M, h \bmod M). \tag{66.5}$$

At the first newly split origin collision, one instead has

$$\boxed{F_N(M-h,h) = F_M(-h,h) + 1} \qquad (1 \le h < M). \tag{66.6}$$

Proof. Replace the value at zero in both V and W by c. The resulting background sequences are exactly periodic with period M. Their quartic sums agree, since each seed summand is repeated s times and s is odd in characteristic two. Restoring the value 1 at zero changes only the summands indexed by E. When its distinct positions remain distinct modulo M, these corrections correspond one to one and agree, proving (66.5).

For $g = M - h$, the two distinct positions h and $-g = h - M$ merge modulo M. All other corrections still match. At the merged seed position the background term is $c^2 V_h V_{-h} = c$, and the restored term is ω; their difference is 1. At each of the two separate positions in $\mathbb{Z}_N$, the

background term is c and the restored term is $cV_hV_{-h} = 1$. Their two equal corrections cancel. This proves (66.6). □

For a full-Q seed of order $M \geq 5$, the inflated phases therefore satisfy (66.1) at every odd multiple $N > M$, but

$$F_N(M-1, 1) = 1. \tag{66.7}$$

Taking either saved order-9 orbit gives such counterexamples for every $N = 3^k$, $k \geq 3$, and more generally for every larger odd multiple of 9. Their first-phase degree remains six. At powers of three they also have $F_N(N/3, 1) = \omega^2$, as follows from (65.2). Thus they are false partial candidates, not new HI certificates.

At order 27, inflating the 12 full-Q order-9 points accounts for 12 of the 18 degree-six solutions in the table. Each passes $F(3, 1) = 0$ by (66.5), and fails $F(8, 1) = 0$ by (66.6). The remaining six partial points fail $F(3, 1) = 0$. This is exactly the filter history recorded by the exhaustive search.

`hi_char2_partial_completion.py` independently reconstructs all saved order-9 and order-27 representative phases, checks their Q equations, and applies (66.4) with multipliers 3, 5 and 9. It verifies the norm and Frobenius laws, every local Q with $1 \leq g, h \leq (M-1)/2$, and the nonzero residual (66.7). All checks pass. They verify the claimed failure of partial sufficiency, not existence at the inflated orders.

66.3. **The counts do not give a degree-six rational iteration.** The numbers $2, 12, 72$ in the first column of the table suggest testing a fixed rational preimage recursion. That precise test fails. Let $E_3, E_9, E_{27} \in \mathbb{F}_4[X]$ be the monic products over the distinct first-phase coordinates of the solutions of (66.1). Their degrees are 2, 12 and 72, respectively; for example $E_3 = X^2 + \omega X + \omega$.

Exact finite obstruction. No degree-six rational map $f = A/B \in \mathbb{F}_4(X)$ has both

$$B^2E_3(A/B) \text{ proportional to } E_9, \qquad B^{12}E_9(A/B) \text{ proportional to } E_{27}. \tag{66.8}$$

Here A and B are coprime with maximum degree six.

`hi_char2_rational_recursion.py` rebuilds these eliminants from fresh exhaustive searches. To exhaust the first identity, choose a root r of E_3 in $\mathbb{F}_{16}$. The factorization of E_9 over this field consists of four irreducible cubics in two fourth-power-conjugate pairs. The polynomial $A + rB$ must choose one factor from each pair. There are four such choices, and five scalar choices modulo $\mathbb{F}_4^\times$, hence 20 rational maps in total. Every map is checked against the first identity, and every one fails the second identity exactly.

The weaker proposal of following just one branch also fails. Over $\mathbb{F}_4$, write $E_9 = b_0b_1$, where the two irreducible degree-six factors have ascending coefficient lists

$$b_0 : [1, \omega^2, 0, \omega^2, 0, 1, 1], \qquad b_1 : [1, \omega, 1, \omega, \omega^2, \omega^2, 1].$$

For either choice b, exactly ten degree-three maps $f = A/B$ satisfy $B^2E_3(A/B) \propto b$: two choices of a cubic factor over $\mathbb{F}_{16}$, times five scalars as above. The program checks all twenty maps. For none of them is $B^6b(A/B)$ proportional to any of the three degree-eighteen factors of E_{27}. Those three factors are precisely the full-Q branches by Section 66.1. Hence no stationary degree-three rational map in this first-phase coordinate follows even one branch through orders 3, 9 and 27. Maps varying with the level, changes of coordinates varying with the level, and multivariable recursions remain open.

This excludes a particular one-variable iteration. It leaves open the coupled trace construction in Section 65, and does not change the missing requirements for infinitude: nonempty full-degree solutions in unbounded order and a uniform proof of the remaining additive equations.

67. Patching the origin does not permit bounded rational ratios

Section 60.4 left open a separately defined exceptional phase. The Frobenius relation closes this escape for rational consecutive-ratio formulas in characteristic two. The proof below is uniform in every odd order; it is not restricted to powers of three.

Let k be an algebraically closed field of characteristic two, let $\omega^2+\omega+1=0$, and put $c=\omega^2$. Suppose

$$W_0=1, \quad W_a\neq 0, \quad W_aW_{-a}=\omega\ (a\neq 0), \quad W_{2a}=W_a^4. \tag{67.1}$$

Write $r_a=W_a/W_{a-1}$, with cyclic indices, and let ζ be a primitive n-th root of unity. Assume that a rational function $R=P/Q$, with $\deg P, \deg Q\le r$, interpolates only

$$R(\zeta^a)=r_a \qquad (2\le a<n), \tag{67.2}$$

with no zero or pole at these sample points. Nothing is prescribed for R at 1 or ζ; in particular, (67.2) allows arbitrary separate treatment of the ratios adjoining the origin.

Theorem 67.1. *If $n>4r+3$, then every nonzero-index phase equals c, and the first phase quadratic has residual*

$$F(1,1)=\sum_m W_m^2W_{1-m}W_{-1-m}=\omega\neq 0. \tag{67.3}$$

Consequently any full-Q solution satisfying (67.1) and (67.2) has $4r\ge n-3$. The coefficients of R may vary with n and may belong to an arbitrarily large extension field.

Proof. Inverse norms first give $r_a=r_{1-a}$ for $a\notin\{0,1\}$. The polynomial

$$z^r\big(P(z)Q(\zeta/z)-Q(z)P(\zeta/z)\big)$$

has degree at most $2r$ and vanishes at the $n-2$ sample points. The assumed inequality therefore proves the rational identity

$$R(z)=R(\zeta/z). \tag{67.4}$$

Frobenius covariance gives an independent ratio identity, valid even at the exceptional indices:

$$r_{2a}r_{2a-1}=r_a^4. \tag{67.5}$$

All three ratios in (67.5) are among the prescribed samples unless $a\in\{0,1,(n+1)/2\}$. Thus the polynomial

$$H(z)=P(z^2)P(\zeta^{-1}z^2)Q(z)^4-Q(z^2)Q(\zeta^{-1}z^2)P(z)^4 \tag{67.6}$$

vanishes at $n-3$ distinct points. It has degree at most $8r$ and contains only even powers of z. Since k is perfect, it is the square of a polynomial of degree at most $4r$. Its square root has the same distinct zeros. Hence $n>4r+3$ implies $H=0$, or

$$R(z^2)R(\zeta^{-1}z^2)=R(z)^4. \tag{67.7}$$

Here is the rigidity in (67.7). Let $D=\operatorname{div}R$, regarded as a finitely supported integer vector on the geometric points of $\mathbb{P}^1$. Both $\pi: z\mapsto z^2$ and $L: z\mapsto \zeta z$ permute these points. Pullback by π has multiplicity two at every point. Taking divisors in (67.7), pushing forward by π, and dividing the integer multiplicities by two gives

$$D+L_*D=2\pi_*D. \tag{67.8}$$

Equip these finitely supported vectors with their ordinary real Euclidean norm; no norm on k is used. The two permutations preserve this norm, so squaring (67.8) gives $\langle D, L_*D\rangle=\|D\|^2$. It follows that $D=L_*D$.

Every zero or pole of R in $k^\times$ would now occur on an entire n-element orbit under multiplication by ζ. But $r<n$, so R has degree less than n and cannot contain such an orbit in either its zero or its pole divisor. Its divisor is supported at 0 and infinity, and therefore $R(z)=\kappa z^s$. Reflection (67.4) forces $s=0$, and (67.7) then gives $\kappa^2=\kappa^4$. Since $\kappa\neq 0$, characteristic two implies $\kappa=1$.

Equation (67.2) now makes $W_1, \dots, W_{n-1}$ equal. Their inverse norm makes this common value the unique square root $c = \omega^2$ of ω. In (67.3), the terms at $m = 0, 1, -1$ sum to $c^2 + c^3 + c^3 = \omega$. The other $n - 3$ equal terms cancel in characteristic two. This proves the theorem. □

`hi_char2_ratio_bound.py` independently reconstructs the saved phases and checks (67.5), reflection, and the constant-phase residual. It then solves the exact rational interpolation problems (67.2), with the denominator required to be nonzero at every sample. The minimum degrees are 0 at order 3, 4 for each order-9 branch, and 12 for each order-27 branch. It also checks the square-root polynomial and its $n - 3$ required zeros in (67.6). These finite checks verify the algebra used in the proof; they do not supply an existence argument.

The theorem concerns rational functions of the cyclic parameter ζ^a. It does not apply to arbitrary rational functions on the elliptic curve of Section 61, or to the double-sine product of Section 51. The variational candidate of Section 53 also remains available at every odd order. Its full additive identities, and infinitude itself, are still unproved.

68. The first quadratic is linear in all lower Frobenius seeds

There is a constructive elimination step that avoids searching the lower norm tori independently. It applies before their inverse norms are imposed, and does not assume any of the other additive equations.

Fix $n = 3^k$ and write $N = 2n/3$. All Frobenius-covariant phases can be placed in $E = \mathbb{F}_{2^{2N}}$, as in Section 58. Fix the top seed $x = W_1$, with $x^{2^N+1} = \omega$. Its iterated fourth powers determine every W_a with $3 \nmid a$; denote these fixed values by U_a. On $H = 3\mathbb{Z}_n$, write the remaining phases as V_t, retaining $V_0 = 1$. For the moment impose Frobenius covariance but allow arbitrary lower values, including zeros.

Theorem 68.1 (Uniform affine elimination)**.** *The first quadratic is*

$$\boxed{F(1,1) = \sum_{t \in H} (a_t V_t^2 + b_t V_t), \qquad a_t = U_{1-t}U_{-1-t}, \quad b_t = U_{1-t}^2 U_{t-2} + U_{-1-t}^2 U_{t+2}.} \tag{68.1}$$

Consequently $F(1,1) = 0$ is an affine $\mathbb{F}_2$-linear system in exactly $N - 2$ lower seed coordinates, with values in $K = \mathbb{F}_{2^N}$. Its constant term is

$$\kappa_x = \omega + \omega^2 (x + \omega/x)^2. \tag{68.2}$$

Thus the equation has the form

$$L_x(v) = \kappa_x, \qquad L_x : \mathbb{F}_2^{N-2} \longrightarrow \mathbb{F}_2^N. \tag{68.3}$$

All solutions, or a certificate of inconsistency, can be obtained by binary linear algebra once x is supplied.

Proof. In $F(1,1) = \sum_m W_m^2 W_{1-m} W_{-1-m}$, split m into its three residue classes modulo 3. For $m \equiv 0$, use $t = m$. For $m \equiv 1$, use $t = 1 - m$; for $m \equiv -1$, use $t = -1 - m$. In each class all the other displayed indices are units modulo 3. This gives (68.1) directly. The only powers of the lower phases are first and second powers, both binary linear.

For $1 \le s < k$, the stratum $3^s \parallel a$ is one doubling orbit of length $2 \cdot 3^{k-s-1}$. Its seed is an arbitrary element of $\mathbb{F}_{2^{4 \cdot 3^{k-s-1}}}$, and the other values are its iterated fourth powers. No norm relation is needed for this assertion. The lower seed space therefore has binary dimension

$$\sum_{s=1}^{k-1} 4 \cdot 3^{k-s-1} = 2(3^{k-1} - 1) = N - 2. \tag{68.4}$$

At $t = 0$, the top norm gives $a_0 = \omega$. Using $U_{\pm 2} = U_{\pm 1}^4$ gives $b_0 = \omega^2 (U_1 + U_{-1})^2$, which proves (68.2).

For completeness, the target field in (68.3) also does not require the lower norms. For arbitrary phase data, reindexing the summation proves

$$F(g,h) = F(-g,h) = F(g,-h) = F(h,g). \tag{68.5}$$

Frobenius covariance gives $F(g,h)^4 = F(2g,2h)$. Since $2^{n/3} \equiv -1 \pmod{n}$, iterating this equality gives $F(1,1)^{2^N} = F(-1,-1) = F(1,1)$. Hence every value of the affine map lies in K, proving (68.3). □

The lower norm equations must still be imposed on the affine solution space. For a seed $y_s = W_{3^s}$, they are

$$y_s^{2^{2\cdot 3^{k-s-1}}+1} = \omega. \tag{68.6}$$

These are quadratic equations in the binary coordinates, because the displayed Frobenius power is binary linear. This yields a precise uniform procedure: choose x on its norm torus, solve (68.3), impose (68.6) on its kernel parameters, and then test the remaining Q and C. It does not assert that a suitable choice of x exists at every order.

`hi_char2_firstq_linear.py` implements this reduction independently of the exhaustive search. It checks (68.1) on lower data not satisfying the norms, checks the target field and affine constant, and solves the linear systems for each saved first phase. The results are:

order	saved branch	lower binary dimension	rank of L_x	norm-compatible lower solutions
3	0	0	0	1
9	0	4	4	1
9	1	4	4	1
27	0	16	16	1
27	1	16	15	1
27	2	16	16	1

The recovered values are compared exactly with every saved lower phase. At order 27, branch 1, the first quadratic alone has two lower solutions; exactly one satisfies all lower norms. Thus even the saved solutions do not justify assuming full rank uniformly.

The program also exhausts all 65 top norm choices at order 9. Of these, 23 give consistent affine systems; imposing the lower norms leaves a total of 24 first-Q candidates, agreeing with the independent exhaustive search. Twelve of these pass the remaining equations, as already certified in Section 58.

At order 81 it samples 64 top norm choices. The 63 nonconstant choices give ranks 51 or 52; nineteen of their affine systems are consistent, but none of their lower solutions satisfies all inverse norms. The constant top choice has rank 2 and a large kernel; its kernel is not enumerated by this check. Such a collapsed top seed cannot give full Q by Section 65. These are finite sample results, not a search of the entire order-81 norm torus. The new result is the uniform linear elimination, while nonempty full solutions at unbounded orders remain to be constructed.

69. Relative traces and a linear number of quadratic orbits

Two further uniform reductions apply to the characteristic-two search. They reduce the equations to be solved; neither asserts that they have solutions at unbounded orders.

69.1. **Pairing each quadratic into a relative trace.** Let $n = 3^k$, $N = 2n/3$, $E = \mathbb{F}_{2^{2N}}$, and $K = \mathbb{F}_{2^N}$. Retain

$$W_0 = 1, \qquad W_a W_{-a} = \omega \quad (a \neq 0), \qquad W_a^4 = W_{2a}, \qquad \omega^2 + \omega + 1 = 0.$$

Write $\bar{z} = z^{2^N}$. Since $2^{n/3} \equiv -1 \pmod{n}$, Frobenius covariance implies $\overline{W_a} = W_{-a}$, including at zero. For $g, h \neq 0$, set

$$T_m = W_m W_{h-m} W_{-m-g} W_{m+g-h}, \qquad \iota(m) = h - g - m.$$

Direct substitution gives $T_{\iota(m)} = \overline{T_m}$. The involution has the unique fixed point $m_0 = (h-g)/2$. If $a = (h-g)/2$ and $b = (h+g)/2$, its summand is

$$T_{m_0} = (W_a W_{-a})(W_b W_{-b}) = \begin{cases} \omega, & h = g \text{ or } h = -g, \\ \omega^2, & h \neq \pm g. \end{cases}$$

Thus, for any set $M_{g,h}$ containing one element of each nontrivial orbit of ι,

$$\boxed{F(g,h) = \epsilon_{g,h} + \mathrm{Tr}_{E/K}\left(\sum_{m \in M_{g,h}} T_m\right), \quad \epsilon_{g,h} = \begin{cases} \omega, & h = \pm g, \\ \omega^2, & h \neq \pm g. \end{cases}} \tag{69.1}$$

There are exactly $(n-1)/2$ terms inside the trace. This proof uses only the displayed norms and involution, so the same pairing identity holds over any characteristic-two field with an involution sending W_a to W_{-a} and fixing ω.

69.2. **Exactly $(n-1)/2$ orbit representatives suffice for Q.** In the Frobenius gauge, the identities

$$F(g,h) = F(-g,h) = F(g,-h) = F(h,g), \qquad F(2g,2h) = F(g,h)^4 \tag{69.2}$$

reduce vanishing of all nonzero-index quadratics to a linear number of residuals. This statement needs no inverse norms.

For a pair (g,h), put $s = \min(v_3(g), v_3(h))$, $M = 3^{k-s}$, and divide both indices by 3^s. At least one resulting index is a unit modulo M; exchange them if necessary and normalize that index to 1. Every unit modulo M is a power of 2. Consequently this normalization preserves vanishing by (69.2). The remaining ratio $t \neq 0$ is identified with $-t$; when t is a unit, it is also identified with t^{-1} by exchanging the two indices. Representatives can therefore be taken as

$$(3^s, 3^s t), \quad 1 \leq t \leq (M-1)/2, \quad t \leq \min(t^{-1} \bmod M, -t^{-1} \bmod M) \text{ if } 3 \nmid t. \tag{69.3}$$

There are $(M/3-1)/2$ nonunit ratio classes. For unit ratios, Burnside's lemma applied to negation and inversion gives

$$\frac{\varphi(M)+2}{4} = \frac{M/3+1}{2}$$

classes: inversion fixes exactly 1 and -1, negation fixes none, and $t^2 = -1$ has no solution modulo 3. The total at this stratum is M/3. Summing over s proves

$$\boxed{\#\{\text{quadratic orbits}\} = \sum_{j=1}^{k} 3^{j-1} = \frac{n-1}{2}.} \tag{69.4}$$

This counts orbits under the stated symmetries; it does not claim algebraic independence of the residuals.

The residual at a stratum of order $M = 3^j$ lies in $\mathbb{F}_{2^{2M/3}}$: iterate (69.2) $M/3$ times and use simultaneous sign reversal. Thus the representative equations can be written using at most

$$\sum_{j=1}^{k} 3^{j-1}\,(2 \cdot 3^{j-1}) = \frac{n^2-1}{4} \tag{69.5}$$

binary coordinate equations. In particular, order 81 has 40 representative residuals, with 1640 such coordinates altogether.

`hi_char2_quadratic_orbits.py` implements (69.3) and verifies every nonzero pair, its exchange, its sign reversal, and its doubling through order 729. Its exact counts are 1, 4, 13, 40, 121, and 364.

69.3. **An exact Boolean search, with independent acceptance.** `hi_char2_sat.py` now encodes the norm equations and the quadratic representatives using CryptoMiniSat's native XOR clauses. The seed coordinates use normal bases, making fourth powers coordinate permutations. Products are simplified using inverse norms, squares, and doubling before Boolean multiplication gates are introduced. Equation (69.1) cuts the quartic summation in half. Full top degree, which is necessary by Section 65, is imposed explicitly.

The default search starts with two quadratics and adds a failed representative whenever a model is found. The `-all-quadratics` option encodes every representative at startup. Any proposed model is checked by separate integer finite-field arithmetic against every Q, and any existence claim additionally requires the independent complete GP cubic verifier. `-skip-cubic` explicitly produces only unverified-for-C full-Q data.

The smaller encoding used 55,892 auxiliary variables for the first two order-27 equations and 981,919 at order 81 in the initial trace runs, down from 165,395 and 2,997,151 in the direct encodings. Canonicalizing the representative indices can slightly change these circuit counts. The complete order-27 representative encoding has 113,644 auxiliary variables. These are implementation sizes, not mathematical bounds on proof complexity.

`hi_char2_sat_checks.py` fixes norm-compatible seed assignments and compares the encoded formula with direct finite-field evaluation at orders 3, 9, 27, and 81. The order-9, -27, and -81 assignments have nonzero quadratic residuals, so these checks also test the formula away from its zero locus. All comparisons pass. Fresh SAT searches at orders 3 and 9 pass every Q and every C independently; their logs are in `hi_char2_sat_results.txt`.

The bounded two-equation order-27 run reached its time limit without a SAT or UNSAT conclusion. Larger searches are exploratory. No order-81 certificate or infinitude theorem follows from this work. A solver timeout is not evidence of nonexistence.

Reproduction with the currently available Python binding is:

```
PYTHONPATH=/tmp/hi_boolean_solver39 /usr/bin/python3.9 hi_char2_sat.py --order 9 --all-quadratic
PYTHONPATH=/tmp/hi_boolean_solver39 /usr/bin/python3.9 hi_char2_sat_checks.py
python hi_char2_quadratic_orbits.py
```

The binding is CryptoMiniSat 5.14.7 for Python 3.9, installed outside the workspace in the displayed temporary directory. The project documentation describes its XOR clause interface. Recreating that temporary installation may be necessary in a later environment.

70. Complete characteristic-two quadratic-to-cubic ideals through 13

Magma gives a second finite completion theorem, now entirely in characteristic two and without the Frobenius ansatz of Sections 58 and 66.

Theorem 70.1 (Finite ideal)**.** *For each $n \in \{3, 5, 7, 9, 11, 13\}$, every characteristic-two phase solution of the inverse norms and all Q satisfies all C. More strongly, each cross-free balanced cubic residual belongs to the ideal generated by the inverse norms and Q over $\mathbb{F}_4$. The quadratic phase algebra is nonzero and has the following lengths, counting any scheme multiplicity:*

n	norm/Q generators used	Groebner basis size	phase algebra length	cross-free cubic orbit representatives
3	2	3	6	1
5	5	10	20	2
7	9	28	42	6
9	14	126	108	14
11	20	194	132	17
13	27	301	156	30

Here the ambient ring is

$$R = \mathbb{F}_4[W_1, \ldots, W_{n-1}], \qquad W_0 = 1,$$

and the ideal contains $W_aW_{-a}-\omega$ for $1\le a\le (n-1)/2$, together with $F(g,h)$ for $1\le g\le h\le (n-1)/2$. No phase variable needs an additional saturation equation: the inverse norms already make every nonzero-index phase a unit. Identity (68.5) proves coverage of the remaining Q; the program also independently reduces every omitted nonzero-index Q to zero.

For a balanced cross-free pair (a,c) of triples, define

$$S_{a,c}=\sum_m\prod_{i=1}^{3}W_{m+a_i}\prod_{j=1}^{3}W_{-m+c_j},\qquad P_{a,c}=\prod_{i,j=1}^{3}W_{a_i+c_j}.$$

The normalized cubic is $\omega^2S_{a,c}=(\omega-1)P_{a,c}$. Since $\omega^2=\omega-1$, its residual is simply $S_{a,c}-P_{a,c}$. The program computes a Groebner basis of I and verifies

$$\mathrm{NormalForm}(S_{a,c}-P_{a,c},I)=0 \tag{70.1}$$

for every cross-free orbit representative from the already verified balanced-index reduction. The remaining cubic cases follow from norms and Q by Sections 34–40. This is ideal membership, not just vanishing on a selected list of roots.

The positive lengths prove that these finite quadratic schemes have points over finite extensions of $\mathbb{F}_4$. Equation (70.1) makes each point a complete cubic certificate. Section 58 therefore supplies complex categorification at these six finite orders. This recovers finite existence without reading any saved HI matrices. The calculation does not impose Frobenius covariance, scalar H, or any predetermined extension degree.

`hi_char2_phase_ideal.py` generates the exact Magma input; the completed output is saved in `hi_char2_phase_ideal_results.txt`. With the user-provided Magma 2.28-4 installation, reproduction is:

```
python hi_char2_phase_ideal.py --source /tmp/hi_char2_phase_ideal.m
/home/tzuchen/Software/Magma/magma -b /tmp/hi_char2_phase_ideal.m
```

The computations use the documented Groebner-basis and normal-form operations. The higher-order calculations have not supplied a completed result. No uniform ideal containment, uniform nonemptiness, or infinitude is asserted. In particular, the proof in this section is specific to characteristic two; transferring a finite algebra-length bound to characteristic zero would require a separate argument.

71. The cubics force a unique Frobenius gauge

The Frobenius gauge is now justified uniformly for every full characteristic-two phase solution. This changes the interpretation of the earlier finite searches: at the cubic level the gauge loses no phase solutions. It also gives an explicit finite etale algebra whose nonvanishing is the remaining existence question.

71.1. **A convolution identity from the diagonal cubics.** Work in characteristic two with the normalized matrix A and $\delta=\omega$. Fix $g\neq 0$. For sequences on $\mathbb{Z}_n$, let $(x*y)_s=\sum_t x_ty_{s-t}$, let e be the delta sequence at zero, and use square brackets to denote pointwise powers. Define

$$u_s=A_{s,g},\quad v_s=A_{g,-s},\quad f_s=A_{s,2g},\quad w=u^{[2]},\quad z=v^{[2]}.$$

The Q column at g is exactly $u*v=e$, so its square gives $w*z=e$. The cubic with indices (g,g,k,k), using $A_{g+k,k}=A_{-k,g}$, is exactly

$$f*z=w. \tag{71.1}$$

Multiplying by w and using $z*w=e$ yields $f=w*w$. In characteristic two all terms of this self-convolution cancel in pairs except the unique term at $t=s/2$. Consequently

$$f_s=w_{s/2}^2=A_{s/2,g}^4.$$

Conversely this identity and Q imply (71.1). The fixed axis entries handle $g = 0$ directly. We have therefore proved the uniform polynomial equivalence

$$\boxed{Q + \{C(g,g,k,k) : g,k \in \mathbb{Z}_n\} \quad \Longleftrightarrow \quad Q + \{A_{2a,2b} = A^4_{a,b} : a,b \in \mathbb{Z}_n\}.} \tag{71.2}$$

This argument uses the order-three symmetry and the normalized matrix equations; it does not use the phase parametrization. It involves no square-root choice or division by an unknown.

There are also exact identities away from Q and C. Set

$$q = u * v - e, \qquad D = f * z - w, \qquad H = f - w * w.$$

Then

$$\boxed{D = H * z + w * q^{[2]}, \qquad H = D * w + f * q^{[2]}.} \tag{71.3}$$

Indeed $w * z = e + q^{[2]}$, and both identities follow by associativity. Thus (71.2) is an equality of the corresponding ideals after Q is imposed, not merely a pointwise implication. The proof works over arbitrary commutative characteristic-two coefficient rings.

71.2. **Unique normalization of the phases.** Assume now the inverse-normalized phase parametrization. Since $\omega \in \mathbb{F}_4$, equation (71.2) also gives $B_{2a,2b} = B^4_{a,b}$. Define

$$r(a) = \frac{W_a^4}{W_{2a}}, \qquad r(0) = 1.$$

For $a + b \neq 0$, substitute $B_{a,a+b} = W_a W_b / W_{a+b}$ into matrix covariance to obtain

$$r(a)r(b) = r(a+b).$$

For $a + b = 0$, the same equality follows from $r(a)r(-a) = \omega^4/\omega = 1$. Thus r is a character of $\mathbb{Z}_n$, in particular $r(a)^n = 1$.

A character change $W'_a = \chi(a)W_a$ leaves A unchanged and changes the ratio to $r'(a) = \chi(a)^2 r(a)$. Since n is odd, squaring is an automorphism of the character group. The unique choice giving $r' = 1$ is

$$\boxed{\chi(a) = r(a)^{-(n+1)/2}, \qquad W'_a = \chi(a)W_a, \qquad (W'_a)^4 = W'_{2a}.} \tag{71.4}$$

All divisions here are by units already supplied by the inverse norms. This normalization is therefore defined over the original coefficient algebra, with no field extension or root choice. It applies to every full cubic phase solution. It also applies whenever Q and just the diagonal cubics hold.

71.3. **An explicit field bound and finite etaleness.** Put $r_n = \mathrm{ord}_n(2)$ and $E_n = \mathbb{F}_{4^{r_n}}$. Iterating (71.4) gives $(W'_a)^{4^{r_n}} = W'_a$. Also $n \mid 4^{r_n} - 1$, so all character values lie in E_n. Hence the original phases, as well as their canonical normalization, lie in E_n. Matrix covariance directly gives the same upper field bound for A, even before a phase parametrization is assumed.

The reasoning is valid in the universal coordinate algebra of the full phase equations. In particular it proves

$$W_a^{4^{r_n}} - W_a = 0 \tag{71.5}$$

there. These polynomials are separable. The full phase coordinate algebra is consequently a quotient of a finite etale algebra over $\mathbb{F}_4$, and is itself finite etale, possibly zero. This supplies a characteristic-two reducedness theorem without assuming nonemptiness. In particular the lengths in Section 70 are also the numbers of geometric points.

For $n = 3^k$, $r_n = 2n/3$. Section 65 shows that the canonical first phase, and already the invariant row entry $R_2 = (W'_1)^{-2}$, generates E_n over $\mathbb{F}_4$. Since R is unchanged by character gauge, every full phase solution at this order has its matrix entries generating exactly

$$\boxed{E_n = \mathbb{F}_{2^{4n/3}}.} \tag{71.6}$$

This is an equality of minimal fields in the stated normalized phase framework, not only an available ambient field.

71.4. **The remaining nonemptiness problem.** Let $\mathcal{Z}_k$ be the full cubic phase scheme at $n = 3^k$, with Frobenius covariance imposed. Equations (71.4)–(71.6) show that there is an integer $b_k \geq 0$ such that

$$\boxed{\mathcal{O}(\mathcal{Z}_k) \cong \left(\mathbb{F}_{2^{4n/3}}\right)^{b_k} \quad \text{as } \mathbb{F}_4\text{-algebras.}} \tag{71.7}$$

For $b_k = 0$ the algebra is zero and the scheme is empty. Every nonempty factor has the full displayed residue field: smaller factors are excluded by the necessary quadratic of Section 65.

The full phase scheme before normalization is $\mu_n \times \mathcal{Z}_k$, by the unique normalization. Since E_n contains all n-th roots of unity, its coordinate algebra is $E_n^{nb_k}$. Thus its geometric length is $n(2n/3)b_k$, while the canonical length is $(2n/3)b_k$.

The exhaustive finite searches and full cubic checks in Section 58 now prove the complete phase classifications

$$b_1 = 1, \qquad b_2 = 2, \qquad b_3 = 3,$$

with canonical lengths 2, 12, 54 and total phase lengths 6, 108, 1458, respectively. These are complete classifications of the characteristic-two solutions in the phase parametrization, not merely searches in an optional Frobenius subfamily. They do not classify arbitrary complex categories or prove that every complex category has good reduction at 2.

For this construction route, infinitude is reduced to showing $b_k > 0$ for unbounded k. Neither finite etaleness nor the prescribed residue field proves that inequality. The arithmetic branch counts of Section 63 suggest values for b_k, including 8 at order 81, but identifying these integers with class numbers remains an unproved additive construction statement.

`hi_char2_cubic_frobenius.py` checks both identities (71.3) on arbitrary norm phases with nonzero Q and Frobenius defects at orders 3, 5, 7, 9, 11, and 13. It separately verifies matrix covariance on every saved characteristic-two branch at 3, 9, and 27, twists each by every cyclic character, and recovers the original phases using (71.4). All exact checks pass; results are saved in `hi_char2_cubic_frobenius_results.txt`.

Infinitude remains unproved. The new result removes the Frobenius-gauge qualification from the full characteristic-two phase search and identifies precisely which finite etale algebras must be shown nonzero.

72. General doubling-orbit searches and complete finite counts through 31

The forced gauge of Section 71 gives a uniform search parametrization for every odd cyclic order, including orders where 2 is not primitive on the units. This leads to new exact workspace certificates at 15, 21, 23, and 31, and complete phase counts at every odd order through 31. These remain finite results.

72.1. **Arbitrary odd orders: paired orbits and norm orbits.** Partition the nonzero indices into doubling orbits. Let O be the orbit of a, of length r. Frobenius covariance says that its seed $x = W_a$ belongs to $\mathbb{F}_{4^r}$, and determines all other entries on O.

If $-O \neq O$, the seed is any nonzero element of $\mathbb{F}_{4^r}$; the opposite orbit is then determined by $W_{-b} = \omega/W_b$. There are exactly $4^r - 1$ choices for this pair of orbits.

If $-O = O$, then r is even and $2^{r/2}a = -a$. The seed must satisfy

$$x^{2^r+1} = \omega, \qquad x \in \mathbb{F}_{2^{2r}}. \tag{72.1}$$

There are exactly $2^r + 1$ choices, because this is a relative norm fiber over $\mathbb{F}_{2^r}$. The scalar $c = \omega^2$ has the required norm, so the fiber is $c\mu_{2^r+1}$.

These choices are independent across orbit pairs. Conversely they produce every inverse-normalized Frobenius-covariant phase sequence. All fields embed in $E_n = \mathbb{F}_{4^{\mathrm{ord}_n(2)}}$. Thus there

is a finite, explicit search containing every full cubic phase solution, with no assumption that its entries have the largest possible field degree.

`hi_char2_general_search.py` implements this parametrization. Its current C++ backend supports at most three independent seeds and binary field degree below 63; these are implementation limits, not mathematical restrictions. A common cyclic parameter group makes all Q evaluations exact XORs of finite-field power-table entries. The Q representatives use sign reversal, exchange, and doubling; no unproved Galois action of other units is used.

72.2. **Completed searches and independent verification.** The following searches are exhaustive in the parametrization above. Every resulting Frobenius orbit passes the independent verifier's complete Q and C loops, including all exceptional indices.

n	ambient binary field degree	independent seed-set sizes	assignments tested	canonical phase points	Frobenius orbits
15	8	255, 17, 5	21,675	24	6
17	16	257, 257	66,049	16	2
19	36	262,145	262,145	36	2
21	12	4,095, 63, 5	1,289,925	48	8
23	22	4,194,303	4,194,303	44	4
25	40	1,048,577, 17	17,825,809	40	2
29	56	268,435,457	268,435,457	60	3
31	10	1,023, 1,023, 1,023	1,070,599,167	60	12

At every listed order other than 29, all orbits have length $\mathrm{ord}_n(2)$. At 29 the lengths are 28, 28, and 4. The smaller orbit must be retained; imposing maximal first phase degree would incorrectly discard valid full solutions.

For example, the order-31 first Q accepts 1,046,076 assignments, whereas all Q accept only 60. At order 29, in contrast, all 60 first-Q assignments already pass the remaining Q and C. No uniform first-Q completion statement is inferred.

By Section 71 these give complete classifications of the normalized characteristic-two **phase points** at these orders, modulo cyclic character gauge. They are not counts of inequivalent fusion categories. The raw phase counts are n times the canonical counts. Together with Sections 58 and 70, complete characteristic-two phase counts and hence complex existence certificates are now available at every odd order $3 \le n \le 31$.

The saved JSON files have names `hi_finite_field_certificates/hi_finite_field_N_char2_general_J.json`. The combined log is `hi_char2_general_search_results.txt`. Reproduction uses:

```
python hi_char2_general_search.py --orders 15 17 19 21 25
python hi_char2_general_search.py --orders 23 31 --max-table 5000000
python hi_char2_general_search.py --orders 29 --max-table 300000000
```

The last run uses about four GiB for its two power tables. The third coefficient table has just one entry after dividing each quadratic by ω^2: only a term with two zero indices needs that coefficient, and its remaining indices are an inverse pair. This optimization was checked against the completed order-15 and order-17 counts and their complete cubic certificates.

72.3. **Completion of the full quadratic ideal at order 17.** The previously running Magma calculation has completed. The full order-17 Q ideal over $\mathbb{F}_4$, with inverse norms and without Frobenius covariance, has a Groebner basis of size 694 and quotient dimension 272. Its source and output are saved as `hi_char2_phase17_length.m` and `hi_char2_phase17_length_results.txt`.

Independently, the exhaustive search above and Section 71 give exactly $17 \cdot 16 = 272$ geometric points of the full C phase scheme. That scheme is finite etale. The surjection from the Q coordinate algebra to the Q-and-C coordinate algebra is therefore between two vector spaces of dimension 272, and is an isomorphism. Equivalently, the 272 distinct cubic

points already exhaust the entire length of the quadratic algebra, leaving no extra points or nonreduced structure. This proves

$$\boxed{Q \Longrightarrow C \quad \text{at } n = 17 \text{ in characteristic two.}} \tag{72.2}$$

It also proves reducedness of that Q algebra. This argument does not infer a characteristic-zero Q-to-C implication from a finite special-fiber dimension; such an inference would need additional control of specialization.

72.4. **The full quadratic ideal also completes at order 15.** A separate Magma run now gives the full order-15 quadratic algebra, again without Frobenius covariance or a prescribed extension field. Its 35 norm/Q generators produce a Groebner basis of size 693, dimension zero, and quotient length 360. The completed basis calculation used 57.570 CPU seconds in the saved run.

Section 72.2 independently gives 24 canonical cubic points and therefore exactly $15 \cdot 24 = 360$ raw cubic points. The same surjection and length comparison as in Section 72.3 proves

$$\boxed{Q \Longrightarrow C \quad \text{at } n = 15 \text{ in characteristic two,}}$$

and proves that its full quadratic algebra is reduced. Combining this with Sections 70 and 72.3 establishes the characteristic-two implication at **every odd order from 3 through 17**. This is a finite implication theorem, not a proof in unbounded order.

The source `hi_char2_phase1519_length.m` performs the order-15 calculation and then starts the same test at 19. Its output is `hi_char2_phase1519_length_results.txt`. The order-19 computation has not yet returned a basis or length at this checkpoint. A length of 684 would match the independently known 36 canonical cubic points, whereas any larger length would require further analysis; neither result is assumed here.

The reduced order-81 SAT run, meanwhile, completed its one-hour budget with neither SAT nor UNSAT. Its log is preserved in `hi_char2_sat81_search_results.txt`. It supplies no existence or nonexistence conclusion.

73. The count must include quadratic overorders

The complete finite counts reveal a more precise arithmetic target. Counting the ray group of just one quadratic order misses genuine solutions at orders 11 and 29.

Let

$$D = n^2 + 4 = F^2 D_0, \qquad K = \mathbb{Q}(\sqrt{D_0}), \qquad d = \frac{n + \sqrt{D}}{2}, \qquad O_F = \mathbb{Z}[d].$$

Here D_0 is fundamental. For every $f \mid F$, write $O_f = \mathbb{Z} + fO_K$, let h_f be its ordinary class number, and set

$$m_f = [O_f^\times : O_F^\times].$$

All these orders contain the norm-minus-one unit d. In particular ordinary and narrow class numbers agree. The symbol + in the ray modulus below means positivity at the real embedding where $d > 0$.

73.1. **A positive arithmetic count, proved independently of HI.** The element d is the smallest unit greater than 1 in O_F. Indeed, for any such unit u, write $u - u' = b\sqrt{D}$, with a nonzero integer b. Since $u > 1$ and $u' = \pm u^{-1}$, b is positive. If the norm is -1, then $u + u^{-1} \geq \sqrt{D} = d + d^{-1}$, giving $u \geq d$. If the norm is +1, then $u - u^{-1} \geq \sqrt{D}$, which gives $u > d$. Thus, if $\epsilon_f > 1$ generates the positive unit group of O_f, then $d = \epsilon_f^{m_f}$.

Also $\gcd(F, n) = 1$. The quotient $O_f/(d-1)O_f$ is therefore $\mathbb{Z}/n\mathbb{Z}$, as is clear first for O_F by setting d equal to 1, and then for every overorder by localization away from its index. The image of ϵ_f in this quotient has order exactly m_f. For $0 < a < m_f$, one has

$$0 < \left| N_{K/\mathbb{Q}}(\epsilon_f^a - 1) \right| < n,$$

where the even-a norm is smaller than $\epsilon_f^a - \epsilon_f^{-a}$, and the odd-a norm equals that number. Such an element cannot lie in an ideal of norm n.

The ray-class exact sequence, with the positivity condition at one real place, consequently gives

$$\left|\mathrm{Cl}_{(d-1),+}(O_f)\right| = \frac{h_f \varphi(n)}{m_f}. \tag{73.1}$$

Define the total arithmetic count

$$\boxed{\mathcal{R}(n) = \varphi(n) \sum_{f|F} \frac{h_f}{m_f} > 0.} \tag{73.2}$$

Every summand is an integer ray-class degree. This positivity is an arithmetic theorem; it does not supply a phase solution. The ray-class formalism for orders is the one used in Sections 32 and 63 and in Kopp–Lagarias, Corollary 6.7.

For explicit evaluation, put $u_f = [O_K^\times : O_f^\times]$ and $m = u_F$. The quadratic order class-number formula gives

$$h_f = \frac{h_1}{u_f} f \prod_{p|f} \left(1 - \frac{(D_0/p)}{p}\right), \qquad m_f = \frac{m}{u_f}.$$

Thus the unit indices cancel inside the sum:

$$\boxed{\mathcal{R}(n) = \frac{h_1 \varphi(n)}{m} \sum_{f|F} f \prod_{p|f} \left(1 - \frac{(D_0/p)}{p}\right).} \tag{73.3}$$

73.2. **Exact finite comparison and the corrected prediction.** The canonical phase counts proved in Sections 58, 70, and 72 are

n	3	5	7	9	11	13	15	17	19	21	23	25	27	29	31
canonical points	2	4	6	12	12	12	24	16	36	48	44	40	54	60	60
$\mathcal{R}(n)$	2	4	6	12	12	12	24	16	36	48	44	40	54	60	60

At 11 the discriminant is $5^2 \cdot 5$. The maximal order contributes $10/5 = 2$ and $\mathbb{Z}[d]$ contributes 10, for a total of 12.

At 29 the discriminant is $13^2 \cdot 5$. The maximal order contributes $28/7 = 4$, while the order of conductor 13 has class number 2 and contributes 56. The observed split into Frobenius orbit lengths $4, 28, 28$ matches these ray degrees. This is an equality of finite counts; it is not by itself a constructed bijection between points and ray classes.

Proposed uniform count, still unproved. The canonical full characteristic-two phase algebra has dimension $\mathcal{R}(n)$ over $\mathbb{F}_4$ for every odd $n \geq 3$. A uniform construction giving even one point for unbounded n would suffice for the user's infinitude goal; proving this entire count formula is stronger than necessary. Its arithmetic right side is already positive at every n.

This corrects the use of a single order class number as the full prediction in nonfundamental discriminants. At $n = 6561 = 3^8$, for example, $F = 5$, $D_0 = 1721869$, and the unit index m is 1. The maximal and conductor-5 class numbers are 96 and 384. Formula (73.2) therefore predicts $b_8 = 96 + 384 = 480$ in the notation of Section 71, rather than 384. No HI solution is claimed at that order. At 81 the discriminant is fundamental, so the prediction remains 432 canonical points, or $b_4 = 8$. At 39 the predicted count is $24 + 96 = 120$; existence and the count there are also not established here.

`hi_char2_ray_count.py` certifies each maximal class group with `bnfcertify`, computes the order unit indices exactly, evaluates (73.1) for all overorders, and compares (73.2) with every completed finite phase count. All comparisons pass. Its additional arithmetic outputs at 39, 81, and 6561 explicitly make no HI assertion.

Infinitude remains unproved. The missing step is an additive construction or lower bound for the full phase algebra, not the existence or positivity of the ray-class inputs.

74. The scalar action, a split quotient, and a stronger finite ray comparison

The count in Section 73 admits a more informative comparison: the complete action of $U_n = (\mathbb{Z}/n\mathbb{Z})^\times$, including its stabilizers. There is also a uniform descent statement for the quotient by this action. These results identify additional structure a construction must preserve; they do not establish that the solution scheme is nonempty in new orders.

74.1. **The quotient by scalar reindexing is split over $\mathbb{F}_4$.** Let Z_n be the full cubic phase scheme in the unique Frobenius gauge of Section 71, and let $\mathcal{A}_n$ be its coordinate algebra over $\mathbb{F}_4$. This is finite étale, possibly the zero algebra. Scalar reindexing acts by

$$(s \cdot W)_a = W_{sa}, \qquad s \in U_n.$$

Reindexing the summation variable proves preservation of all Q and C; the inverse norms and canonical gauge are preserved as well.

Theorem 74.1. *On Z_n, field Frobenius $W \mapsto W^4$ is exactly the action of $2 \in U_n$. Consequently*

$$\boxed{\mathcal{A}_n^{U_n} \cong \mathbb{F}_4^{c_n}} \tag{74.1}$$

for an integer $c_n \geq 0$. It counts scalar orbits of canonical phase points over the algebraic closure. In particular every such orbit is defined over $\mathbb{F}_4$, even when no point in that orbit is $\mathbb{F}_4$-rational.

Proof. Section 71 gives the coordinate identities $W_a^4 = W_{2a}$. Therefore, for every polynomial function f with coefficients in $\mathbb{F}_4$, one has $f^4 = 2 \cdot f$ in $\mathcal{A}_n$. An invariant f satisfies $f^4 = f$. The invariant algebra is finite and reduced, and every residue field is consequently $\mathbb{F}_4$. This proves (74.1). Equivalently, on geometric points, Frobenius acts within each scalar orbit. No averaging by $|U_n|$ is used; that number is even and is not invertible here. □

The same finite étaleness and split-quotient conclusions hold for the **full normalized matrix scheme** defined by the order-three symmetry, Q and C, without requiring a phase parametrization or nonzero generic entries. Indeed (71.2) was proved at that matrix level. Each coordinate satisfies $A_{a,b}^{4^{rn}} = A_{a,b}$, so this matrix scheme too is a quotient of a finite étale scheme; on its coordinate algebra fourth power is scalar reindexing by 2. This is a characteristic-two rigidity statement for the normalized matrix equations. It is not a nonemptiness theorem or an assertion that all matrix solutions have the phase form.

More explicitly, let $H \leq U_n$ be the stabilizer of one point and set

$$r_H = \min\{r > 0 : 2^r \in H\}.$$

Its scalar orbit is the transitive U_n-set U_n/H. Frobenius has cycles of length r_H on this orbit, so its finite étale algebra is

$$\boxed{\left(\mathbb{F}_{4^{r_H}}\right)^{[U_n:H]/r_H}}. \tag{74.2}$$

Thus stabilizers and their multiplicities determine both the complete scalar action and the field-factor decomposition. This is stronger than knowing the total number of geometric points.

There is a small uniform restriction on H: $-1 \notin H$. Otherwise the inverse norm gives $W_a^2 = \omega$, hence $W_a = \omega^2$ for every nonzero a. The residual $F(1,1)$ is then $\omega \neq 0$, contradicting Q. In particular H has odd order when n is an odd prime power, because U_n is cyclic with unique involution -1. This does not classify the possible stabilizers.

The quotient in (74.1) is a quotient of the displayed matrix/phase scheme. No extra claim about descent of associators, or about equivalence classes of all fusion categories, is needed for this statement.

74.2. **The arithmetic action and its exact finite match.** Retain $D = n^2 + 4 = F^2 D_0$, O_f, h_f, and m_f from Section 73. Let

$$H_f = \langle \epsilon_f \bmod (d-1) \rangle \leq U_n, \qquad |H_f| = m_f.$$

The one-positive-place ray-class exact sequence gives

$$1 \longrightarrow U_n/H_f \longrightarrow \mathrm{Cl}_{(d-1),+}(O_f) \longrightarrow \mathrm{Pic}(O_f) \longrightarrow 1. \tag{74.3}$$

Hence, as a U_n-set, this ray group is a disjoint union of h_f copies of U_n/H_f. The choice of an origin in each copy is not canonical. Since $D_0 \equiv 5 \pmod 8$, the prime 2 is inert in the quadratic field; it is coprime to F and to the odd ray modulus. Its Artin element is the local class of the positive generator 2. Thus Frobenius over the residue field $\mathbb{F}_4$ acts by the same element $2 \in U_n$. This uses the ray-class convention already fixed in Sections 63 and 73 and the exact sequence for orders in Kopp–Lagarias, Corollary 6.7.

The strengthened proposed correspondence is therefore

$$\boxed{Z_n(\overline{\mathbb{F}}_2) \;\cong\; \coprod_{f|F} \coprod_{j=1}^{h_f} U_n/H_f} \quad \text{as } U_n\text{-sets}, \tag{74.4}$$

with Frobenius acting by 2 on both sides. It would imply the count of Section 73, the predicted residue fields, and $c_n = \sum_{f|F} h_f$. Formula (74.4) is **not proved uniformly**.

Exact finite result. Formula (74.4), as an abstract equivariant bijection, holds at every odd $3 \leq n \leq 31$. The numbers of scalar orbits are

n	3	5	7	9	11	13	15	17	19	21	23	25	27	29	31
scalar orbits	1	1	1	2	2	1	3	1	2	4	2	2	3	3	2

All stabilizers are trivial except the following two single orbits:

n	H	orbit size
11	$\{1,3,4,5,9\} = \langle 4 \rangle$	2
29	$\{1,7,16,20,23,24,25\} = \langle 24 \rangle$	4.

These are exactly the maximal-order unit images in the two $\mathbb{Q}(\sqrt{5})$ cases. At 31, for example, the twelve Frobenius orbits of length 5 combine into two scalar orbits of length 30, rather than twelve different scalar orbits.

`hi_char2_scalar_action.py` reconstructs the canonical phases from the independently verified row certificates, generates every Frobenius conjugate, and checks closure under every unit reindexing. It computes the actual stabilizer subgroups, not only their orders. On the arithmetic side it certifies the maximal class number and computes the unit images exactly. Matching transitive-orbit stabilizers and multiplicities proves the claimed abstract finite bijections. It does not construct a Stark unit or a distinguished assignment of phase solutions to ideal classes. The additional arithmetic outputs at 39, 81, and 6561 remain predictions without HI existence assertions.

In particular (74.1) does not turn the problem into finding a rational point on a known nonempty scheme: the integer c_n can still be zero without contradicting any of its uniform proof. A nonzero map from a constructed ray-class family to the phase scheme, or another uniform positive lower bound, remains missing. **Infinitude is not established.**

75. A fixed-period lens product for the uniform analytic candidate

The Shintani candidate of Section 51 has a second exact expression, with one pair of positive periods for each n. This connects it directly to the lens quantum dilogarithm, and recovers the exceptional phase at zero from the same product. The additive HI identities remain unproved.

75.1. **The product and its origin value.** Put

$$d = \frac{n + \sqrt{n^2+4}}{2}, \quad \omega_1 = d, \quad \omega_2 = d^{-1}, \quad \omega_1 - \omega_2 = n,$$

and write $[b]_n \in \{0, \dots, n-1\}$ for the least residue. Extend the notation for the double sine by homogeneity in its two periods.

Theorem 75.1. *For every odd $n \geq 3$ and $1 \leq a < n$, the positive number q_a of (51.3) is*

$$\boxed{q_a = \prod_{j=0}^{n-1} \mathrm{Sin}_2\Big(\frac{a + \omega_2 + j\omega_1 + [-a-1-j]_n\omega_2}{n}; \omega_1, \omega_2\Big).} \tag{75.1}$$

The same displayed product at $a = 0$ equals d. Thus, defining $q_0 = d$, the single formula

$$W_a = -(-1)^{a(a+n)/2}\sqrt{q_a/d}, \qquad 0 \leq a < n, \tag{75.2}$$

recovers all phases of Section 51, including $W_0 = -1$.

Every double-sine argument in (75.1) is in its positive strip. Indeed the numerator is positive and is at most $a + (n-1)\omega_1 + n\omega_2 < n(\omega_1 + \omega_2)$, since $a < n < \omega_1$.

Here is the change of variables proving (75.1). In the original basis of Section 23, let

$$\tau = d^{-1}, \quad P = \begin{pmatrix} 1 & n \\ n & n^2+1 \end{pmatrix}, \quad \mathbf{r} = \frac{a}{n}\begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Then $P\tau = \tau$, and $\mathbf{r} - P\mathbf{r} = (-a, -(n+1)a)^{\mathsf{T}}$. The positive characteristic here is the inverse label to the negative characteristic used for q_a. Its Shintani cocycle has absolute value q_a^{-1}, by Section 51 and its inverse symmetry.

Use the unnormalized lens function γ_M with the conventions $M = \left(\begin{smallmatrix} -p & -s \\ k & -r \end{smallmatrix}\right)$ of Sarkissian–Spiridonov, Section 2. For P these parameters are

$$k = n, \quad p = -1, \quad r = -n^2 - 1, \quad s = -n.$$

Before scaling the periods, take $(\Omega_1, \Omega_2) = (d^2, 1)$. Its denominator argument is $(\mu + m)/n$. Substitution of the characteristic gives

$$m = -(n+1)a - 1, \qquad \mu = 1 + a(\tau + n) = 1 + ad. \tag{75.3}$$

Indeed $(\mu + m)/n = a(\tau - 1)/n$, and the numerator argument is $(m+1)P\tau + (\mu + m)/(n(n\tau + n^2 + 1))$, modulo the integer a. These are exactly the two arguments in the Shintani cocycle ratio. The equality holds first in the upper half-plane, with the periods and μ varying as prescribed by τ, and then by analytic continuation at the RM point. It does not cancel two divergent products on the unit circle.

The unnormalized lens function is periodic in m, so reduce $m \equiv -a - 1 \pmod{n}$. Replacing $r = -n^2 - 1$ by -1 changes its base parameter τ by an integer n and leaves the unnormalized function unchanged. The corresponding modular matrix is $L_n = \left(\begin{smallmatrix} 1 & 0 \\ n & 1 \end{smallmatrix}\right)$. The finite-product identity for this function has summation pairs (j, γ) satisfying $j + \gamma \equiv m \pmod{n}$. Taking its absolute value at positive real periods gives

$$q_a^{-1} = \prod_{j=0}^{n-1} \mathrm{Sin}_2\Big(\frac{1 + ad + jd^2 + [-a-1-j]_n}{n}; d^2, 1\Big)^{-1}.$$

Scaling both periods and arguments by d^{-1} proves (75.1). This remains the order-$\mathbb{Z}[d]$ candidate of Section 51; no maximal-order substitution is being made.

For the origin assertion, the arguments in (75.1) become $\omega_2 + j$. Reflection pairs j with n-j for $1 \leq j < n$, because $\omega_1 + \omega_2 = n + 2\omega_2$. Their factors cancel. The remaining factor is

$$\mathrm{Sin}_2(\omega_2; \omega_1, \omega_2) = \sqrt{\omega_1/\omega_2} = d,$$

which proves the asserted origin value. □

75.2. **An explicit sign and the remaining integral problem.** The normalization in the cited paper can also be made explicit here. For $0 \le m < n$, its normalized lens gamma for L_n satisfies

$$\Gamma_{L_n}(\mu, m; \omega_1, \omega_2) = (-1)^{m(m+3)/2} \prod_{j=0}^{n-1} \mathrm{Sin}_2\left(\frac{\mu + j\omega_1 + [m-j]_n\omega_2}{n}; \omega_1, \omega_2\right)^{-1}. \tag{75.4}$$

In particular (75.1) is an evaluation along the discrete line $\mu = a + \omega_2,\ m = [-a-1]_n$, with a specified sign, rather than an unspecified normalization of the lens function.

For completeness the phase cancellation is an elementary calculation. Set $S_m = \sum_{j=0}^{n-1} j[m-j]_n$. Splitting the sum at j=m gives

$$S_m = \frac{n(n^2-1)}{6} + \frac{nm(n-m-2)}{2}.$$

The difference between the sum of the n Bernoulli polynomials in the product and $1/n$ times the unshifted Bernoulli polynomial is $-(n-1)^2/(2n) + 2S_m/n^2$. Adding the normalization phase $(n-1)(n-5)/(12n) + (n+1)m(m+2)/(2n)$ leaves $m(m+3)/2$, proving (75.4). This calculation is independent of the RM equation and of the continuous argument μ.

The two elementary period shifts also give a useful equation along the sampling line. With $x = (\mu + m\omega_2)/(n\omega_2)$, their quotient is

$$\frac{\Gamma_{L_n}(\mu + n, m)}{\Gamma_{L_n}(\mu, m)} = \frac{\sin(\pi x)}{\sin(\pi(x - \mu + \omega_2))}. \tag{75.5}$$

Here one first shifts by ω_1 and then back by ω_2; the two changes of the discrete index cancel. The difference between the two sine arguments is $\mu - \omega_2$, since $\omega_1 - \omega_2 = n$ and $\omega_1\omega_2 = 1$. At $\mu = \omega_2 + a$, with integer a and nonzero denominator, the ratio is $(-1)^a$. At the exceptional point $(\mu, m) = (\omega_2, -1)$, the ratio is a removable 0/0: its limit is $\omega_1/\omega_2 = d^2$. Thus the sampling line has a specific exceptional multiplier. The shift formula alone supplies no finite-sum identity for its values.

The published lens identities include a continuous integral in addition to the finite sum. Applying the six-factor identity to the HI candidate still requires a reduction of that integral and a treatment of the square roots in (75.2). Sarkissian–Spiridonov, Sections 3–4. There is a concrete mismatch in the direct substitution: three factors with shifts a_i and three with shifts c_j on our line have continuous parameter sum

$$6\omega_2 + \sum_i a_i + \sum_j c_j.$$

The HI balance makes the integer sum a multiple of n, whereas the six-factor integral balance requires $\omega_1 + \omega_2 = n + 2\omega_2$. Their difference cannot vanish for any choice of integer lifts, because $4\omega_2$ is irrational. The discrete parameter sum is $-6 \pmod n$, while the same integral requires $-2 \pmod n$, which also fails at every odd $n \ge 3$. Thus the direct line evaluation does not apply that theorem. This calculation does not exclude a transformation with additional factors, a different integral identity, or a suitable finite invariant subspace of the integral operator.

`hi_shintani_lens.py` checks the Bernoulli/sign calculation exactly off the RM locus. It also compares the two independently evaluated positive double-sine products and their inverse norms numerically, and recovers the origin value. These checks verify the implementation of the proved change of variables; they do not replace the missing additive proof. **Infinitude remains unproved.**

76. The Shintani product is the square of one double sine

The square-root difficulty left in Section 75 can be removed uniformly. The candidate phases themselves, with their fixed signs, are values of one ordinary double sine. Its meromorphic extension has an exact jump on integer translates that accounts for the exceptional origin.

76.1. **A product identity away from the RM specialization.** Write $S(z;u,v) = \mathrm{Sin}_2(z;u,v)$, in the convention of Section 51. For every integer $N \geq 1$, $b > 0$, and $0 < v < b+1$, one has

$$\boxed{\prod_{j=0}^{N-1} S(b+v+j;b+N,b) = \frac{S(v;b,1)}{S(v;b+N,1)}.} \tag{76.1}$$

All the arguments are in their positive convergence strips.

Proof. Insert the integral (51.8), in its homogeneous two-period form, on both sides. In the sum of its hyperbolic-sine terms use

$$\sum_{j=0}^{N-1} \sinh((N-2v-2j)t/2) = \frac{\sinh(Nt/2)}{\sinh(t/2)} \sinh((1-2v)t/2),$$

and

$$\frac{\sinh(Nt/2)}{\sinh((b+N)t/2)\sinh(bt/2)} = \coth(bt/2) - \coth((b+N)t/2).$$

Expanding the two hyperbolic sines on the right of (76.1) gives this same expression; their cosh terms cancel. The linear subtraction terms also agree, by $N/(b(b+N)) = 1/b - 1/(b+N)$. The convergent logarithmic integrals are therefore equal. Positivity proves the stated identity without a root-of-unity ambiguity. □

76.2. **The formula with one factor per phase.** Retain $d = (n+\sqrt{n^2+4})/2$, and put

$$\lambda = \frac{d}{d+1} = \frac{d-1}{n}.$$

Theorem 76.1. *At every odd $n \geq 3$, the numbers of Sections 51 and 75 satisfy*

$$\boxed{q_a = S(1+\lambda a;d,1)^2, \qquad 0 \leq a < n,} \tag{76.2}$$

with $q_0 = d$. Consequently the candidate phases are exactly

$$\boxed{W_a = -\frac{(-1)^{a(a+n)/2}}{\sqrt{d}} S(1+\lambda a;d,1), \qquad 0 \leq a < n.} \tag{76.3}$$

In particular there is no square root of a special-function value in this expression. The positive constant $\sqrt{d}$ and the displayed elementary signs fix its normalization.

Proof. Let $1 \leq a < n$, $b = d^{-1}$, and $v = a/(d+1)$. In (75.1), the arguments with $0 \leq j < n-a$ are $b+v+j$; the other a arguments are $b+v+j+b$. Shifting those latter factors back by b gives

$$q_a = \frac{\prod_{j=0}^{n-1} S(b+v+j;d,b)}{\prod_{k=1}^{a} 2\sin(\pi(k-v)/d)}. \tag{76.4}$$

Use (76.1), $d = b+n$, and homogeneity to replace the numerator by $S(dv;d,1)/S(v;d,1)$. The a shifts by the period 1 give

$$\frac{1}{\prod_{k=1}^{a} 2\sin(\pi(k-v)/d)} = \frac{S(a+1-v;d,1)}{S(1-v;d,1)}.$$

Here $a+1-v = 1+dv$. Reflection and the period-d shift imply $S(v;d,1)S(1-v;d,1) = 2\sin(\pi v)$, while the period-1 shift gives $S(dv;d,1) = 2\sin(\pi v)S(1+dv;d,1)$. Substitution into (76.4) proves (76.2), since $dv = \lambda a$.

At a=0 use $S(1;d,1)=\sqrt{d}$. Every factor in (76.2) is positive, so taking the positive square root recovers (76.3) exactly. □

The inverse norms now follow immediately from reflection: $1+\lambda(n-a)=d-\lambda a$. Combining this with the opposite signs at a and n-a gives $W_aW_{n-a}=-d^{-1}$. This proves the normalization, but still does not prove any of the missing additive equations.

76.3. **A meromorphic interpolation and an exact integer jump.** Define the meromorphic function

$$f_n(z)=-d^{-1/2}e^{\pi iz(z+n)/2}S(1+\lambda z;d,1).$$

It agrees with W_a at $a=0,\dots,n-1$. The two elementary shifts and reflection of the double sine prove

$$\boxed{\frac{f_n(z+n)}{f_n(z)}=-e^{\pi inz}\frac{\sin(\pi z/(d+1))}{\sin(\pi\lambda z)},} \tag{76.5}$$

and

$$\boxed{f_n(z)f_n(-z)=d^{-1}e^{\pi iz^2}\frac{\sin(\pi\lambda z)}{\sin(\pi z/(d+1))}.} \tag{76.6}$$

These are identities of meromorphic functions, with removable values interpreted by limits.

Since $\lambda+1/(d+1)=1$, for every nonzero integer a the sine ratio in (76.5) is $(-1)^{a+1}$. Its prefactor cancels this sign. At zero the ratio is $1/d$, and hence

$$f_n(a+n)=f_n(a)\ (a\in\mathbb{Z}\setminus\{0\}),\quad f_n(0)=-1,\quad f_n(n)=d^{-1}.$$

Equivalently, the entire integer sequence obeys

$$\boxed{f_n(a+n)-f_n(a)=(1+d^{-1})\,\mathbf{1}_{a=0}.} \tag{76.7}$$

No nonzero integer makes either sine vanish, because the two slopes are irrational. Thus this argument includes negative integer a, without omitting other exceptional residues. Equation (76.6) likewise gives $f_n(a)f_n(-a)=-d^{-1}$ at every nonzero integer, and the limiting value 1 at zero.

This gives a precise possible starting point for a contour or periodization argument: the finite matrix phases are one period of a specified meromorphic function, whose integer period defect is exactly one delta term. Such an argument still has to prove the finite sums Q and C. The interpolation identities do not imply those sums by themselves. The direct balancing calculation of Section 75 concerned the lens integral; (76.3) removes its square-root issue by passing to an ordinary double sine, but supplies no completed integral-to-finite-sum reduction.

`hi_shintani_single.py` verifies (76.1) numerically away from the RM specialization, compares (76.2) with the earlier continued-fraction product, and checks Q and all cross-free cubic orbits numerically at the requested finite orders. These numerical checks support the implementation of the proved formulas. **The additive identity, and therefore infinitude, remains unproved.**

77. The full lens product factors, and its contour boundary terms

The simplification in Section 76 extends off the sampling line. This gives an explicit integrand for further work with the lens beta identity. It also makes two limitations of a direct contour argument precise.

77.1. **Two factors at every continuous argument.** Set $b = d^{-1}$, and for $0 \le m < n$ define

$$P_n(\mu, m) = \prod_{j=0}^{n-1} S\Big(\frac{\mu + jd + [m-j]_n b}{n}; d, b\Big).$$

Then, as an identity of meromorphic functions of μ,

$$\boxed{P_n(\mu, m) = S\Big(\frac{d\mu + m}{n}; d, 1\Big)\, S\Big(\frac{d(n-m) - \mu}{n}; d, 1\Big).} \tag{77.1}$$

The product identity itself works for every integer $n \ge 1$. We retain odd $n \ge 3$ for the HI application.

Proof. Put $a = n - m - 1$ and $v = (\mu + (m-n)b)/n$. Initially choose real $0 < v < 1$, so all the factors below are in their positive strips. The arguments with $j \le m$ are $b + v + j$, while those with $j > m$ are $b + v + j + b$. The same shifts as in (76.4) give

$$P_n(\mu, m) = \frac{S(dv; d, 1)}{S(v; d, 1)} \frac{S(a + 1 - v; d, 1)}{S(1 - v; d, 1)}.$$

Since $S(v)S(1-v) = 2\sin(\pi v)$ and $S(dv) = 2\sin(\pi v)S(1+dv)$, this equals $S(1+dv)S(n-m-v)$. The two arguments are respectively $(d\mu+m)/n$ and $(d(n-m)-\mu)/n$. Meromorphic continuation proves (77.1) at all other arguments. □

Consequently the normalized lens gamma of Section 75 has the formula

$$\Gamma_L(\mu, m) = \frac{(-1)^{m(m+3)/2}}{S((d\mu + m)/n; d, 1) S((d(n-m) - \mu)/n; d, 1)}. \tag{77.2}$$

Here m is the representative $0, \dots, n-1$; one must retain the prescribed discrete normalization when changing representatives.

The two arguments coincide precisely on $\mu + m = d - 1$. At $\mu = a + b$, $m = n - a - 1$, both equal $1 + da/(d+1)$. Thus (76.2) is the restriction of (77.1) to this diagonal, including its removable endpoint at a=0.

There is also an exact normalization at the pole used in beta-integral residue arguments. The period-1 shift and $S(1; d, 1) = \sqrt{d}$ give $S(z; d, 1) \sim 2\pi z/\sqrt{d}$ at zero. Reflection gives $S(d; d, 1) = 1/\sqrt{d}$. Therefore

$$P_n(\mu, 0) \sim \frac{2\pi\mu}{n}, \qquad \boxed{\operatorname*{Res}_{\mu=0} \Gamma_L(\mu, 0) = \frac{n}{2\pi}.} \tag{77.3}$$

This agrees with the general normalization in the residue computation preceding the theorem of Sarkissian–Spiridonov, Section 3. The present derivation uses only the two ordinary factors.

77.2. **Integer values and derivatives have different shift defects.** Let $f = f_n$ be (76.3)'s meromorphic interpolation, and retain $\nu = 1/(d+1)$, $\lambda = 1 - \nu$. Although f has the single integer-value defect (76.7), its derivatives have additional terms at every nonzero integer a:

$$\boxed{f'(a+n) - f'(a) = \pi\big(in + \cot(\pi\nu a)\big) f(a).} \tag{77.4}$$

For the second logarithmic derivative one similarly has

$$\left(\frac{f'}{f}\right)'(a+n) - \left(\frac{f'}{f}\right)'(a) = \pi^2(\lambda - \nu)\csc^2(\pi\nu a). \tag{77.5}$$

These formulas involve no choice of logarithm.

Proof. Logarithmically differentiate the meromorphic ratio (76.5). Its first derivative is

$$\pi i n + \pi\nu \cot(\pi\nu z) - \pi\lambda \cot(\pi\lambda z).$$

At a nonzero integer, $\cot(\pi\lambda a) = -\cot(\pi\nu a)$, and the ratio itself is 1. This proves (77.4). A further derivative, using $\csc^2(\pi\lambda a) = \csc^2(\pi\nu a)$, gives (77.5). Irrationality of the slopes ensures all these evaluations are regular. At zero the removable ratio gives instead $f'(n) = -f'(0)/d + \pi i n/d$. □

Thus a contour with multiple poles at the integers cannot treat the derivative contributions as periodic up to the single origin term. Equations (77.4)–(77.5) supply the missing terms explicitly.

77.3. **A growth obstruction for the direct vertical integral.** Rescale the periods to

$$A = d + 1, \qquad B = 1 + d^{-1}, \qquad A - B = n, \quad AB = A + B =: Q_*.$$

Homogeneity gives $S(1 + \lambda z; d, 1) = S(B + z; A, B)$. The usual positive-period asymptotic is

$$\log|S(u + iy; A, B)| = -\frac{\pi|y|}{AB}\left(u - \frac{A+B}{2}\right) + o(1) \quad (y \longrightarrow \pm\infty), \tag{77.6}$$

for fixed real u. For these quadratic-irrational periods this follows directly from the convergent upper-half-plane expansion in Koyama–Kurokawa, Theorem 2, and real conjugation gives the lower half-plane. Their double-sine convention is the same as ours.

Let $a_1, \ldots, a_r, c_1, \ldots, c_r$ be real numbers with $\sum_i a_i + \sum_j c_j = 0$, and set $s = \sum_i a_i$. The balanced product

$$\Phi_r(z) = \prod_{i=1}^{r} f(z + a_i) \prod_{j=1}^{r} f(c_j - z)$$

then satisfies the uniform formula

$$\boxed{\log|\Phi_r(x + iy)| = \frac{\pi r n}{Q_*}|y| - 2\pi(rx + s)y - r\log d + o(1).} \tag{77.7}$$

Indeed, the elementary Gaussian factors contribute $-2\pi(rx + s)y$. The sum of the real sine arguments minus their half-period centers is $2rB - r(A + B) = -rn$, proving the remaining exponential term by (77.6).

At every fixed x, at least one vertical end has positive exponential growth. Multiplication by an exponential $e^{2\pi i t z}$, with real t, cannot change the sum of the two growth rates. Thus this balanced product does not by itself give an absolutely convergent vertical integral.

Even the simple-pole kernel $1/\sin(\pi z)$ is insufficient for both the quadratic and cubic products, r=2 and r=3. It subtracts $\pi|y|$ from (77.7), whereas

$$\frac{2n}{Q_*} > 1 \qquad (n \geq 3).$$

For this last inequality, $Q_* = \sqrt{n^2 + 4} + 2 < 2n$, as is seen by squaring $\sqrt{n^2 + 4} < 2n - 2$. Higher-order poles can supply sufficient decay, but introduce the derivative terms in Section 77.2. This rules out the indicated direct absolutely convergent integrals, not deformed contours, subtraction procedures, or other reductions of the beta identity.

77.4. **Repairing the beta balance with four explicit shifts.** There is nevertheless a convergent ordinary beta integral containing the desired six sine factors. Write $S_*(x) = S(x; A, B)$. Let $a_1, a_2, a_3, c_1, c_2, c_3$ satisfy

$$\sum_i a_i + \sum_j c_j = n. \tag{77.8}$$

Initially impose $a_1, a_2, c_1, c_2 < 0$ and $a_3, c_3 < A$, so the six gamma parameters used below are positive. Then

$$\boxed{\begin{aligned}&\int_{-i\infty}^{i\infty} \frac{\prod_{i=1}^{3} S_*(B+z+a_i)S_*(B-z+c_i)}{\prod_{i=1}^{2} 2\sin(\pi(z+a_i)/B)\,2\sin(\pi(c_i-z)/B)}\,\frac{dz}{i\sqrt{AB}}\\ &\quad=\frac{2\sin(\pi(A-a_3-c_3)/B)}{\prod_{i,j=1}^{2} 2\sin(\pi(a_i+c_j)/B)}\prod_{i,j=1}^{3} S_*(B+a_i+c_j).\end{aligned}} \tag{77.9}$$

Removable factors on either side are interpreted by continuation of the whole expression. More general parameters require the usual continued contour separating the two families of gamma poles.

Proof. Put $\gamma_*(x) = 1/S_*(x)$. Apply the ordinary three-plus-three hyperbolic beta integral, the k=1 case of Sarkissian–Spiridonov, equation (69), to the parameters

$$(-a_1, -a_2, A-a_3), \qquad (-c_1, -c_2, A-c_3).$$

Their sum is $2A - n = A + B$, exactly the required balance. Reflection and the period-A shift give

$$\gamma_*(A-x) = S_*(B+x), \qquad \gamma_*(-x) = -\frac{S_*(B+x)}{2\sin(\pi x/B)},$$

and

$$\gamma_*(2A-x) = 2\sin(\pi(A-x)/B)S_*(B+x).$$

The four shifted factors on the left give the four sine denominators; their four minus signs cancel. On the right, the pairs with i,j at most 2 give four denominators and four cancelling minus signs. The pair (3,3) gives the displayed numerator, while the other four pairs are unshifted. This proves (77.9). □

For these real parameters its integrand has modulus asymptotic to a constant times $e^{-2\pi|\operatorname{Im} z|}$. Indeed, (77.8) changes the six-sine growth rate to $2\pi n/(AB)$, and the four denominators subtract $4\pi/B$; their difference is exactly -2π, since $AB = A + B$.

The normalization also agrees exactly with that of the desired finite identity. Define the canonical cyclic samples

$$V_a = (-1)^{a(a+n)/2} S_*(B+a) = -\sqrt{d}\,W_a, \qquad 0 \le a < n.$$

For balanced residues with no $a_i + c_j = 0$ modulo n, the cross-free HI cubic identity is equivalent to

$$\boxed{\frac{1}{\sqrt{AB}} \sum_{m\in\mathbb{Z}_n} \prod_{i=1}^{3} V_{m+a_i} \prod_{j=1}^{3} V_{c_j-m} = \prod_{i,j=1}^{3} V_{a_i+c_j}.} \tag{77.10}$$

This follows by substituting $W = -V/\sqrt{d}$ into the beta form of C, and using $(d+1)/\sqrt{d} = \sqrt{AB}$. Equation (77.10) is the **unproved target**, not a consequence of (77.9). The sine denominators and numerator in (77.9), the passage from the continuous contour to canonical samples, and the origin corrections still have to be accounted for. The quadratic and collision equations also remain to be established uniformly. The four-shift calculation supplies a correctly balanced integral for that work.

`hi_shintani_beta.py` checks the normalization of (77.9) independently by numerical integration at n=3 and $a = c = (-1/2, -1/2, 5/2)$. With 64 Gauss–Legendre nodes on each half of the truncated contour, the integral and product agree to 4.9×10^{-20}. This is a numerical check of the continuous identity; it does not verify an integral-to-finite-sum reduction.

`hi_shintani_meromorphic.py` independently compares the n-factor lens product with (77.1) off the diagonal, checks (76.5)–(76.7) at complex and integer arguments, and recovers (77.3) by

a symmetric numerical limit. The saved runs at n=3,7,11 all pass. No finite additive equation is deduced from these meromorphic formulas; **infinitude remains unproved**.

78. Rational monodromy of the quadratic and cubic kernels

The same two rational functions control the n-shift of all the double-sine factors in a summation kernel. Their degrees depend on the number of factors, rather than on n.

78.1. **A rational function of degree r.** Retain $A = d + 1$, $B = 1 + d^{-1}$, so $A - B = n$, $AB = A + B$, and $A^{-1} + B^{-1} = 1$. Put

$$F(z) = S(B + z; A, B) = S(1 + z/B; d, 1).$$

The period shifts give the meromorphic identity

$$\frac{F(z+n)}{F(z)} = \frac{\sin(\pi z/A)}{\sin(\pi z/B)}. \tag{78.1}$$

For r=2 or r=3 choose integer lifts a_i, c_j satisfying

$$\sum_i a_i + \sum_j c_j = (r-2)n, \qquad \Psi_r(z) = \prod_i F(z + a_i) \prod_j F(c_j - z).$$

The r=2 balance is that of the quadratic kernel; the r=3 balance is the one used in (77.8). Define

$$q = e^{2\pi i/d}, \qquad \alpha_i = e^{2\pi i a_i/A}, \qquad \beta_j = e^{2\pi i(c_j - n)/A},$$

$$P(X) = \prod_i (1 - \alpha_i X), \qquad D(X) = \prod_j (1 - X/\beta_j), \qquad H(X) = P(X)/D(X).$$

The balance and $n/A = 1 - d^{-1}$ imply

$$\prod_i \alpha_i \prod_j \beta_j = q^2, \qquad H(0) = 1, \qquad H(\infty) = q^2. \tag{78.2}$$

For $x = e^{2\pi i z/A}$, $y = e^{-2\pi i z/B}$, and $t = e^{2\pi i z}$, one has $x = ty$ and

$$\boxed{\frac{\Psi_r(z+n)}{\Psi_r(z)} = \frac{H(x)}{H(y)}.} \tag{78.3}$$

Proof. Apply (78.1) to the positive arguments and its inverse to the negative arguments. Write each sine as a difference of exponentials. The facts that the lifts are integers and $A^{-1} + B^{-1} = 1$ identify the two sets of exponential parameters. For example, $e^{2\pi i a_i/B} = \alpha_i^{-1}$. The constant left by these conversions is removed by (78.2), giving (78.3). All steps are meromorphic identities, so the formula extends across removable zero-over-zero expressions. □

At a regular integer m, $x = y$, and the shift ratio is one. The exceptional integers are among $-a_i$ and $c_j - n$, where the origin jump of an individual factor can occur. In particular, this statement does not discard the canonical-sample corrections of Sections 76–77.

78.2. **A nonsingular matrix of size two or three.** Define the symmetric Bezout polynomial

$$\mathcal{B}(X, Y) = \frac{P(X)D(Y) - P(Y)D(X)}{X - Y} = \sum_{i,j=0}^{r-1} b_{ij} X^i Y^j.$$

It has degree at most r-1 in each variable. Equation (78.3) gives the exact divided difference

$$\boxed{\frac{\Psi_r(z+n) - \Psi_r(z)}{e^{2\pi i z} - 1} = \Psi_r(z) \frac{y\,\mathcal{B}(x, y)}{D(x)P(y)}.} \tag{78.4}$$

The matrix $\mathsf{B}_r = (b_{ij})$ satisfies

$$\det \mathsf{B}_r = (-1)^{r(r-1)/2} \operatorname{Res}(P, D), \qquad \operatorname{Res}(P, D) = \frac{\prod_{i,j}(1 - \alpha_i \beta_j)}{(\prod_j \beta_j)^r}. \tag{78.5}$$

The second equality follows by evaluating D at the roots α_i^{-1} of P, with multiplicities; the first is the Bezout determinant identity. The exact polynomial checks in the script below verify the determinant signs for both r=2 and r=3.

If all $a_i + c_j \neq 0 \pmod n$, these determinants are nonzero. Indeed, $\alpha_i \beta_j = 1$ would imply $(a_i + c_j - n)/A \in \mathbb{Z}$; since A is irrational, this forces $a_i + c_j = n$, contrary to the cross-free condition. Thus H has degree exactly r and B_r has rank r. Repeated lifts within one of the two lists cause no difficulty.

At every nonexceptional integer m, differentiating (78.3) yields

$$\frac{\Psi_r'(m+n)}{\Psi_r(m+n)} - \frac{\Psi_r'(m)}{\Psi_r(m)} = 2\pi i \, x \frac{H'(x)}{H(x)}, \qquad x = e^{2\pi i m/A}. \tag{78.6}$$

Correspondingly, the removable value of (78.4) is $\Psi_r(m) x H'(x)/H(x)$. This is a weighted value, not the unweighted summand required by Q or C.

For r=2, write $P = 1 - pX + sX^2$ and $D = 1 - \rho X + \tau X^2$. Then explicitly

$$\mathcal{B}(X, Y) = (\rho - p) + (s - \tau)(X + Y) + (p\tau - \rho s) XY.$$

Away from the diagonal, the relation $H(X) = H(Y)$ is the graph of the Möbius involution

$$Y = -\frac{(\rho - p) + (s - \tau)X}{(s - \tau) + (p\tau - \rho s)X}.$$

For r=3 the corresponding relation has bidegree at most (2,2). No nonsingularity or genus claim about that curve is needed here.

For a quadratic with nonzero external residues g,h, one can take $a = (0, g-h)$ and $c = (h, -g)$. For the cubic, use the balanced lifts of (77.10). Thus the matrix sizes in (78.4) are uniformly two and three. The Gaussian signs and exceptional origin values must still be restored when passing from Ψ_r to the canonical finite products of V or W.

In particular, dividing by $H(x) - H(y)$ to create poles at integer z also introduces zeros of $\mathcal{B}(x, y)$ into the denominator. Their contributions cannot be omitted without an additional argument. Equations (78.3)–(78.6) reduce the analytic shift problem; they do not prove the finite sums.

`hi_shintani_monodromy.py` verifies the generic Bezout identities symbolically for r=2,3 and independently checks (78.3)–(78.4) using double-sine evaluations at two complex arguments, for both kernels at n=3,7,11. The largest numerical discrepancy in the saved 50-digit run is below 2×10^{-49}. The identities above have direct proofs; the numerical run checks their implementation and conventions. **Infinitude remains unproved.**

79. Canonical balanced lifts remove all signs from the cross-free targets

The origin jump in Section 76 does not require corrections inside the desired cross-free sums if their integer representatives are chosen appropriately. Such representatives always exist. Together with the earlier collision reduction, this gives two finite double-sine identities whose uniform proof would suffice for existence.

79.1. **The canonical-lift lemma.** Let $p_1, \dots, p_r, q_1, \dots, q_r \in \mathbb{Z}_n$, with r=2 or r=3, satisfy

$$\sum_i p_i = \sum_j q_j, \qquad p_i \neq q_j \quad \text{for all } i, j.$$

Here p are the zeros of the positive-index factors, so $a_i = -p_i$, and q are the zeros of the negative-index factors, so $c_j = q_j$.

Lemma 79.1. *After a common cyclic translation of p and q, and, only if necessary for r=3, interchanging the two lists, there are integer lifts with*

$$\boxed{-(n-1) \le a_i \le 0, \qquad 0 \le c_j \le n-1, \qquad \sum_i a_i + \sum_j c_j = (r-2)n.} \tag{79.1}$$

These operations preserve the finite sum: translation reindexes its summation variable, while interchange also reverses that variable. The cross-sum product for r=3 is preserved as well.

Proof. Use representatives in $\{0, \dots, n-1\}$ and set

$$k(t) = \frac{\sum_j [q_j - t]_n - \sum_i [p_i - t]_n}{n}, \qquad t \in \mathbb{Z}_n.$$

These are integers in $\{-(r-1), \dots, r-1\}$, and their average over t is zero. Their successive changes are

$$k(t+1) - k(t) = \#\{j : q_j = t\} - \#\{i : p_i = t\}. \tag{79.2}$$

The two counts cannot both be nonzero by the cross-free condition.

For r=2, suppose no k(t) is zero. Then all k(t) are in $\{-1, 1\}$, so all jumps in (79.2) are even. Each list must therefore consist of two copies of one point, say p and q. The balance gives $2p = 2q \pmod n$. Since n is odd this forces p=q, a contradiction. Thus k(t)=0 occurs.

For r=3, at least one point in either list has odd multiplicity, so (79.2) has an odd jump. Hence some k(t) is odd. Its bound implies k(t)=1 or -1. Interchanging the lists changes its sign, so k(t)=1 is obtainable. In either case put $a_i = -[p_i - t]_n$, $c_j = [q_j - t]_n$. □

79.2. **Exact cancellation of the interpolation signs.** For the lifts in (79.1) and every $m = 0, \dots, n-1$, all arguments $m + a_i$ and $c_j - m$ lie in $[-(n-1), n-1]$. In this interval the integer interpolation f from Section 76 equals the canonical phase $W_{[u]_n}$, including at u=0. No argument is a positive multiple of n. The same holds for every nonzero cross argument $a_i + c_j$.

Write $F(u) = S(B + u; A, B)$ as in Section 78 and put

$$E = \frac{1}{2} \sum_i (a_i^2 + c_i^2 + n(a_i + c_i)) \in \mathbb{Z}.$$

Since $V_{[u]_n} = e^{\pi i u(u+n)/2} F(u)$ on this integer interval, direct expansion gives

$$\prod_i V_{m+a_i} \prod_j V_{c_j - m} = (-1)^E \Psi_r(m). \tag{79.3}$$

Indeed, the remaining exponent is $rm^2 + (2\sum_i a_i - (r-2)n)m$, an even integer for every integer m because n is odd. Thus the sign is independent of m.

For r=3 define

$$J = \frac{1}{2} \sum_{i,j} (a_i + c_j)(a_i + c_j + n).$$

If $s = \sum_i a_i$, $t = \sum_j c_j$, so s+t=n, then

$$J - E = \sum_i (a_i^2 + c_i^2) + st + n^2 \equiv 0 \pmod 2.$$

Here the sum of the squares is odd, st is even, and n^2 is odd. Consequently

$$\prod_{i,j} V_{a_i + c_j} = (-1)^E \prod_{i,j} F(a_i + c_j). \tag{79.4}$$

The two signs cancel exactly in (77.10).

79.3. **Two explicit unweighted identities suffice.** For each odd n, with $A = d + 1$, $B = 1 + d^{-1}$, the analytic existence problem is now reduced to the following statements for integer lifts satisfying (79.1) and $a_i + c_j \neq 0$:

$$\boxed{\sum_{m=0}^{n-1} \prod_{i=1}^{2} F(m + a_i)F(c_i - m) = 0 \quad (\textstyle\sum_i (a_i + c_i) = 0),} \tag{79.5}$$

and

$$\boxed{\frac{1}{\sqrt{AB}} \sum_{m=0}^{n-1} \prod_{i=1}^{3} F(m + a_i)F(c_i - m) = \prod_{i,j=1}^{3} F(a_i + c_j) \quad (\textstyle\sum_i (a_i + c_i) = n).} \tag{79.6}$$

Equations (79.5) and (79.6) are conjectural uniform identities. Their equivalence to the required quadratic and cross-free cubic identities is proved by the lemma and (79.3)–(79.4). The inverse norms and origin normalization are already proved in Section 76. Once (79.5) holds, Section 34.2–34.3 supplies every collision and boundary cubic by algebraic identities. Hence proving both displayed sums for unbounded odd n would imply the requested categorification through Section 52, without first assuming Stark algebraicity. Conversely, a complete solution by this analytic candidate must satisfy these two sums.

This also explains exactly what remains in the continuous beta route. The canonical choice removes the Gaussian signs and the origin corrections from the finite target. The extra trigonometric factors and the continuous contour in (77.9) still need to be converted into (79.6); the lift lemma does not perform that conversion. The rational shift formulas of Section 78 apply to both targets.

`hi_shintani_canonical.py` exhaustively checks the lift and sign identities for every balanced cross-free pair of multisets at n=3,5,7,9,11. At n=11 this covers 330 quadratic pairs and 3520 cubic pairs; 198 cubic pairs use the reversal allowed in the lemma. Independent numerical double-sine evaluations check (79.5)–(79.6) on selected representatives at n=3,7,11, with normalized residuals below 1.6×10^{-40} in the saved run. These finite checks support the formulas; they do not prove the conjectural identities for unbounded n. **Infinitude remains unproved.**

79.4. **The exact scope of a period-shift recurrence.** Reflection gives the useful quotient form

$$F(z)F(n - z) = 1, \qquad \Psi_r(z) = \prod_i \frac{F(z - p_i)}{F(z + n - q_i)}, \qquad p_i = -a_i,\ q_i = c_i.$$

For the canonical lifts, this kernel has no poles on the closed vertical strip $0 \leq \operatorname{Re} z \leq n$. Indeed, the zeros and poles of F are respectively

$$-B - jA - kB, \qquad A + jA + kB \qquad (j, k \geq 0).$$

The positive-argument factors $F(z - p_i)$ first have poles at $z = p_i + A > n$, while the negative-argument factors $F(q_i - z)$ have all their poles at $z = q_i - A - jA - kB < 0$. All these poles are real. Thus the finite-sum contour problem can be considered in a strip where the summation kernel itself is holomorphic.

However, a finite sequence of ordinary period shifts cannot connect distinct integer residue classes:

$$\boxed{(\mathbb{Z}A + \mathbb{Z}B) \cap \mathbb{Z} = n\mathbb{Z}.}$$

To prove this, write $jA + kB = jn + (j + k)B$ and use the irrationality of B. Adding reflection only changes an integer argument to one of the form $\pm u + jn$. Therefore direct use of shifts and reflection gives the norm and translate laws already proved, but does not by itself provide a recurrence between arbitrary labels in the finite target. An additive summation or integral

identity is still needed to bridge that step. This statement concerns finite compositions of these operations; it does not exclude limits or other identities of S.

79.5. **Available mathematical software.** In addition to PARI/GP and the Python tools used above, this machine has Magma at `/home/tzuchen/Software/Magma/magma`, Macaulay2, Singular, Prover9, Vampire, Julia with TensorCategories and Oscar, and Mathematica with Anyonica. The inspected package files identify TensorCategories 0.7.2 and Anyonica 0.9.20.1; Oscar 1.8.1 and 1.8.2 are both present in the Julia package cache. Julia's `v1.12` environment lists TensorCategories as a direct dependency. Anyonica is installed at `/home/tzuchen/.Mathematica/Paclets/Repository/Anyonica-0.9.20.1`. Its `FusionRingHI` constructs the fusion ring; its existence does not itself provide associators. TensorCategories includes pentagon checking and saved order-3 Haagerup examples. These capabilities are available for independent categorical checks of future constructions. Native FLINT 3.4.0 is also available (`pkg-config -modversion flint`). The user requests notification whenever a useful package cannot be found.

Files from this exploration

- `hi_char2_phase1519_length.m`, `hi_char2_phase1519_length_results.txt`: completed full quadratic-algebra length at 15 and the continuing independent length test at 19, without Frobenius assumptions.
- `hi_shintani_canonical.py`, `hi_shintani_canonical_results.txt`: exhaustive integer checks of the canonical-lift and sign lemmas, and numerical checks of the two finite targets in Section 79.
- `hi_shintani_monodromy.py`, `hi_shintani_monodromy_results.txt`: symbolic Bezout and resultant identities, and independent numerical checks of the uniform rational shift formulas in Section 78.
- `hi_shintani_beta.py`, `hi_shintani_beta_results.txt`: numerical normalization check of the correctly balanced continuous beta integral in Section 77.4.
- `hi_shintani_meromorphic.py`, `hi_shintani_meromorphic_results.txt`: independent complex-argument factorization, shift, jump, and residue checks for Sections 76–77.
- `hi_shintani_single.py` and `hi_shintani_single_results.txt` The one-double-sine phase formula, off-RM product comparisons, and independent numerical Q/C checks for Section 76.
- `hi_shintani_lens.py` and `hi_shintani_lens_results.txt` The fixed-period lens product, intrinsic origin normalization, and exact Bernoulli/sign check for the analytic construction of Section 75.
- `hi_char2_scalar_action.py` and `hi_char2_scalar_action_results.txt` Complete scalar stabilizers and Frobenius-factor comparisons with the quadratic-overorder ray actions at all odd orders through 31.
- `hi_char2_scalar_small_search_results.txt` Reconstructed full certificates at 5, 7, 11, and 13 used in the scalar comparison. Its maximal-order-only count at 11 is distinguished from the correct sum over both overorders in Sections 73 and 74.
- `hi_char2_general_search.py` and `hi_char2_general_search_results.txt` General doubling-orbit parametrization and exhaustive complete phase classifications at 15, 17, 19, 21, 23, 25, 29, and 31.
- `hi_char2_ray_count.py` and `hi_char2_ray_count_results.txt` Certified sums of ray-class degrees over quadratic overorders, with exact comparison to all completed counts through 31.
- `hi_char2_phase17_length.m` and `hi_char2_phase17_length_results.txt` Magma Q-algebra length 272, proving full characteristic-two completion at 17 by comparison with the independently classified cubic scheme.

– `hi_char2_sat81_search_results.txt` The completed one-hour SAT attempt, with no SAT/UNSAT conclusion.
– `hi_char2_cubic_frobenius.py` and `hi_char2_cubic_frobenius_results.txt` Exact off-solution convolution identities and unique Frobenius-gauge recovery across every character twist of all saved finite branches.
– `hi_char2_phase_ideal.py` and `hi_char2_phase_ideal_results.txt` Complete characteristic-two Q-to-C ideal containments and nonzero phase-algebra lengths at every odd order through 13.
– `hi_char2_quadratic_orbits.py` and `hi_char2_quadratic_orbits_results.txt` Exact coverage checks for the uniform linear count of Q orbits through 729.
– `hi_char2_sat.py`, `hi_char2_sat_checks.py`, and their result files Native-XOR searches, exact off-solution arithmetic checks through 81, and complete independent finite certificates at 3 and 9.
– `hi_char2_firstq_linear.py` and `hi_char2_firstq_linear_results.txt` Exact linear elimination of lower Frobenius seeds from the first Q, recovery of the saved branches, and finite tests through order 81.
– `hi_char2_ratio_bound.py` and `hi_char2_ratio_bound_results.txt` Exact checks of the uniform patched rational-ratio obstruction and minimum interpolation degrees for the saved certificates.
– `hi_stark_seedless.gp` and `hi_stark_seedless_results.txt` Construction without HI seeds and exact complete Izumi certificates for $n = 7, 11, 13, 17, 29$, including exact identification of the earlier $n = 13, 17$ polynomials.
– `hi_stark19_certificate.gp`, `hi_stark25_certificate.gp`, `hi_stark199_certificate.gp`, and `hi_stark_verify_certificate.py` Saved exact matrix data and an independent existence verifier.
– `hi_stark19_results.txt`, `hi_stark199_construction_results.txt`, `hi_stark199_cubic_results.txt`, and `hi_certificate_verification_results.txt` Complete construction, polynomial-identification, and verification logs.
– `hi_stark_prime_power.gp`, `hi_stark25_artin.gp`, `hi_stark_single_root.gp`, `hi_stark25_construction_results.txt`, and `hi_stark25_verification_results.txt` Prime-power one-root construction, the exact conjugate-prime twist at order 25, and its complete independent verification.
– `hi_single_root_checks.gp`, `hi_single_root_results.txt`, `hi_prime_power_signs.py`, and `hi_prime_power_signs_results.txt` Exact single-root reconstruction and sign-selection checks, and independent off-solution tests of the linear sign identity.
– `hi_stark_cubic_batches.py` Exact integer-polynomial cubic verification in disjoint local batches.
– `hi_ray_unit_search.gp`, `hi_ray_unit_search_results.txt`, `hi_ray_unit199_results.txt`, and `hi_ray_unit_explicit_results.txt` Unit-lattice selection with exact full Izumi certificates at eight prime orders; numerical shortest-shell selection is distinguished from exact acceptance.
– `hi_pentagon_operator.gp`, `hi_pentagon_operator.py`, and their result files Exact operator certificates and numerical indexing checks for the uniform fifth-order reformulation of the cubic equations.
– `hi_shintani_ghost.py` and `hi_shintani_ghost_results.txt` Explicit double-sine construction with fixed signs and high-precision numerical checks; no algebraic seeds or arithmetic reciprocity are assumed.
– `hi_cubic_quartic.py` and `hi_cubic_quartic_results.txt` Exact off-solution checks of the uniform cubic-to-quadratic and cubic-to-quartic residual identities.

- `hi_column_barrier.py` and `hi_column_barrier_results.txt` Convex construction with a uniform existence and algebraicity proof for one complete quadratic column; further Izumi checks are numerical.
- `hi_column_trace.py`, `hi_column_trace_exact.py`, and their result files Numerical scalar and period-40 trace tests, plus exact scalar-completion certificates at orders 3, 5, and 7 using finite-algebra rank comparisons.
- `hi_paley_obstruction.py` and `hi_paley_obstruction_results.txt` A symbolic obstruction to an infinite two-value quadratic-character family, with complete exact Q and C certificates for its remaining orders 3 and 11.
- `hi_finite_field_transfer.py` and `hi_finite_field_transfer_results.txt` Off-solution transfer identities and complete characteristic-2 checks of the corrected pentagon criterion, rank-three defect, and order 30.
- `hi_finite_field_certificate.py`, `hi_finite_field_certificate_results.txt`, and `hi_finite_field_char2_results.txt` Independent complete finite-field certificates, supporting categorical lifting rather than only modular rank bounds.
- `hi_char2_frobenius.py`, `hi_char2_frobenius_results.txt`, and `hi_finite_field_certificates/*.json` Exhaustive Frobenius-covariant searches at 3, 9 and 27, followed by independent checks of every cubic for all six Frobenius representatives.
- `hi_char2_elliptic.py` and `hi_char2_elliptic_results.txt` A fixed-curve rational-function construction with complete finite certificates at 3, 9 and 27, and exhaustive rejection of its proposed recurrence at order 81.
- `hi_char2_ray_indices.gp`, `hi_char2_twisted_norm.py`, and their result files Certified conjugate-prime and Frobenius relations, and exact norm compatibility across the saved order-9 and order-27 branches.
- `hi_char2_small_field_obstruction.py` and its result file Exact off-solution checks for the uniform four-element-field and quotient inflation obstructions, and the complete F16 search at order 43.
- `hi_char2_field_tower.py` and its result file Exact checks for the uniform first-phase degree theorem and the irreducible cubic construction with prescribed normalized relative norm.
- `hi_char2_partial_completion.py`, `hi_char2_rational_recursion.py`, and their result files Exact inflation identities and false partial solutions, plus the exhaustive rejection of stationary degree-six and single-branch degree-three rational preimage recursions.
- `hi_column_fullq.py` and `hi_column_fullq_results.txt` The complete order-13 quadratic-to-cubic implication by matching a full quadratic modular upper bound with an exact cubic lower bound.
- `hi_column_circuits.py` and `hi_column_circuits_results.txt` Uniform prime-order critical-algebra multiplication matrices and exact scalar-completion certificates through orders 11 and 13.
- `hi_column_arrangement.gp` and `hi_column_arrangement_results.txt` Exact hyperplane-arrangement counts supporting the uniform classification of all one-column solutions.
- `hi_column_counterexample.py` and `hi_column_counterexample_results.txt` Exact algebraic obstruction to extending an arbitrary one-column solution to all columns.
- `hi_explicit_ray_unit.gp` and `hi_explicit_ray_unit_results.txt` Exact checks of the uniform integral ray-unit construction and its nonzero diagonal quadratic residual.
- `hi_ray_norm.gp` and `hi_ray_norm_results.txt` Ray-field norm elements constructed without Stark or HI seeds; no additive Izumi claim.

– `hi_row11_verify.py`, `hi_row11_homogeneous_basis.txt.gz`, `hi_row11_basis.txt`, and `hi_row11_verification_results.txt` Exact homogeneous Hilbert comparison and the complete order-11 quadratic-to-cubic certificate.
– `hi_diagonal_counterexample.py`, `hi_diagonal_kummer.gp`, and their `_results.txt` files An exact physical obstruction to the diagonal-only quadratic reduction, and its failure of the one-root condition.
– `hi_row_ideal.py` and `hi_row_ideal_results.txt` Exact quadratic-to-cubic checks in the quotient by cyclic character gauge.
– `hi_amplitude_ideal.py` and `hi_amplitude_ideal_results.txt` Exact varying-amplitude elimination and algebraic branch lengths at orders 5 and 7.
– `hi_design_alltop.py` and `hi_design_alltop_results.txt` Weighted-design reformulation and checks of the uniform Alltop rigidity obstruction.
– `hi_beta_reductions.py`, `hi_quadratic_ideal.py`, `hi_composite_signs.py`, and their corresponding `_results.txt` files Reproducible beta, ideal-membership and composite sign checks.
– `hi_pell_ray.gp` and `hi_pell_ray_results.txt` Exact fixed-field Lucas-family ray arithmetic and maximal-order RM characteristics. These arithmetic tests do not assert HI existence.
– `hi_stark19_partial_results.txt` Exact field and row-root checkpoint at $n = 19$; full matrix verification was not completed in the bounded run.
– `hi_phase_reductions.py` and `hi_phase_reductions_results.txt` Reproducible cocycle, binary sign, and Fourier-reduction checks.
– `z17_haagerup_izumi_minpolys.sage` Exact reconstructed $\mathbb{Z}_{17}$ minimal-polynomial data.
– `z17_stark_test.gp` Reproducible exact ray-group, Stark-polynomial, field-isomorphism, discriminant, signature, and short-orbit-relation checks for $p = 17$.
– `z7_stark_test.gp` Exact derivation from Evans–Gannon's $\mathbb{Z}_7$ solution and the full Stark-orbit comparison.
– `z11_stark_test.gp` Reproducible checks for the two $p = 11$ conductors and branch fields.
– `z13_stark_test.gp` High-precision $p = 13$ reconstruction and exact ray-field, Stark-orbit, discriminant, and Frobenius-(2) checks.
– `z19_ray_test.gp` Exact $p = 19$ class group, ray group, subgroup fields, discriminants, and full Stark-field checks.
– `z19_stark_test.gp` High-precision two-branch reconstruction, exact Hilbert-class factorization and full-ray-field checks, plus the numerical Stark-quotient search.
– `haagerup_izumi_class_field_stark_notes.md` This handoff.